A Physicist's Companion to Gauge-Covariant Variational Free Energy

Intuition, Notation, and an Honest Status Map for the Single-Point Theory and Its Supplements

Robert C. Dennis

Independent Researcher

Date: September 2026

Contents

How to Read This Companion **1**
What this companion is . . . 1
Two questions about every result . . . 1
Reading the technical documents: the house style decoded . . . 2
A notation survival kit . . . 5
Reading paths . . . 6

The Theory at a Glance **7**
The theory in one page . . . 7
Where the chapters stand . . . 7
The headline claims . . . 9
What a physicist should take away . . . 11
How far this companion was checked . . . 12

I The Main Manuscript **13**

1 Program, Scope, and Standing Conventions **14**
1.1 The question and the answer . . . 14
1.2 The picture . . . 14
1.3 Objects and notation . . . 15
1.4 The formulas, explained . . . 16
1.5 The standing conventions, and which of them matter . . . 17
1.6 What the manuscript declines to claim . . . 18
1.7 The program's commitments . . . 18
1.8 Open ends . . . 19

2 Geometry at One Base Point **20**
2.1 The question and the answer . . . 20
2.2 The picture . . . 21
2.2.1 One fiber, many frames . . . 21
2.2.2 A link is an automorphism of the shared fiber . . . 21
2.2.3 Pure gauge, frame alignment, and Wilson loops . . . 22
2.2.4 Three kinds of agreement . . . 23
2.2.5 Compact and noncompact structure groups . . . 24
2.3 Objects and notation . . . 24
2.4 The formulas, explained . . . 26

2.5 What is proved, and how solid it is . . . 28
2.6 Open ends . . . 30

3 Probability Semantics at c_* 31
3.1 The question and the answer . . . 31
3.2 The picture . . . 31
3.2.1 Laws first, because the theory needs degenerate laws . . . 32
3.2.2 Local beliefs do not fix the population law . . . 32
3.2.3 Three levels of description . . . 33
3.2.4 Auxiliary variables are free only when exact . . . 34
3.3 Objects and notation . . . 34
3.4 The formulas, explained . . . 35
3.5 What is proved, and how solid it is . . . 37
3.6 Toy versus general . . . 39
3.7 Open ends . . . 39

4 The Fixed Normalized Generative Law 41
4.1 The question and the answer . . . 41
4.2 The picture . . . 42
4.2.1 Three graphs, not one . . . 42
4.2.2 Records close loops, and the evidence is the partition function . . . 42
4.2.3 Why the loop is not a potential between latents . . . 42
4.2.4 The typing prohibition . . . 43
4.2.5 Passive re-expression moves the whole joint . . . 44
4.3 Objects and notation . . . 44
4.4 The formulas, explained . . . 46
4.5 What is proved, and how solid it is . . . 47
4.6 Open ends . . . 49

5 One Evidence Bound and Local-Collective Free Energy 51
5.1 The question and the answer . . . 51
5.2 The picture . . . 52
5.3 Objects and notation . . . 53
5.4 The formulas, explained . . . 54
5.4.1 The bound, its energy form, and the singular branch . . . 55
5.4.2 Correlated recognition and total correlation . . . 55
5.4.3 Restriction, and what ELBO values cannot compare . . . 56
5.4.4 Observations as records . . . 56
5.4.5 Local free energy, exact block updates, and the potential identity . . . 56
5.4.6 Accounting, and links as random variables . . . 57
5.4.7 How model parameters move . . . 58
5.5 What is proved, and how solid it is . . . 58

5.6 Open ends . . . 59

6 Attention and Selective Communication 61
6.1 The question and the answer . . . 61
6.2 The picture . . . 62
6.3 Objects and notation . . . 63
6.4 The formulas, explained . . . 64
6.4.1 The softmax, the recognition row, and the temperature . . . 64
6.4.2 Receiver occupancy and the event law . . . 65
6.4.3 The reduced potential, its curvature, and row flows . . . 65
6.4.4 Communication kernels, factored maps, and loops . . . 66
6.4.5 Comparison maps and the dot product . . . 66
6.4.6 The exact receiver update . . . 67
6.4.7 The transformer correspondence and heads . . . 67
6.4.8 What survives an interface . . . 68
6.4.9 What the chapter establishes about the query map, the key map, and the matrix link . . . 69
6.5 What is proved, and how solid it is . . . 69
6.6 Toy versus general . . . 71
6.7 Open ends . . . 71

7 Fiber Information Geometry and Inference Histories 73
7.1 The question and the answer . . . 73
7.2 The picture . . . 74
7.3 Objects and notation . . . 75
7.4 The formulas, explained . . . 76
7.4.1 Fiber tensors, and why the metric is a choice . . . 76
7.4.2 Relative entropy to second order, and the cost of a holonomy . . . 76
7.4.3 Coarse observation: the Fisher-loss theorem and its limits . . . 77
7.4.4 Configurations, lifts, and why marginals do not suffice . . . 78
7.4.5 Natural-gradient histories: the orbit depends on the mobility . . . 79
7.4.6 Fisher duration, and why it is not called time . . . 80
7.4.7 Record clocks, semiconjugacy, and depth . . . 81
7.5 What is proved, and how solid it is . . . 82
7.6 Toy versus general . . . 83
7.7 Open ends . . . 84

8 Two-Timescale Dynamics 86
8.1 The question and the answer . . . 86
8.2 The picture . . . 87
8.3 Objects and notation . . . 87
8.4 The formulas, explained . . . 89

8.4.1 Descent with any mobility, and when it is local . . . 89
8.4.2 Fast and slow blocks, the envelope identity, and the reduced flow . . . 89
8.4.3 The rate-separated flow, and what is not proved about it . . . 89
8.4.4 The four-cycle: slow models need slow links . . . 90
8.4.5 Parents: the prepared parent and the refresh hypothesis . . . 91
8.5 What is proved, and how solid it is . . . 92
8.6 Toy versus general . . . 93
8.7 Open ends . . . 94

9 Coarse Channels 96
9.1 The question and the answer . . . 96
9.2 The picture . . . 96
9.3 Objects and notation . . . 98
9.4 The formulas, explained . . . 99
9.4.1 Channels, contraction, and recovery . . . 99
9.4.2 The exact coarse free energy . . . 100
9.4.3 The pointwise parent datum . . . 100
9.4.4 Three operations . . . 101
9.4.5 Symmetry and scale diagrams . . . 101
9.4.6 Labeled families and Fisher information . . . 102
9.5 What is proved, and how solid it is . . . 103
9.6 Open ends . . . 105

10 Gauge-Retained Network State and Fixed Sectors 107
10.1 The question and the answer . . . 107
10.2 The picture . . . 108
10.3 Objects and notation . . . 109
10.4 The formulas, explained . . . 110
10.4.1 Depth maps and the rooted datum . . . 110
10.4.2 The holonomy-conditioned marginal law . . . 111
10.4.3 Structural fixed sectors . . . 112
10.4.4 Cut links . . . 113
10.4.5 The frame sector . . . 114
10.4.6 The operational record . . . 114
10.5 What is proved, and how solid it is . . . 114
10.6 Toy versus general, and the noncompact target . . . 116
10.7 Open ends . . . 117

11 Exact Agent-Network Coarse-Graining 119
11.1 The question and the answer . . . 119
11.2 The picture . . . 120
11.3 Objects and notation . . . 120

11.4 The formulas, explained . . . 122
11.4.1 The effective action is a conditional partition function . . . 122
11.4.2 Perturbing the fine action: cumulants and contraction . . . 122
11.4.3 Interaction coordinates . . . 123
11.4.4 The exact step and the native growth bound . . . 123
11.4.5 Residuals, invariant sectors, and the finite-network obstruction . . . 124
11.4.6 Hyperedges and bridge kernels . . . 125
11.4.7 Meta-attention . . . 125
11.4.8 Path laws and memory . . . 126
11.4.9 The retained-state table and what "exact" means . . . 127
11.5 What is proved, and how solid it is . . . 127
11.6 Open ends . . . 129

12 The Multiscale Graph VFE and Closing the Tower **131**
12.1 The question and the answer . . . 131
12.2 The picture . . . 132
12.3 Objects and notation . . . 132
12.4 The formulas, explained . . . 134
12.4.1 One free energy and the structural channel . . . 134
12.4.2 Adaptive reversal and the cost of dropping the graph datum . . . 134
12.4.3 The hierarchical decomposition and the row free energies . . . 135
12.4.4 Random partitions and the two normalizations . . . 135
12.4.5 Coarse links . . . 136
12.4.6 How a parent enters its children's prior . . . 136
12.4.7 Closing the tower and the three loci . . . 136
12.5 What is proved, and how solid it is . . . 137
12.6 Open ends . . . 139

13 Formation Witnesses **141**
13.1 The question and the answer . . . 141
13.2 The picture . . . 142
13.3 Objects and notation . . . 144
13.4 The formulas, explained . . . 145
13.4.1 The carrier . . . 146
13.4.2 One normalized experiment and its three laws . . . 146
13.4.3 Marked parents and the discarded loss . . . 147
13.4.4 Kinetics and learning . . . 147
13.4.5 The regime . . . 148
13.4.6 The theorem, conclusion by conclusion . . . 149
13.4.7 The Gaussian core and whether the doubling is needed . . . 150
13.4.8 The four-cycle witness . . . 150
13.4.9 The gated family and depth one . . . 152

13.4.10 The authoring rules R13 to R16 in physics terms . . . 152
13.5 What is proved, and how solid it is . . . 154
13.6 Toy versus general . . . 155
13.7 Open ends . . . 156

14 Interpretation and Its Limits **158**
14.1 The question and the answer . . . 158
14.2 The picture . . . 159
14.2.1 Scaffolding, idle posits, and what survives a change of frame . . . 159
14.2.2 Participation and why it does not yet distinguish the program . . . 160
14.2.3 The four-cycle: a ring of agents choosing whom to listen to . . . 161
14.3 Objects and notation . . . 163
14.4 The formulas, explained . . . 164
14.4.1 Energy, invariant law, and learning rule . . . 164
14.4.2 Where the probability of formation comes from . . . 165
14.4.3 The row flow, its saddle, and the imprint . . . 165
14.4.4 What the corollary and the tracking bound add . . . 166
14.4.5 Persistence, and why it is equilibrium . . . 167
14.4.6 The early first-block null . . . 167
14.4.7 Conditioning, and what a vanishing discarded term means . . . 167
14.4.8 The activation clock . . . 168
14.5 What is proved, and how solid it is . . . 169
14.6 Toy versus general . . . 171
14.6.1 Mechanism and small-network specifics . . . 171
14.6.2 What the four-cycle says about waiting out the transient . . . 172
14.7 Open ends . . . 173

15 Related Work **175**
15.1 The question and the answer . . . 175
15.2 The picture . . . 175
15.3 Objects and notation . . . 177
15.4 The formulas, explained . . . 178
15.5 What is proved, and what is standard . . . 178
15.6 Standard versus new . . . 180
15.7 Open ends . . . 180

II The Supplements **182**

16 Supplement G, Part 1: The Gaussian Realization and Its Geometry **183**
16.1 Exponential families and the operator layer (G1) . . . 183
16.1.1 The question and the answer . . . 183
16.1.2 The picture . . . 184

16.1.3 Objects and notation . . . 184
16.1.4 The formulas, explained . . . 185
16.2 The Gaussian realization at the base point (G2) . . . 185
16.2.1 The question and the answer . . . 185
16.2.2 The picture . . . 186
16.2.3 Objects and notation . . . 186
16.2.4 The formulas, explained . . . 187
16.3 The interaction family (G3) . . . 187
16.3.1 The question and the answer . . . 188
16.3.2 The picture . . . 188
16.3.3 Objects and notation . . . 189
16.3.4 The formulas, explained . . . 189
16.4 Recognition restrictions (G4) . . . 191
16.4.1 The question and the answer . . . 191
16.4.2 The picture . . . 191
16.4.3 Objects and notation . . . 192
16.4.4 The formulas, explained . . . 192
16.5 Fisher geometry and gauge invariants (G5) . . . 193
16.5.1 The question and the answer . . . 193
16.5.2 The picture . . . 193
16.5.3 Objects and notation . . . 194
16.5.4 The formulas, explained . . . 194
16.6 What is proved, and how solid it is . . . 195
16.7 Toy versus general, and what this part implies for $\mathrm{GL}(K)$ networks at large N . . . 197
16.8 Open ends . . . 198
16.9 Verdict . . . 199

17 Supplement G, Part 2: Gaussian Coarse-Graining 200
17.1 Score Tangents and the Hermite Spectrum (G6) . . . 200
17.1.1 The question, the answer, and the picture . . . 200
17.1.2 Objects and formulas . . . 201
17.2 Aggregation and Kron Reduction (G7) . . . 202
17.2.1 The question, the answer, and the picture . . . 202
17.2.2 Objects and formulas . . . 203
17.3 Nonflat Links, Relaxation, and Quotient Volume (G8) . . . 205
17.3.1 The question, the answer, and the picture . . . 205
17.3.2 Objects and formulas . . . 206
17.4 The Operator Arrow (G9) . . . 207
17.4.1 The question, the answer, and the picture . . . 207
17.4.2 Objects and formulas . . . 208
17.5 Reciprocal Pairs, Anchored Stars, and Loop Holonomy (G10) . . . 209
17.5.1 The question, the answer, and the picture . . . 209

17.5.2 Objects and formulas . . . 209
17.6 The Decision Metatheory (GA) . . . 211
17.6.1 The question, the answer, and the picture . . . 211
17.6.2 Objects and formulas . . . 211
17.7 What is proved, and how solid it is . . . 212
17.8 Toy versus general, and the 512-agent target . . . 214
17.9 Open ends . . . 215
17.10Verdict . . . 216

18 Supplement B: Base Transport, Infinite Networks, and the Lattice Cluster 217
18.1 The question and the answer . . . 217
18.2 Objects and notation . . . 218
18.3 Part B-I: base transport and multi-point designs . . . 219
18.3.1 Frames, connections, and cross maps . . . 219
18.3.2 Four kinds of flatness . . . 219
18.3.3 When is a graph link a base transport? . . . 220
18.3.4 Finite designs and shared links . . . 220
18.3.5 Base-direction information geometry . . . 221
18.3.6 Section-space histories . . . 222
18.3.7 Covers, availability, and descent . . . 222
18.3.8 How Part B-I connects to the base-manifold picture . . . 223
18.4 Part B-II: infinite networks and Wilsonian limits . . . 223
18.4.1 What Wilsonian words require, and why finite depth has none . . . 223
18.4.2 Beta functions and fixed points . . . 225
18.4.3 The homogeneous Gaussian hierarchy . . . 225
18.4.4 Two-index limits, and what the absence of universality means . . . 226
18.4.5 The numerical statements . . . 226
18.5 Part B-III: the lattice cluster . . . 226
18.5.1 The model and the flow . . . 226
18.5.2 Pressure, label selection, and the specific VFE . . . 227
18.5.3 Thermodynamic Fisher geometry . . . 228
18.5.4 The typing question and the thermodynamic-limit reading . . . 228
18.5.5 What is new, and the classical antecedents . . . 229
18.6 What is proved, and how solid it is . . . 229
18.7 Conditions that matter . . . 231
18.8 Open ends . . . 232
18.9 Verdict . . . 233

III Back Matter 234

Glossary 235

Geometry and gauge at one base point . . . 235
Probability, the joint law, and the evidence bound . . . 236
Attention and selective communication . . . 237
Information geometry and dynamics . . . 237
Coarse-graining . . . 238
Formation . . . 239
The Gaussian realization . . . 240
Base transport and the lattice cluster . . . 241

Open Ends of the Theory **243**
The parent's own dynamics . . . 243
Formation that is not supplied . . . 243
Selecting structure . . . 244
Convergence and time scales . . . 244
Foundations . . . 244
Contact with the world . . . 245
The supplements . . . 245
Questions the documents do not pose . . . 245

How to Read This Companion

What this companion is

The theory lives in three technical documents: the main manuscript, cited here as MT, which develops everything at one base point c_*; Supplement G, the multivariate-Gaussian realization; and Supplement B, which holds base transport, infinite networks, and the lattice cluster, all deferred until the single-point theory is settled. Together they run to more than six hundred pages. They are written for correctness under adversarial checking, not for a first reading, and they are the only authority. This companion proves nothing. It explains what the technical documents say, in the language of a physicist, and it says plainly what kind of knowledge each main result is and whether the theory delivers what its own words promise.

When the technical documents prove something, the companion states it and cites it. A citation such as MT Corollary 5.7 refers to the main manuscript, and statements of the supplements carry their own letters (Proposition G4.2, Theorem B6.1). In the PDF these references are links that open the technical document at the cited statement, provided the technical PDFs sit in the neighboring folder where they are built. The companion's chapters 1 to 15 mirror the fifteen chapters of the main manuscript one to one; chapters 16 and 17 cover Supplement G and chapter 18 covers Supplement B. The companion has no numbered statements of its own, so every theorem, proposition, or lemma number you meet here points into a technical document.

Each companion chapter follows one template: the question the technical chapter answers and its answer in plain language; the physical picture, with the nearest physics analogue graded as exact, structural, or loose; a decoder for the chapter's notation and jargon; the key formulas explained term by term with their limiting cases; a scorecard of the main results; a box that sorts the chapter's hypotheses into those that matter and those that are hygiene; the open ends; and a verdict. Where a proof would help intuition, the companion gives a sentence or two of informal reason and labels it as such.

Two questions about every result

The first question is what kind of knowledge a result is. Nearly every theorem-type statement in the main manuscript carries the tag ESTABLISHED, meaning it is proved there. The tag says nothing about how general the result is, how much it depends on a special instance, or whether it rests on an assumption that might fail in the cases the program cares about. The companion adds that information with one badge per result.

Table 1: Status badges used in this companion.

Badge	Meaning
SOLID	Proved in the technical documents, or a checked textbook result, at the generality the theory relies on. Established negative results (refutations and no-go statements) are solid. When the mathematics is textbook and the novelty is only the typing, the companion says "standard in substance".
TOY	Proved, but only on a named small or special instance: the four-cycle, a pair, three binary agents, a scalar or abelian case, a particular constant schedule. The mechanism is not established beyond that instance.
COND	Proved under a hypothesis that could fail in the models the program cares about and is not shown to hold there. The companion names the hypothesis.
NUM	Supported only by a named computation.
OPEN	An open problem or a conjecture.
VAGUE	A reading, an interpretation, or a programmatic statement with no precise mathematical statement behind it, or a construction whose meaning is not pinned down.
CHOICE	A definition or standing hypothesis that is a substantive modeling commitment, the place where physics enters, as opposed to notation.

The second question is whether the theory does what it says. Each companion chapter ends with a verdict on the technical chapter's own claims, taken from its opening paragraph and from the reading map of the introduction, measured against what its statements actually prove.

Table 2: Delivery verdicts.

Verdict	Meaning
DELIVERED	The words match the mathematics.
NARROWER THAN IT SOUNDS	The claim is proved, but in a setting narrower than the headline wording suggests: a supplied hierarchy, a single small network, a particular family.
PARTLY DELIVERED	Part of the claim is proved and part is open.
NOT DELIVERED	The claim is posed but not proved; the technical document usually says so itself.

A result that is correct and standard is not a defect, and the companion says "standard in substance" as information, not as criticism: a physicist gains from knowing that an identity is the Gibbs variational principle in new notation, because then the physicist already knows what it can and cannot do.

Reading the technical documents: the house style decoded

The technical documents are written under an authoring specification whose rules explain most of what makes them hard to read. Knowing the rules lets you skip what is bookkeeping and slow down where a qualifier carries content.

Status tags and scope paragraphs. Every lemma, proposition, theorem, corollary, and claim-bearing remark carries exactly one tag: ESTABLISHED (proved in the document), STANDARD (a

known result, cited with a locator), or NUMERICAL (supported only by a named computation). Definitions, standing hypotheses, conjectures, and open problems carry their status in the heading. After its opening, a chapter may add one scope paragraph, tagged NOT-CLAIMED, listing what it declines to claim; in practice every chapter has one. Declining is not refuting: a declined statement may be true, and a refuted statement appears instead as an established negative result. The scope paragraphs are the fastest way to find out what a chapter does not do, and the companion's verdicts lean on them.

Defaults, and why the prose is full of qualifiers. Chapter 1 of the main manuscript fixes defaults once for the whole document (MT Section 1.4): measurable spaces are standard Borel, laws and kernels are normalized and measurable, a coarse channel is a Markov kernel that does not read the recognition law and leaves observations alone, admitted observations have finite positive evidence, structural data are parameters unless a law for them is declared, and the interaction graph is the support of the source prior, complete unless masked. A statement mentions only departures from these defaults, so a hypothesis that appears inside a statement is a departure and worth reading. Three other sources of qualifiers are worth knowing. The documents work with arbitrary laws on arbitrary spaces rather than Gaussians, which forces extended-real values (a relative entropy may be $+\infty$) and almost-sure qualifiers. They keep apart objects that coincide in the Gaussian or flat cases but not in general: three description levels (sample, law point, configuration), three transports (frame comparison, graph link, base parallel transport), three kinds of agreement between agents, three coarse operations (Markov pushforward, energy precomposition, Galerkin restriction), and five attention objects (the source prior, the generative and the recognition attention rows, the receiver occupancy, and the event law). Keeping them apart is deliberate: each pair can differ outside the Gaussian or flat case, and a theorem about one does not transfer to the other. And the scope rules described next constrain what may be called formation, selection, or recursion.

The Curie rule (R13): what may be called selection by learning. Curie's principle says that symmetric causes cannot produce less symmetric effects when the effect is uniquely determined by the cause. The rule applies it to learning. Let H be the symmetries shared by the generative model with its records, the parameterization, and the kinetics, including the kinetic metric and the proposal mechanism (H may be a group of graph automorphisms or a residual gauge group). A claim that learning creates or selects a structure that H does not fix must name the H-breaking datum it amplifies (the initial law, an initial parameter, the model, or the records) and must compare with the frozen-learning arm on the same data. A claim that learning selects a structure that H does fix must name the symmetry it restores, with the frozen arm at the non-invariant initial value as its null. A claim of selection with no breaking datum must name the hypothesis its mechanism violates and declare the mechanism as part of the model: a rule that reads realized paths, records, or a finite ensemble; selection among non-unique optimizers, critical points, or branches; non-uniqueness at masked entries, which amounts to edge creation; or a non-equivariant metric, mobility, or proposal mechanism. This is substantive, not bookkeeping: an equivariant law-level learning rule whose coupled law and parameter dynamics is unique cannot break a symmetry that its initial law and initial parameter share. In physics terms (the companion's reading, not part of the rule),

spontaneous symmetry breaking needs non-uniqueness, which in statistical mechanics arrives with the thermodynamic limit. The four-cycle results of chapter 14 are written under this rule, which is why they speak of amplifying an initial asymmetry rather than of spontaneous selection.

Recursion versus a supplied-scale sequence (R14). A two-stage claim declares a handoff time, the first- and second-stage events with their blocks, one statistic of the handoff state, and a class of handoffs fixed in advance that also contains, with probability bounded below, handoffs of histories in which the first stage did not form. If the second-stage event occurs with probability tending to one uniformly over the handoff class, the sequence is a supplied-scale sequence, not recursion, whatever differences appear earlier. Otherwise the second coalescence counts as recursive only if, after first-stage formation, the second-stage event has probability bounded below, its block is a union of at least two first-stage blocks, and the declared statistic exceeds its value after a counterfactual handoff by a margin bounded below. A margin in the other direction, where the first stage delays or suppresses the second, is recorded as first-stage dependence, not recursion. In physics terms (again the companion's reading), a supplied-scale sequence is a hierarchy built into the interaction strengths through which the dynamics relaxes in order, and the rule separates a genuine cascade from the successive freezing of modes whose stiffnesses were prescribed. It matters most in chapter 13.

Coherence episodes (R15): what counts as formation. A block counts as having formed only if two clauses hold. The restoring clause names the interaction that holds the block coherent against thermal noise at the stated resolution, with restoring rate times squared resolution divided by temperature tending to infinity, and the rate is evaluated on the actual law of the assignment sheets or on the landscape marginalized over sources, not on a sheet held fixed by hand. The horizon clause requires the coherence to be a dynamical event rather than a property of equilibrium: either the observation horizon is short compared with the stationary recurrence time of the event, or the event is more probable than it is in the stationary state by a margin. Coherence that is restored but merely reflects equilibrium (a strongly bound pair in thermal equilibrium is always coherent) is called restored equilibrium coherence; coherence with no restoring interaction is coincidental. In physics terms, the rule separates ordering produced by the dynamics from equilibrium order and from fluctuations.

Settling conditions (R16). Every open problem and conjecture states a class of models and two outcomes, a proof or a counterexample within that class, and neither outcome may already be met by an existing statement or by a trivial construction. This is why the open problems read like contracts. It keeps open items falsifiable, at the price of posing them in narrower classes than the questions that motivate them.

No Wilsonian vocabulary in the main manuscript. The main manuscript constructs exact finite-step coarse-graining with no rescaling, so it does not use the words relevant, irrelevant, fixed point, or beta function except in one pointer to Supplement B and when describing other authors' work. The asymptotic apparatus (rescalings, spectra, fixed points, universality) lives in

Supplement B. When the companion uses renormalization-group language, it says which of the two it means.

Labels and names. Labels are frozen when a statement changes type, so a label beginning `thm:` may belong to a corollary and `conj:phil-contrast-formation` is a theorem. Trust the printed heading. Words such as agent, meta-agent, parent, and block are naming decisions: the documents make no claim of agency, and population laws are not agents.

A notation survival kit

The notation table of the main manuscript (MT Appendix D) is complete but long. The following symbols carry most of the load; the glossary at the end of the companion (Glossary) covers the jargon.

Term or symbol	Plain meaning	Physics analogue
c_*	The single base point (context) at which everything happens.	A zero-dimensional base; one spacetime point.
G, $\mathscr{P}_G$	Structure group and principal bundle; at c_* the fiber is a G-torsor (a copy of G with no preferred identity).	Gauge group; the space of frames at a point.
ρ_b, ρ_m	Representations of G on the belief and model sample spaces; they may differ.	Matter representations.
q_i^b, q_i^m	Agent i's belief law and model law (probability laws on the fibers).	A mixed state of a matter field at site i.
u_i	Agent i's frame, one for both channels at c_* (a gauge choice).	A local gauge fixing at site i.
$\mathfrak{I}$, e, $\bar{e}$	Interaction multigraph, an oriented edge copy, its reversal.	The lattice, with multiple links allowed.
Θ_e^x	Independent group-valued link on edge copy e for channel $x \in \{b, m\}$, with $\Theta_{\bar{e}} = \Theta_e^{-1}$.	Link variable U_ℓ of lattice gauge theory.
$\mathrm{Hol}^x(p)$	Holonomy: product of links around a closed walk p.	Wilson loop before the trace.
K_a, $\mathfrak{A}$	A normalized record kernel (an interaction or observation factor) and the set of them.	A factor of the Boltzmann weight; a local term of the action.
$\mathbb{P}_{\theta,V}$	The one normalized joint generative law of all agents and records.	The Gibbs measure e^{-S}/Z.
$\mathbb{Q}_{V,o,X}$	The one recognition law of the whole population (a coupling of the agents' local laws).	The variational trial state.
$\mathbf{\Pi}_{\theta,V,o,X}$	The exact posterior given observation o and structure X.	The exact Gibbs state conditioned on the data.

Term or symbol	Plain meaning	Physics analogue
$\mathcal{L}_V$, $\mathcal{F}_V = -\mathcal{L}_V$	Evidence lower bound and variational free energy of the population.	Minus the variational free energy; the Gibbs–Bogoliubov functional.
π_{ij}, β_{ij}, τ_i	Source prior, attention weight of receiver i on source j, attention temperature.	Prior weight, Boltzmann weight over sources, temperature.
S_e^Q, S_e^K, L_e	Query and key comparison maps and the composite communication map of an edge.	None standard; linear maps between internal spaces.
C_A, ℓ	Coarse channel that makes parent A from its child block; coarse-graining depth.	Blocking kernel; step index of a renormalization without rescaling.
T, ε	Temperature and small parameters of the formation families (chapters 13 and 14).	Temperature; small expansion parameters.

Reading paths

For a first pass, read The Theory at a Glance and the verdict box at the end of each chapter. For the physical core of the theory, read chapters 2, 5, 6, 9, 11, 13 and 14 in that order. For what the theory says about Gaussian networks with $\mathrm{GL}(K)$ links at large size, read chapters 2, 10, 16 and 17. For the state of the renormalization program at one base point, read chapters 9, 11, 12 and 18. The open ends of the whole theory are collected in Open Ends of the Theory.

The Theory at a Glance

The theory in one page

At one base point c_*, a population of agents is a lattice gauge theory on a graph whose site variables are probability laws. Each agent carries a belief law and a model law on spaces where a structure group G acts, and a frame saying in which gauge those laws are written; each edge carries an independent group-valued link, so products of links around loops carry holonomy. Agents interact through normalized record kernels. From these local data the theory builds one normalized joint law, one recognition law for the whole population, and one evidence bound, which is the Gibbs variational principle applied to that joint. Attention is a posterior over which neighbor a receiver listens to, and selective communication adds query, key, and value maps between the agents' internal spaces. Fisher geometry turns free-energy descent into oriented histories with an information clock, once a metric is chosen, and a fast-slow split separates beliefs from models.

Coarse-graining is a Markov kernel from a block of agents to a parent. Applied to the joint, the posterior, and the recognition law at once, it preserves the evidence and splits the free energy exactly into the parent's free energy plus the information discarded inside the block. The parent's law therefore always exists, with an exact effective action. What coarse-graining does not give is a closed flow on a finite set of couplings, a thermodynamic limit, or a proof that the parent evolves as an autonomous process. Formation of meta-agents is proved in two settings. In a family with a hierarchy of interaction strengths supplied by hand, exact parents track a prepared state as it relaxes through the hierarchy. On a four-agent ring, a pair forms by feedback between attention and belief relaxation, and slow learning selects one of the two pairings by amplifying an asymmetry of the initial law. Formation that the agents' coupled inference produces on its own, beyond one level and beyond four agents, is open.

Where the chapters stand

Figure 1 shows the chapters in dependency order, colored by verdict. No chapter is "not delivered": every chapter proves what its statements say, and the open problems are posed as open. Where a verdict is "narrower than it sounds", the gap is between an opening paragraph or a title and the theorems, and the companion chapter names the sentence. The theory's unproved ambitions sit in its open problems, which Open Ends of the Theory collects; the two "not delivered" rows of the table below are two of those ambitions, which the technical documents themselves pose as open.

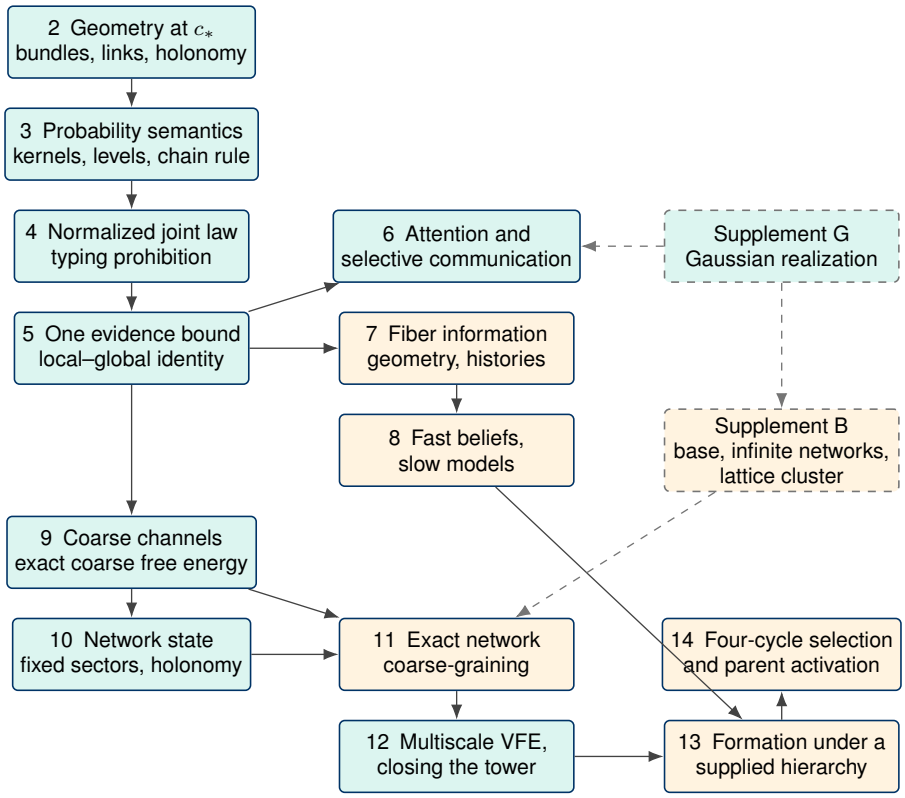


Figure 1: How the chapters depend on one another. Fill color gives the companion's verdict on each technical chapter's own claims: green, delivered as worded; orange, delivered but narrower than the wording. Dashed boxes are the supplements, on which no main-manuscript result depends. Chapter 1 is the program statement, whose verdict is "partly delivered" (chapter 1), and chapter 15 is related work.

The headline claims

Claim (where)	Verdict	In one sentence
One evidence bound for the whole population, tight exactly at the posterior (chapter 5)	DELIVERED	The Gibbs variational principle for a correctly normalized joint, written in the extended reals; standard in substance, new in its typing.
Local objectives are exact conditional pieces of the collective one (chapter 5)	DELIVERED	The chain rule for relative entropy with the rest of the population held fixed; the "exact potential game" needs singleton blocks with product profiles.
Everything is gauge covariant at one base point (chapters 2 and 4)	DELIVERED	Evidence and bound are invariant under passive frame changes when kernels are covariant; links carry per-edge twist classes and loop holonomy.
Attention is inference (chapter 6)	DELIVERED	The softmax is Bayes' rule for a declared latent pointer; the transformer's free value and output maps are replaced by a tied value map in the exact update.
Information geometry and inference histories (chapter 7)	NARROWER THAN IT SOUNDS	Fisher contraction and the information clock are proved; they cover a meta-agent only when its law is a parameter-independent Markov image, and every orbit rests on a chosen metric.
Collective free energy does not factor through marginals (chapter 7)	DELIVERED	An exact refutation by a three-agent parity witness.
Fast beliefs, slow models (chapter 8)	NARROWER THAN IT SOUNDS	The reduced slow flow is the kinematics of adiabatic elimination; that the actual flow tracks it is proved only on the four-cycle's held sheets.
Coarse channels preserve evidence and split the free energy exactly; a parent law always exists (chapter 9)	DELIVERED	The relative-entropy chain rule for one channel pushing three laws; exact, static, and with no closure or selected channel.
Holonomy decides what a coarse description can retain (chapter 10)	DELIVERED	Fixed sectors and a holonomy-conditioned parent score; for generic $\mathrm{GL}(K)$ links the Gaussian score is infinite on any block with a loop.

Claim (where)	Verdict	In one sentence
Exact coarse-graining of the agent network (chapter 11)	NARROWER THAN IT SOUNDS	The effective action exists for every channel, but closure keeps the complete laws, the coupling calculus excludes Gaussian $\mathrm{GL}(K)$ couplings, everything is finite-network, and there is no thermodynamic limit.
One free energy for the multiscale tower; parents influence children (chapter 12)	DELIVERED	As accounting and typing; parent influence is proved as a typed possibility, not as dynamics.
Successive meta-agents form under free-energy descent with learning (chapter 13)	NARROWER THAN IT SOUNDS	Exact parents track linear relaxation through a hierarchy supplied by hand; attention and learning play no role in the coherence conclusion.
A pair forms endogenously on a four-cycle (chapter 13)	DELIVERED TOY	One pairing episode on four agents, seeded by the random initial assignment, with no learning.
Slow learning selects a pairing (chapter 14)	DELIVERED TOY	A law-level theorem on the four-cycle that amplifies an asymmetry of the initial law; not spontaneous symmetry breaking.
Parents activated after formation behave well (chapter 14)	DELIVERED TOY	On the four-cycle, activation at a deterministic time after the fast modes thermalize works; activation at the first completion is refuted.
A parent evolves as an autonomous process (chapter 11)	NOT DELIVERED	Posed as the defining open problem of the phase, MT Open Problem 11.38.
Collectives form through the agents' coupled inference, recursively (chapters 1 and 13)	NOT DELIVERED	The program aim; posed as MT Open Problem 13.25 and MT Open Problem 12.27.
Gaussian realization: aggregation, Kron reduction, holonomy (chapters 16 and 17)	DELIVERED	Aggregation is the exact algebraic-multigrid coarse operator; Kron closure is classified; the interaction family is declared, not derived, and complex Gaussians are not treated.
Asymptotic renormalization and the lattice cluster (chapter 18)	NARROWER THAN IT SOUNDS	Finite depth gives no asymptotics; the $\mathbb{Z}_2$ lattice cluster is a genuine constructive result whose sites are not yet typed as agents.

What a physicist should take away

The foundations are right, and mostly standard in substance. The evidence bound, the local objectives, gauge covariance, and exact coarse-graining are proved at full generality, and most of them are known identities (the Gibbs variational principle, the chain rule for relative entropy, change of variables, pushforward densities) written in a typed language. The typing is where the value lies: it separates objects that coincide only in Gaussian or flat cases, and it prevents collective objects from being sums that merely look global. The one foundational gap with consequences is that belief-alignment divergences, the attention energies the program relies on, are law-point energies without a normalized prior over law points (MT Open Problem 3.23), so they are not yet terms of one evidence bound at that level (chapter 3).

The renormalization at one base point is exact but not closed. Every coarse channel produces an exact parent law and an exact effective action. There is no closed truncation, no flow on a finite set of couplings (a finite-network obstruction rules one out across reducing steps), no thermodynamic limit, and no autonomous parent dynamics. None of the main manuscript's open problems takes the limit of many agents, so the thermodynamic-limit view of renormalization at c_* is not yet posed there (chapter 11, Open Ends of the Theory).

Formation is proved only where it is supplied or small. The finite-depth theorem shows that exact parents survive a relaxation through a supplied hierarchy of stiffnesses; it is a statement about tracking, not about emergence. The endogenous results live on four agents, where selection by learning amplifies an initial asymmetry, as Curie's principle requires of a unique law-level dynamics. By the small-network rule these results ground the mechanisms; they neither establish nor exclude formation at scale, and they say nothing for or against phase transitions, which need the many-agent limit (chapters 13 and 14).

Noncompact groups meet specific friction. For the program's target of hundreds of agents with $\mathrm{GL}(K,\mathbb{R})$, $\mathrm{GL}(K,\mathbb{C})$, or $\mathrm{SU}(N)$ links, the documents record several boundaries. A normalized law on noncompact links is not constructed (chapter 2). The holonomy-conditioned Gaussian parent score is infinite for generic $\mathrm{GL}(K)$ loops (chapter 10). The coupling calculus of exact network coarse-graining is proved only in a bounded tier that excludes Gaussian $\mathrm{GL}(K)$ couplings (chapter 11). Gauge absorption of Gaussian gains is proved on forests and pairs, with only sign conditions on cycles, and complex Gaussians are not treated (chapter 16). No symmetric rule that ignores a reference point averages frames in $\mathrm{GL}^+(K)$, so a meta-agent's frame must be chosen with a reference (chapter 10). $\mathrm{SU}(N)$, being compact, avoids the first two.

What is new. Beyond the typing, the new mathematics is concentrated in a few places: the per-edge twist invariants that a shared fiber gives links at one point (chapter 2), the holonomy-conditioned fixed sectors and parent score (chapter 10), the parity no-go for factoring collective free energy (chapter 7), the Kron closure classification (chapter 17), the formation machinery that carries

exact parents through a relaxation (chapter 13), and the four-cycle selection and activation-clock results (chapter 14).

How far this companion was checked

The companion's chapters were drafted by Claude Opus 5.5 against the technical statements. A second model, Claude Fable 5.1, checked the reading guide, chapters 1, 5, 6, 9, 10, and 18, and the open ends against the technical documents, and its findings were applied: none of them changed a verdict, and several corrected paraphrases that were stronger than their sources. The remaining chapters (2 to 4, 7, 8, 11 to 17), this overview, and the glossary were not checked by a second model. Their drafters verified every cited label and recomputed many numerical witnesses, and the central claim of chapter 13, that attention and learning play no role in the coherence conclusion of the formation theorem, was checked against its source. Read the unchecked chapters with that in mind, and trust the technical statement over the companion wherever they differ.

Part I

The Main Manuscript

Chapter 1

Program, Scope, and Standing Conventions

1.1 The question and the answer

The main manuscript asks two questions. What collective inference is available to a population of agents, each a local section of a belief bundle and a model bundle of one principal bundle, all placed at one base point c_* and interacting through independent links under one normalized joint law? And which parts of that inference survive when agents are coarse-grained exactly into meta-agents? Its answer, announced in the opening of MT Chapter 1, has four parts. The whole population has one evidence bound, whose gap vanishes exactly when the recognition law equals the posterior (MT Corollary 5.7, MT Theorem 5.8). Each block of agents has a local objective, a conditional free energy, such that when the block's conditional law is replaced with the rest of the population held fixed, the change of the collective free energy equals the outside-averaged change of the block's local objective (MT Theorem 5.23); holding the outside marginal fixed is load-bearing. Coarse channels preserve the evidence and split the free energy exactly (MT Theorem 9.6), with an exact effective action for what they retain (MT Theorem 11.3). And a finite-depth family of models forms successive meta-agents under free-energy descent with learning, in a stated scope (MT Theorem 13.17). The program-aim section announces two further results on a homogeneous four-cycle: seeded formation from every Gaussian initial law with source learning idle (MT Proposition 14.4), and law-level selection of one pairing by slow source learning for almost every Gaussian initial mean (MT Theorem 14.5, a theorem despite its label). The chapter also names what is not done: how a parent's own dynamics are determined by the network it summarizes is MT Open Problem 11.38, which it calls the defining open problem of the present phase.

1.2 The picture

Think of a lattice gauge theory on a graph whose sites all sit at one spacetime point. Each site carries an agent. What the agent carries is not a field value but two probability laws, a belief q_i^b and a model q_i^m, on spaces on which the structure group G acts through representations ρ_b and ρ_m; the agent also carries a frame u_i, a point of the one G-torsor over c_* that says in which gauge the agent's laws are written. Each edge of the interaction graph carries a group element Θ_e for each channel, an independent variable rather than something computed from the frames, so products

of links around loops (holonomies) can be nontrivial. Agents interact through normalized record kernels K_a, which play the role of local factors of a Boltzmann weight.

From these local data the manuscript builds three global objects, in this order: one normalized joint generative law $\mathbb{P}_{\theta,V}$ of everything, one recognition law $\mathbb{Q}_{V,o,X}$ of the whole population, and one evidence bound $\mathcal{L}_V$. The order matters to the author, and it is the first commitment of the program: every collective object is a complete probability object, constructed from the agents' mechanisms and records or selected explicitly, never an extra agent added to the population and never a sum of local objectives that happens to look like a global one.

Physics analogue (exact for the variational principle, structural for the gauge part)

The evidence identity below is the Gibbs variational principle of statistical mechanics: for any trial state q, $\log Z = -F[q] + D_{\mathrm{KL}}(q\|p)$ with $F[q] = \mathbb{E}_q[\text{energy}] - \text{entropy}(q)$. Here the "energy" is $-\log$ of the joint generative density and Z is the evidence. The gauge part is lattice gauge theory over a zero-dimensional base: frames are local gauge choices, links are link variables, holonomies are untraced Wilson loops. Two differences matter. The site variables are probability laws on representation spaces, not field values, and the links need not carry a kinetic (plaquette) term; whether they are fixed parameters or dynamical variables is declared model by model.

1.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Agent $\mathcal{A}^i = (\mathcal{C}_i; q_i^b, q_i^m, u_i)$	A belief section, a model section, and one common frame (MT Definition 2.8); at one base point, two laws and a frame.	A site with its matter content and gauge choice.
Common frame u_i	One frame for both channels; a gauge choice, not an assumption (MT Proposition 2.7).	Gauge fixing at a site.
Cross morphism Φ	A map from belief fibers to model fibers (or back); extra structure, not implied by the bundles (MT Definition 2.9).	A coupling between two matter species.
Record kernel K_a	A normalized conditional law on the scope of an interaction or observation.	A local factor of the Boltzmann weight.
Evaluator	The map that turns a model presentation m_i into a generative kernel; a model is not itself a law.	The map from coupling constants to a Hamiltonian.
Constructed joint $\mathbb{P}_{\theta,V}$	The normalized product of all kernels (MT Proposition 4.4).	The Gibbs measure.
Selected coupling $\mathbb{Q}_{V,o,X}$	One joint recognition law whose marginals are the agents' local laws, chosen by a declared measurable rule (MT Definition 3.15).	A correlated trial state; not a product state unless chosen so.

Term or symbol	Plain meaning	Physics analogue
Posterior slice $\mathbf{\Pi}_{\theta,V,o,X}$	The exact posterior given the observation o and the structural data X.	The exact conditioned Gibbs state.
Structural data X	Graph, masks, and other structure, held as parameters unless a law for them is declared.	Quenched background.

A notation trap: the manuscript's T_{ij} is a frame comparison, not a temperature or a transfer matrix, and the formation chapters write their temperature as T. The letter π is both the bundle projection and, with indices, the source prior of attention.

1.4 The formulas, explained

The architecture display MT (1.1) is a flow chart:

$$\{\mathcal{A}^i, G_{\theta,i}, \mathbb{Q}_{i,o,X}, K_a\}_{i\in V, a\in\mathfrak{A}} \longrightarrow \big(\mathbb{P}_{\theta,V} \text{ by construction},\ \mathbb{Q}_{V,o,X} \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}) \text{ by selection}\big) \longrightarrow \mathcal{L}_V.$$

On the left are the local data: agents, their generative mechanisms, their local recognition laws, and the record kernels. The joint law is built by multiplying normalized kernels; the population's recognition law is chosen among the couplings Cpl of the local laws, the joint laws with those marginals. The selection step is not trivial. Once two agents have non-degenerate laws, the marginals do not determine the joint, and the correlations the selection puts in are exactly what a mean-field (product) choice leaves out.

The one identity MT (1.2),

$$\log p_{\theta,V}(o \mid X) = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + D_{\mathrm{KL}}\big(\mathbb{Q}_{V,o,X} \big\| \mathbf{\Pi}_{\theta,V,o,X}\big),$$

reads as follows. The left side is the log evidence, the log partition function of the system conditioned on the observed data. The first term on the right is the evidence lower bound, minus the variational free energy of the population's recognition law. The second term is the relative entropy from that recognition law to the exact posterior: nonnegative, zero exactly when the two agree as measures. Two consequences are spelled out in the chapter. Getting every single-agent marginal right does not close the gap when the posterior is correlated (MT Theorem 5.8); the missing piece is the correlation structure. And the identity does not change when agents re-express their laws in other frames, provided the kernels transform covariantly (MT Proposition 4.9, MT Corollary 4.10): the evidence, like a partition function, is gauge invariant. A subtler point concerns restricted families: if the posterior is excluded from the recognition family, every individual gap is positive, but the best gap over the family may still be zero, approached without being attained (MT Section 5.4).

1.5 The standing conventions, and which of them matter

MT Section 1.4 fixes defaults for the whole manuscript, so that later statements mention only departures. The reading guide lists them. Three conventions are physics rather than hygiene.

The first is that the interaction graph at c_* is the support of the source prior of attention, a structural parameter that is the complete graph unless masked. Gibbs attention weights are then strictly positive on every allowed edge whose comparison cost is finite (the extended-cost rows of MT Proposition 12.9 may put zero weight on an infinite-cost edge), and all dynamic structure is carried by attention weights and link values. Creating or deleting edges is not part of the model; it is MT Open Problem 6.9. Loosely, this fixes the set of possible couplings the way a canonical ensemble fixes the particle number: only coupling strengths and link values move, never the graph. Induced coarse multigraphs are derived objects.

The second is that three transports must stay distinct. A frame comparison T_{ij} relates two frames in the same fiber: because all agents sit over the same point, their frames are points of one torsor and differ by a unique group element, with no path involved (MT Proposition 2.5). A link Θ_e is an independent variable on an edge (MT Definition 2.11); setting every link equal to the frame comparison gives frame-aligned, holonomy-free links, and general links carry holonomy that no frame change removes. Parallel transport along a curve in the base is not used at all. Graph links are therefore not base transport, and graph holonomy is neither the holonomy of a base connection nor a topological property of the bundle.

The third is that "coarse" has distinct meanings that the manuscript keeps apart: a Markov channel, which discards sample information while preserving normalization; a hard restriction, which substitutes a lower-dimensional configuration into an energy and defines a new model; and parameter aggregation inside a closed family. MT Section 9.4, where the three are defined, calls them Markov pushforward, energy precomposition, and Galerkin restriction (the compression of an operator, which is how aggregation acts on a Gaussian precision). They can agree in special realizations (Gaussian ones are worked out in Supplement G), and a theorem about one does not transfer to another.

Conditions: which ones matter

Standard Borel spaces, measurability of laws and kernels, and the finite positive evidence of admitted observations are hygiene; every model the program uses satisfies them. Absolute continuity of the recognition law with respect to the posterior is deliberately not a default: it is clause (H3) of MT Hypothesis 5.3, and when it fails the bound is $-\infty$. This matters in practice, because a recognition law supported on a lower-dimensional set (for example, one that identifies agents exactly) is singular and costs an infinite free energy. The decision that structural data are parameters unless a law is declared matters too: every statement about random graphs, random partitions, or random memberships needs that law to be named.

1.6 What the manuscript declines to claim

The scope paragraph of MT Section 1.4 declines several claims for the whole manuscript. No index of the theory is identified with physical time: not the inference step, the flow parameter, the coarse-graining depth, or the Fisher duration. The chapter states what such an identification would need: a named target clock, a calibration map with an uncertainty model, causal and compositional consistency, and evidence against alternative parameterizations. The base is not spacetime and carries no Lorentzian structure. Agents, laws, and meta-agents are mathematical objects, and calling a node an agent is a naming decision. No canonical rule for choosing a recognition coupling, a partition, or a membership is asserted. No renormalization semigroup with rescaling, fixed points, or beta functions is constructed in the main manuscript; that apparatus is in Supplement B (chapter 18). Identifying any limiting object with a physical law is posed as MT Open Problem 14.7. Each of these is declined, not refuted.

1.7 The program's commitments

Result	What it says	Status	Scope and caveats
MT Section 1.1	Every collective object is a complete probability object built from the agents' mechanisms and records or selected explicitly.	CHOICE	Methodological; it excludes mean-field sums that only look global.
MT Section 1.1	Every statement about a parent holds for the normalized parent law that coarse-graining produces.	CHOICE	Rules out "effective" parents defined by hand.
MT Section 1.1	Formation of a parent is a theorem about a declared family, with its scope stated.	CHOICE	The formation results are held to this; see chapters 13 and 14.
MT Section 1.4	One base point; no base maps, neighborhoods, or connection descent.	CHOICE	A phase decision; base transport is deferred to Supplement B.
MT Section 1.4	Interaction graph is the support of the source prior, complete unless masked; no edge creation.	CHOICE	Edge creation is MT Open Problem 6.9.
MT Section 1.4	Frame comparison, graph link, and base transport are distinct.	CHOICE	Links are independent variables; holonomy is network-loop holonomy.

1.8 Open ends

The chapter names one open problem as the defining one of the phase: how a parent's dynamics follow from the network it summarizes (MT Open Problem 11.38). The coarse-graining of chapters 9 and 11 produces the parent's law at each instant exactly; what is missing is a theorem that this law evolves as an autonomous process of its own. The other open ends of the program are collected in Open Ends of the Theory.

Does it deliver what it claims? PARTLY DELIVERED

The chapter claims four results and names one open problem. Three of the four are proved as stated: the one evidence bound, the local-global potential identity, and exact coarse-graining with an effective action. "Exact" there means an identity for one channel, not a closed renormalization flow; the chapter does not claim a flow, and chapter 11 explains what is and is not closed. The fourth, formation, is proved in a narrower sense than the program aim suggests: the finite-depth theorem relaxes a prepared state through a hierarchy of interaction strengths supplied by hand, and the endogenous witness, seeded formation, and the pairing-selection theorem all live on the four-cycle (chapters 13 and 14); the selection theorem is the strongest learning result, a law-level statement for almost every initial mean, and it amplifies an asymmetry of the initial law rather than creating one. The program aim that parents act back on their children through normalized laws is set up (chapter 12) but not closed, because the parent's own dynamics is the open problem the chapter names. The chapter is candid about this; its opening states the scope of the formation theorem and names the open problem. The program-aim paragraph, which speaks of collectives forming through the agents' coupled inference, describes the goal rather than the delivered result.

Chapter 2

Geometry at One Base Point

2.1 The question and the answer

The technical chapter asks what geometric data an agent carries at the single base point c_* at which the whole main manuscript works, and which of those data are invariant, meaning unchanged by any change in how the data are written down. It gives three answers. The group of genuine symmetries at c_*, the active gauge group, is one copy of the structure group G sitting inside the much larger group $\prod_{i \in V} G$ of independent per-agent frame changes, which the chapter calls passive (MT Proposition 2.6). Every independent link between two agents carries a conjugacy class that neither passive nor active changes can alter, and the holonomy of a closed loop of agents is the telescoped product of the per-edge twists whose classes these are (MT Proposition 2.12); the classes alone do not determine the holonomy. Three notions of agreement on a block of agents, frame agreement, agreement of an equivariant statistic such as the mean, and law coherence, are logically distinct, and the chapter proves exactly which implications hold among them (MT Proposition 2.18). These objects are used downstream: passive re-expression is the transformation under which the generative law must be covariant (chapter 4), link-transported comparisons are what attention compares (chapter 6), and holonomy together with the trivialization criterion becomes the fixed sectors of coarse channels (chapter 10).

Is the principal- and associated-bundle machinery doing real work at one base point? Only as typing. At c_* the principal bundle enters only through its fiber $(\mathscr{P}_G)_{c_*}$, which is a G-torsor, that is, a copy of G with no preferred identity element; the belief and model bundles enter only through their fibers, which are G-spaces of probability laws; and a section enters only through its value at c_*, which is a law. Nothing in the chapter uses a connection, a base curve, the topology of $\mathscr{P}_G$, or the manifold structure of the context space $\mathcal{C}$, and the manuscript says so itself: graph flatness is a property of the links alone, whatever the topology of $\mathscr{P}_G$ (MT Remark 2.15), and graph-link holonomy is not the holonomy of any base connection (MT Section 1.4). What the bundle language contributes is a clean separation between coordinates (frames, link coordinates, sample coordinates) and frame-independent objects (link maps, points of the associated fibers), and both unfamiliar statements of the chapter come out of that separation. The link sector itself is lattice gauge theory on the interaction multigraph, with the lattice vertex law, Wilson loops, and the lattice criterion for pure gauge; what is added is a group-valued frame at every agent, all frames lying in one shared fiber. Transport along the base, which would make the bundle language indispensable, is deferred to Supplement B.

2.2 The picture

2.2.1 One fiber, many frames

All agents sit at the same base point, so their frames u_i are points of one and the same fiber. Pick any reference point σ of that fiber and each frame becomes a group element U_i through $u_i = \sigma \cdot U_i^{-1}$; any two frames are then compared by $T_{ij} = U_i U_j^{-1}$, which does not depend on σ (MT Proposition 2.5). A useful picture is a set of observers in one laboratory, each with private coordinate axes. An observer who changes axes, a passive re-expression g_i, changes every description: sample coordinates are multiplied by $R_i^x = \rho_x(g_i)$, laws are pushed forward, the comparisons become $g_i T_{ij} g_j^{-1}$, and nothing in the laboratory has moved. Rotating the laboratory itself, an active transformation, that is, an automorphism of the fiber that commutes with the right action of G, is a single operation, which observer i records as φ_i and observer j as the conjugate $\varphi_j = T_{ij}^{-1} \varphi_i T_{ij}$. The active transformations therefore form a copy of G, of dimension $\dim G$ rather than $|V| \dim G$, and a tuple that changes the frame of one agent alone is not active when G is nontrivial and there are at least two agents (MT Proposition 2.6).

Physics analogue (structural)

In lattice gauge theory a gauge transformation has two readings that give the same group: a change of local trivialization at every site, and a vertical automorphism of the bundle over the lattice. Both are $\prod_x G$, because every site x has its own fiber and nothing but the link variable relates the fibers at two sites; Elitzur's theorem [28] concerns this local group. At c_* the two readings come apart. The passive group of frame changes is still $\prod_{i \in V} G$, and it acts on link coordinates by exactly the lattice formula $\Theta_e \mapsto g_i \Theta_e g_j^{-1}$ (MT (2.28)). The automorphisms form only one copy of G, because the agents are not sites with fibers of their own: they are frames in one shared fiber, and every automorphism of the whole bundle restricts at c_* to one element of that copy (MT Proposition 2.6). Lattice sites correspond to base points, not to agents at one base point; which reading a lattice model of the supplements has is left open there (Open Problem B7.1). The correspondence is structural: the same transformation formula for the links, a different automorphism group.

2.2.2 A link is an automorphism of the shared fiber

A link $\Theta_e^x \in G$ on the oriented edge copy $e : j \to i$ defines the link map $\mathrm{t}_e^x(u_j \cdot g) = u_i \cdot \Theta_e^x g$, a right-equivariant bijection of the one fiber onto itself that does not depend on the frames used to write it (MT Proposition 2.12). An equivariant self-map of a torsor is, once a reference point is chosen, left multiplication by one group element, here the per-edge twist $\mathring{\Theta}_e^x = U_i^{-1} \Theta_e^x U_j$. Passive re-expressions leave every twist unchanged, a change of reference point conjugates all twists by one common element, and an active transformation does the same. Each edge copy therefore carries the invariant conjugacy class $[\mathring{\Theta}_e^x]$, and the full invariant content of the links is the twist tuple modulo one simultaneous conjugation. In lattice gauge theory a link is a map between two different

fibers and has no conjugacy class; here all agents share a fiber, so a link is a symmetry of that fiber and has one.

In lattice language, and informally, the twist is the link measured against the direct frame comparison. The comparison T_{ij} transforms under passive re-expression exactly like a link from j to i, so $T_{ij}^{-1}\Theta_e^x$ is the holonomy of a two-step loop based at j: out to i along the link, back along the frame comparison. It is conjugate to the twist. The per-edge class is thus a Wilson loop, of a loop that closes through the frame comparison instead of through another link, and pure lattice gauge theory has no such loops because it has no frame comparison between sites. The same reading removes an apparent clash with lattice intuition. A single link coordinate between two distinct agents carries no invariant by itself, since the re-expression $g_i = e_G$, $g_j = \Theta_e^x$ sends it to e_G (MT Remark 2.15), exactly as in lattice gauge theory. The invariant belongs to the pair formed by the link and the frames.

Whether a model can see these extra invariants depends on whether it reads the frames. Informally: if a model satisfies the kernel-covariance hypothesis of chapter 4 with parameters that the re-expression leaves unchanged, and none of its kernels reads the frames, then two link configurations related by the lattice vertex law, with the frames held fixed, have the same evidence, by the covariance of the joint (MT Proposition 4.9). Such a model sees only the Wilson-loop data, and the running example below is of this kind. A mechanism that contains a map shared by all agents and referred to their frames reads the twists directly; the shared query and key reference maps of attention are of this kind (MT Table 6.1). The per-edge classes are physical to the extent that the model contains frame-referenced structure.

Physics analogue (structural)

Write the frames as group elements U_i against one reference point. A passive re-expression acts by $U_i \mapsto g_i U_i$, a change of reference point by $U_i \mapsto U_i g_0$, and the twist $U_i^{-1}\Theta_e U_j$ is invariant under the first and conjugated by the second. These are the field content and the symmetry pattern of a lattice gauge–Higgs model whose Higgs field is group valued (for $SU(2)$, a fundamental doublet of fixed length written as a matrix): the twist is the link in unitary gauge, and the single remaining global group acts on unitary-gauge links by conjugation [31, 59]. Supplement B spells out this reading for $G = \mathbb{Z}_2$, identifying the matter sign with the frame coordinate and the dressed link sign with the twist (Open Problem B7.1). The correspondence is structural and kinematic only. The frames are structural data at c_*, not a field that is summed over, so nothing about gauge–Higgs dynamics, phase structure, or complementarity carries over.

2.2.3 Pure gauge, frame alignment, and Wilson loops

Links are graph flat on a block I when every closed internal walk has trivial product, equivalently when they are a coboundary $\Theta_e^x = V_i V_j^{-1}$ for vertex potentials V_i, unique up to one right multiplication per connected component (MT Proposition 2.13, MT Definition 2.14). This is pure gauge. Links are frame aligned when $\Theta_e^x = T_{ij}$, the canonical pure-gauge configuration supplied by the frames

themselves; then every twist is e_G and every link map is the identity of the fiber. Frame alignment implies graph flatness and not conversely, and on a tree every configuration is graph flat while its per-edge classes can all be nontrivial (MT Remark 2.15). In lattice terms: on a tree all links are pure gauge, as in the maximal-tree gauge fixing [17], and here they still carry twists relative to the frames.

2.2.4 Three kinds of agreement

On a connected block I the chapter separates three conditions (MT Definition 2.17). Frame agreement is graph flatness of the links on I, a condition on the links alone. Agreement of an equivariant statistic, with the mean as the basic case, asks that transported statistics match on every internal edge, $\mu_i = \rho_x(\Theta_e^x)\mu_j$, a condition on links and means. Law coherence asks that whole laws match, $q_i = \widehat{\rho}_x(\Theta_e^x)q_j$, a condition on links and laws. The relations are proved in MT Proposition 2.18. Law coherence holds exactly when the base law q_{i_0} is fixed by every loop holonomy based at i_0 and the other laws are its transports, and mean agreement holds exactly when the base mean lies in the common fixed space of the represented holonomies and the other means are its transports. Law coherence implies agreement of every equivariant statistic, and the converse fails for the mean. Frame agreement neither implies nor is implied by either of the other two. With several independent cycles, a condition checked on one loop is necessary and not sufficient.

The distinctions matter because each notion is a candidate criterion for when agents form a collective, and the three give different verdicts. Frame agreement without law coherence is a block with no flux whose agents simply disagree. Law coherence without frame agreement is a block with nontrivial flux whose laws agree anyway, because the flux lies in the stabilizer of the common law: isotropic Gaussians around the running example's loop, or, for $G = \mathrm{GL}(2,\mathbb{R})$, the law $\mathcal{N}(0,\Sigma)$ transported around a loop whose holonomy $\Sigma^{1/2}R(0.7)\Sigma^{-1/2}$ is not orthogonal and fixes no nonzero vector. Mean agreement without law coherence is a block whose covariances disagree. The coarse-graining chapters use the third notion: they score a block by its divergence from holonomy-fixed parent laws, and an attained zero score means exactly that the transported member laws coincide in one such law, which makes them law coherent (MT Theorem 10.9). A collective criterion built on frame agreement would be too strong for that purpose, and one built on means too weak. None of the three selects a partition into meta-agents, which is part of MT Open Problem 9.19.

Physics analogue (exact)

Part (i) of MT Proposition 2.18 is the graph version of the holonomy principle of differential geometry: a covariantly constant section through a given fiber element exists exactly when that element is fixed by the holonomy group. Law coherence is a covariantly constant section of the law bundle over the block, with the links playing the connection.

Physics analogue (structural)

A covariantly constant Higgs field forces the holonomy of the gauge field into the stabilizer of the Higgs value, the unbroken subgroup, while flux inside that subgroup is allowed. Law coherence with nontrivial holonomy has the same pattern with a probability law in place of the Higgs value: flux is allowed exactly inside the law's stabilizer. No symmetry breaking is claimed by the chapter.

2.2.5 Compact and noncompact structure groups

Nearly everything in the chapter holds for an arbitrary Lie group G, and the trivialization criterion holds for an arbitrary group, so the frame comparisons, the single active copy of G, the per-edge classes, and the three agreements hold verbatim for $\mathrm{GL}(K,\mathbb{R})$, $\mathrm{GL}(K,\mathbb{C})$, and $SU(N)$. The exception is the construction of a probability law on the links from loop penalties (MT Remark 2.16). For a unitary representation the penalty $1-\mathrm{Re}\,\mathrm{Tr}\,\mathrm{U}/n$ of a represented loop holonomy U is Wilson's plaquette action [81] normalized per dimension; it lies in $[0,2]$, and for compact G the Boltzmann weight of finitely many such penalties against the product Haar probability measure on the links has a normalizer between $e^{-2\sum_p c_p}$ and 1. This covers $SU(N)$ and $SO(K)$ with unitary representations. For $\mathrm{GL}(n,\mathbb{R})$ with $n \geq 2$ the same remark proves a no-go: no norm on $\mathfrak{gl}(n,\mathbb{R})$ is invariant under conjugation by all of $\mathrm{GL}(n,\mathbb{R})$. An invariant nonnegative penalty vanishing exactly at the identity exists once a symmetric positive-definite fiber metric at the base vertex is declared and made to transform covariantly, which is additional data, playing the role a metric plays in reducing a general linear group to an orthogonal one. Whether any such penalty gives a normalizable law on GL-valued links is declined in the chapter's scope paragraph.

The gap is not a technicality. Informally, the Haar measure of a noncompact group is infinite, and a penalty that depends only on loop holonomies leaves every tree direction of the link configuration unpenalized, so the naive Boltzmann weight already fails to integrate on a graph with a single edge. A proper reference that is not Haar would normalize such a law at the cost of the invariance the penalty was built to have; the manuscript makes that trade once, for the Gaussian presentations of its formation model, as recorded in MT Open Problem 3.23. Two further facts bear on the target group. Links are structural parameters by default, not random variables (MT Section 1.4), so the missing GL link law blocks only models that integrate over links. And for nondegenerate Gaussian laws under independent $\mathrm{GL}(K)$ links, law coherence around a loop requires the holonomy to be conjugate into $\mathrm{O}(K)$, which forces $|\det| = 1$ on the loop and so holds only on a Lebesgue-null set of link configurations (MT Corollary 10.11, discussed in chapter 10); for the target group, law coherence of Gaussian beliefs around a loop is nongeneric.

2.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Base point c_*; context space $\mathcal{C}$	The single point of the context manifold at which everything happens; $\mathcal{C}$ itself is idle in this chapter.	One spacetime point, or one lattice site
Principal bundle $\mathscr{P}_G$; its fiber at c_*	At c_* only the fiber matters: a G-torsor, a copy of G without a preferred identity.	Space of frames at one point
Frame u_i; reference point σ; U_i	Agent i's chosen point of the shared fiber; against a reference σ it is the group element U_i with $u_i = \sigma \cdot U_i^{-1}$.	Local frame; group-valued Higgs field in the analogy above
Passive re-expression g_i; passive vertex group $\prod_{i\in V} G$	Agent i replaces its frame by $u_i \cdot g_i^{-1}$; coordinates change, nothing else does.	Local gauge transformation read as a change of trivialization
Active gauge group $\mathcal{A}_{c_*}$; local functions φ_i	Automorphisms of the shared fiber; one copy of G, written in agent i's frame as φ_i.	One global symmetry transformation
Frame comparison T_{ij}	The unique group element with $u_j = u_i \cdot T_{ij}$, equal to $U_i U_j^{-1}$.	Canonical pure-gauge link on the complete graph
Sample fibers K, M; representations ρ_b, ρ_m	Spaces of belief samples and model presentations, with G acting; the two actions may be inequivalent, unfaithful, and of different dimension.	Matter representations
Law fibers $\mathcal{B}_b$, $\mathcal{B}_m$; pushforward action $\widehat{\rho}_x$	Families of probability laws on the sample fibers, on which G acts by pushforward.	A nonlinear target space whose points are laws
Associated bundles $\mathcal{E}_b$, $\mathcal{E}_m$	Bundles whose fiber at c_* is a space of laws, identified with $\mathcal{B}_x$ by each frame.	Matter bundle, with laws in place of field values
Agent $\mathcal{A}^i = (\mathcal{C}_i; q_i^b, q_i^m, u_i)$	A frame plus a belief law and a model law; at c_* only their values matter.	None; a naming decision
Cross morphisms Φ, $\widetilde{\Phi}$	Fiberwise maps from belief laws to model laws and back; extra structure, not supplied by the bundle.	None
Multigraph $\mathfrak{I} = (V, E, \bar{})$; edge copy $e : j \to i$; reversal $\bar{e}$	Oriented edges between agents, parallel copies allowed; every copy has a reversed partner.	Lattice with multiple bonds
Link Θ_e^x; vertex law	A group element per edge copy and channel, independent of the frames; transforms by $g_i \Theta_e^x g_j^{-1}$.	Link variable U_{xy}
Link map $\mathfrak{t}_e^x$	The frame-independent automorphism of the fiber that the link defines.	None in pure lattice gauge theory
Per-edge twist $\mathring{\Theta}_e^x$	$U_i^{-1} \Theta_e^x U_j$, the link written against the single reference point; its conjugacy class is invariant.	Link in unitary gauge
Path product Θ_p^x; holonomy $\mathrm{Hol}^x(p)$	Ordered product of links along a walk; for a closed walk, a group element covariant under conjugation.	Wilson loop, before taking the class

Term or symbol	Plain meaning	Physics analogue
Graph flat; jointly trivializing	Trivial holonomy on every internal loop; in both channels at once.	Pure gauge; zero field strength
Frame aligned	Links equal the frame comparisons, so all twists are trivial.	The canonical pure-gauge background
Law coherent	Laws on a block equal each other's transports along every internal edge.	Covariantly constant matter field
s-agreeing; mean agreement	An equivariant statistic, such as the mean, agrees after transport.	Agreement of an order parameter

Several words and glyphs trip a physicist. "Gauge" appears with two qualifiers: the passive group is the local one and the active group is global, and "gauge invariant" without qualifier means invariant under both. "Holonomy", "flat", and "twist" are graph-link notions: a flat link configuration says nothing about the curvature of any connection on $\mathcal{C}$, and a twist is not a twist of the bundle, which over one point is always trivial. "Common frame" means one frame shared by an agent's belief and model channels (MT Proposition 2.7), not one frame shared by all agents. The orientation runs from source to receiver: $e : j \to i$ carries j's coordinates into i's, a path product is written so that its rightmost link acts first, and T_{ij} transforms like a link on $e : j \to i$. T_{ij} is never a temperature or a transfer matrix, a hat on ρ means pushforward, and a sans-serif G is a metric, not the group. Calling a section-bearing object an agent is a naming decision with no claim of agency attached (MT Section 1.4).

2.4 The formulas, explained

Frames and comparisons. The fiber at c_* is a right G-torsor, and the chapter's first relations are

$$u_i = \sigma \cdot U_i^{-1}, \qquad u_j = u_i \cdot T_{ij}, \qquad T_{ij} = U_i U_j^{-1}, \qquad T'_{ij} = g_i T_{ij} g_j^{-1}, \qquad U'_i = g_i U_i$$

(MT Proposition 2.5). The reference point σ is an arbitrary origin, and T_{ij} does not depend on it. The identities $T_{ii} = e_G$, $T_{ji} = T_{ij}^{-1}$, $T_{ij}T_{jl} = T_{il}$ make (T_{ij}) a pure-gauge configuration on the complete graph. The re-expression $g_i = U_i^{-1}$ gives every agent the frame σ and all $T_{ij} = e_G$; this equal-frame gauge is always reachable at c_*, and the running example uses it.

The active image. An automorphism F of the fiber, written in agent i's frame as $F(u_i) = u_i \cdot \varphi_i$, satisfies $\varphi_j = T_{ij}^{-1}\varphi_i T_{ij}$ (MT (2.12)), so the whole tuple is fixed by any one entry. In coordinates it acts like the passive re-expression $g_i = \varphi_i$; the difference is that the active map moves the configuration while the passive one relabels it.

Links, the vertex law, and holonomy. For each edge copy $e : j \to i$ and channel $x \in \{b, m\}$,

$$\Theta^x_{\bar{e}} = (\Theta^x_e)^{-1}, \qquad \Theta^{x\prime}_e = g_i \Theta^x_e g_j^{-1}, \qquad \mathrm{Hol}^x(p) = \Theta^x_{e_0}\Theta^x_{e_1}\cdots\Theta^x_{e_{r-1}}, \qquad \mathrm{Hol}^x(p)' = g_{i_0}\mathrm{Hol}^x(p)g_{i_0}^{-1},$$

the last two for a closed walk based at i_0 (MT Definition 2.11). These are the lattice formulas term for term. The two channels carry separate link families, and the common-link case $\Theta^b_e = \Theta^m_e$ is a declared specialization.

Twists. With $u_i = \sigma \cdot U_i^{-1}$,

$$\mathring{\Theta}^x_e = U_i^{-1}\Theta^x_e U_j, \qquad \mathrm{Hol}^x(p) = U_{i_0}\mathring{\Theta}^x_{e_0}\mathring{\Theta}^x_{e_1}\cdots\mathring{\Theta}^x_{e_{r-1}}U_{i_0}^{-1}$$

(MT (2.32), MT (2.33)). The twist is the link rewritten against the single reference point σ; frame factors cancel in pairs along a closed walk. For abelian G each twist is itself invariant (Supplement B's $\mathbb{Z}_2$ reading calls it the dressed link sign), and on a tree the twists are the entire invariant content.

The trivialization criterion. For group elements Ξ_e on the internal edge copies of a block,

$$\begin{aligned} & \Xi_e = V_i V_j^{-1} \text{ for every internal } e : j \to i \\ \iff \quad & \Xi_{e_0}\Xi_{e_1}\cdots\Xi_{e_{r-1}} = e_G \text{ for every closed internal walk} \end{aligned}$$

(MT (2.34)). The first line is pure gauge and the second is triviality of all Wilson loops. The proof uses only the group law, so it holds for $\mathrm{GL}(K)$, for its subgroups, and for represented links, with no compactness or topology.

Loop penalties. For a represented loop holonomy $\mathsf{U} = \rho_x(\mathrm{Hol}^x(p))$ of dimension n,

$$\frac{1}{2n}\,\mathrm{Tr}\big((I_n - \mathsf{U})^*(I_n - \mathsf{U})\big) = 1 - \frac{1}{n}\,\mathrm{Re}\,\mathrm{Tr}\,\mathsf{U} \in [0,2] \qquad (\mathsf{U}\text{ unitary}),$$
$$\mathrm{Tr}\big((\mathsf{G}^{\mathrm{fib}})^{-1}(I_n - \mathsf{U})^\top \mathsf{G}^{\mathrm{fib}}(I_n - \mathsf{U})\big) \qquad (\mathsf{U}\text{ real}).$$

The first is Wilson's action per loop, vanishing exactly at $\mathsf{U} = I_n$; for $U(1)$ it is $1 - \cos\phi$. It is invariant because a passive re-expression conjugates U by the unitary $\rho_x(g_{i_0})$. The second is the general linear substitute (MT Remark 2.16); it is invariant only because the fiber metric $\mathsf{G}^{\mathrm{fib}}$ is declared to transform by $(R^x_{i_0})^{-\top}\mathsf{G}^{\mathrm{fib}}(R^x_{i_0})^{-1}$, so it is a covariant field at the base vertex, not a constant.

Agreement. Law coherence and its mean-level shadow read

$$q_i = \widehat{\rho}_x(\Theta^x_e)q_j \text{ on each internal } e : j \to i, \qquad \mu_{i_0} \in \bigcap_{p \text{ closed at } i_0} \ker\big(\rho_x(\mathrm{Hol}^x(p)) - \mathrm{id}\big)$$

(MT (2.35), MT (2.37)). The first is a parallel section of the law bundle over the block; the second says the base mean survives every loop. For a nontrivial planar rotation loop the kernel is $\{0\}$ and every agreeing mean is zero.

Cross morphisms. A cross morphism has representatives transforming by $\Phi'_i = \widehat{\rho}_m(g_i) \circ \Phi_i \circ \widehat{\rho}_b(g_i)^{-1}$, and one map Φ_0 serves in every frame exactly when it intertwines the two actions (MT Proposition 2.10). In the running example's setting, with $SO(3)$ acting on $\mathbb{R}^3$ in the belief channel

and trivially on $\mathbb{R}$ in the model channel, the only equivariant linear map $\mathbb{R}^3 \to \mathbb{R}$ is zero by Schur's lemma, while pushforward of laws by the norm, $\beta \mapsto (|\cdot|)_{\#}\beta$, is an equivariant map of laws: law-level equivariance is weaker than linear equivariance.

The running example. Take $G = SO(3)$, ρ_b the defining representation on $\mathbb{R}^3$, ρ_m trivial on $\mathbb{R}$, three agents on a triangle with $e_1 : 2 \to 1$, $e_2 : 3 \to 2$, $e_3 : 1 \to 3$, all frames equal, and belief links

$$\Theta^b_{e_1} = R_z(\tfrac{\pi}{2}), \qquad \Theta^b_{e_2} = R_x(\tfrac{\pi}{2}), \qquad \Theta^b_{e_3} = I_3, \qquad \mathrm{Hol}^b(p) = R_z(\tfrac{\pi}{2})R_x(\tfrac{\pi}{2}), \qquad \mathrm{Tr}\,\mathrm{Hol}^b(p) = 0$$

(MT Section 2.5, MT (2.39)). The per-edge classes are the rotation angles $\pi/2$, $\pi/2$, 0, and the holonomy is a rotation by $2\pi/3$, here about the axis $(1, 1, 1)$. Replacing the second quarter turn by one about the same axis as the first gives a rotation by π with the same three classes, which is the concrete reason the class tuple alone does not determine the holonomy while the tuple modulo one simultaneous conjugation does. The loop is not graph flat, yet the isotropic laws $\mathcal{N}(0, s^2 I_3)$ are law coherent on it, because every rotation fixes them. The one record of the example has success probability $\exp(-\frac{1}{2}|k_1 - \rho_b(\Theta^b_{e_1})k_2|^2)$, a transported discrepancy whose length no passive re-expression changes (MT (2.40)). In lattice terms, this is a frustrated plaquette with flux $2\pi/3$ and a matter law that does not feel the flux.

2.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
Agents and law bundles: MT Definition 2.4, MT Definition 2.8	An agent is a frame in the one fiber plus a belief law and a model law; the channels may carry inequivalent representations.	CHOICE	Standard construction; the commitment is laws valued in G-spaces at one fiber.
Frame comparisons: MT Proposition 2.5	T_{ij} is unique, a cocycle, and transforms by $g_i T_{ij} g_j^{-1}$.	SOLID	Any Lie group; torsor algebra.
Active image: MT Proposition 2.6	Fiber automorphisms form one copy of G inside $\prod_i G$, proper when $G \neq \{e_G\}$ and $\lvert V\rvert \geq 2$.	SOLID	Any Lie group; standard in substance.
Common frame: MT Proposition 2.7	Separate belief and model frames reduce to one by a re-expression.	SOLID	Harmless because nothing reads the relative frame.
Cross morphisms: MT Proposition 2.10	Gauge law of representatives; intertwiners are exactly the frame-independent ones; not every cross morphism is induced; Schur vanishing does not remove cross morphisms.	SOLID	Their existence is additional structure.

Result	What it says	Status	Scope and caveats
Independent links: MT Definition 2.11	One group element per oriented edge copy and channel, independent of the frames.	CHOICE	Structural parameters by default.
Twist invariance: MT Proposition 2.12	Each edge carries an invariant twist class; link sets are actively related exactly when their twist tuples are simultaneously conjugate.	SOLID	Any Lie group; standard in substance; an invariant of links with frames.
Trivialization: MT Proposition 2.13	Coboundary exactly when all closed-walk products are trivial; potentials unique up to one right factor per component.	SOLID	Any group; pure gauge exactly when Wilson loops are trivial.
Strictness of flatness: MT Remark 2.15	Frame alignment is strictly stronger than graph flatness; one link coordinate has no invariant; trees can carry nontrivial classes.	SOLID	General non-implications on minimal graphs.
Compact loop penalty: MT Remark 2.16	For unitary ρ_x, $1 - \mathrm{Re}\,\mathrm{Tr}\,\mathrm{U}/n \in [0,2]$ is invariant; for compact G the penalty law against product Haar probability has normalizer in $[e^{-2\sum_p c_p}, 1]$.	COND	Needs G compact, ρ_x unitary: holds for $SU(N)$, $SO(K)$; fails for $\mathrm{GL}(K,\mathbb{R})$, $\mathrm{GL}(K,\mathbb{C})$.
No invariant norm for GL: MT Remark 2.16	No norm on $\mathfrak{gl}(n,\mathbb{R})$, $n \geq 2$, is $\mathrm{GL}(n,\mathbb{R})$-conjugation invariant; a covariant fiber metric restores an invariant penalty.	SOLID	Negative result; the metric is extra data.
Link law for noncompact G (scope paragraph)	Normalizability of a loop-penalty link law for noncompact or general linear G.	OPEN	Declined, with no ledger entry; the program's target case.
Three agreements: MT Proposition 2.18	Holonomy characterizations; law coherence implies statistic agreement, not conversely; frame agreement independent; one loop insufficient; $\mathrm{GL}(2,\mathbb{R})$ witness with non-orthogonal holonomy.	SOLID	Any G; witnesses refute general implications.
Agreement and partitions (scope paragraph)	Whether law coherence bears on selecting a partition into meta-agents.	OPEN	Part of MT Open Problem 9.19.

Conditions: which ones matter

Three conditions matter for the models the program cares about. Compactness of G and unitarity of the representation are what make the loop-penalty link law normalizable; they hold for $SU(N)$ and fail for $\mathrm{GL}(K,\mathbb{R})$ and $\mathrm{GL}(K,\mathbb{C})$, and the chapter supplies no substitute. The existence of cross morphisms is additional structure: the bundles do not supply maps between

belief laws and model laws, and with irreducible inequivalent linear representations no nonzero equivariant linear map exists. The single frame shared by the two channels of an agent is justified only because no energy, kernel, or evaluator reads the relative frame between them (MT Proposition 2.7); a model that read it would need both frames back. The common-link specialization $\Theta_e^b = \Theta_e^m$ is an optional modeling choice, not a consequence.
The rest is hygiene here: the context space is a second countable Hausdorff manifold (idle at c_*), representations are measurable, sample fibers are standard Borel, and the smooth tier (MT Hypothesis 2.3), which excludes infinite-dimensional law fibers, matters only for the fiber geometry of chapter 7.

The chapter has no toy results: its small graphs and its $SO(2)$, $SO(3)$, and $\mathrm{GL}(2,\mathbb{R})$ instances are counterexamples to general implications, solid at full generality, or illustrations.

2.6 Open ends

Two questions are left unsettled. The first is whether a normalizable law on links built from loop penalties exists for a noncompact or general linear structure group. The chapter declines the claim and records no open problem or settling condition for it; for the $\mathrm{GL}(K)$ target, links therefore remain structural parameters, and the theory supplies no Wilson-type prior to integrate them against. The second is whether any of the three agreements bears on choosing a partition of agents into meta-agents, part of MT Open Problem 9.19, which is settled by a relabeling-natural, gauge-compatible functional of the laws, not monotone under refinement, with a nondegenerate optimizer, or by a theorem that no such selector exists in a stated class. Connections, base transport, and descent are out of scope by design rather than open.

Does it deliver what it claims? DELIVERED

The chapter says that it determines what geometric data an agent carries at c_* and which are invariant: the active gauge group is one copy of G inside the passive vertex group, every link carries an invariant conjugacy class with loop holonomy the telescoped product of per-edge twists, and three agreements are distinct; the introduction's reading map repeats this. Every one of these statements is proved, for an arbitrary Lie group, with short proofs. The mathematics is standard in substance: torsor algebra, automorphisms of a torsor, degree-one nonabelian cohomology of a graph, and the holonomy principle on a graph. What is new is the typing, and it does real work in one place: because all agents share one fiber, links become automorphisms of that fiber and acquire per-edge invariants that pure lattice gauge theory lacks, visible to a model only through frame-referenced structure. No connection, curvature, or bundle topology enters, and the text does not claim otherwise. The gap that matters for the program is the declined normalizability of loop-penalty link laws for noncompact G: for $\mathrm{GL}(K,\mathbb{R})$ the chapter supplies an invariant penalty only after adding a covariant fiber metric, and no normalized link law.

Chapter 3

Probability Semantics at c_*

3.1 The question and the answer

The technical chapter asks on which measurable objects, and in which order, every probabilistic statement at c_* rests. Its answer is a discipline. Probability laws and Markov kernels are the primitive objects. Densities are Radon–Nikodym derivatives against reference measures fixed once for a whole family of laws, generative and recognition alike. Logarithms of density ratios are written only where the numerator is absolutely continuous with respect to the denominator, and relative entropy is $+\infty$ everywhere else (MT Definition 3.10). The stated main results are that a family of mixed laws on the real line has a common dominating measure exactly when the union of its atoms is countable (MT Theorem 3.5), the relative-entropy chain rule in $[0, +\infty]$ that every later decomposition uses (MT Lemma 3.12), and three levels of description, sample, law point, and configuration, at which a divergence between laws is not an energy of samples (MT Proposition 3.20) and a point-mass recognition of a law point has infinite cost (MT Proposition 3.21). Along the way the chapter types the population recognition law as a selected coupling of the agents' local laws and prices restricted access. The generative law of chapter 4 is built on these spaces, the evidence bound of chapter 5 is an application of the chain rule, and attention (chapter 6) and formation (chapter 13) are typed by the levels.

Which of this is substance and which is hygiene? Most of the chapter is measure theory done carefully so that later identities hold without case distinctions, and a physicist can take it on trust. Three things are substantive. The population recognition law is not determined by the agents' local beliefs and must be selected, and that selection is where recognition-level correlations between agents live. Restricting what each agent can see has an evidence price that no selection removes. And the level typing shows that a relative entropy between two agents' beliefs, the building block of the program's attention free energies, is not the expectation of any energy of samples, so its place inside one evidence bound depends on a prior over beliefs that the theory has not constructed (MT Open Problem 3.23). A fourth item sits on the border: families of laws with moving atoms, and noiseless records with a continuum of values, have no density model, and the evidence bound for such records is open (MT Open Problem 3.9).

3.2 The picture

3.2.1 Laws first, because the theory needs degenerate laws

A physicist's default is to write a probability as a density and a free energy as an integral of $q \log(q/p)$. The chapter refuses that default for a concrete reason: the models it must cover include point-mass model presentations (the definite case $q_i^m = \delta_{m_i}$ of MT Definition 3.3), laws carried by lower-dimensional sets such as proper affine subspaces, and laws with atoms whose locations move with a parameter. None of these has a density against Lebesgue measure, and some families of them have no common reference at all. A sigma-finite measure has at most countably many atoms, so the moving-atom family $\mu_t = \frac{1}{2}\mathcal{N}(0,1) + \frac{1}{2}\delta_t$, $t \in \mathbb{R}$, has no common dominating measure, and neither does any uncountable family of Dirac laws on an uncountable space (MT Theorem 3.5, MT Remark 3.6). Working with measures and setting relative entropy to $+\infty$ off absolute continuity keeps every identity meaningful in the extended reals: the evidence bound of a singular recognition law is $-\infty$ rather than undefined. This is the whole content of "kernels before densities". It costs nothing when densities exist, and it keeps the degenerate cases inside the theory when they do not.

Physics analogue (exact)

Conditioning on an event of probability zero is not determined by the joint law: the posterior at a single observation depends on the version of the conditional one uses, which is the Borel–Kolmogorov paradox. The chapter's regular-conditional convention (MT Hypothesis 3.8) therefore asserts every fixed-observation statement only for almost every observation, and its witness (MT Remark A.5) is an instance of the paradox: two versions of one joint density on $\mathbb{R}^2$ give different posteriors at $o = 0$.

3.2.2 Local beliefs do not fix the population law

Each agent has a local recognition law $\mathbb{Q}_{i,o,X}$ on its own block, computed from what it can access. The population recognition law is a law on all blocks at once whose agent marginals are the local laws, that is, a member of the coupling class $\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$. The class is never empty and is convex (MT Proposition 3.14), and it is a single point only when at most one local law is non-Dirac (MT Proposition 3.17). As soon as two agents are uncertain, their beliefs underdetermine the joint recognition law, and the theory selects one by a declared rule (MT Definition 3.15). The product coupling is one admissible selection, not the default; copulas [56], shared latent variables, ordered conditionals, or graphical families supply others. The coupling is a modeling input, not an output of the local beliefs, and the manuscript asserts no canonical selector.

Restricted access sharpens the point (MT Proposition 3.18). An access map limits what each local law may depend on, but a selector that reads all records can still make the population law depend on data that no agent sees, so a claim of decentralized inference also needs a selector indexed by the agents' accesses alone. And restricting access costs evidence. When an agent cannot distinguish two observations whose posteriors differ, its single local law sits at a positive divergence from at least one of them; in the chapter's two-agent binary witness the worst-case gap is at least the Chernoff information between the two posterior marginals, about 0.0885 nats [15, Section 11.9].

Physics analogue (structural)

The product coupling is the naive mean-field trial state of the Gibbs–Bogoliubov variational principle, and correlated couplings are richer trial states with the same one-body marginals. The identification is exact for the product case; for the others it is structural, since a coupling class fixes the one-body marginals and leaves every higher correlation free.

3.2.3 Three levels of description

The chapter distinguishes three kinds of latent variable (MT Definition 3.19). At the sample level the latent state is the population sample $y = ((k_i, m_i))_{i \in V}$, the agents' microstates, and an agent's belief is the marginal on k_i of the recognition law, a law over microstates. At the law-point level the beliefs themselves are latent: the latent state of agent i is a point $\mathfrak{q}_i^b$ of the associated law fiber, a probability law treated as a random variable, and a recognition law is a law over laws. At the configuration level the latent state is a presentation w, for instance a mean and a covariance, together with a sample y drawn from the law that w presents, the two linked by a presentation kernel. The levels define different optimization problems, and the manuscript requires every energy to name its level.

The reason is MT Proposition 3.20. An energy D of samples enters a free energy as $\mathbb{E}_Q D$, which is affine in the recognition law. A transported relative entropy $D_{\mathrm{KL}}(q_i \| \widehat{\rho}_x(\Theta_e^x) q)$ is strictly convex in the source law q along mixtures, because $-\log$ is strictly convex, so no sample energy can match it at two source laws and one strict mixture of them. A divergence between two agents' beliefs is a functional of the ensemble, like an entropy, and not a Hamiltonian term. The consequence runs through the program. The belief-alignment terms $D_{\mathrm{KL}}(q_i \| (\Theta_e)_{\#} q_j)$ that weight attention rows (MT (12.12)) are law-point energies, and they become terms of one evidence bound only if beliefs are latent law points under a normalized prior, which is MT Open Problem 3.23.

MT Proposition 3.21 closes the obvious shortcut. If law points are latent and their prior has no atom at the law point one wants, a recognition law that puts all its mass on that law point has infinite relative entropy to the posterior and an evidence bound of $-\infty$. A row or an update computed at fixed law points is therefore not a recognition optimum at the law-point level. With law points latent it is a generative posterior quantity conditional on them; with law points declared as parameters it is a function of parameters whose change is an M-coordinate subject to the acceptance rule of MT Definition 5.28. Both readings are legitimate, and neither is an agent's recognition of its own belief.

Physics analogue (structural)

The entropy $-\int q \log q$ is a functional of the ensemble and not the average of any state-independent microscopic energy: it is strictly concave in q, while every average $\int qH$ is linear. The cross-entropy between two agents' beliefs has the same character in its second argument, and that is all MT Proposition 3.20 uses. The point-mass result has an exact counterpart: a trial state concentrated on one point of a continuum has entropy $-\infty$ relative to a continuous reference.

3.2.4 Auxiliary variables are free only when exact

The configuration level is reached from the sample level by a lift (MT Proposition 3.22). Draw configuration coordinates from a kernel $\kappa^{\mathrm{cfg}}(y, dw)$ given the sample: the evidence does not change, and a lifted recognition law pays, relative to the sample-level bound of its sample marginal, the expected divergence between its conditional law of w given y and κ^{cfg}. The lift never raises the bound, and it leaves a finite bound unchanged exactly when recognition uses κ^{cfg} itself as its conditional.

Physics analogue (exact)

Introducing an auxiliary variable with its exact conditional law, as in a Hubbard–Stratonovich or data-augmentation step, leaves the partition function unchanged, and a variational ansatz on the enlarged space pays the conditional relative entropy for any mismatch. Both statements are the chain rule of MT Lemma 3.12.

3.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Sample blocks $\mathsf{Y}_i = \mathsf{K}_i \times \mathsf{M}_i$; latent space Y	Agent i's belief-sample and model-presentation coordinates; their product over agents.	Single-site and full configuration space
Record $a \in \mathfrak{A}$; scope ∂a; record space O_a	An observation factor attached to a set of agents; a single-agent observation has singleton scope.	Measured quantity on a bond or plaquette
Structural configuration $X \in \mathsf{X}$	The multigraph, links, frames, and mechanism parameters, held fixed by declaration.	Lattice, couplings, background fields
Access map Acc_i	What agent i may use from the records and the structure.	A local observer's data
Local recognition kernel $\mathbb{Q}_i$; law $\mathbb{Q}_{i,o,X}$	Agent i's approximate posterior on its own block, computed from its access.	One-body trial state
Population recognition law $\mathbb{Q}_{V,o,X}$	A joint approximate posterior on all blocks, selected to have the local laws as marginals.	Many-body trial state
Coupling class $\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$	All joint laws with the given agent marginals.	Fréchet class of joint laws
Selection rule S	A declared measurable rule picking one coupling.	Choice of variational ansatz
Recognition marginals $q_i^{b;o,X}$, $q_i^{m;o,X}$	The belief and model parts of the local law; the values of agent i's sections at c_*.	One-body reduced laws
Generative law $\mathbb{P}_{\theta,V}(\cdot \mid X)$	The constructed joint of latents and records (chapter 4).	Prior times likelihood; Boltzmann weight
Posterior slice $\mathbf{\Pi}_{\theta,V,o,X}$	The exact conditional law of latents at observation o.	Gibbs state conditioned on data

Term or symbol	Plain meaning	Physics analogue
Evidence $p_{\theta,V}(o \mid X)$	The marginal density of the records.	Partition function
References ν^{Y}, ν^{O}, ν_{mix}	Fixed measures against which densities are taken; ν_{mix} adds countably many atoms to Lebesgue measure.	Phase-space measure
Relative entropy $D_{\mathrm{KL}}(Q\|P)$	Kullback–Leibler divergence, set to $+\infty$ off absolute continuity.	Free-energy excess over equilibrium
Presentation m_i; evaluator ev_i	A model coordinate that the evaluator turns into a generative kernel; not itself a law.	Coupling constants of a local Hamiltonian
Law point $\mathfrak{q}_i^x$; block $\mathsf{Y}_i^{\mathrm{law}}$	A law treated as a latent random variable.	A one-body distribution treated as a field
Configuration space $\mathsf{Y}^{\mathrm{cfg}}$; pres; κ^{cfg}	Presentations such as mean and covariance; the map to law points; the kernel linking them to samples.	Auxiliary-field representation
Regular set $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$	Observations of full probability with finite positive evidence, where the posterior formula is used.	None
Extended reals	Values in $[-\infty,+\infty]$; bounds may be $-\infty$ and divergences $+\infty$.	None

Several terms carry nonstandard meaning. "Configuration" is used twice, and the two uses must not be merged: the structural configuration X collects graph, links, frames, and parameters, while the configuration level of description has latent presentations $w \in \mathsf{Y}^{\mathrm{cfg}}$. "Kernel" always means a Markov kernel, a measurable family of probability laws, never an integral-operator kernel or a positive-definite kernel. "Record" is an observation factor with a scope, a hyperedge on the agents, and never a memory. "Recognition law" is the approximate posterior of variational inference, and the exact conditional is called the posterior slice. "Evidence" is the marginal likelihood, which a physicist should read as a partition function. "Presentation" is a parameter-like coordinate of a generative factor and has nothing to do with a group presentation. "Version" means a representative up to a null set. "Selected" means chosen by a declared rule, not derived.

3.4 The formulas, explained

Relative entropy. The convention of MT (3.17) is

$$D_{\mathrm{KL}}(Q\|P) = \begin{cases} \displaystyle\int \log\Big(\frac{dQ}{dP}(y)\Big) Q(dy), & Q \ll P, \\ +\infty, & \text{otherwise}, \end{cases}$$

the divergence of Kullback and Leibler [48] with the extension of Csiszár [20]. The negative part of the integrand has integral at most $1/e$, so the first branch is a well-defined element of $[0,+\infty]$. The typical infinite case in the program is a Gaussian with singular covariance measured against

one with nonsingular covariance; the typical finite case is two nondegenerate Gaussians. Whether $Q \ll P$ can be read off from densities: against a common reference it holds exactly when Q puts no mass where the density of P vanishes (MT Proposition 3.11).

The mixed reference. When the atoms of a family of mixed laws on $\mathbb{R}$ form the countable set $\{a_1, a_2, \ldots\}$,

$$\nu_{\mathrm{mix}} = \mathrm{Leb} + \sum_{n \geq 1} 2^{-n} \delta_{a_n}$$

dominates the whole family with one parameter-independent measure (MT (3.10)). If the atoms move over an uncountable set, no such measure exists, and the chapter then declares no density model on that coordinate. For learning this is concrete: a law with an atom whose location is a learned parameter cannot have its densities compared across parameter values against one reference.

The chain rule and its corollaries. For laws on a product with first marginals Q_1, P_1 and conditional kernels $Q_{2|1}, P_{2|1}$,

$$D_{\mathrm{KL}}(Q \| P) = D_{\mathrm{KL}}(Q_1 \| P_1) + \int D_{\mathrm{KL}}\big(Q_{2|1}(y_1, \cdot) \big\| P_{2|1}(y_1, \cdot)\big) Q_1(dy_1) \quad \text{in } [0, +\infty]$$

(MT (3.20)). This is the relative-entropy chain rule, standard in substance [15, Section 2.5]; the chapter's version needs no densities and no finiteness hypothesis, and it does not depend on the chosen conditional versions. The same lemma gives four further tools (MT Lemma 3.12). Measured against a product reference, a joint law's divergence splits into its total correlation $D_{\mathrm{KL}}(Q \| \bigotimes_b Q_b)$ plus the marginal divergences; physically, the connected correlations separate from the one-body terms. With the first marginal fixed, the divergence is minimized by copying the reference's conditional. Divergence cannot increase under a Markov kernel, which is the data processing inequality (clause (d), also cited as MT Lemma 3.12). And divergence is unchanged by a bimeasurable bijection, which is why frame changes preserve every divergence without any density hypothesis.

The evidence price of restricted access. In the witness of MT Proposition 3.18, agent 1 sees only its own record, so its local law $\mathrm{Ber}(t)$ is the same at the observations $(1, 0)$ and $(1, 1)$, whose posterior marginals for y_1 are $\Pi_1^{(0)} = (37/89, 52/89)$ and $\Pi_1^{(1)} = (13/161, 148/161)$. Then

$$\max_{o_2 \in \{0,1\}} D_{\mathrm{KL}}\big(\mathbb{Q}_{V,(1,o_2),X} \big\| \mathbf{\Pi}_{\theta,V,(1,o_2),X}\big) \geq \max_{o_2} D_{\mathrm{KL}}\big(\mathrm{Ber}(t) \big\| \Pi_1^{(o_2)}\big) \geq \sup_{\xi \in [0,1]} \big(-\log Z_\xi\big) = 0.08849\ldots$$

(MT (3.31)), with $Z_\xi = \sum_{y_1} \Pi_1^{(0)}(y_1)^\xi \Pi_1^{(1)}(y_1)^{1-\xi}$. The first step is data processing onto agent 1's coordinate, and the second is the Chernoff bound: a single law cannot be close to two distinct targets at once, and the best compromise is the Chernoff point. The bound is attained. The equality condition of the evidence bound, recognition equal to the posterior, lies outside the access-restricted class at one of the two observations.

Transported divergences are not sample energies. For a receiver law q_i and two source laws p_1, p_2 with finite transported divergences whose transported densities differ on a set of positive q_i-measure,

$$D_{\mathrm{KL}}\big(q_i \| \widehat{\rho}_x(\Theta_e^x)(sp_1 + (1-s)p_2)\big) < sD_{\mathrm{KL}}\big(q_i \| \widehat{\rho}_x(\Theta_e^x)p_1\big) + (1-s)D_{\mathrm{KL}}\big(q_i \| \widehat{\rho}_x(\Theta_e^x)p_2\big)$$

for $0 < s < 1$ (MT Proposition 3.20). The left side is what the divergence does along a mixture of source laws, and the right side is what any energy averaged over samples must do. Besides finiteness, the hypotheses exclude only the case in which the two transported laws have equal densities almost everywhere under the receiver law.

Point masses on law points. At the law-point level, with a posterior dominated by the prior (true when every record kernel has a density against a fixed reference) and a law point $\bar{\mathfrak{q}}$ that is not an atom of the prior,

$$D_{\mathrm{KL}}\big(\delta_{\bar{\mathfrak{q}}} \big\| \mathbf{\Pi}^{\mathrm{law}}_{\theta,V,o,X}\big) = +\infty$$

(MT (3.33)). The singleton $\{\bar{\mathfrak{q}}\}$ is a posterior-null set carrying all of the recognition mass. With an atom at $\bar{\mathfrak{q}}$ the divergence can be finite; the proposition concerns non-atomic priors, which is the situation for any continuous family of candidate beliefs.

The configuration lift. For a lifted recognition law $\mathbb{J}$ with sample marginal $\bar{\mathbb{Q}}$,

$$\mathcal{L}_V^{\mathrm{ext,cfg}}(\mathbb{J}; X, o) = \mathcal{L}_V^{\mathrm{ext}}(\bar{\mathbb{Q}}; X, o) - \mathbb{E}_{\bar{\mathbb{Q}}}\Big[D_{\mathrm{KL}}\big(\mathbb{J}(\cdot \mid Y) \big\| \kappa^{\mathrm{cfg}}(Y, \cdot)\big)\Big]$$

(MT (3.34)). The first term is the bound of the sample-level problem, the second is the price of any mismatch between recognition's conditional over configurations and the lift kernel, and both live in the extended reals.

3.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
Kernels first; levels: MT Definition 3.7, MT Definition 3.19	Laws and kernels are primitive, densities are derivatives against family-wide references, and every energy names its level: sample, law point, or configuration.	CHOICE	A typing discipline; it costs nothing when densities exist.
Common domination: MT Theorem 3.5	A family of mixed laws on $\mathbb{R}$ without singular-continuous parts has a sigma-finite common dominating measure exactly when its atoms are countable; moving atoms are not dominated.	SOLID	Standard in substance (a sigma-finite measure has countably many atoms).

Result	What it says	Status	Scope and caveats
Dirac families: MT Remark 3.6	An uncountable family of Dirac laws on an uncountable standard Borel space has no common sigma-finite dominating measure.	SOLID	Standard.
Chain rule and corollaries: MT Lemma 3.12	Chain rule, product-reference split, fixed-marginal minimization, data processing, and bijection invariance, all in $[0, +\infty]$.	SOLID	Standard in substance; proved in MT Appendix A without density or finiteness hypotheses.
Couplings exist: MT Proposition 3.14	The coupling class is nonempty and convex; the product coupling is a measurable kernel.	SOLID	Standard.
Marginals do not fix the joint: MT Proposition 3.17	The coupling class is a singleton exactly when at most one marginal is non-Dirac.	SOLID	Standard in substance (Fréchet classes).
Coupling selection: MT Definition 3.15	The population recognition law is chosen by a declared measurable rule inside the coupling class.	CHOICE	No canonical selector; equivariance of the selector under frame changes is declined.
Access restriction, part (i): MT Proposition 3.18	Access maps constrain local laws, not the population law; decentralization also needs an access-indexed selector.	SOLID	A general no-go shown by counterexample.
Access restriction, part (ii): MT Proposition 3.18	Every access-restricted recognition law has worst-case gap at least $0.08849\ldots$ nats, and the bound is attained.	TOY	Two binary agents with one prior; the constant is instance specific.
Law-point energy: MT Proposition 3.20	A transported relative entropy is strictly convex in the source along mixtures, so no sample energy reproduces it.	SOLID	Needs finite divergences and transported laws that differ on the receiver's support.
Point-mass cost: MT Proposition 3.21	Recognition by a point mass at a non-atomic law point has infinite divergence and bound $-\infty$; rows at fixed law points are posterior quantities or parameter functions.	SOLID	Needs the posterior dominated by the prior, true for record kernels with densities.
Configuration lift: MT Proposition 3.22	Lifting leaves the evidence unchanged and lowers the bound by the expected conditional divergence.	SOLID	Standard in substance (augmentation and chain rule).
Law-point prior: MT Open Problem 3.23	A normalized generative law over law points for a general class.	OPEN	Needed for transported divergences to be evidence-bound terms.

Nondominated likelihoods: MT Open Problem 3.9	Evidence bound and coarse channels for record families with no common dominating measure.	OPEN	Covers noiseless records with a continuum of values.

Conditions: which ones matter

Four conditions can fail in the models the program cares about. Common domination fails for families with moving atoms and for deterministic records with a continuum of values, and then the density formulas, and everything that uses them, are not available (MT Open Problem 3.9). Absolute continuity of recognition with respect to the posterior is not assumed; its failure is the singular branch on which the bound is $-\infty$, and it genuinely occurs for degenerate Gaussian beliefs and point masses. The coupling selection is a modeling commitment whose choice changes every collective quantity, and its equivariance under frame changes is not claimed. The access structure, when imposed, carries an evidence price.

The rest is hygiene a physicist may skip on first reading: standard Borel spaces, sigma-finite references, joint measurability of Radon–Nikodym derivatives and of relative entropies in their parameters (proved once in MT Appendix A), the almost-every-observation convention for posteriors, and bookkeeping in the extended reals. The restriction of the common-domination theorem to laws without singular-continuous parts is also hygiene: no model in the program uses such laws.

3.6 Toy versus general

The only toy result is the quantitative half of the access-restriction proposition. Its mechanism is visible in the proof and has a general form: data processing onto the coordinates an agent can see, followed by the Chernoff bound for one law facing two targets. Nothing in that argument depends on binary variables or on two agents, so the mechanism may well survive in large networks with many agents sharing an access class, but the manuscript states and proves it only for the displayed instance, and the constant $0.08849\ldots$ and the optimizers $\xi \approx 0.548$ and $t \approx 0.783$ are specific to that prior and those record accuracies. The qualitative half, that access restriction constrains local laws and not the population law, is a general negative statement proved by counterexample.

3.7 Open ends

Two open problems are recorded. The first is a normalized prior over law points (MT Open Problem 3.23). Law fibers carry no canonical sigma-finite reference, since the space of all laws on a sample fiber is infinite dimensional, and the Haar measure of a noncompact G is infinite, so a Haar prior on a free orbit does not normalize. The problem is settled, for a class of law points, by a sigma-finite reference on the law-point or presentation space, a finite normalizer for the exponentiated law-point energies, the transformation law of the prior under the active gauge action, record kernels that read law points, and a compatibility relation between the law-point posterior and

the sample-level recognition marginals. Until then, the transported divergences MT (12.12) enter as quantities conditional on law points, as functions of parameters, or at the configuration level under the one proper prior the formation model supplies. The second is the treatment of record families with no common dominating measure (MT Open Problem 3.9): the joint law and its posterior exist, but the likelihood and possibly the evidence density do not. It is settled by a treatment of the evidence bound, its local disintegrations, and the coarse channels at the level of joint measures and regular conditional laws in which the exact identities of the later chapters are proved, or by a record family for which they provably have no measure-level counterpart.

The scope paragraph also records three limits. Injectivity of a model evaluator is not assumed, and no regularity of the space of presentations modulo equal evaluated kernels is supplied, so presentations are kept as they are; this is bookkeeping for overparameterized models. The selector's equivariance under frame re-expression is not claimed; this is substantive, because a recognition procedure rerun in new frames need not return the transformed coupling, and the invariance results of chapter 4 are stated for a coupling transported as a whole.

Does it deliver what it claims? DELIVERED

The chapter says that laws and kernels come first and densities follow against family-wide references, so that the evidence bound becomes an identity in the extended reals, and names as its main results common domination exactly when atoms are countable, the chain rule in $[0, +\infty]$, and three levels at which a law-point energy is not a sample energy and point-mass recognition of law points has infinite cost; the reading map adds that it proves the chain rule used everywhere and distinguishes the three levels. All of this is proved at the generality the theory uses. Most of it is standard measure theory and information theory in careful form, and a physicist loses nothing by treating it as such. The substantive content is elsewhere and is also delivered: local beliefs do not determine the population recognition law, so a coupling must be selected; restricted access has a provable evidence price; and a divergence between beliefs is not a sample energy. The last result exposes a gap that the chapter names rather than closes: without a normalized law-point prior, the belief-alignment divergences at the center of the program's attention free energies are not terms of one evidence bound at the law-point level.

Chapter 4

The Fixed Normalized Generative Law

4.1 The question and the answer

The technical chapter asks how one normalized joint law of latent samples and records is built at c_*, and what a passive re-expression of the frames does to it. The construction is directed. Structural data fix a directed acyclic graph Γ on the agents. Each agent has a model factor, the law of its model presentation given its parents' models, and a state factor, the law of its belief sample given its parents' beliefs and its own model; composed, they form the agent's mechanism, and composing the mechanisms along a topological order of Γ gives the latent law (MT Definition 4.1, MT Definition 4.2). Records are then attached as conditionally independent children of the latents, each with an arbitrary scope. The chapter proves that the result is normalized for every finite record hypergraph, cyclic or not (MT Proposition 4.4); that locally normalized reciprocal couplings between latents can fail to give any joint (MT Proposition 4.6); and that, under a covariance hypothesis on the kernels, a passive re-expression pushes the whole joint forward, so the evidence and the bound of a transported recognition law are unchanged (MT Proposition 4.9, MT Corollary 4.10). It also states, as a standing hypothesis, the typing prohibition: no generative factor may read a recognition law or the posterior (MT Hypothesis 4.3). The evidence of this joint is bounded in chapter 5, and attention and coarse channels act on it in chapter 6 and chapter 9.

What is nontrivial about normalizing on graphs with cycles? Mathematically, nothing: the proof integrates out the records first, each has unit mass, and what remains is a directed composition of normalized kernels. This is the standard fact that a Bayesian network is normalized whatever the undirected loops of its moral graph [49]. The content lies in what the construction requires. Generative influence among latents must be acyclic, and every interaction that closes a loop must be written as a record, a normalized law of an observed quantity given the latents it reads. The alternative, reciprocal couplings written directly between latents, has a parameter-dependent normalizer that can be infinite even when every local piece is normalized, which is what the Gaussian counterexample shows. The chapter chooses the directed route because its normalization does not depend on the parameters, and the price is that loop interactions enter only through conditioning on records.

4.2 The picture

4.2.1 Three graphs, not one

A physicist will be tempted to see one interaction graph. There are three, with different roles. The directed acyclic graph Γ says who generates whom among latents. The interaction multigraph $\mathfrak{I}$ of chapter 2 carries the links. The record hypergraph, formed by the scopes ∂a, says which latents each observation reads. The running example makes the distinction concrete: Γ is a tree in which agent 1 is the parent of agents 2 and 3, $\mathfrak{I}$ is the triangle, and two records with scopes $\{1,2\}$ and $\{2,3\}$ close the latent tree into the triangle. Cycles live in the record hypergraph; Γ never has any.

4.2.2 Records close loops, and the evidence is the partition function

When each record kernel has the form $K_a(X, y_{\partial a}, do_a) = \exp(-E_{a,o}(y_{\partial a}))\nu_a(do_a)$, the joint density is the latent density times $\exp(-\sum_a E_{a,o})$ (MT (4.12)). Combined with the posterior formula of MT Hypothesis 3.8, this says that the posterior is a Gibbs measure whose reference is the directed prior and whose Hamiltonian is the sum of the record energies, and that its partition function is the evidence. A binary record whose success probability is $\exp(-E)$ with $E \geq 0$, observed at success, contributes exactly the Boltzmann factor $\exp(-E)$ to the posterior, and the running example uses this device for its loop-closing records. Normalization of the joint is automatic; what can fail is finiteness and positivity of the evidence, and the manuscript admits only observations with finite positive evidence (MT Section 1.4). One physical consequence is easy to miss. Sampling the latents from the generative model ignores the records altogether, because the records are children of the latents, not constraints on them; the loop interaction shapes the latents only after conditioning on the observed records.

Physics analogue (exact)

Prior times likelihood, normalized by the evidence, is a Gibbs measure with the prior as reference and the record energies as Hamiltonian; the evidence is its partition function. This is Bayes' rule and needs no further argument.

4.2.3 Why the loop is not a potential between latents

The undirected alternative writes the joint as a product of potentials on agents, edges, and records, divided by a partition function Z_X, and it is legitimate whenever $0 < Z_X < \infty$ (MT Definition 4.5). The trouble is that Z_X depends on the parameters and nothing local controls it. Two scalar agents with normalized Gaussian site potentials and the positive, finite, continuous edge potential $e^{\mathrm{c}k_1k_2}$ give the quadratic form with matrix $\mathrm{W_c}$, whose smaller eigenvalue $1 - \mathrm{c}$ is not positive for $\mathrm{c} \geq 1$, and the integral diverges (MT Proposition 4.6). Each conditional $k_1 \mid k_2$ and $k_2 \mid k_1$ is a perfectly normalized Gaussian, and still there is no joint.

Physics analogue (exact)

This is the Gaussian spin model beyond its critical coupling: once the ferromagnetic coupling exceeds the single-site stiffness, the quadratic form loses positivity along the uniform mode and the partition function diverges. The counterexample is the two-site case of the same integral.

A related temptation is to append an energy U to the free energy of a joint that does not contain it. The chapter's remark computes the result (MT Remark 4.7): for U bounded below, $\mathcal{F}_V[Q; X, o]+\mathbb{E}_Q U$ equals the free energy of the tilted joint $Z_U^{-1}e^{-U}\mathbb{P}_{\theta,V}$ minus $\log Z_U$ (MT (4.18)). Because $\log Z_U$ does not depend on Q, recognition updates under the appended objective are those of the tilted model; because it depends on (θ, X), comparisons across parameter values, and hence learning, are off by $\log Z_U$ unless it is restored.

4.2.4 The typing prohibition

The standing hypothesis MT Hypothesis 4.3 states that every generative factor and record kernel is a function only of the parameters θ, the structural data X, and its displayed latent coordinates. No generative kernel takes as an argument the recognition law, one of its marginals or parameters, or the posterior. In the two-argument form $\mathcal{F}_V(\mathbb{Q}; \theta)$ of the free energy [55], every evaluation holds the generative argument fixed while the recognition argument varies, which excludes the composite $\mathbb{Q} \mapsto \mathcal{F}_V(\mathbb{Q}; \theta(\mathbb{Q}))$ inside one evaluation. The chapter's example of what is forbidden is a transition built from a live peer recognition marginal, such as agent i's prior set to the current transported belief of agent j. What is allowed is coupling through latents (agent i's state factor reads agent j's sample), a parameter that depends on the observation or on an earlier recognition law provided it is held fixed while the bound is evaluated, slow evolution of an agent's model as recognition dynamics of the model channel or as accepted M-coordinates, each accepted value defining a new fixed joint, and loops of inference updates under a fixed joint.

Why forbid it? The manuscript's answer is MT Proposition 5.11: a family of joints indexed by recognition laws satisfies the evidence identity for each member separately, but the change of the bound between two members mixes a change of evidence with a change of gap, and the family designates no joint as the model. A companion witness shows that the stronger failures one might expect, such as evidence that must vary along the family, need not occur (MT Proposition 5.12). The prohibition is therefore a commitment to having one target, not the avoidance of a contradiction.

Physics analogue (structural)

In naive mean-field theory for the Ising model, the Gibbs–Bogoliubov functional $\mathbb{E}_Q H - TS[Q]$ over product trial laws uses the fixed Hamiltonian $H = -J\sum_{\langle ij\rangle} s_i s_j$, which reads samples. Scoring Q against the effective Hamiltonian $H_{\text{eff}}(s; m) = -J\sum_i\big(\sum_{j\sim i} m_j\big)s_i$ with the current magnetizations $m = \mathbb{E}_Q s$ inserted counts every bond twice, which is why the mean-field free energy carries its familiar correction $+J\sum_{\langle ij\rangle} m_i m_j$. Holding m fixed during each variation and updating it between variations gives the same stationarity equations. The first procedure is the

composite the typing prohibition excludes; the second is the admissible parameter held fixed while the bound is evaluated. The illustration is the companion's, not the manuscript's.

4.2.5 Passive re-expression moves the whole joint

A passive re-expression acts on latent samples by the block map $R = \bigoplus_i (R_i^b \oplus R_i^m)$, leaves the records alone, and transforms the frames and links in X by the vertex law. The kernel-covariance hypothesis (MT Hypothesis 4.8) asks that re-expressed parameters (θ', X') exist for which every factor is the pushforward of the old one and every record kernel reads the rotated latents as it read the old ones. Under it, the new joint is the pushforward of the old one as a measure, with no density hypothesis, the observation marginals agree, and the evidence is unchanged for almost every observation (MT Proposition 4.9). Relative entropy is invariant under bijections, so a recognition law transported as a whole keeps its divergence to the transported posterior and its extended bound, singular branch included (MT Corollary 4.10).

Physics analogue (exact)

This is a change of integration variables in a partition function with gauge-invariant observables: the records are the observables, and the Jacobian $|\det R|^{-1}$ appears in densities but cancels from every ratio, which is why the measure-level statement needs no density hypothesis.

4.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Structural data X; parameters θ	X holds the directed graph, the multigraph with links, and the frames; θ holds the numerical parameters that learning may move; the split is declared.	Lattice and background fields; couplings
Directed acyclic graph Γ; parents $\mathrm{pa}(i)$	Who generates whom among latents.	A causal ordering; none in equilibrium statistical mechanics
Model factor $P^m_{\theta,i}$	Law of agent i's model presentation given its parents' models.	Hierarchical prior on couplings
State factor $P^k_{\theta,i}$; evaluated kernel K^X_{i,m_i}	Law of agent i's belief sample given its parents' beliefs and its own model; the same kernel indexed by the model sample.	Conditional Boltzmann weight at fixed couplings
Agent mechanism $G_{\theta,i}$	Model factor then state factor, one normalized kernel from the parents' block to the agent's block.	Local stochastic update rule
Root; law bridge $\widetilde{\Phi}_i^{\mathrm{gen}}$	An agent without parents; the belief law induced by a law over its model.	Posterior-predictive law

Term or symbol	Plain meaning	Physics analogue
Directed latent law $\mathbb{P}^{\mathsf{Y}}_{\theta,V}$	Ordered composition of the mechanisms along a topological order.	Bayesian-network prior
Record kernel K_a; energy $E_{a,o}$	Normalized law of record a given the latents in its scope; its negative log density at the observed value.	Likelihood factor; bond energy of the posterior
Joint $\mathbb{P}_{\theta,V}(dy, do \mid X)$	The directed latent law times the record kernels.	Prior times likelihood
Undirected model; potentials ψ; Z_X	A Gibbs model on agents, edges, and records with its partition function.	Boltzmann distribution
Appended energy U; tilted joint; Z_U	An energy added to a free energy without being part of the joint; the joint it would define; its normalizer.	A potential added without renormalizing
Typing prohibition	No generative kernel reads a recognition law, its marginals or parameters, or the posterior.	Hamiltonian independent of the trial state
Two-argument free energy $\mathcal{F}_V(\mathbb{Q};\theta)$	Free energy as a function of recognition law and parameters, varied one at a time.	Gibbs–Bogoliubov functional
M-coordinate; accepted value	A change of generative parameters accepted between evaluations.	M-step of expectation maximization
Maps R, T; gauge inert	The passive action on latents, and on latents and records; records are unchanged.	Change of variables; gauge-invariant observables
Kernel covariance	The model's coordinate form is compatible with re-expressed parameters.	Gauge covariance of the action
Reference factor j_R	Radon–Nikodym factor of the reference under R; $\lvert\det R\rvert^{-1}$ for invertible linear blocks.	Jacobian
Transported coupling $R_{\#}\mathbb{Q}_{V,o,X}$	A recognition law moved as a whole to the new frames.	Gauge-transformed trial state

The word "fixed" in the chapter title means held fixed while the evidence bound is evaluated; the joint changes between evaluations when an M-coordinate is accepted, and each accepted value defines a new joint. "Model" has two meanings: the agent's model channel (m_i and its law q_i^m), and the generative model $\mathbb{P}_{\theta,V}$. The model and state factors are generative factors and are not components of the agent of chapter 2, even though they carry the agent's index; the agent's belief q_i^b is a recognition object. "Cycle" refers to the record hypergraph; Γ is acyclic by definition. "Covariance" in the hypothesis name means gauge covariance, not a second moment. The evidence invariance holds for almost every observation, not pointwise.

4.4 The formulas, explained

The agent mechanism.

$$G_{\theta,i}\big(d(k_i, m_i) \mid y_{\mathrm{pa}(i)}, X\big) = P^{m}_{\theta,i}(dm_i \mid m_{\mathrm{pa}(i)}, X) K^{X}_{i,m_i}(k_{\mathrm{pa}(i)}, dk_i)$$

(MT (4.4)). The model presentation m_i is drawn first from its parents' models, and the belief sample k_i then from its parents' beliefs using the kernel that m_i evaluates to. At a root the first factor is a proper law and the second a bridge from the agent's own model to its belief; averaging the bridge over a model law s gives the belief law $\widetilde{\Phi}^{\mathrm{gen}}_i(s)$ (MT (4.5)), which may be declared to represent the cross morphism of chapter 2. At a nonroot the evaluated kernel is indexed by the parent beliefs and is not a map of law fibers.

The joint and its normalization.

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X) \prod_{a \in \mathfrak{A}} K_a(X, y_{\partial a}, do_a), \qquad \mathbb{P}_{\theta,V}(\mathsf{Y} \times \mathsf{O} \mid X) = 1$$

(MT (4.10), MT (4.11)). The latent law is the ordered composition of the mechanisms along a topological order (MT (4.6)), and each record kernel has unit mass in its own coordinate, so integrating the records first removes the product whatever the overlaps of the scopes. Each record is owned once by the joint, while the local free energies of chapter 5 include every factor incident to a coordinate, as a bond energy is counted once in the Hamiltonian but enters the local field of both of its sites.

The posterior as a Gibbs measure. Rewriting MT (4.12) with the posterior formula of MT Hypothesis 3.8, on the regular set of observations,

$$\mathbf{\Pi}_{\theta,V,o,X}(dy) = \frac{\exp\big(-\sum_a E_{a,o}(y_{\partial a})\big)\mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X)}{p_{\theta,V}(o \mid X)},$$
$$p_{\theta,V}(o \mid X) = \int \exp\Big(-\sum_{a \in \mathfrak{A}} E_{a,o}(y_{\partial a})\Big)\mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X).$$

With no records the posterior is the prior and the evidence is 1. Cycles in the scopes make the posterior's factor graph loopy, which matters for computing it and not at all for its normalization.

The running example. Agent 1 is the root, with $k_1 \mid m_1 \sim \mathcal{N}(0, e^{-m_1} I_3)$; agents 2 and 3 are its children, with $k_2 \mid k_1, m_2 \sim \mathcal{N}(\rho_b(\Theta^b_{\bar{e}_1})k_1, e^{-m_2} I_3)$ and $k_3 \mid k_1, m_3 \sim \mathcal{N}(\rho_b(\Theta^b_{e_3})k_1, e^{-m_3} I_3)$, and independent standard normal model coordinates. The record on $\{2, 3\}$ has success probability $\exp(-\frac{1}{2}|k_2 - \rho_b(\Theta^b_{e_2})k_3|^2)$. At the conditional means of the children its residual is

$$\rho_b(\Theta^b_{\bar{e}_1})k_1 - \rho_b(\Theta^b_{e_2}\Theta^b_{e_3})k_1 = \rho_b(\Theta^b_{e_1})^{-1}\big(I_3 - \rho_b(\mathrm{Hol}^b(p))\big)k_1,$$

whose length is $\sqrt{3}$ times the component of k_1 orthogonal to the rotation axis of the holonomy (the running-example paragraph after MT Proposition 4.4). A record that closes the loop measures how far going around the loop rotates the root's state; the records see the holonomy through an invariant, and with trivial holonomy the residual vanishes.

The Gaussian counterexample.

$$\psi_i(k_i, m_i; X) = (2\pi)^{-1/2} e^{-k_i^2/2}, \qquad \psi_{12}(y_1, y_2; X) = e^{\mathrm{c} k_1 k_2}, \qquad \mathrm{c} \geq 1,$$
$$\mathsf{W}_\mathrm{c} = \left(\begin{smallmatrix} 1 & -\mathrm{c} \\ -\mathrm{c} & 1 \end{smallmatrix}\right), \qquad \operatorname{spec} \mathsf{W}_\mathrm{c} = \{1 - \mathrm{c}, 1 + \mathrm{c}\}$$

(MT (4.16), MT (4.17)). The product of potentials is $\exp(-\frac{1}{2} k^\top \mathsf{W}_\mathrm{c} k)$ up to a constant; in the sum and difference coordinates the integral factorizes, and the sum direction, with eigenvalue $1 - \mathrm{c} \leq 0$, diverges. For $0 \leq \mathrm{c} < 1$ the same model is a legitimate joint.

The appended energy.

$$\mathcal{F}_V[Q; X, o] + \mathbb{E}_Q U = \widetilde{\mathcal{F}}_V[Q; X, o] - \log Z_U, \qquad Z_U = \int e^{-U(y)} \mathbb{P}_{\theta,V}(dy, do \mid X) \in (0, \infty)$$

(MT (4.18)). The left side is what one computes when an energy is simply added; the right side is the free energy of the tilted joint, which does contain U, corrected by its normalizer. The identity holds on the classical domain with $\mathbb{E}_Q |U| < \infty$.

Covariance. Under kernel covariance,

$$\mathbb{P}_{\theta',V}(\cdot \mid X') = T_\# \mathbb{P}_{\theta,V}(\cdot \mid X), \qquad p_{\theta',V}(o \mid X') = p_{\theta,V}(o \mid X),$$
$$p_{\theta',V}(o, y' \mid X') = |\det R|^{-1} p_{\theta,V}(o, R^{-1} y' \mid X),$$

the second for almost every o and the third for invertible linear blocks with Lebesgue references (MT (4.27), MT (4.28), MT (4.31)). For orthogonal representations $|\det R| = 1$ and densities simply move; for $\mathrm{GL}(K)$ they pick up the Jacobian, while measures, evidence, and divergences do not. The belief and model channels may have different dimensions, and the reference must be the same within each channel. The root law bridges transform as cross-morphism representatives do, which ties the generative construction to the geometry of chapter 2. The corollary then gives

$$D_{\mathrm{KL}}(\mathbb{Q}'_{V,o,X'} \| \mathbf{\Pi}_{\theta',V,o,X'}) = D_{\mathrm{KL}}(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}), \qquad \mathcal{L}'^{\mathrm{ext}}_V(\mathbb{Q}'_{V,o,X'}; X', o) = \mathcal{L}^{\mathrm{ext}}_V(\mathbb{Q}_{V,o,X}; X, o)$$

for $\mathbb{Q}'_{V,o,X'} = R_\# \mathbb{Q}_{V,o,X}$ (MT (4.32)), the recognition law being selected before the re-expression and transported as a whole.

4.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
Directed construction: MT Definition 4.1, MT Definition 4.2	Latents are generated along a declared acyclic graph, model before belief within each agent; the split between structural data and parameters is declared.	CHOICE	Excludes reciprocal latent influence; loops must be records.
Normalization: MT Proposition 4.4	Latent composition is a probability kernel; with records conditionally independent given the latents, the joint is normalized for every finite record hypergraph, and its density is the latent density times $\exp(-\sum_a E_{a,o})$.	SOLID	Standard in substance (Bayesian networks with observed children).
Local normalization fails: MT Proposition 4.6	Normalized agent potentials and a positive, finite, continuous edge potential need not give a finite partition function.	SOLID	A general negative statement; the two-site Gaussian witness is textbook.
Undirected alternative: MT Definition 4.5	A Gibbs model with its partition function is a legitimate joint whenever $0 < Z_X < \infty$.	CHOICE	Admissible, but the finiteness check is parameter dependent and no general criterion is given.
Appended energy: MT Remark 4.7	Adding a bounded-below energy to a free energy equals the free energy of the tilted joint minus $\log Z_U$.	SOLID	Elementary; the consequence is for learning, since $\log Z_U$ depends on (θ, X).
Typing prohibition: MT Hypothesis 4.3	No generative factor or record kernel reads a recognition law, its marginals or parameters, or the posterior; the composite $\mathcal{F}_V(\mathbb{Q}; \theta(\mathbb{Q}))$ is excluded inside one evaluation.	CHOICE	A standing hypothesis; it admits fixed parameters updated between evaluations.
No distinguished target: MT Proposition 5.11, MT Proposition 5.12	Recognition-indexed joints obey the identity member by member but designate no target; the stronger failures need not occur.	SOLID	Justifies the prohibition as a choice of one target, not as avoidance of inconsistency.
Kernel covariance: MT Hypothesis 4.8	Re-expressed parameters exist for which every factor and record kernel transforms covariantly.	CHOICE	Excludes mechanisms compatible with no re-expression of their parameters.
Joint covariance: MT Proposition 4.9	The re-expressed joint is the pushforward of the old one; evidence is invariant for almost every observation; densities pick up j_R, equal to $\lvert\det R\rvert^{-1}$ for linear blocks; root bridges transform as cross-morphism representatives.	SOLID	Standard in substance (change of variables).

Result	What it says	Status	Scope and caveats
Bound invariance: MT Corollary 4.10	For a coupling transported as a whole, the posterior transforms by pushforward and the divergence and extended bound are unchanged.	SOLID	A coupling reselected by a rule in the new frames is not covered; selector equivariance is declined.

Conditions: which ones matter

Conditions that matter: the latent graph Γ must be acyclic, which excludes reciprocal generative influence and forces loop interactions into records. Records must be conditionally independent given the latents; correlated record noise has to be modeled by an added latent coordinate. Each record kernel must be a normalized law of its record for every latent value, which is what makes normalization automatic and what an arbitrary pairwise potential is not. The typing prohibition and kernel covariance are modeling commitments, the first excluding peer-belief priors, the second excluding mechanisms whose coordinate form fixes a frame. Finite positive evidence at an admitted observation is a standing default; it is automatic for binary records whose success probabilities are positive on a set of positive prior mass, and must be checked for density records. For the undirected alternative, finiteness of Z_X is a parameter-dependent condition that can fail.

Hygiene: measurability of the kernels and of their composition, Tonelli's theorem, jointly measurable log densities for the density formula, and the reference condition $R_{\#}\nu^{\mathsf{Y}} \ll \nu^{\mathsf{Y}}$ for the density form of covariance, automatic for invertible linear blocks with Lebesgue reference.

The chapter has no toy results: the two-site Gaussian counterexample refutes a general implication, and the running example is an illustration.

4.6 Open ends

The chapter records no open problem of its own, and its scope paragraph declines two things. Cross morphisms supply no generative factor, and a model with factors in both directions between the belief and model channels needs its own normalized joint, which the chapter does not construct; the directed order, model before belief, is part of the construction rather than a convenience. No quotient of the parameter space by passive re-expression is formed, and no parameter count subtracts the dimension of the vertex group, because its action can have nontrivial stabilizers. A physicist used to counting physical degrees of freedom as total minus gauge should note that the theory does not make this count. Two further limits follow from the construction without being flagged as open. Reciprocal couplings between latents are admitted only through the undirected model with a verified finite partition function, and no general criterion for that finiteness is supplied. And the one way around the typing prohibition that keeps a single target, promoting beliefs to latent law points so that a generative factor may read them, requires the normalized law-point prior that is MT Open Problem 3.23.

Does it deliver what it claims? DELIVERED

The chapter says that it builds one normalized joint law of latent samples and records at c_*, normalized for every finite record hypergraph, cyclic ones included; that local normalization of reciprocal latent conditionals supplies no joint; and that under kernel covariance the evidence and the bound of a transported coupling are invariant. The reading map adds that it states the typing prohibition. All of this is proved or stated as written. The normalization and covariance results are standard in substance: a Bayesian network with observed children is normalized whatever the loops of its moral graph, and a change of variables with gauge-invariant observables preserves evidence and relative entropy. The substance is in the commitments the chapter makes explicit: generative influence is acyclic, loop interactions enter as records and therefore act only after conditioning, reciprocal latent couplings need a separate finiteness check that can fail, and no factor may read a recognition law. The last commitment is justified as a choice of one target rather than by an inconsistency in the alternative, which is what the cited propositions actually show. Two narrowings should be kept in view: "cycles" means cycles of records, not of generative influence, and the invariance of the bound is for a recognition law transported as a whole, not one recomputed by a selection rule in new frames.

Chapter 5

One Evidence Bound and Local-Collective Free Energy

5.1 The question and the answer

The foundations chapters (chapters 3 and 4) built two objects and kept them apart: the constructed joint $\mathbb{P}_{\theta,V}(dy, do \mid X)$, one normalized law over every latent coordinate and every record of the population, normalized on any record hypergraph, cycles included (MT Proposition 4.4); and the recognition law $\mathbb{Q}_{V,o,X}$, one joint law over the same latents, selected as a coupling of the agents' local recognition laws and correlated in general (MT Definition 3.15). The technical chapter, MT chapter 5, asks what exact relation binds the two, and how an agent's or a block's objective relates to the population's.

The answer has two parts. The first is one number for the whole population,

$$\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \log p_{\theta,V}(o \mid X) - D_{\mathrm{KL}}\big(Q \| \mathbf{\Pi}_{\theta,V,o,X}\big),$$

defined for every probability law Q on the latent space, equal to the log evidence exactly when Q is the posterior as a measure, and equal to $-\infty$ when Q puts mass where the posterior puts none (MT Definition 5.6, MT Corollary 5.7). The second is the local–global potential identity (MT Theorem 5.23): hold the law of everything outside a block B fixed and change only the block's conditional recognition law; the collective free energy then changes by exactly the outside average of the change of a local free energy, the free energy of the block's own inference problem with its surroundings frozen. For single agents and product recognition laws this makes the local objectives an exact potential game with the collective free energy as potential (MT Corollary 5.24). The bound is a definition; every local object is the chain rule for relative entropy (MT Lemma 3.12) applied to it. The chapter also prices the error of summing local entropies (total correlation), states what ELBO values cannot compare, types observations as records, counts shared records correctly, treats links as random variables, and fixes how parameters may move. The bound is the objective of chapters 6 to 8, and chapters 9 and 12 carry it along evidence-preserving channels. The scope paragraph declines four claims: existence of a maximizer over a restricted family, tightness for any restricted family, convergence of the alternating scheme, and that a block with its local free energy is a meta-agent.

Is this more than the Gibbs variational principle? The bound is not. Reading $-\log p_{\theta,V}(o, y \mid X)$ as an energy at unit temperature, $\mathcal{L}_V^{\mathrm{ext}} = -\mathcal{F}$ and $\log Z = -\mathcal{F}[Q] + D_{\mathrm{KL}}(Q\|\mathbf{\Pi})$ is the Gibbs variational principle, the inequality behind the Gibbs–Bogoliubov–Feynman bound. What the chapter adds is typing and bookkeeping: the target is one fixed joint, never a family indexed by Q; singular trial laws get the value $-\infty$ instead of an undefined expression; the recognition law is a correlated coupling, and the error of treating it as a product is named and signed; and interactions enter as records, so cyclic interaction graphs sit inside a normalized directed joint. Each is careful, and each is standard in substance.

Is the potential identity more than block coordinate ascent? In mechanism, no. With a product baseline and product laws, each agent's best response is the textbook mean-field update [10], and the potential-game statement says these best responses descend one functional. The identity is more general in one respect: the block law may be any conditional kernel $r_B(dy_B \mid y_{-B})$, so it covers correlated recognition and exact block updates (a heat-bath step in law). It says nothing about simultaneous updates of several correlated blocks, about equilibria, or about convergence.

What do observations as interaction records buy? A uniform type. Environmental observations and agent–agent interactions are both records with a scope and a normalized kernel; pair interactions become likelihoods of observed records, which puts a cyclic Hamiltonian inside a normalized joint and turns its partition function into the evidence; and an environment can be rewritten as a boundary node emitting records (MT Theorem 5.17). That rewriting is exact for the joint law, not for variational inference over the node's internal randomness (MT Remark 5.19).

What converges? Evidence values, when each accepted parameter step follows a full exact posterior refresh (MT Proposition 5.30): classical EM monotonicity. Nothing is proved about the parameters (MT Open Problem 5.32), and finite natural-gradient steps can lower the bound (MT Proposition 5.31).

5.2 The picture

Read the constructed joint as a statistical-mechanics model on a hypergraph. The latent baseline $\mathbb{P}^{\mathsf{Y}}_{\theta,V}$ is the a priori measure on configurations, correlated in general by the directed latent model of chapter 4. Each record a touches a set of agents, its scope ∂a, and contributes a Boltzmann factor $e^{-E_{a,o}(y_{\partial a})}$, with $E_{a,o}$ the negative log density of the record at its observed value. The sum $H_o = \sum_a E_{a,o}$ is a Hamiltonian with one term per record, pair records being bonds. The posterior slice is the Gibbs measure of H_o and the evidence is its partition function (MT (5.12), MT (5.13)):

$$\mathbf{\Pi}_{\theta,V,o,X}(dy) = \frac{e^{-H_o(y)}\mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy)}{p_{\theta,V}(o \mid X)}, \qquad p_{\theta,V}(o \mid X) = \int e^{-H_o} d\mathbb{P}^{\mathsf{Y}}_{\theta,V}.$$

Physics analogue (exact)

Gibbs variational principle. For every trial state $\mathcal{F}_o^{\mathrm{ext}}(Q) \geq -\log p_{\theta,V}(o \mid X)$, with gap $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X})$ and equality exactly at the Gibbs state; under absolute continuity to the baseline, finite relative entropy, and an integrable negative energy part, $\mathcal{F}_o^{\mathrm{ext}}(Q) = \mathbb{E}_Q H_o + D_{\mathrm{KL}}(Q\|\mathbb{P}_{\theta,V}^{\mathsf{Y}})$. This is the variational characterization of $-\log Z$ behind mean-field methods [75], with relative entropy to the a priori measure in place of Shannon entropy. MT Corollary 5.7 is this statement for every Q in the extended reals. A point mass on a continuous configuration space has free energy $+\infty$, as a delta-function state of a continuous system has entropy $-\infty$; that is the singular branch.

Fix the state y_{-B} outside a block B. Records wholly outside B become constants; records touching B define a conditional Gibbs measure on the block with its own partition function $Z_B(y_{-B})$.

Physics analogue (exact)

Finite-volume Gibbs specification. The local posterior $\mathbf{\Pi}_{\theta,V,o,X,B}(dy_B \mid y_{-B}) \propto e^{-H_{B,o}(y_B;y_{-B})}\mathbb{P}_{\theta,V,B}^{\mathsf{Y}}(dy_B \mid y_{-B})$ of MT Lemma 5.20 is the Gibbs specification in volume B with boundary condition y_{-B} [33], with the baseline's conditional law as a priori measure; Z_B is the finite-volume partition function, and the local free energy of MT Definition 5.21 is the finite-volume free energy of a trial block state. The identification holds at every finite N.

Treat each agent as a player lowering its own local free energy, averaged over what the others believe. Because each player's cost changes by exactly the change of one collective functional, unilateral improvement is collective improvement.

Physics analogue (exact)

Mean-field coordinate ascent, exact on product profiles. With a product baseline $\bigotimes_i \varrho_i$ and product laws, minimizing agent i's averaged local free energy gives the familiar $r_i \propto \varrho_i \exp(-\mathbb{E}_{Q_{-i}} H_{\{i\},o})$ [10], and the potential-game corollary says these updates descend one functional. On product profiles the correspondence is exact; the identity is strictly more general, because it also covers correlated block conditionals. The update is the companion's orientation; the technical chapter does not display it.

5.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
$\mathbb{P}_{\theta,V}(dy, do \mid X)$, constructed joint	One law for all latent states and records	A priori measure times Boltzmann factors
$\mathbb{P}_{\theta,V}^{\mathsf{Y}}$, latent baseline	Law of the latents before records; may correlate agents	A priori measure

Term or symbol	Plain meaning	Physics analogue
Record a, scope ∂a, kernel K_a, energy $E_{a,o}$	A datum attached to a set of agents, its likelihood, and its negative log density	Bond or plaquette term
L_o, $H_o = -\log L_o$	Product of record likelihoods; total interaction energy	Boltzmann weight; Hamiltonian
$p_{\theta,V}(o \mid X)$; $\mathbf{\Pi}_{\theta,V,o,X}$	Evidence; posterior slice at the observed records	Partition function; Gibbs state
$\mathbb{Q}_{V,o,X}$, recognition coupling	Trial joint law over all agents' latents, correlated in general	Trial state
$\mathrm{Cpl}(\cdot)$, coupling	Set of joint laws with given marginals	None (not a coupling constant)
$\mathcal{L}_V^{\mathrm{ext}}$, $\mathcal{F}_V^{\mathrm{ext}}$	ELBO and free energy for every Q, values $\pm\infty$ allowed	Minus the trial free energy
Singular branch	Q charges a posterior-null set; bound $-\infty$	Delta-function state of a continuum
(H1)–(H4)	Regular posterior version, finite positive evidence, absolute continuity, log-integrability	Hygiene, except (H3)
$\mathrm{TC}(Q)$	Relative entropy of Q from the product of its block marginals	Multi-information
Restriction to $\mathcal{Q}'$	Best bound over an imposed recognition family	Variational ansatz
Block B, outside $-B$	Any set of agents and its complement	Finite volume and exterior
Z_B, $\mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid y_{-B})$, $\mathcal{F}_{B,o}^{\mathrm{ext}}$	Local evidence, local posterior, local conditional free energy	Finite-volume partition function, specification, free energy
E-coordinate; M-coordinate, accepted	Exact block posterior update; parameter step that does not lower the bound	Heat-bath update in law; EM M-step
Typing prohibition	No generative factor reads the recognition law or the posterior	A Hamiltonian independent of the trial state

Several words carry nonstandard meanings. A *coupling* is a joint law with prescribed marginals, never an interaction strength. A *record* is any observed datum on a set of agents, interactions included. *Local* means block-conditional: at one base point there is no spatial locality, and a block is any set of agents. The *potential* of a potential game is game-theoretic [53]. *Free energy* is dimensionless, $\mathcal{F} = -\mathcal{L}$ at unit temperature. *Slice* means conditioning at the observed value, which is why posteriors hold "for almost every o". The posterior $\mathbf{\Pi}$ is unrelated to the attention prior π_{ij} of chapter 6, and the Hamiltonian H_o to its message spaces $H_e^{\pm}$.

5.4 The formulas, explained

5.4.1 The bound, its energy form, and the singular branch

The extended bound $\mathcal{L}_V^{\mathrm{ext}}(Q) = \log p_{\theta,V}(o \mid X) - D_{\mathrm{KL}}(Q\|\mathbf{\Pi}) \in [-\infty, \log p_{\theta,V}(o \mid X)]$ (MT (5.15)) never forms $\infty - \infty$: the evidence term is finite by (H2) and relative entropy lies in $[0, +\infty]$. Under absolute continuity with respect to the baseline, finite relative entropy, and an integrable negative part of the energy,

$$\mathcal{F}_o^{\mathrm{ext}}(Q) = D_{\mathrm{KL}}\big(Q\|\mathbb{P}_{\theta,V}^{\mathsf{Y}}\big) + \mathbb{E}_Q H_o$$

(MT (5.19)): a complexity cost measured against the prior plus the expected misfit to the records. With no records the free energy is minimized at the prior with value zero; at $Q = \mathbf{\Pi}$ it equals $-\log p_{\theta,V}(o \mid X)$.

The singular branch is where (H3) fails: Q charges a posterior-null set and the bound is $-\infty$, for example a law on a proper affine subspace when the posterior has a Lebesgue density. A proper closed support alone does not suffice: the uniform law on $[0, \frac{1}{2}]$ is at divergence $\log 2$ from the uniform law on $[0, 1]$ (MT Proposition 5.4). For this program the branch is not a technicality: point-mass recognition of continuous law points lives there, which decides the status of the softmax over current beliefs in chapter 6, and so does smooth recognition of an environment's internal randomness (section 5.4.4).

The classical form $\log p = \mathbb{E}_{\mathbb{Q}}[\log p_{\theta,V}(o, Y \mid X) - \log q(Y)] + D_{\mathrm{KL}}(\mathbb{Q}\|\mathbf{\Pi})$ (MT (5.22)) holds under (H1)–(H4), with equality exactly at the posterior as a measure; equal coordinate marginals are not enough (MT Theorem 5.8). The extended form exists because the classical split can be undefined where the bound is not: a Cauchy trial law against a Gaussian posterior, or an infinite-entropy countable law that is its own posterior (MT Proposition A.7). In the program's Gaussian models (H4) holds, and these witnesses are heavy-tailed pathologies.

5.4.2 Correlated recognition and total correlation

For any law Q of blocks and any reference laws ϱ_j,

$$D_{\mathrm{KL}}\Big(Q \,\Big\|\, \bigotimes_j \varrho_j\Big) = \mathrm{TC}(Q) + \sum_j D_{\mathrm{KL}}(Q_j\|\varrho_j), \qquad \mathrm{TC}(Q) = D_{\mathrm{KL}}\Big(Q \,\Big\|\, \bigotimes_j Q_j\Big)$$

in $[0, +\infty]$ (MT Corollary 5.1). The total correlation vanishes exactly for product laws and is $+\infty$ for two perfectly correlated continuous copies. As a standard orientation, a Gaussian with scalar blocks and covariance Σ has $\mathrm{TC} = \frac{1}{2}\log(\prod_j \Sigma_{jj}/\det\Sigma)$. If each agent reports its own entropy and the population adds them, it computes a pseudo-ELBO with

$$\widetilde{\mathcal{L}} = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + \mathrm{TC}(\mathbb{Q}_{V,o,X})$$

(MT Proposition 5.2): the overstatement is exactly the total correlation, and the result need not bound the evidence. Since local marginals do not determine the coupling once two are non-Dirac (MT Proposition 3.17), an exact bound needs the coupling.

5.4.3 Restriction, and what ELBO values cannot compare

For any family $\mathcal{Q}'$, $\sup_{\mathcal{Q}'} \mathcal{L}_V^{\mathrm{ext}} = \log p_{\theta,V}(o \mid X) - \inf_{\mathcal{Q}'} D_{\mathrm{KL}}(Q\|\mathbf{\Pi})$ (MT Proposition 5.9): the best bound in an ansatz is the evidence minus a reverse information projection, larger families do at least as well, and attainment is not required. Excluding the posterior does not make the optimized gap positive, and product families are not convex, so strict convexity gives no uniqueness on them (MT Remark 5.10).

Two negative results fix what ELBO values mean. Across unrelated models the difference of optimized bounds is an evidence difference minus a tightness difference, and binary witnesses show its sign orders neither models nor recognition families (MT Proposition 5.13): comparing variational free energies of two Hamiltonians computed with different ansatzes says nothing about their true free energies. And if the joint moves with the trial law, each Q has its own evidence and the family designates no model (MT Proposition 5.11); the witness of MT Proposition 5.12 shows such dependence need not be inconsistent. The typing prohibition (MT Hypothesis 4.3) therefore rests on the absence of a selected target, not on a derived contradiction. Its physical content: a self-consistent-field Hamiltonian $H[Q]$ is not one variational problem, and self-consistency may appear only as stationarity of one fixed functional.

5.4.4 Observations as records

The record hypothesis (MT Hypothesis 5.5) gives every record a density $e^{-E_{a,o}}$ against one dominating measure. A pair energy $E(y_i, y_j) \geq 0$ enters, for example, as a binary success record observed as a success with probability e^{-E}. This is how a Boltzmann factor on a cyclic graph becomes a likelihood in a normalized directed joint; it is a modeling commitment that coupling energies are data. The environment theorem is the randomization lemma [44]: every observation kernel is the output $O = F(Y, U)$ of a node with private uniform randomness U, and conversely (MT Theorem 5.17). Given o, however, the posterior of (Y, U) sits on the level set $\{F(y, u) = o\}$, a null curve in the Gaussian example, so every smooth trial law on (Y, U) has infinite cost, and a finite problem integrates U out or restricts recognition to the fiber $\{F(y, u) = o\}$, where the cost is that of the induced law on Y (MT Remark 5.19). The information a record carries is $I(Y; O) = \int D_{\mathrm{KL}}(P(dY \mid o)\|P^Y)P^O(do)$, and a statistic S of the record preserves the posterior exactly when $I(Y; O \mid S) = 0$ (MT Proposition 5.18), standard facts [15] stated in $[0, +\infty]$.

5.4.5 Local free energy, exact block updates, and the potential identity

The local conditional free energy of a trial block law is

$$\mathcal{F}_{B,o}^{\mathrm{ext}}(r_B; y_{-B}) = -\log Z_B(y_{-B}) + D_{\mathrm{KL}}\big(r_B(\cdot \mid y_{-B})\|\mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid y_{-B})\big)$$

(MT (5.44)): the block's log partition function with the boundary frozen plus its recognition error. It contains every record incident to the block and no other. For $B = V$ it is the collective free energy;

for a block touched by no record, $Z_B = 1$. The underlying fact is the chain rule

$$D_{\mathrm{KL}}\big(Q\|\mathbf{\Pi}\big) = D_{\mathrm{KL}}\big(Q_{-B}\|\mathbf{\Pi}_{-B}\big) + \mathbb{E}_{Q_{-B}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{-B})\|\mathbf{\Pi}_B(\cdot \mid Y_{-B})\big)$$

(MT (5.46)) for $Q = Q_{-B} r_B$; informally, with Q_{-B} fixed only the second term moves. For two trial laws with the same outside marginal and finite divergence,

$$\mathcal{F}_o^{\mathrm{ext}}(Q') - \mathcal{F}_o^{\mathrm{ext}}(Q) = \mathbb{E}_{Q_{-B}}\big[\mathcal{F}_{B,o}^{\mathrm{ext}}(r'_B; Y_{-B}) - \mathcal{F}_{B,o}^{\mathrm{ext}}(r_B; Y_{-B})\big]$$

(MT Theorem 5.23). The fixed outside marginal is load-bearing: when two correlated blocks change at once, each moves the other's outside marginal. Zeroing the conditional term is the exact block E-coordinate: among laws with outside marginal R, the divergence is minimized, uniquely when finite, by R times the exact local posterior (MT Corollary 5.22). This is a heat-bath update in law, not a mean-field update, and a damped version is also monotone, so sequential sweeps do not raise the free energy. For singletons and product profiles with finite divergence and $\mathbb{E}_{Q_{-i}}|\log Z_{\{i\}}| < \infty$, the outside-averaged local free energies make the agents an exact potential game in the sense of Monderer and Shapley [53] (MT Corollary 5.24), with no equilibrium or convergence consequence drawn.

5.4.6 Accounting, and links as random variables

With a product baseline $\bigotimes_i \varrho_i$,

$$\mathcal{F}_o^{\mathrm{ext}}(Q) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\varrho_i) + \mathbb{E}_Q \sum_a E_{a,o}(Y_{\partial a}), \qquad \sum_{i\in V} H_{\{i\},o}(y_i; y_{-i}) = \sum_a |\partial a| E_{a,o}(y_{\partial a})$$

(MT Corollary 5.25). In the first identity each record appears once and mean field is the case $\mathrm{TC} = 0$, not a premise. The second says that single-agent local energies count a record of scope k exactly k times: local free energies are coordinate potentials, not shares, the double counting that region-based approximations correct with counting numbers [85]. A single-count additive split exists by peeling agents off in a chosen order, each term conditioned on the agents before it.

If link values Θ are latent, with prior $Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}} \nu^{\mathscr{T}}$ for a probability reference $\nu^{\mathscr{T}}$, and recognition $R(d\Theta)Q_\Theta(dy)$,

$$\mathcal{F} = \mathbb{E}_R \mathcal{F}_{\mathrm{matter}}(Q_\Theta; o, \Theta) + \mathbb{E}_R \mathscr{S} + D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \log Z_{\mathscr{S}}$$

(MT Proposition 5.26): the matter free energy at frozen links, averaged, plus the expected link action, the link entropy cost, and the link normalizer. Structural links contribute only the matter term. The form is that of lattice gauge theory on the agent graph with links integrated against an action [16], no action being specified. It needs a normalizable link prior; for a noncompact group such as $\mathrm{GL}(K, \mathbb{R})$, Haar measure is not a probability reference.

5.4.7 How model parameters move

A parameter step is accepted when it does not lower the bound at fixed recognition (MT Definition 5.28); the test needs no evidence value, only $\mathbb{E}_{\mathbb{Q}} \log(p_{\theta^{\mathrm{new}}}/p_{\theta^{\mathrm{old}}})$ (MT Corollary 5.29). If recognition equals the old posterior,

$$\log p_{\theta^{\mathrm{new}},V}(o \mid X) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{new}}}(\mathbb{Q}) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{old}}}(\mathbb{Q}) = \log p_{\theta^{\mathrm{old}},V}(o \mid X)$$

(MT Proposition 5.30), the EM monotonicity of Dempster et al. [22] in the free-energy reading of Neal and Hinton [55]. Block E-coordinates keep only the bound nondecreasing; the transfer to the evidence needs the full refresh. A finite natural-gradient step is a proposal: on a quadratic model, from a point displaced along the largest-curvature direction of the preconditioned spectrum, with $d_{\max}$ the corresponding eigenvalue, a step longer than $2/d_{\max}$ lowers the bound (MT Proposition 5.31), so a line search or trust region is needed whatever the preconditioner.

5.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Corollary 5.7, MT Theorem 5.8	One bound for every Q; gap $D_{\mathrm{KL}}(Q\|\mathbf{\Pi})$; equality exactly at the posterior; classical form under (H1)–(H4)	SOLID	Standard in substance (Gibbs variational principle); equal marginals do not give equality
MT Proposition 5.4	Laws charging posterior-null sets have bound $-\infty$	SOLID	Closed support alone is not enough
MT Proposition 5.2	Summing local entropies overstates the bound by exactly the total correlation	SOLID	Standard chain rule
MT Proposition 5.9, MT Proposition 5.13	Best bound in a family is evidence minus a reverse projection; ELBO values of unrelated models order nothing	SOLID	The second is an established negative result
MT Hypothesis 4.3	Generative factors may not read the trial law or the posterior	CHOICE	Justified by MT Proposition 5.11, which is SOLID
MT Hypothesis 5.5	Every datum, interactions included, is a dominated record on a scope	CHOICE	Coupling energies are observed data
MT Theorem 5.17	Observation kernels are environment nodes $F(Y,U)$	SOLID	Standard randomization lemma; not for inference over U
MT Corollary 5.22	Exact and damped block updates do not raise the free energy	SOLID	Sequential only

Result	What it says	Status	Scope and caveats
MT Theorem 5.23	Fixed outside marginal: collective change equals averaged local change	SOLID	Chain rule; finite divergences
MT Corollary 5.24	Agents form an exact potential game	SOLID	Singletons, product profiles only; no equilibrium claim
MT Corollary 5.25	Single-count ledger; singleton energies count record a exactly $\lvert\partial a\rvert$ times	SOLID	Product baseline (used in (i)–(ii))
MT Proposition 5.26	Latent links add action, link entropy, normalizer	SOLID	Normalizable link prior needed
MT Proposition 5.30, MT Proposition 5.31	Evidence monotone under full refresh; long natural-gradient steps can lower the bound	SOLID	Standard; values only, not parameters
MT Open Problem 5.32, MT Open Problem 5.14	Parameter convergence; a criterion inside one bound selecting a recognition structure	OPEN	See section 5.6

The chapter has no TOY results. Its small examples (binary moving-target and model-comparison witnesses, Cauchy and countable integrability witnesses, the uniform diagonal) are counterexamples, SOLID at full generality as refutations.

Conditions: which ones matter

Conditions that matter. Absolute continuity (H3) separates the finite from the singular branch, and it fails for point-mass recognition of continuous latents and for smooth recognition of an environment's randomness, both of which the program meets. Coupling energies must be negative log densities of records with a common dominating measure. The typing prohibition rules out self-consistent-field constructions inside the joint. The potential identity needs the outside marginal fixed and finite divergences; the game needs product profiles. Evidence monotonicity needs a full exact refresh, which no recognition family excluding the posterior can perform, so with restricted families only bound monotonicity survives. Latent links need a normalizable prior, which the Haar measure of a noncompact group is not.
Hygiene. Standard Borel spaces, regular conditional versions (H1), finite positive evidence (H2), joint measurability, almost-everywhere qualifications on local posteriors, and log-integrability (H4) hold in the Gaussian and finite-state models of the program.

5.6 Open ends

Convergence of the alternating scheme (MT Open Problem 5.32) is open for models with finitely many latent states, parameters in an open cone, C^2 joint densities, and an exact M-step with a unique maximizer. The classical conditions [84] bear on isolated stationary points, which this class does not supply: the evidence may be constant along a curve of parameters. It is settled

by checkable conditions, including compact upper level sets of the evidence, met by a model of parameter dimension at least two with a non-discrete stationary set, under which every orbit converges to one stationary point; or by a model with compact upper level sets whose orbit has more than one limit point. For block refreshes the question is weaker still, because a null step is accepted.

An intrinsic criterion (MT Open Problem 5.14) is open: nothing internal to one evidence bound at a fixed model is known to select a nondegenerate recognition structure, such as a partition of agents. Gap minimization selects the largest family, replacing the joint gives no common evidence, and evidence-preserving channels order bounds without selecting a partition. It is settled by a functional on channels or partitions, relative to one joint and not monotone toward an endpoint, with a nondegenerate optimizer, or by a proof of endpoint degeneracy for a stated class. For the program, the free energy alone does not say which agents belong together; formation (chapter 13) is posed through declared families and dynamics instead.

Does it deliver what it claims? DELIVERED

The technical chapter says it gives one extended evidence bound for every recognition law, a local–global potential identity under which the local objectives form an exact potential game, observations as interaction records, and the rules for model updates; it declines existence, tightness, convergence, and meta-agent claims. The proved statements deliver this. The bound is the Gibbs variational principle in the extended reals, the potential identity is the relative-entropy chain rule, the environment theorem is the randomization lemma, and evidence monotonicity is EM monotonicity. None of these is new mathematics; each is a known identity applied with care, and the contribution is typing discipline: one fixed joint, correlated recognition with its total-correlation cost, a value on the singular branch, and explicit negative results on what ELBO values can compare. One headline sentence is broader than its statement: the potential game holds for single agents with product recognition laws, while the block identity behind it covers correlated conditionals one block at a time. There is no parameter convergence, no criterion that selects a partition, and no statement about simultaneous updates.

Chapter 6

Attention and Selective Communication

6.1 The question and the answer

The technical chapter, MT chapter 6, asks which agent a receiver listens to and what it hears. For the first, attention is a posterior over a latent pointer. Each receiving row i carries a categorical label J_i naming one incoming edge copy, with a structural prior row π_i whose support is its source set; the union of supports is the interaction graph, and a mask is a zero prior weight. The row's record has likelihood proportional to $\exp[-D_{ij}(y)/\tau_i]$ when $J_i = j$, with D_{ij} a comparison energy and τ_i a temperature, so Bayes' formula gives the softmax

$$\beta^P_{ij}(y) = \frac{\pi_{ij} e^{-D_{ij}(y)/\tau_i}}{\sum_k \pi_{ik} e^{-D_{ik}(y)/\tau_i}}$$

(MT Proposition 6.2). A recognition row held constant in the latent state, as a transformer computes one row per pass, is optimally a different softmax, over averaged energies, at an exact expected relative-entropy cost (MT Proposition 6.3). For the second, communication is a composed Markov kernel, encoder then transmission then decoder [65], covariant under frame changes and separate recoordinatizations of each message space (MT Corollary 6.11). On the linear tier the composite $\mathsf{L}_e = \mathsf{L}_e^{\mathrm{dec}} \mathsf{L}_e^{\mathrm{tr}} \mathsf{L}_e^{\mathrm{enc}}$ transforms like a link but may be singular (MT Proposition 6.13, MT Proposition 6.14). For a Gaussian comparison record the exact receiver update has the transformer shape, but its value map is fixed by the query map, the key map, the precision, and the receiver covariance, its residual is precision-weighted, and its weights use a predictive covariance (MT Proposition 6.19).

The row calculus feeds chapter 8, the event law is what chapter 11 coarse-grains, and comparison records supply endpoint energies to chapter 10. The scope paragraph declines three claims: that transformer attention as a whole is derived (feed-forward blocks and layer normalization have no counterpart), that declared per-head rows follow from a label model, and that an interface, its rank, or its coding establishes capacity, reconstruction of the sender's belief, or shared meaning. Open are multi-head label models, edge creation and deletion, and formation with evolving interfaces.

Is attention derived or declared? The softmax is derived; the model yielding it is declared. Given a pointer with a prior and a record exponential in a comparison energy, the softmax is Bayes' formula, exactly as mixture responsibilities are [9]; attention as a latent alignment variable is known in machine learning [23]. Not derived: why the joint should contain a pointer, why its record is exponential in D_{ij}/τ_i, or which comparison energy to use. The chapter shows the form is inhabited

by normalized records (a binary success record, a Gaussian comparison record). For this program one more caveat matters: the softmax over divergences between *current beliefs* is the generative posterior row with the law points held fixed, not a recognition optimum of a finite bound, because point-mass recognition of a law point has infinite cost against any non-atomic law-point prior; and the normalized law-point prior that would make such divergences evidence-bound terms at all is open (MT Corollary 6.17, MT Proposition 3.21, MT Open Problem 3.23).

Why so many attention objects? The prior π, the generative row $\beta^P(y)$, the recognition row β^Q, the receiver occupancy α, and the event law $\eta = \alpha\beta$ are laws of different things; β^Q and β^P differ by an exact Jensen gap, and coarse-graining acts on η, not on rows. The table in section 6.3 sets them side by side.

What is the temperature? A parameter of the record's likelihood, not a heat-bath temperature: it sets how sharply the row concentrates, it weights the prior relative entropy in the row functional, and in the Gaussian comparison record only precision over temperature enters (section 6.4.1).

The query map, the key map, and the matrix link? The chapter fixes their types, transformation laws, and linear-algebraic limits, shows how the independent link enters the key map, and, in one model, ties the value map to the query and key maps. It selects, learns, or derives none of them (section 6.4.9).

6.2 The picture

A receiver has a hidden pointer; its attention record is evidence about which source the pointer names, and D_{ij} measures how badly source j explains that record. The posterior over the pointer is a Gibbs distribution over sources with degeneracies π_{ij}, energies D_{ij}, and temperature τ_i.

Physics analogue (exact)

Canonical ensemble over sources, and mixture responsibilities. The row functional $\sum_j \beta_{ij} D_{ij} + \tau_i D_{\mathrm{KL}}(\beta_i \| \pi_i)$ is energy minus temperature times entropy relative to the degeneracies; its unique minimizer is the Gibbs row, and its minimum is τ_i times the row free energy $-\log \sum_j \pi_{ij} e^{-D_{ij}/\tau_i}$ (MT (6.38)). The same formula gives mixture responsibilities [9]. Its natural-gradient flow on the simplex is the replicator equation (MT Corollary 6.7), in the sense of Amari [2].

Physics analogue (exact)

Kalman update with unknown measurement origin. In the Gaussian comparison-record model, the receiver's mean given the pointer is a Kalman update with gain $\Sigma_i S_i^{Q\top} (\Sigma_i^{\mathrm{pred}})^{-1}$ (part (c) of MT Proposition 6.19), the pointer posterior weights sources by their innovation likelihoods, and the posterior mean averages the per-source updates: the structure of data association in tracking.

Physics analogue (structural)

Mean-field Ising free energy. Integrating the pointer out leaves a log-sum-exp potential; in the witness of MT Proposition 6.8 it is $(\omega+1)x^2/(2\tau) - \log\cosh(\delta x/\tau)$, the Curie–Weiss form, whose curvature at the symmetric point changes sign when $\delta^2 > \tau(\omega+1)$. With one latent variable and no thermodynamic limit, this is a bifurcation of posterior modes, not a phase transition.

Physics analogue (structural)

General linear link fields. L_e transforms as $\mathsf{L}'_e = \mathsf{g}_i \mathsf{L}_e \mathsf{g}_j^{-1}$, like a hopping term's link, without being a group element, as in linear lattice gauge theory [77], whose compact-group and continuum conclusions the technical chapter says do not follow here. The trace of a loop product of communication maps is invariant, like a Wilson loop, but is not the holonomy of the independent links (MT Proposition 6.14).

6.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Row i or (i,h); source set $\mathcal{N}_i$	A slot selecting one source (a head row shares agent i's state); the support of its prior	Neighborhood; mask
J_i; π_{ij}	Latent pointer; its structural prior, fixed here	Hidden assignment; degeneracy
$D_{ij}(y)$; τ_i; c_i	Recognition-free comparison energy; temperature; source-independent record factor	Energy; temperature; prefactor
$\beta^P_{ij}(y)$; β^Q_{ij}	Posterior row at latent state y; recognition row, constant in y	Gibbs occupation at fixed y; mean-field occupation
α^x_i; $\eta^x_{ij} = \alpha^x_i \beta^x_{ij}$	Receiver occupancy; law of a (receiver, source) event; mark x is P (generative at y), Π (posterior-averaged), or Q (recognition)	None; joint bond occupation
E^v_e, N^v_e, D^v_e; K^v_e	Encoder, transmission, decoder kernels; composed value kernel	Shannon channel
$\mathsf{L}^{\mathrm{enc}}_e$, $\mathsf{L}^{\mathrm{tr}}_e$, $\mathsf{L}^{\mathrm{dec}}_e$, L_e	Linear factors and their composite $E_j \to E_i$	General linear link
$\mathsf{M}^{\mathrm{val}}$, $\mathsf{M}^{\mathrm{out}}$, U_i	Value extraction, output placement, agent frame	W_V, W_O
$F_e = \mathsf{L}_e \Theta_e^{-1}$; L_p	Receiver processing after the link; loop product of communication maps	Adjoint field; Wilson-loop-like, not holonomy
H^{cmp}_e, S^Q_e, S^K_e; d_e	Comparison space, query and key maps; KL between their images of the two beliefs	W_Q, W_K as measurements
V_e; W_e, $\widetilde{W}_i = W_i/\tau_i$	Quadratic disagreement; comparison precision; tempered precision	Elastic energy; stiffness

Term or symbol	Plain meaning	Physics analogue
Comparison record at $o_i = 0$	Pseudo-observation clamping agreement of query and chosen key	Soft constraint as data
Σ_i^+, Σ_i^{pred}, $\mathsf{L}_e^{\mathrm{val}}$; R_e	Posterior and predictive covariances, value map; Σ_i^{-1}-minimal right inverse of S_i^Q	Kalman covariances, gain times observation map; metric pseudo-inverse

Glyphs collide. τ_i is neither time nor the Langevin temperature T of the formation chapters. The posterior $\mathbf{\Pi}$, the role mark Π, and the source prior π_{ij} are different objects; V is the agent set and V_e an energy; E_i is a sample fiber, $E_{a,o}$ a record energy, E_e^v an encoder, $\mathbb{E}$ an expectation; L_o, L_e, and $\mathcal{L}$ are a likelihood, a communication map, and the ELBO. The *link* is always the invertible group-valued edge variable Θ_e of MT Definition 2.11; L_e, which the technical chapter calls factored matrix communication, is not a link in that sense. The value *kernel* K_e^v is a Markov kernel; the value *map* $\mathsf{L}_e^{\mathrm{val}}$ is a matrix. An *event* is a sampled (receiver, source) edge. "Derived", "declared", "absent" in the transformer table are standings, not proof statuses.

Object	Law of	Depends on	Why it differs from the others
π_i	The pointer before any record	Structure	Its support is the interaction graph; it is the prior in Bayes' formula and never reads recognition.
$\beta_i^P(y)$	The pointer given state and record	y	Exact posterior at a state; its average over y is not a softmax of averaged energies.
β_i^Q	The pointer under recognition	Averaged energies	Differs from β^P by the Jensen gap; equals β^P at the mean state when energy differences are affine.
α_i^x	Which receiver an event belongs to	A declared law	Not supplied by the per-receiver model; a structural selection law or the marginal of a single joint edge label.
η_{ij}^x	The (receiver, source) event	α, β	What coarse-graining pushes forward; rows are recovered by disintegration where $\alpha > 0$, and a row alone is not recursive state.

6.4 The formulas, explained

6.4.1 The softmax, the recognition row, and the temperature

The declared model (MT Definition 6.1) adds independent pointers with priors π_i and gives row i the record density $L_i(o_i \mid y, J_i = j) = \mathrm{c}_i(o_i, y)\exp[-D_{ij}(y)/\tau_i]$ (MT (6.2)). Under exclusivity (no other factor reads a pointer) the posterior factors by rows into the softmax $\beta_{ij}^P(y)$, strictly positive exactly on the source set (MT Proposition 6.2). With nonnegative energies, an included self edge with $D_{ii} = 0$, and equal priors, no source outweighs the self edge: any energy that vanishes on self-comparison, such as a divergence of a belief from itself, makes self-attention maximal under a uniform prior. One label per row also fixes occupancy: every receiver listens to exactly one source, and attending to none needs a null label or another joint label model.

Restricting recognition to rows independent of y, row i contributes

$$\mathcal{F}_i^{\mathrm{att}}(\beta_i^Q) = D_{\mathrm{KL}}(\beta_i^Q \| \pi_i) + \frac{1}{\tau_i}\sum_j \beta_{ij}^Q \mathbb{E}_{\mathbb{Q}_Y} D_{ij}, \qquad \beta_{ij}^{Q\star} \propto \pi_{ij} e^{-\mathbb{E}_{\mathbb{Q}_Y} D_{ij}/\tau_i}$$

(MT (6.7)), a Gibbs row over averaged energies. With R_i^Q the row free energy at averaged energies and $R_i^P(y)$ that at y, the price of the constant row is

$$R_i^Q - \mathbb{E}_{\mathbb{Q}_Y} R_i^P(Y) = \mathbb{E}_{\mathbb{Q}_Y} D_{\mathrm{KL}}\big(\beta_i^{Q\star} \| \beta_i^P(Y)\big) \geq 0$$

(MT Proposition 6.3), zero exactly when every difference $D_{ij} - D_{ik}$ is almost surely constant. The reason, informally, is that the row free energy is concave in the energies. This is the mean-field decoupling of pointer and state, with an exact gap. If energy differences are affine in a statistic of the state, $\beta^{Q\star}$ is the posterior row at the recognition mean (MT (6.12)). For transported quadratic energies $\frac{1}{2} r_{ij}^\top W_i r_{ij}$, $r_{ij} = y_i - \rho_b(\Theta_{ij}^b) y_j$, with uncorrelated receiver and source, it is the row at the mean with source j's logit lowered by $\mathrm{Tr}(W_i \rho_b(\Theta_{ij}^b)\,\mathrm{Cov}(Y_j)\rho_b(\Theta_{ij}^b)^\top)/(2\tau_i)$: uncertain sources are penalized. Computing the softmax at the mean state, as a transformer does, is exact in the affine case, and for transported quadratic energies with uncertain sources it is not. Without the restriction the optimal pointer law is the product of posterior rows, and correlated pointer laws pay a total-correlation term.

The temperature scales the record's exponent: as $\tau_i \to 0$ the row concentrates on minimal energy (hard attention), and as $\tau_i \to \infty$ it returns to the prior; in the row functional it weights the prior relative entropy, the canonical-ensemble role. In the Gaussian comparison record the noise covariance is $\tau_i W_i^{-1}$, so τ_i is a noise scale and only $\widetilde{W}_i = W_i/\tau_i$ enters: temperature and precision are not separately identifiable there. Nothing fixes τ_i; the transformer's $1/\sqrt{\dim H^{\mathrm{cmp}}}$ is the declared choice $\varsigma^2 \tau_i = \sqrt{\dim H_i^{\mathrm{cmp}}}$ for key covariance $\varsigma^2 I$ (MT Table 6.1).

6.4.2 Receiver occupancy and the event law

A row is a conditional law of source given receiver, so coarse-graining needs a receiver law; the event law $\eta_{ij}^x = \alpha_i^x \beta_{ij}^x$ supplies it (MT Definition 6.4). The per-receiver model does not determine α, which enters either as a structural selection law fixed before observation or as the receiver marginal of a single joint edge label in a different model (MT Definition 6.5); the two give different coarse rows (MT proposition 11.31 has masses $1/2$ against $3/4$). With uniform rows and no self edges, blocks of size n among N agents have $\beta_{II}^{\mathrm{c}} = (n-1)/(N-1)$ and $\beta_{IJ}^{\mathrm{c}} = n/(N-1)$ (MT Corollary 6.6): a coarse block listens to itself even when no fine agent does.

6.4.3 The reduced potential, its curvature, and row flows

Summing the pointer out, or optimizing a constant row, leaves $R_i(x) = -\log \sum_j \pi_{ij} e^{-E_{ij}(x)/\tau_i}$, with

$$\nabla R_i = \frac{1}{\tau_i}\sum_j \beta_{ij} \nabla E_{ij}, \qquad \nabla^2 R_i = \frac{1}{\tau_i}\sum_j \beta_{ij} \nabla^2 E_{ij} - \frac{1}{\tau_i^2}\,\mathrm{Cov}_{\beta_i}(\nabla E_i)$$

(MT Proposition 6.8). The gradient is the frozen-weight gradient (the weights' own dependence drops out, as in the Hellmann–Feynman theorem); the Hessian is frozen-weight curvature minus the variance of the sources' forces, the fluctuation term of any $-\log Z$. The correction scales as τ_i^{-2} against τ_i^{-1}, so low temperature favors loss of convexity. In the witness (scalar latent with baseline $\mathcal{N}(0, \tau/\omega)$, sources ± 1, record $\mathcal{N}(x - j\delta, \tau)$) the curvature at the origin is $(\omega + 1 - \delta^2/\tau)/\tau$; at $(\omega, \delta, \tau) = (1, 2, 1)$ the frozen value is 2 and the reduced value -2, though prior and energies are convex.

At fixed state recognition the row follows the replicator equation with cost $\mathsf{c}_{ij} = \log(\beta^Q_{ij}/\pi_{ij}) + \mathbb{E}_{\mathbb{Q}_Y} D_{ij}/\tau_i$ and dissipation $-\mathsf{r}_i \operatorname{Var}_{\beta^Q_i}(\mathsf{c}_{ij}) \le 0$ (MT Corollary 6.7); a moving state adds a driving term of no definite sign. Each weight is its initial value times a finite exponential, so no support changes at finite time: edges are neither created nor destroyed (MT Open Problem 6.9).

6.4.4 Communication kernels, factored maps, and loops

An interface on $e : j \to i$ is a triple of kernels E^v_e, N^v_e, D^v_e composed into the value kernel K^v_e (MT Definition 6.10); message recoordinatizations cancel, and $K^{v\prime}_e(s', \cdot) = (\mathsf{g}_i)_\# K^v_e(\mathsf{g}_j^{-1} s', \cdot)$ (MT Corollary 6.11). On the linear tier (MT Definition 6.12) the composite obeys $\mathsf{L}'_e = \mathsf{g}_i \mathsf{L}_e \mathsf{g}_j^{-1}$, message bases do not enlarge the structure group, rank is at most the smallest factor rank, and a distinguished sending subspace needs a metric (MT Proposition 6.13). The shared-head case $\mathsf{L}_e = U_i \mathsf{M}^{\mathrm{out}} \mathsf{M}^{\mathrm{val}} U_j^{-1}$ is a transformer's value and output maps in agent frames; a sum of heads factoring through spaces H_h has rank at most $\sum_h \dim H_h$.

Given an invertible link, $\mathsf{L}_e = F_e \Theta_e$ with $F'_e = \mathsf{g}_i F_e \mathsf{g}_i^{-1}$; this exists for every choice of link, so communication does not recover the link (MT Proposition 6.14). The reverse copy is specified separately and need not be an inverse, an adjoint, or of equal rank. A loop product L_p is conjugated at the base vertex even when singular, so rank, trace, and characteristic polynomial are invariant, but it records processing and loss of distinguishability, not holonomy: two sites with trivial holonomy already give round trips that keep one direction and discard the other, and parallel $\mathrm{GL}(2)$ copies give nontrivial holonomy with arbitrary singular communication. The shared-head family contains no Θ_e, so for a loop of length r its product is $U_{i_0}(\mathsf{M}^{\mathrm{out}}\mathsf{M}^{\mathrm{val}})^r U_{i_0}^{-1}$, which depends on the loop only through its length: shared value and output maps are blind to the links around a loop.

6.4.5 Comparison maps and the dot product

Comparison is separate from content: query and key maps $S^Q_e : E_i \to H^{\mathrm{cmp}}_e$, $S^K_e : E_j \to H^{\mathrm{cmp}}_e$ and the role divergence $d_e = D_{\mathrm{KL}}((S^Q_e)_\# q_i \| (S^K_e)_\# q_j)$ (MT Definition 6.15), a cospan that maps nothing back into the receiver. Reference maps $S^Q_{\mathrm{ref}} U_i^{-1}$, $S^K_{\mathrm{ref}} U_j^{-1}$ reproduce projected query–key comparison; with independent links the key can be $S^K_{\mathrm{ref}} U_i^{-1} \Theta_e$, which enters only through the per-edge twist $U_i^{-1} \Theta_e U_j$ (MT Proposition 2.12) and reduces to the reference form at trivial twist. The divergence is an energy of law points, admissible only with law points latent; then the softmax over role divergences is the posterior row at a latent configuration, and the recognition optimum reduces to it for a point-mass recognition law, whose free energy is infinite when the baseline gives the point zero mass (MT Corollary 6.17, MT Proposition 3.21). For Gaussian law points with a common

projected key covariance Σ^K,

$$d_{ij} = d_i^\circ - (\mu_i^Q)^\top (\Sigma^K)^{-1} \mu_j^K + \tfrac{1}{2} (\mu_j^K)^\top (\Sigma^K)^{-1} \mu_j^K$$

(MT (6.39)): a dot-product logit plus a key-norm bias. The derived logit is minus a squared query–key distance; it equals the transformer's dot product only for constant whitened key norms or a compensating prior. The natural gradient of a quadratic disagreement V_e fixes one receiving map $\mathsf{G}_i^{-1} S_e^{Q\top} W_e S_e^K$ (MT Proposition 6.16), so a declared attention insertion descends that energy only when its value map has this form. When content also depends on the pointer, its conditional relative entropy changes the optimal routing (MT Corollary 6.18); exact inference routes on content too, the transformer on query and key only.

6.4.6 The exact receiver update

The comparison record $o_i \mid k_i, J_i = j, y_{-i} \sim \mathcal{N}(S_i^Q k_i - S_e^K k_j, \tau_i W_i^{-1})$ is observed at $o_i = 0$ (MT (6.41)): a pseudo-observation clamping agreement of the receiver's query with the chosen key, with density $\mathrm{c}_i \exp[-V_e/\tau_i]$ at zero. Let the receiver's conditional law given all else be $\mathcal{N}(\mu_i, \Sigma_i)$, let the copies share the query map and precision, and let this record be the only factor reading the pointer. With $\Sigma_i^+ = (\Sigma_i^{-1} + S_i^{Q\top} \widetilde{W}_i S_i^Q)^{-1}$, $\Sigma_i^{\mathrm{pred}} = \widetilde{W}_i^{-1} + S_i^Q \Sigma_i S_i^{Q\top}$, and $r_{ij} = S_i^Q \mu_i - S_e^K k_j$,

$$w_{ij} \propto \pi_{ij} e^{-\frac{1}{2} r_{ij}^\top (\Sigma_i^{\mathrm{pred}})^{-1} r_{ij}}, \qquad \mathbb{E}[k_i \mid y_{-i}, o] = \Sigma_i^+ \Sigma_i^{-1} \mu_i + \sum_j w_{ij} \mathsf{L}_e^{\mathrm{val}} k_j, \qquad \mathsf{L}_e^{\mathrm{val}} = \Sigma_i^+ S_i^{Q\top} \widetilde{W}_i S_e^K$$

(MT Proposition 6.19). The weights integrate the receiver state out, so they use the record noise plus the receiver's own projected uncertainty; $\Sigma_i^+ \Sigma_i^{-1} \mu_i = (I - \Sigma_i^+ S_i^{Q\top} \widetilde{W}_i S_i^Q) \mu_i$ is the receiver's mean shrunk by the record's information, in place of an identity skip connection; and $\mathsf{L}_e^{\mathrm{val}}$ is a Kalman gain times the key map. In the scalar example with sources at 0 and 2, $w_{i2} = e^{-1}/(1 + e^{-1})$ while the posterior row at the mean gives $e^{-2}/(1 + e^{-2})$, and the value map is $1/2$. For a full-row-rank query map, as the record noise vanishes, $\mathsf{L}_e^{\mathrm{val}} \to R_e S_e^K$, the key map followed by the metric right inverse of the query map (Proposition G2.19). The update is exact for one head, one receiver, and fixed source states: a heat-bath step on one coordinate, not joint inference over all agents.

6.4.7 The transformer correspondence and heads

The technical correspondence table (MT Table 6.1) gives each row a standing; the last column is the companion's reading.

Transformer	Theory counterpart	MT	Companion reading
Token position	Agent; embedding as a latent coordinate	declared	A naming choice.
Query, key projections	Reference maps; role divergence; dot product plus key-norm bias	derived	Logit form derived for Gaussian law points with common key covariance; the maps are declared.

Transformer	Theory counterpart	MT	Companion reading
One key map for all edges	Links enter only through the twist; shared map is trivial twist	derived	Correct; nontrivial twist and holonomy have no counterpart, and nothing is said about their effect.
Softmax row	Posterior row of the pointer; at a latent configuration over law-point divergences	derived	From a declared exponential likelihood; over current beliefs it is a posterior at fixed law points, never a finite-cost recognition optimum against a non-atomic law-point prior.
Mask	Zero prior weight	derived	Immediate.
Temperature, $1/\sqrt{d}$	τ_i; scaled dot product one choice	declared	Free; not identifiable from precision in the Gaussian record.
Value, output, residual	Mean operator declared; comparison-record update fixes value map and residual weight	derived	What is derived is a different, tied value map and a shrinkage residual; the transformer's free value and output maps are declared.
Multiple heads	Declared per-head rows	absent	Per-head softmaxes are exact at a fixed receiver state and are not the integrated label posterior once the shared state is integrated out.
Feed-forward, layer norm	None	absent	None.

The table has no row for stacking layers or for training. On heads: the declared mean operator (MT (6.36)) is multi-head attention in operator form with declared rows. Two heads with their own pointers and comparison records have factorized pointer posteriors at fixed receiver state, but not after the state is integrated out; in the scalar instance with sources at 0 and a the odds ratio between agreeing and disagreeing label pairs is $e^{a^2/3}$ (MT Proposition 6.20). The general Gaussian criterion is the log cross ratio $-(\mathsf{a}_j - \mathsf{a}_{j'})^\top \Lambda_{12}^{\mathrm{rec}} (\mathsf{b}_l - \mathsf{b}_{l'})$ (MT Remark 6.21); correlated projected queries alone do not force dependence.

6.4.8 What survives an interface

Information about the source cannot grow along encoder, channel, and decoder given receiver side information (MT Proposition 6.24). Agreement on communicated coordinates leaves a private conditional divergence, $D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q_M\|p_M) + \int D_{\mathrm{KL}}(q(\cdot \mid M=z)\|p(\cdot \mid M=z))q_M(dz)$ (MT Proposition 6.25), so agents agreeing on everything communicated may disagree by any amount. For surjective linear measurements M_j, M_i, M that retain some coordinates (interfaces), a linear map descends to the retained coordinates exactly when $\mathsf{L}_e \ker M_j \subseteq \ker M_i$, a linear flow exactly when its generator preserves $\ker M$, a moving measurement exactly when $\dot{M}_t + M_t \mathcal{A}_t$ vanishes on $\ker M_t$, and a low-rank interface is not by itself a Markov blanket (MT Proposition 6.26). The flow condition is the linear analogue of exact lumpability [46].

6.4.9 What the chapter establishes about the query map, the key map, and the matrix link

Established: the types of S_e^Q and S_e^K (MT Definition 6.15); the transformation law of S_e^Q under frame change and comparison recoordinatization, stated in Supplement G (Proposition G2.19), with the law of S_e^K following the same pattern by the reference-map form and the kernel-level law MT Corollary 6.11; the types and transformation laws of L_e, with message coordinates internal (MT Proposition 6.13); the entry of the independent link into attention through the key map $S_{\mathrm{ref}}^K U_i^{-1}\Theta_e$, only via its per-edge twist, so that a shared key map is the trivial-twist case; the separation of the matrix link from the gauge link, $\mathsf{L}_e = F_e\Theta_e$ with Θ_e unrecoverable, loop products that are not holonomy, and holonomy independent of communication rank; the tie between value and comparison, the receiving map $\mathsf{G}_i^{-1}S_e^{Q\top}W_eS_e^K$ of a quadratic disagreement and the exact value map $\Sigma_i^+S_i^{Q\top}\widetilde{W}_iS_e^K$ of the Gaussian comparison-record model; and linear limits (rank bounds, descent conditions, rank not being capacity).

Not established: any rule that selects or learns S_e^Q, S_e^K, or L_e; the technical chapter states that nothing there selects the content maps. MT Definition 6.23 lists what a learned or random interface must carry (every varying normalizer in its score, or a normalized prior and declared dynamics with a regular metric) and constructs no learning rule. A declared content map L_e and the comparison-induced value map are two mechanisms the chapter does not unify; Supplement G gives L_e its own linear Gaussian record (Proposition G2.20). Evolving interfaces are MT Open Problem 6.27. The linear statements are on real vector fibers and use no compactness, so they cover $\mathrm{GL}(K,\mathbb{R})$ links in the defining representation; complex fibers are not treated separately.

6.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Definition 6.1	Attention is a latent pointer with a structural prior and an exponential record	CHOICE	Exclusivity; one label per row fixes occupancy
MT Proposition 6.2	Posterior row is the softmax; constant-row optimum is a softmax over averaged energies	SOLID	Standard in substance (Bayes)
MT Proposition 6.3	Exact Jensen-gap cost of a constant row	SOLID	Zero only for constant energy differences
MT Definition 6.4, MT Definition 6.5	Occupancy and event law are declared	CHOICE	The two choices give different coarse rows
MT Corollary 6.6, MT Corollary 6.7	With uniform rows and occupancy, a fine no-self convention is not preserved under blocking; row flow is replicator	SOLID	Flow never changes the support

Result	What it says	Status	Scope and caveats
MT Proposition 6.8	Reduced curvature is frozen curvature minus force variance; convexity can be lost	SOLID	Witness one-dimensional (section 6.6)
"Seed of symmetry breaking" (MT summary)	The curvature correction drives later instability	VAGUE	No statement behind it
MT Definition 6.10, MT Definition 6.12, MT Definition 6.15	Kernels, factored maps, query and key maps, mean operator	CHOICE	All declared; nothing selects them
MT Corollary 6.11, MT Proposition 6.13, MT Proposition 6.14	Covariance; rank bounds; $\mathrm{L}_e = F_e\Theta_e$; loops are not holonomy	SOLID	Elementary; small examples are refutations
MT Proposition 6.16, MT Corollary 6.17	Receiving map of a disagreement; softmax over role divergences; dot product plus key-norm bias	SOLID	Common key covariance; point-mass recognition singular
MT Proposition 6.19	Tied value map, shrinkage residual, predictive weights	SOLID	Gaussian comparison record; one head; fixed sources
MT Proposition 6.20, MT Remark 6.21	Shared receiver state couples head pointers; exact criterion	SOLID	Refutes per-head factorization
MT Proposition 6.24, MT Proposition 6.25, MT Proposition 6.26	Data processing; private divergence; descent conditions	SOLID	Standard in substance
MT Open Problem 6.9, MT Open Problem 6.22, MT Open Problem 6.27, MT Open Problem 3.23	Edge kinetics; head recursion; moving interfaces; law-point prior	OPEN	See section 6.7

Conditions: which ones matter

Conditions that matter. Label exclusivity makes the posterior factor by rows; the exponential record form produces the softmax; one label per row forces a declared α; the constant-row restriction carries an exact cost. Energies must not read recognition, so divergences between beliefs need latent law points and hence the open law-point prior. The receiver update needs a Gaussian conditional receiver law, a shared query map and precision, and fixed sources; the dot product needs a common key covariance; the link factorization needs invertible links; the interaction graph is fixed.

Hygiene. Finite energies with finite recognition means, measurability, standard Borel spaces, C^2 energies for the curvature formula, and versions chosen off null sets hold in the program's Gaussian and finite models.

6.6 Toy versus general

The small examples (the one-dimensional curvature witness, the scalar receiver and two-head instances, the two-site loop, the parallel $\mathrm{GL}(2)$ copies) each refute a general implication, and as refutations they are SOLID. The mechanism-level content, that integrating out a pointer subtracts a force variance from the curvature and that a shared receiver state couples heads, is proved in general form. The threshold $\delta^2 = \tau(\omega + 1)$ and the odds ratio $e^{a^2/3}$ belong to their instances, and nothing here shows that attention-induced loss of convexity yields symmetry breaking in networks of hundreds of agents with $\mathrm{GL}(K)$ links, or how strong head coupling is at scale; those consequences are conjectural until proved there.

6.7 Open ends

Edge creation and deletion (MT Open Problem 6.9): the graph is the prior's support and the replicator flow preserves supports. The problem fixes a threshold rule (delete a copy whose recognition weight falls below a threshold, create one whose averaged comparison energy falls below a threshold, renormalize) and asks for a normalized joint with the edge set as a structural coordinate under which the rule's master equation does not increase the collective free energy, or a proof that no joint of that class works for any such rule. In plain terms: is there a free energy that edge kinetics descend?

Label-level multi-head attention (MT Open Problem 6.22): heads sharing a receiver state have coupled pointers, the event law has no head index, and under exact coarse-graining the rows a parent must read through one coordinate multiply with depth unless a head-merging channel, with its own conditional relative entropy, is declared. The problem asks for an exact recursion with bounded head count in the coupled class, or a block whose posterior mean no comparison-record model with fewer heads than rows reproduces.

Formation with evolving interfaces (MT Open Problem 6.27): for Gaussian agents whose query and key maps are fixed-rank interfaces learned slowly at mobility ϵ_{M}, with noisy mean dynamics at temperature T and pointers redrawn from posterior rows, does a finite-dimensional Markov parent reproduce the path law of the communicated coordinates with vanishing path relative entropy on horizons of order $1/\epsilon_{\mathrm{M}}$, uniformly in small T? Known are the full-dimension alternative and the pointwise descent criterion for a supplied interface path.

The law-point prior (MT Open Problem 3.23) belongs to the probability chapter but bears on this one: without it, divergences between beliefs are separately declared potentials rather than evidence-bound terms. Even with it, a softmax computed at fixed law points is a generative posterior quantity

or a function of parameters, not a recognition optimum, because point-mass recognition of a non-atomic law point has infinite cost (MT Proposition 3.21).

Does it deliver what it claims? DELIVERED

The technical chapter says that attention enters the joint as a latent source label whose generative posterior is the softmax row, that the optimal constant recognition row is a different softmax at an exact cost, that communication is a normalized encoder–channel–decoder kernel, and that the exact receiver update of a comparison record fixes the value map by the query and key maps; it declines to claim that transformer attention as a whole, per-head rows, or channel capacity is derived. The proved statements deliver exactly this. The substance is standard: the softmax is Bayes' formula for a declared pointer, the receiver update is Gaussian conditioning with a Kalman gain, the row flow is the replicator equation. Specific to the theory are where the independent link enters (the key map, through its twist), the separation of communication maps from links and of loop products from holonomy, and the finding that exact inference ties the value map to the query and key maps, shrinks the residual by precision, and weights by predictive likelihood. One table standing is generous: the "derived" value-and-residual row derives a tied substitute, not the transformer's free value and output maps. The chapter does not select or learn the query, key, or communication maps, and gives no label-level multi-head model, no edge kinetics, and no formation with moving interfaces; the softmax over current beliefs is a posterior at fixed law points rather than a recognition optimum, and the law-point prior that would make belief divergences evidence-bound terms is open.

Chapter 7

Fiber Information Geometry and Inference Histories

7.1 The question and the answer

Chapter 5 left each agent at c_* as a pair of points on law fibers, a belief law q_i^b and a model law q_i^m, with one collective free energy for the population. MT Chapter 7 adds what a dynamics and a comparison of scales need: a geometry on the law fibers, and a rule that turns a free energy into a path with a length. It asks how the collective free energy selects an inference history and what duration the Fisher geometry assigns to it, and, for meta-agents, what happens to both when a block of agents is replaced by a coarser description.

The answer has five parts. The Fisher metric and the cubic Amari–Chentsov tensor of each law fiber are gauge invariant, hence frame-independent at c_* (MT Proposition 7.3), but the Fisher metric is chosen: none of the axioms that single it out is imposed. A Markov kernel that does not depend on the parameter replaces each score by its conditional expectation given the coarse record and loses Fisher information by exactly the expected conditional covariance of the score (MT Theorem 7.8), the data-processing inequality for Fisher information with its defect written out. A free energy selects a path only after a metric or mobility is chosen; the steepest-descent ray then defines an oriented orbit whose Fisher arc length, the information clock, is reparameterization invariant (MT Theorem 7.36). The image of a history under one parameter-independent kernel, such as the law of a meta-agent when that law is such an image, accumulates no more duration (MT Corollary 7.42), while a coarse orbit recomputed from its own free energy is comparable with the fine one only through an oriented semiconjugacy, whose time change cancels from every duration (MT Theorem 7.47, MT Theorem 7.49). Finally, a three-agent parity witness shows that the collective free energy and its Fisher metric are not functions of single-agent laws (MT Proposition 7.28, MT Corollary 7.25), and two finite theorems show that naturality under local kernels forces the product law and that, once correlated refinements are admitted, no reference-free selector exists (MT Theorem 7.20, MT Theorem 7.21).

The chapter consumes the collective free energy and the local–global identity of chapter 5. Chapter 8 uses its mobility and clock results, chapter 9 the Fisher-loss theorem, and chapters 11 and 12 the coarse configuration map and record clocks. Most of the mathematics is standard (Fisher information under Markov kernels, arc length, gradient flows with a mobility, Łojasiewicz finite

length); the chapter's contribution is the typing at c_* and a counterexample at each boundary where a standard statement stops applying to the program's objects.

7.2 The picture

The Fisher metric measures distinguishability: to second order the relative entropy between p and $p+dp$ is $\frac{1}{2}g^F(dp,dp)$. For an exponential family $p_\lambda \propto e^{\lambda\cdot A(y)}$ the score in direction λ_a is the centered observable $A_a - \langle A_a\rangle$ and the Fisher metric is $\mathrm{Cov}(A)$, the static susceptibility matrix up to a factor of temperature. Most of the chapter consists of statements about fluctuations of scores.

Physics analogue (exact)

Fisher loss under coarse observation is the law of total covariance. In an exponential family the fine information is $\mathrm{Cov}(A)$, the coarse one $\mathrm{Cov}(\mathbb{E}[A \mid \bar{Y}])$, and the loss $\mathbb{E}\,\mathrm{Cov}(A \mid \bar{Y})$: the fluctuation of the conjugate observable inside a coarse cell, invisible to a probe that sees only the record $\bar{Y}$. For a stochastic Kadanoff block rule applied to spins in a field, the Fisher information of the coupling βh seen through the block spins is the variance of the conditional mean magnetization. MT Theorem 7.8 holds for every regular family, and it requires a blocking rule that does not depend on the parameter.

A free energy alone does not define a dynamics: in overdamped physics $\dot{x} = -\mathsf{M}\nabla\mathcal{F}$, the potential fixes the equilibria and the mobility the paths. The natural gradient [3] takes a chosen metric as inverse mobility, and no fluctuation–dissipation relation ties that mobility to anything in the model; it is a modeling choice, and an anisotropic one can change the orbit (section 7.4.5).

Physics analogue (structural)

The Fisher duration of a path of laws is the thermodynamic length of Crooks [18]; for Gibbs laws and paths in natural parameters the mathematical object is the same. What does not transfer is its physical use, bounds on dissipation for driving at finite speed, which needs a protocol and a physical time. The chapter supplies neither and declines to call its duration time (section 7.4.6).

Physics analogue (structural)

A loop holonomy $\exp(t\xi)$ near the identity acting on an agent's law costs relative entropy $\frac{1}{2}t^2\mathfrak{k}_q(\xi,\xi)$, where $\mathfrak{k}_q(\xi,\eta) = g^F_q(\xi_{\mathcal{B}}(q),\eta_{\mathcal{B}}(q))$ vanishes exactly on the isotropy algebra of the law (MT Corollary 7.7). This has the algebraic form of the gauge-boson mass matrix $M^2_{ab} \propto \langle T_a v, T_b v\rangle$, which vanishes on the unbroken generators: both pull a positive form back along an orbit map. The correspondence is one of form; no mass, link dynamics, or symmetry breaking is claimed.

Physics analogue (exact)

At its center the parity witness (MT Definition 7.22) is the Gibbs law of six Ising spins $\sigma = (-1)^y$ with a single six-spin coupling, $\frac{1}{64}(1 + \bar{\varkappa}\prod\sigma) \propto e^{J\prod\sigma}$ with $\tanh J = \bar{\varkappa}$. Every correlation of fewer than six spins vanishes, so a mean-field description from single-spin magnetizations reproduces all marginals and misses the coupling; the free energy and Fisher metric do not (section 7.4.4). The identification is exact at the center only.

7.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Law fiber $\mathcal{B}_x$, $x \in \{b, m\}$	Manifold of laws an agent's belief or model can take at c_*	Statistical manifold of one site
Score $\mathbf{s}_Z$	Derivative of $\log p$ along Z; mean zero	Centered conjugate observable
Fisher metric g^F	Covariance of scores	Susceptibility matrix
Amari–Chentsov tensor T^F	Third moment of scores	Third cumulant; not a torsion
$\mathfrak{k}_q$, $\mathfrak{g}_q$	Cost of moving a law along a group direction; directions that do not move it	Mass matrix; unbroken generators
Markov coarse morphism $\mathscr{K}$, kernel K	Coarse observation by one fixed, frame-covariant stochastic rule	Stochastic block-spin map
Vertical Fisher defect $\Delta_F^{\mathscr{K}}$	Fisher information lost by $\mathscr{K}$ on the fibers at c_*	Within-block fluctuation
Configuration manifold $\mathcal{Q}_i$; metric G, weights w_b, w_m	Belief and model laws of an agent together; chosen metric on them	Configuration space; inverse mobility
Exact lift ι_i, extraction ext_i	Supplied rule building a joint recognition law from a configuration, and its left inverse	Closure relation
G_J, G_w, $\mathfrak{K}$, $\mathfrak{C}$	Joint Fisher metric; weighted marginal metrics; genuinely joint residual; signed cross term	Higher-order versus pairwise correlations
Natural selector	Rule from marginals to a joint law commuting with every admitted kernel	Closure; mean field when forced
$\varkappa$, $\bar{\varkappa} = \varkappa/64$	Parity amplitude; six-bit correlation at the center	Six-spin coupling
History; natural-gradient history	Oriented curve modulo reparameterization; orbit of $-\mathrm{grad}^{\mathsf{G}}\mathcal{F}$	Trajectory without time label; overdamped path
Fisher duration $\mathfrak{t}$	Fisher arc length from a chosen origin (the information clock)	Thermodynamic length
Orthogonal clock $\mathfrak{t}^\perp$	Unit-rate clock whose level sets are orthogonal to descent	Synchronous time slicing

Term or symbol	Plain meaning	Physics analogue
F_ℓ; oriented semiconjugacy $(\mathsf{F}_\ell, \mathsf{a}_\ell)$	Fine-to-meta-agent configuration map; it carries the fine flow onto the coarse flow up to an increasing time change	Exact dynamical coarse-graining
Depth ℓ	Number of coarse-graining steps	Blocking steps; not time

"Natural" has two unrelated senses in this chapter: Amari's metric-raised gradient, and a selector that commutes with a class of maps in the category-theoretic sense. "Metric" is a chosen Riemannian metric on laws or configurations, never a spacetime metric. "History" is an orbit with its parameterization forgotten; "duration" and "clock" mean Fisher arc length, not time. "Vertical" means along the fibers over the single base point, so no connection enters. A "lift" reconstructs a joint law from marginal data; it is not the horizontal lift of a connection. Glyphs collide across adjacent chapters: par is the parity character here and the parent map in chapter 8, and a is a rate factor here and a proposal weight there. $\mathfrak{K}$ (residual tensor), $\mathfrak{k}_q$ (orbit form), $\mathscr{K}$ (coarse morphism), and K (its kernel) are four different objects, and "semiconjugacy" is the dynamical-systems term.

7.4 The formulas, explained

7.4.1 Fiber tensors, and why the metric is a choice

$$g^F_{x,p}(Z_1, Z_2) = \mathbb{E}_p[\mathbf{s}_{Z_1}\mathbf{s}_{Z_2}], \qquad T^F_{x,p}(Z_1, Z_2, Z_3) = \mathbb{E}_p[\mathbf{s}_{Z_1}\mathbf{s}_{Z_2}\mathbf{s}_{Z_3}].$$

The metric is the covariance of the score and the cubic tensor its third moment, the second and third cumulants of conjugate observables in an exponential family. Both are preserved by the gauge action because that action is a pushforward by a map of samples (MT Proposition 7.3). Invariance of the metric alone would not suffice: a Möbius isometry of the normal family's hyperbolic Fisher metric preserves g^F and not T^F, so a symmetry of the metric that does not relabel samples is not a statistical symmetry. The Fisher metric itself is not derived. MT Proposition 7.4 collects the uniqueness theorems of Čencov type [73], and the scope paragraph states that none is imposed; on configurations the metric carries channel weights $w_b, w_m > 0$ that nothing fixes (MT Definition 7.13), and a rate ratio enters the same way in chapter 8. Every orbit and duration below inherits these choices.

7.4.2 Relative entropy to second order, and the cost of a holonomy

$$D_{\mathrm{KL}}\big(q \| \widehat{\rho}(\exp t\xi)q\big) = \frac{t^2}{2}\mathfrak{k}_q(\xi,\xi) + O(t^3), \qquad \mathfrak{k}_q(\xi,\eta) = g^F_q\big(\xi_{\mathcal{B}}(q), \eta_{\mathcal{B}}(q)\big).$$

This is MT Corollary 7.7, resting on the standard fact [4] that the Hessian of relative entropy is the Fisher metric (MT Proposition 7.6). The velocity $\xi_{\mathcal{B}}(q)$ of the law under the subgroup generated by ξ has squared Fisher length $\mathfrak{k}_q(\xi,\xi)$; the null space is the isotropy algebra $\mathfrak{g}_q$, on which the left side

vanishes for every t. For a Gaussian $\mathcal{N}(\mu, \Sigma)$ with $\mathrm{GL}(K)$ acting linearly on samples the velocity is $(\xi\mu, \xi\Sigma + \Sigma\xi^\top)$, so $\mathfrak{g}_q$ consists of the ξ with $\xi\mu = 0$ and $\xi\Sigma + \Sigma\xi^\top = 0$; for a centered isotropic law it is all of $\mathfrak{so}(K)$, so a rotation holonomy is invisible to that agent while a squeeze is always detected. In the running three-agent loop this is the second-order form of law coherence up to holonomy stabilizers (MT Definition 2.17).

7.4.3 Coarse observation: the Fisher-loss theorem and its limits

For a regular family $\{P_\lambda\}$ of laws of Y and a Markov kernel K producing a record $\bar{Y}$, the same kernel for every λ, MT Theorem 7.8 states

$$\bar{\mathbf{s}}_{\lambda_0}(\bar{Y}) = \mathbb{E}\big[\mathbf{s}_{\lambda_0}(Y) \mid \bar{Y}\big], \qquad \mathsf{I}_Y(\lambda_0) - \mathsf{I}_{\bar{Y}}(\lambda_0) = \mathbb{E}\,\mathrm{Cov}\big(\mathbf{s}_{\lambda_0}(Y) \mid \bar{Y}\big) \succeq 0.$$

The coarse score is the best estimate of the fine score from the record, and the second identity is the law of total covariance for a mean-zero score. Equality in a direction z holds exactly when $z^\top \mathbf{s}_{\lambda_0}$ is a function of $\bar{Y}$. A bijective K loses nothing; a K that ignores its input loses everything; discarding an independent component loses exactly its Fisher information. No common dominating measure is needed [62]. This is the Fisher counterpart of the data-processing inequality [15], standard in substance. Equality at one parameter is not recovery: MT Remark 7.9 discards a Bernoulli bit whose score vanishes at $\lambda = 0$ but which depends on λ at second order, and no fixed kernel reconstructs the fine family near $\lambda = 0$.

The one substantive hypothesis is parameter independence of the kernel, and the manuscript shows that natural coarse maps violate it. A parameter-dependent channel adds a term to the record score (MT (7.62)), and then the record can move while the fine law stands still: a constant input fed to a channel emitting a bit with success probability $(1 + e^{-t})^{-1}$ has zero fine speed and record speed $(2\cosh(t/2))^{-1}$ (MT Corollary 7.42). Gaussian Galerkin aggregation, substituting $y = Sc$ into an energy with precision Λ, is the pushforward by $P_\Lambda = (S^\top \Lambda S)^{-1} S^\top \Lambda$ (Proposition G7.5). On a fiber of fixed precision this is one kernel, the theorem applies, and the loss vanishes exactly on directions in $\mathrm{range}(S)$. Across precisions one kernel exists exactly when $\mathrm{range}(\Lambda S)$ does not change, and the supplement exhibits a change of one self term inside a cluster that changes it, so along histories with such moves the theorem does not apply. No parameter-independent kernel sends two normal laws to their joint mean-and-variance barycenter (Proposition G8.5), and fitted approximations or generative–recognition mismatches qualify only when a kernel is exhibited (MT Definition 7.10). A fixed stochastic blocking rule can only lose distinguishability; a state-adapted rule is outside the theorem, and outside it a record can register motion the fine law does not have.

On fibers the statement becomes the vertical defect $\Delta_F^{\mathscr{K}} = g^F - (T\mathscr{K})^* \bar{g}^F \succeq 0$ with $\Delta_F^{\mathscr{K}}(Z, Z) = \mathbb{E}\,\mathrm{Var}(\mathbf{s}_Z \mid \bar{Y})$ (MT Corollary 7.11), and defects compose as a cocycle, $\Delta_F^{\mathscr{K}_{12} \circ \mathscr{K}_{01}} = \Delta_F^{\mathscr{K}_{01}} + (T\mathscr{K}_{01})^* \Delta_F^{\mathscr{K}_{12}}$: losses along a chain add, each pulled back to the finest level (MT Theorem 7.12). The cubic defect has no sign, so coarse-graining gives no monotonicity for T^F.

7.4.4 Configurations, lifts, and why marginals do not suffice

An agent's configuration $Q_i = (q_i^b, q_i^m)$ has squared speed $w_b g_b^F(\dot{q}_i^b, \dot{q}_i^b) + w_m g_m^F(\dot{q}_i^m, \dot{q}_i^m)$ (MT Definition 7.13); on the population the metric is either the block sum, which treats agents as independent, or the pullback of a joint Fisher metric along an exact lift (MT Definition 7.17). The lift is a smooth map from agent i's configurations to conditional recognition kernels of a block given the rest, with an extraction map that recovers the configuration; it carries every correlation the configuration does not determine. With the outside marginal fixed, the lifted objective differs from the collective free energy by a constant (MT Theorem 5.23), so both have one natural-gradient ray for one metric. The substance is that the lift must be supplied and that different lifts give different geometry. MT Proposition 7.18 quantifies this. With joint score $\mathbf{s}$ and embedded marginal scores $\mathbf{s}_i = \mathbb{E}[\mathbf{s} \mid Y_i]$,

$$\mathsf{G}_{\mathrm{J}} - \sum_i w_i \mathsf{G}_i = \mathfrak{K} - \mathfrak{C} + \sum_i (1 - w_i)\mathsf{G}_i, \qquad \mathfrak{K} = \mathbb{E}[\mathbf{r}\mathbf{r}^\top], \quad \mathbf{r} = \mathbf{s} - \sum_i \mathbf{s}_i, \quad \mathfrak{C} = \sum_{i \neq j} \mathbb{E}[\mathbf{s}_i \mathbf{s}_j^\top].$$

The residual score $\mathbf{r}$ is the part of the joint score that is not a sum of single-block scores, so $\mathfrak{K} \succeq 0$ measures genuinely collective information; $\mathfrak{C}$ collects overlaps between blocks' marginal scores and has no sign; the last term is weight mismatch. For independent blocks $\mathfrak{K} = \mathfrak{C} = 0$ and unit weights are forced. For dependent blocks no inequality holds either way (a correlated bivariate normal gives a difference of both signs), and one extraction can have right inverses with different pullback metrics ($\lambda_1 \mapsto (\lambda_1, 0)$ and $\lambda_1 \mapsto (\lambda_1, \lambda_1)$ induce $d\lambda_1^2$ and $2d\lambda_1^2$ on $\mathcal{N}((\lambda_1, \lambda_2), \mathrm{I}_2)$).

The parity witness isolates a pure higher-order dependence. Six bits, one per agent $i \in \{1, 2, 3\}$ and channel $x \in \{b, m\}$, with success probabilities $\lambda_{i,x}$, get the lift

$$p_\varkappa(y; \lambda) = \prod_{(i,x)} \lambda_{i,x}^{y_{i,x}} (1 - \lambda_{i,x})^{1 - y_{i,x}} + \varkappa (-1)^{\sum y_{i,x}} \prod_{(i,x)} \lambda_{i,x}(1 - \lambda_{i,x}), \qquad |\varkappa| < 1,$$

a product law plus a parity term whose amplitude factor keeps the law positive (MT Definition 7.22). Summing out any bit kills the parity term, so every proper marginal is the product marginal (MT Theorem 7.23). The lift is a smooth right inverse of the one-bit marginal map, equivariant under complementing both bits of an agent and under agent permutations, while a single-bit complement flips the sign of $\varkappa$. At the center, with $\bar{\varkappa} = \varkappa/64$, MT Corollary 7.25 gives

$$\mathsf{G}_{\mathrm{J}} = \frac{4}{1 - \bar{\varkappa}^2} \mathrm{I}_6, \qquad \mathsf{G}_{\mathrm{J}} - \sum_{(i,x)} \mathsf{G}_{i,x} = \frac{4\bar{\varkappa}^2}{1 - \bar{\varkappa}^2} \mathrm{I}_6.$$

Pairs of bits are independent, so $\mathfrak{C} = 0$ and the excess is all $\mathfrak{K}$; weighted marginal metrics match only with the $\varkappa$-dependent weight $1/(1 - \bar{\varkappa}^2)$, and since the one-bit families do not depend on $\varkappa$, no rule that reads them returns every joint metric. For the free energy an engineered hyperedge record $K_{\mathrm{hyp}}(1 \mid y) = \frac{1}{2}(1 + \bar{\varkappa}(-1)^{\sum y})$ with uniform prior has the center law as exact posterior (MT Definition 7.27); the posterior attains $\mathcal{F}_1 = \log 2$, while the uniform law, the product of the same one-bit marginals, has $\mathcal{F}_1(\mathrm{Unif}) - \log 2 = -\frac{1}{2}\log(1 - \bar{\varkappa}^2)$ (MT Proposition 7.28). The two laws share all one-bit data and the evidence and differ in free energy. With $\varkappa$ promoted to a coordinate the

joint rank is seven and the marginal rank six, with $\partial_{\varkappa}$ invisible (MT Corollary 7.26); combined with the finite-family clause of MT Theorem 7.14, a history moving only $\varkappa$ has positive joint Fisher speed while every single-agent clock stands still. The numbers are small by construction ($|\bar{\varkappa}| < 1/64$ gives a Fisher excess below 10^{-3} and a free-energy excess of at most about 1.22×10^{-4} nats), so the witness is qualitative. Its message for meta-agents is that a law assembled from constituents' single-agent laws cannot see such a dependence, while the collective free energy and joint Fisher metric can.

Could a canonical rule supply the dependence? A selector maps one-coordinate marginals to a joint law with those marginals, and is natural on a class of kernels if it commutes with them (MT Definition 7.19). Naturality under all tensor products of single-coordinate kernels forces the product law, because kernels from a one-point set prepare each coordinate independently (MT Theorem 7.20); a larger class admits a natural selector exactly when every added kernel preserves products. Admitting two kernels that split a fair bit into correlated bits, with $\mathrm{c} = \frac{1}{3}$ and $\mathrm{c} = \frac{1}{2}$, leaves no natural selector, since both give fair marginals with different joints (MT Theorem 7.21). Both results are elementary; their force lies in the admitted class. What replaces an absolute selector appears only in an unnumbered paragraph: relative to a positive reference law, a relative-entropy minimizer over a convex moment-constrained family is unique when it exists, and the reference conditional completion of a coarse marginal follows from the chain rule (MT Lemma 3.12); both take the reference as extra data. This is maximum entropy relative to a prior measure [41], standard. In physics terms, without a Hamiltonian the only closure that commutes with independent local relabelings is mean field, and a correlated closure needs a reference measure. For meta-agents the dependence among constituents is supplied by the generative model or a declared lift, never manufactured from configurations, in line with the standing refusal of a canonical coupling selector (MT Section 1.4).

7.4.5 Natural-gradient histories: the orbit depends on the mobility

$$\mathsf{G}\big(\operatorname{grad}^{\mathsf{G}} \mathcal{F}, Z\big) = d\mathcal{F}[Z], \qquad \dot{Q}(t) = -\mathsf{a}(t) \operatorname{grad}^{\mathsf{G}} \mathcal{F}\big(Q(t)\big), \quad \mathsf{a}(t) > 0.$$

The gradient raises the index of $d\mathcal{F}$ with the metric, $\mathsf{G}^{-1}\nabla\mathcal{F}$ in coordinates, and the arbitrary speed a makes the pair an oriented orbit rather than a trajectory (MT Definition 7.30). A scalar mobility only reparameterizes orbits; an anisotropic one can bend them: for $\mathcal{F} = \frac{1}{2}(z_1^2 + z_2^2)$ ordinary descent runs along straight lines, descent with mobility $\operatorname{diag}(1, 2)$ along the parabolas $z_2 = z_2^0 (z_1/z_1^0)^2$ (MT Proposition 7.31), and descent orbits are not geodesics. For the program this means that the critical points are metric independent but which one an orbit from a given start reaches is a property of the orbit, and nothing in the chapter shows it is mobility independent. Every dynamics the program runs carries a chosen mobility: the four-cycle kinetics of chapters 8 and 13 use identity mobility with supplied rates, which is the Fisher natural gradient for the unit-variance belief means but not for the model means, whose Fisher rate would be $\frac{1}{2}$ (MT Proposition 8.6).

The remaining statements keep the objects well defined. Noncritical orbits are embedded, with the free energy a strictly decreasing coordinate (MT Proposition 7.32). A semidefinite metric cannot be inverted silently (MT Proposition 7.33); a quotient must come first (MT Theorem 7.14). For softmax

logits the pulled-back Fisher tensor $\mathrm{diag}(p) - pp^\top$ has null direction $\mathbf{1}$, and the Moore–Penrose solution picks the gauge $\sum_j z_j = 0$. For a free energy invariant under a group action the gradient is orthogonal to the orbits, so histories never move along gauge directions, and for a free, proper, isometric action they descend to the quotient with the same duration (MT Proposition 7.39). Explicit Euler steps below $2/\mathsf{c}_\mathsf{G}$ descend and stay within $O(\Delta t)$ of the orbit on compact intervals without lying on it (MT Remark 7.34).

7.4.6 Fisher duration, and why it is not called time

$$\dot{\mathfrak{t}} = \sqrt{\mathsf{G}(\dot{Q},\dot{Q})}, \qquad \mathfrak{t}_{Q,t_0}(t) = \int_{t_0}^{t} \dot{\mathfrak{t}}(t')dt',$$

$$\frac{d\mathfrak{t}}{dt} = \mathsf{a}\|\mathrm{grad}^{\mathsf{G}}\mathcal{F}\|_{\mathsf{G}}, \qquad \frac{d\mathcal{F}}{d\mathfrak{t}} = -\|\mathrm{grad}^{\mathsf{G}}\mathcal{F}\|_{\mathsf{G}}.$$

The clock is arc length from a chosen origin (MT Definition 7.35). Along a natural-gradient history the clock rate is the speed factor times the gradient norm, and the free-energy decrease per unit clock is the gradient norm alone, a property of the orbit. The clock is invariant under orientation-preserving reparameterization and stops on critical or null segments (MT Theorem 7.36). For a block-diagonal metric the collective clock satisfies $\max_i \mathfrak{t}_i \le (\sum_i \mathfrak{t}_i^2)^{1/2} \le \mathfrak{t} \le \sum_i \mathfrak{t}_i$. In the unit-variance location family the path $\mu(t) = e^{-t}$ has duration 1 over infinite parameter time, and a realized update is at least as long as the Fisher distance between its endpoints (MT Remark 7.37). Total duration is finite for a gradient history with an accumulation point at which the objective is real analytic, by the Łojasiewicz argument of Absil et al. [1], and infinite for a smooth example that spirals toward a circle (MT Proposition 7.38).

A clock with level sets orthogonal to the flow is stronger. With $\mathsf{n} = \|\mathrm{grad}^{\mathsf{G}}\mathcal{F}\|_{\mathsf{G}}$ and $\boldsymbol{\alpha} = -d\mathcal{F}/\mathsf{n}$, an orthogonal clock solves $d\mathfrak{t}^{\perp} = \boldsymbol{\alpha}$; since $d\boldsymbol{\alpha} = \mathsf{n}^{-2} d\mathsf{n} \wedge d\mathcal{F}$, it exists exactly when n is locally a function of $\mathcal{F}$ and the periods of $\boldsymbol{\alpha}$ vanish (MT Theorem 7.40). For $\mathcal{F} = z_1 z_2$ on the first quadrant none exists on any open set, although a unit-speed clock built from the cross-section $z_1 z_2 = 1$ exists on the whole quadrant (MT Proposition 7.41).

Physics analogue (structural)

With the level sets of $\mathcal{F}$ labeled by $\tau = -\mathcal{F}$, $\boldsymbol{\alpha} = \mathsf{n}^{-1} d\tau$: the arc length between neighboring slices is a lapse n^{-1} times $d\tau$. A time function whose differential is the unit normal one-form exists exactly when the lapse is constant on each slice, the same Frobenius-type condition as for a synchronous slicing in general relativity, here in Riemannian signature and with no causal content.

The manuscript refuses to call the duration physical time, and the refusal is correct. Its standing scope (MT Section 1.4) says an identification would need a named target clock, a calibration map with an uncertainty model, causal and compositional consistency, and discriminating evidence; none is supplied. The chapter's own results add reasons: the duration depends on the chosen metric and origin, stops at critical points, is not additive over agents, generally has no orthogonal

(simultaneity-like) version, and runs slower for a meta-agent than for its constituents by an unknown ratio. MT Proposition 7.54 separates coarse-graining depth ℓ (a discrete index), the orbit parameter t (a choice), and the duration $\mathfrak{t}^{(\ell)}$ (arc length for a chosen metric and origin), and notes that even a strictly increasing duration need not be a regular coordinate ($z = t^2$ gives $\mathfrak{t} = t^2$). "Clock" here is a name for arc length.

7.4.7 Record clocks, semiconjugacy, and depth

Applied at every instant of a history P_t observed through one fixed kernel, the Fisher-loss theorem gives, for the record history $\bar{P}_t = P_t K$,

$$\dot{\mathfrak{t}}_P^2 - \dot{\mathfrak{t}}_{\bar{P}}^2 = \mathbb{E}_t \operatorname{Var}_t\big(\mathbf{s}_t(Y) \mid \bar{Y}\big) \geq 0, \qquad \operatorname{len}(\bar{P}) \leq \operatorname{len}(P)$$

(MT Corollary 7.42), with equality on an interval of finite length exactly when the fine score is a function of the record at almost every instant. This covers a meta-agent exactly when its law is the image of the same fine history under one fixed kernel, which natural Gaussian meta-agent maps can fail (Proposition G8.5). A coarse configuration map acting channelwise through Markov morphisms with equal weights gives the configuration-level version (MT Corollary 7.51).

A meta-agent that runs its own natural-gradient dynamics on its own free energy has a recomputed orbit, not an image. The chapter compares the two through

$$T_Q \mathsf{F}_\ell \mathscr{V}_\ell(Q) = \mathsf{a}_\ell(Q) \mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q), \qquad \mathsf{a}_\ell > 0$$

(MT Definition 7.45): the pushed fine velocity is a positive multiple of the coarse velocity. Then fine orbits map onto coarse orbits with time change $\psi_Q(t) = \int_0^t \mathsf{a}_\ell$ (MT Theorem 7.47), and the coarse duration of the image is the intrinsic length of the coarse orbit arc, independent of a_ℓ (MT Theorem 7.49). A constant map satisfies the relation trivially, so a nonconstant shared history is a separate requirement (MT Proposition 7.48); with $\mathsf{F}_\ell = \arctan$ on $\mathbb{R}$ both flows are complete and only part of the coarse orbit is reached; and the coarse duration is no larger than the fine one exactly when $\|T\mathsf{F}_\ell \mathscr{V}_\ell\| \leq \|\mathscr{V}_\ell\|$ along the orbit (MT Remark 7.50), which fiberwise Fisher contraction does not give by itself, because configuration weights are separate data and a recomputed orbit is not an image.

MT Proposition 7.53 says where the relation holds. Equal objectives do not suffice, since a change of metric breaks it. It holds for surjective submersions that are functionally compatible, $\mathcal{F}_\ell = \chi_\ell \circ \mathcal{F}_{\ell+1} \circ \mathsf{F}_\ell$ with $\chi_\ell' > 0$, and horizontally conformal with dilation d_ℓ, with $\mathsf{a}_\ell = \chi_\ell' \mathsf{d}_\ell^2$; quotients by free, proper, isometric symmetries of the objective have $\mathsf{a}_\ell \equiv 1$. Compatibility forces the fine objective to be constant along the fibers of the coarse map, so the route misses maps that discard directions on which the fine objective depends: when the coarse objective is the fiberwise infimum with unique fiberwise minimizers, compatibility with $\chi_\ell = \mathrm{id}$ holds on no open set, and in the manuscript's example it holds for no χ_ℓ. The chapter declines to assert a semiconjugacy for such maps and routes the comparison through the reduced slow flow of chapter 8, a conjugacy on the

graph of conditional minimizers (MT Proposition 8.4); whether the actual flow follows that graph is MT Open Problem 8.7.

7.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Proposition 7.3, MT Corollary 7.7	Fisher and Amari–Chentsov tensors are frame independent at c_*; a small group move costs $\frac{1}{2}t^2\mathfrak{k}_q(\xi,\xi)$, null on the isotropy algebra.	SOLID	Standard in substance; needs an action by sample maps.
MT Definition 7.13	The configuration metric, channel weights, and the Fisher metric itself are chosen.	CHOICE	Uniqueness theorems exist and are not imposed (MT Proposition 7.4).
MT Theorem 7.8	Coarse score is the conditional expectation of the fine score; Fisher loss is the expected conditional covariance.	SOLID	Parameter-independent kernel; equality is local sufficiency, not recovery; defects form a cocycle (MT Theorem 7.12).
MT Theorem 7.14	Null space of a pulled-back Fisher tensor; positive quotient bundle on constant-rank strata.	SOLID	Local; a global quotient needs a Hausdorff leaf space (open).
MT Definition 7.17	Dependence between agents enters through a supplied lift.	CHOICE	Different lifts give different metrics (MT Proposition 7.18, SOLID).
MT Theorem 7.20, MT Theorem 7.21	Local naturality forces the product law; two correlated split kernels leave no natural selector.	SOLID	Negative, elementary, relative to the admitted class; the reference-relative replacement is an unnumbered remark.
MT Theorem 7.23	A positive smooth symmetric correlated right inverse of the one-bit marginal map exists.	TOY	Three agents, binary channels, finite symmetry group, $\lvert\bar{\varkappa}\rvert < 1/64$.
MT Proposition 7.28, MT Corollary 7.25	Free energy and joint Fisher metric are not functions of single-agent marginals (and evidence).	SOLID	A refutation by a toy counterexample; effect second order in $\bar{\varkappa}$.
MT Proposition 7.31	Scalar mobility keeps orbits; anisotropic mobility, including a rate ratio, can change them.	SOLID	Orbit selection depends on the chosen metric.
MT Theorem 7.36, MT Theorem 7.40	Duration is reparameterization invariant; an orthogonal clock exists iff the gradient norm is locally a function of the free energy.	SOLID	Standard in substance; metric and origin chosen; orthogonal clocks fail generically.

Result	What it says	Status	Scope and caveats
MT Corollary 7.42	A same-path image under one fixed kernel runs no faster, by the exact conditional-variance defect.	SOLID	Covers a meta-agent only if its law is such an image; fails for parameter-dependent channels.
MT Theorem 7.47, MT Theorem 7.49	If the coarse map intertwines descent fields up to a positive rate, orbits correspond and the rate cancels from durations.	COND	Hypothesis: the semiconjugacy, not shown for any information-discarding meta-agent map.
MT Proposition 7.53	Semiconjugacy holds for compatible conformal submersions and symmetry quotients; for a fiberwise-infimum coarse objective, compatibility holds on no open set.	SOLID	The positive part excludes the maps formation needs.

Conditions: which ones matter

Conditions that can fail in the program's cases: parameter independence of the coarse kernel along the whole history (fails for Galerkin aggregation when $\mathrm{range}(\Lambda S)$ moves, for joint barycenters, for parameter-dependent channels); the chosen metric, weights, and rate ratio, which orbits and durations inherit; the supplied lift; the semiconjugacy, a hypothesis for every information-discarding map; a Hausdorff leaf space for global quotients; an accumulation point for finite total duration; and regularity (MT Hypothesis 7.1), which excludes singular strata, so that coarse families of mixture type, singular where components coincide, need checking. Hygiene a physicist can take for granted: standard Borel spaces, differentiability in quadratic mean, integrability of scores to third order, domination for differentiating divergences, C^1 metrics, C^2 objectives, local Lipschitz coarse fields, and real analyticity for finite duration, which Gaussian relative entropies and log-sum-exp attention terms satisfy in natural coordinates.

7.6 Toy versus general

Most small examples in the chapter are planar refutations with general logical force; they show which implications fail, not what large networks do. The parity witness has a general negative part, that no rule reading single-agent marginals returns the free energy or joint Fisher metric for every law, and a TOY positive part, the existence of a smooth symmetric correlated lift for three agents with binary channels. Its mechanism-level content, conjectural at large N, is that dependences of higher order than any retained marginal enter the collective free energy and joint Fisher metric through $\mathfrak{K}$ while staying invisible to marginal-based coarse descriptions and single-agent clocks; how large $\mathfrak{K}$ is at $N \geq 512$ in the program's generative models, and whether such dependences are generic there, is not addressed. The constants $\frac{65536}{16383}$ and $\frac{4}{16383}$, the bound $|\bar{\varkappa}| < 1/64$, the

ranks seven and six, and the 57-dimensional fiber are small-N specifics. The witness's hypotheses (finite alphabets, a discrete symmetry group) exclude continuous law fibers with connected groups such as $\mathrm{GL}(K,\mathbb{R})$, $\mathrm{GL}(K,\mathbb{C})$, or $\mathrm{SU}(N)$, and whether correlated lifts equivariant under a connected group exist there is open (MT Open Problem 7.29). Everything else (tensor descent, Fisher loss, record clocks, clock invariance, the orthogonal-clock criterion, the semiconjugacy calculus) is general and independent of N; its hypotheses admit nonabelian and noncompact groups acting by sample maps, such as linear $\mathrm{GL}(K)$ on Gaussian samples.

7.7 Open ends

MT Open Problem 7.29 asks, for a stated regular family and a stated class of morphisms containing non-product-preserving kernels, for a smooth right inverse of the marginal map natural under that class, and for correlated lifts equivariant under a connected structure group on continuous fibers; a construction or no-go theorem settles each. MT Open Problem 7.15 asks for a checkable criterion under which the null foliation of a pulled-back Fisher tensor has a Hausdorff leaf space, and for block data on the quotient fixed by a typing of agents and channels, since the law and its Fisher tensor alone select no agent splitting (MT Proposition 11.46). MT Open Problem 7.43 asks for the ratio of an N-constituent meta-agent's duration to its constituents' collective duration, as a function of N and of the correlation of constituent velocities, in the coherent and incoherent regimes. The physicist's expectation, that coherent motion survives coarse-graining while incoherent motion is suppressed with N, is exactly what is unproved, and the problem warns that a divergence computation does not settle it because expected arc-length ratios are not square roots of expected divergence ratios. Finally, for information-discarding maps the comparison depends on MT Open Problem 8.7 (section 8.7); a semiconjugacy for such maps is declined without an open problem of its own.

Does it deliver what it claims? NARROWER THAN IT SOUNDS

Claimed. The opening says that a chosen metric turns the collective free energy into an oriented orbit with a reparameterization-invariant information clock; that a parameter-independent Markov channel loses Fisher information by an exact conditional-covariance defect, so that a same-path Markov image such as the law of a meta-agent accumulates no more duration; that a recomputed coarse orbit compares with the fine one only through an oriented semiconjugacy whose rate cancels from durations; and that a parity witness shows the free energy and its Fisher tensor are not functions of marginal laws.

Delivered. Each statement is proved at its stated generality. Much is standard mathematics in new typing (Fisher data processing, arc length, gradient flows under a mobility, Łojasiewicz), with a counterexample at each boundary. The parity witness and the selection theorems deliver their negative claims exactly.

Gap. "Such as the law of a meta-agent" outruns the theorems: the record clock covers a meta-agent only when its law is a parameter-independent Markov image of the fine history,

which Galerkin aggregation across precisions and joint barycenters are not. For meta-agents that run their own dynamics on information-discarding maps, which formation needs, the chapter's sufficient route to a semiconjugacy does not apply (for a fiberwise-infimum coarse objective it holds on no open set), none is asserted, and the comparison passes to a reduced flow whose tracking is open. The meta-agent clock ratio is open. Every orbit and duration rests on a chosen metric and, for collectives, a supplied lift; neither the orbit nor the clock is a property of the free energy alone.

Chapter 8

Two-Timescale Dynamics

8.1 The question and the answer

MT Chapter 8 asks what the collective free energy supplies for separating fast inference of beliefs from slower evolution of models, priors, and parents. A physicist expects adiabatic elimination: a small rate ratio ε, a reduced slow dynamics with the fast variables at their conditional optimum, and a theorem that the full dynamics follows it as $\varepsilon \to 0$. The chapter proves the first two in general. The third holds only on held matching sheets of a four-agent toy, where the chapter verifies the hypotheses of Tikhonov's theorem; elsewhere it is open.

The results are these. Any positive-definite block mobility gives descent, and in a product recognition family an agent descending its own local free energy moves along a block of collective descent (MT Proposition 8.1). The envelope identity gives the reduced slow flow as the natural gradient of the free energy minimized over the fast block (MT Lemma 8.3). The rate-separated flow agrees with the reduced flow on the graph of conditional minimizers, where projection to the slow coordinates is a conjugacy (MT Proposition 8.4); this is the route chapter 7 uses for information-discarding maps. On the four-cycle the model channel lags the belief channel on an interval of order $1/\varepsilon_{\mathrm{m}}$ exactly when the links are slow too (MT Proposition 8.6). In a finite discrete setting, parent-sensitive sweeps descend (MT Theorem 8.10), the parent is Markov under instantaneous within-parent refresh (MT Theorem 8.11), and it has memory under partial refresh (MT Remark 8.12).

The free energy chooses none of the split, the rate ratio, or the mobilities, and a rate ratio can change the orbit (MT Proposition 7.31). The scope paragraph declines tracking beyond the held sheets, tracking by sampled dynamics with source jumps, persistence of a belief-coherent, model-incoherent pair under such sampling at the formation temperature, convergence of the alternating scheme, a uniform rate for the parent sweep, and closed dynamics for a static parent datum. The chapter uses chapters 5 to 7 and feeds chapters 11 and 13. The finite-depth formation family realizes one scalar slow parameter, with dynamics MT (13.25), not slow per-agent model laws, so the belief–model split is instantiated only on the four-cycle, with supplied rates.

8.2 The picture

The picture is Born–Oppenheimer: fast electrons relax at fixed nuclear positions, and the nuclei move on the resulting surface. Beliefs play the electrons; models and generative parameters play the nuclei.

Physics analogue (structural)

$F_{\mathrm{red}}(s) = \min_f F(f, s)$ plays the Born–Oppenheimer surface [12], and the envelope identity $dF_{\mathrm{red}} = \partial_s F(f^\star(s), s)$ is the Hellmann–Feynman statement for a variational ground state: the slow force is the explicit partial derivative at the conditional optimum, because the implicit variation of the fast variables drops out at a stationary point. What does not transfer: the adiabatic theorem that justifies Born–Oppenheimer and its nonadiabatic corrections; the chapter proves only the zeroth-order reduced dynamics. F_{red} is a minimum over a recognition family, not a partition function; this is zero-temperature elimination, not integrating out.

Physics analogue (structural)

The rate-separated flow is a singularly perturbed gradient system, and Tikhonov's theorem [69] is the standard justification of adiabatic elimination: if the frozen fast subsystem has a uniformly stable equilibrium and the fast start lies in its basin, the slow variables converge to the reduced flow as $\varepsilon \to 0$; Fenichel's theory [30] adds a nearby invariant slow manifold. The chapter states these hypotheses in MT Open Problem 8.7 and verifies them only on held four-cycle sheets; sampled dynamics with source jumps lie outside such a theorem.

Two features have no textbook counterpart. When the kinetic metric couples beliefs and models, as non-product recognition makes it do, "make the belief block fast" is ambiguous: slowing the model rows of the velocity and rescaling the metric by a congruence give different slow sets, and only the second keeps a gradient structure (MT Remark 8.5). On the four-cycle the picture is lattice gauge theory on a ring: each residual is a neighbor difference minus a link, and both channels are compared through the same link, so a fast link cancels a slow channel's mismatch and the fast beliefs follow it. Model disagreement measured through a link is link relative and survives only if the link is slow too.

Physics analogue (structural)

The refresh $\mathsf{D} = \mathsf{CJ}$ is the discrete Nakajima–Zwanzig projector onto conditional equilibrium [54, 86]. Projected dynamics has a memory term unless the unresolved part is reset at every step; a partial reset leaves memory, and the exact discrete recurrence is MT Theorem 11.39 (chapter 11). The prepared-parent theorem is a lumpability statement [46] in intertwining form, valid from conditional-equilibrium starts.

8.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Fast block f; slow block $s = (s_{\mathrm{m}}, \theta)$	Declared fast coordinates (beliefs, source rows) and slow ones (model recognition coordinates, generative parameters)	Electrons and nuclei
$f^\star(s)$, $\mathcal{S}^\star$	Conditional minimizer over the fast block; its graph	Slaved variables; adiabatic surface
$F_{\mathrm{red}}(s)$	Free energy minimized over the fast block	Born–Oppenheimer surface
G_{f}, G_{s}, ε, G_ε	Block metrics; rate ratio; $\varepsilon\mathsf{G}_{\mathrm{f}} \oplus \mathsf{G}_{\mathrm{s}}$	Mobilities; singular-perturbation parameter
Rate-separated, reduced flow	Natural gradient for G_ε; natural gradient of F_{red}	Singularly perturbed system; adiabatic dynamics
Row-rated field, congruence rescaling	Two ways to make a block fast when the metric couples blocks	Choice of coordinates for elimination
Accepted M-coordinate	Parameter step that does not lower the evidence bound (MT Definition 5.28)	Generalized M step
Four-cycle: $\bar{k}$, $\bar{m}$, $\breve{\omega}$; r_i^b, r_i^m	Belief means, model means, link translations on a ring; neighbor difference minus link	Matter and link fields; covariant difference
S, $S^{(0)}$, T	Attention assignment; one held perfect matching; temperature	Quenched bonds; temperature
ε_{m}, $\varepsilon_{\breve{\omega}}$	Supplied rates of model means and links	Mobilities
ξ, $z = \mathrm{par}(\xi)$	Fine state and its parent (complete decorated block label)	Microstate, macrostate
M, C, J, $\mathsf{D} = \mathsf{CJ}$, $\mathsf{T} = \mathsf{MD}$	Metropolis move; projection to parent; posterior lift given parent; refresh; one sweep	Update; lumping; disaggregation; projector
$\overline{\mathsf{T}} = \mathsf{JMC}$; prepared start $\nu\mathsf{J}$	Parent transition kernel; fine law in conditional equilibrium within parents	Lumped chain; local equilibrium
D_ω; $D_N^{(1)}$	Refresh with probability $\omega < 1$; cost of a first-order Markov parent	Finite separation; memory

The chapter's "slow manifold" is Tikhonov's and Fenichel's attracting set; the graph $\mathcal{S}^\star$ is different and at finite ε generally not invariant, so the chapter's exactness statements hold on it, not near it. "Rate" is a chosen mobility factor, "declared" means chosen, and "exact route" means the conjugacy on $\mathcal{S}^\star$, not a tracking theorem. Kernels act on row vectors from the right, so $\mathsf{T} = \mathsf{MD}$ means move, then refresh. The sweep T and temperature T are different glyphs; ω is the refresh fraction, and the four-cycle links, written ω in chapter 13, are $\breve{\omega}$ here; par is the parent map (the parity character in chapter 7); the rate ratio ε is unrelated to the formation regime parameter.

8.4 The formulas, explained

8.4.1 Descent with any mobility, and when it is local

$$\frac{d}{dt}F = -\sum_{i\in V'}(\nabla_{\lambda_i}F)^\top \mathsf{G}_i^{-1}(\nabla_{\lambda_i}F) \le 0.$$

Each moving block $i \in V'$ descends the collective free energy $F(\lambda)$ with its own positive-definite metric, rates absorbed (MT Proposition 8.1); the chain rule gives the sign. The content is locality. In a product family $Q_\lambda = \bigotimes_i Q_{i,\lambda_i}$ the partial gradient $\nabla_{\lambda_i}F$ equals the gradient of agent i's outside-averaged local free energy, so selfish local descent is collective descent, an exact potential game (MT Theorem 5.23), and the block Fisher metrics make it the Fisher natural gradient. Outside product families moving λ_i changes the others' marginals, the lift of chapter 7 is needed, and the Fisher metric has off-diagonal blocks (for a bivariate normal location family with correlation c_2, the entry $-\mathrm{c}_2/(1-\mathrm{c}_2^2)$). "Each agent follows its own natural gradient" is the collective natural gradient only when recognition factorizes.

8.4.2 Fast and slow blocks, the envelope identity, and the reduced flow

$$F(f,s) = -\log p_{\theta,V}(o \mid X) + D_{\mathrm{KL}}\big(Q_{f,s_{\mathrm{m}}} \| \mathbf{\Pi}_{\theta,V,o,X}\big), \qquad s = (s_{\mathrm{m}}, \theta),$$
$$F_{\mathrm{red}}(s) = \min_f F(f,s), \qquad dF_{\mathrm{red}}(s) = \partial_s F\big(f^\star(s), s\big), \qquad \dot{s} = -\mathsf{G}_{\mathrm{s}}(s)^{-1}\partial_s F\big(f^\star(s), s\big).$$

The evidence term depends only on the generative parameter θ; the gap term depends on everything. The fast block carries belief recognition coordinates and attention source rows, the slow block model recognition coordinates and θ, which moves the joint law between evaluations and never inside one (MT Hypothesis 4.3); the assignment to blocks is declared (MT Definition 8.2). MT Lemma 8.3 gives the second line: at the conditional minimizer $\partial_f F = 0$, so the implicit dependence of $f^\star$ drops out, and the reduced flow is the G_{s}-natural gradient of F_{red}. A C^1 branch exists where the fast Hessian is positive definite, and on a compact fast domain with unique minimizer the identity holds without differentiability of $f^\star$ [21]. In the limiting case where the slow block is θ alone and the fast family contains the exact posterior, the gap vanishes at the minimum, $F_{\mathrm{red}}(\theta) = -\log p_{\theta,V}(o \mid X)$, and the reduced flow is gradient ascent on the log evidence, a continuous-time exact EM.

8.4.3 The rate-separated flow, and what is not proved about it

$$\varepsilon \dot{f} = -\mathsf{G}_{\mathrm{f}}^{-1}\partial_f F, \qquad \dot{s} = -\mathsf{G}_{\mathrm{s}}^{-1}\partial_s F.$$

This is the natural gradient for $\mathsf{G}_\varepsilon = \varepsilon\mathsf{G}_{\mathrm{f}} \oplus \mathsf{G}_{\mathrm{s}}$, so it descends for every ε, and its orbit can depend on ε (MT Proposition 8.4). On $\mathcal{S}^\star$ the fast component vanishes and the slow component equals the reduced flow; the reduced field lifted to $\mathcal{S}^\star$ is tangent to it, and projection to the slow coordinates is a conjugacy with rate one, which is MT (8.8); the G_ε-length of the lifted orbit tends to the slow

length. The proposition does not say that a solution started off $\mathcal{S}^\star$ approaches it and follows the reduced flow; that is Tikhonov's content and is not proved at this generality.

MT Remark 8.5 shows why care is needed. Off $\mathcal{S}^\star$ the slow velocity need not be parallel to the reduced gradient, and at finite ε the graph is not invariant. With a correlated metric the naive construction fails: for $F = [(f-s)^2 + s^2]/2$, whose conditional minimizer is $f = s$, and a metric with unit diagonal and off-diagonal $\mathrm{c}_2 \in (0,1)$, slowing the model row of the natural-gradient velocity gives the fast critical set

$$\Big\{ f = \frac{1+2\mathrm{c}_2}{1+\mathrm{c}_2}\, s \Big\},$$

attracting at rate $1/(1-\mathrm{c}_2)$ and not the conditional minimizers; at $\mathrm{c}_2 = \frac{1}{2}$ and $\varepsilon < 7 - 4\sqrt{3} \approx 0.072$ this field increases F at an explicit point, so it is the natural gradient of no metric. The congruence rescaling $\Delta_\varepsilon \mathsf{G} \Delta_\varepsilon$, $\Delta_\varepsilon = \mathrm{diag}(\sqrt{\varepsilon}, 1)$, descends for every ε and recovers the conditional minimizers at $\varepsilon = 0$. With correlated metrics a two-timescale model must state how the rate ratio enters the metric; this is a modeling requirement, since correlated belief–model metrics are what non-product recognition produces.

8.4.4 The four-cycle: slow models need slow links

On the ring of MT Definition 13.21, agents have belief laws $\mathcal{N}(\bar{k}_i, 1)$ and model laws $\mathcal{N}(\bar{m}_i, \frac{1}{2})$, edges e_i from i to $i+1$ carry link translations $\breve{\omega}_i$, and the residuals are $r_i^b = \bar{k}_{i+1} - \bar{k}_i - \breve{\omega}_i$ and $r_i^m = \bar{m}_{i+1} - \bar{m}_i - \breve{\omega}_i$. Each receiver attends one neighbor; the energy U_S collects attended edge costs and temperature terms; the matching $S^{(0)}$ selects e_0 and e_2. MT Proposition 8.6 runs

$$d\mathbf{x} = -\big(\partial_{\bar{k}} U_S, \varepsilon_{\mathrm{m}} \partial_{\bar{m}} U_S, \varepsilon_{\breve{\omega}} \partial_{\breve{\omega}} U_S\big)\, dt + \sqrt{2\mathrm{T}}\big(dW^{\bar{k}}, \varepsilon_{\mathrm{m}}^{1/2} dW^{\bar{m}}, \varepsilon_{\breve{\omega}}^{1/2} dW^{\breve{\omega}}\big),$$

with the assignment redrawn at a rate-one clock. The rates are supplied; a Fisher natural gradient would move model means at rate $\frac{1}{2}$.

Part (i) holds the matching. The mean then follows the noiseless rate-separated flow with $\varepsilon = \varepsilon_{\mathrm{m}}$, fast block $\bar{k}$, slow block $(\bar{m}, \breve{\omega})$. On each selected edge, in the difference coordinates $(\bar{k}_i^-, \bar{m}_i^-, \breve{\omega}_i)$, the Hessian block is

$$\begin{pmatrix} 4+3\mathrm{T} & -2\mathrm{T} & -2\sqrt{2} \\ -2\mathrm{T} & 4+3\mathrm{T} & -2\sqrt{2} \\ -2\sqrt{2} & -2\sqrt{2} & 4+\mathrm{T} \end{pmatrix},$$

with fast curvature $4 + 3\mathrm{T} \geq 4$ uniformly in temperature, so with equal model and link rates the Tikhonov hypotheses hold on that block with constant 4. On the pair means $\bar{k}_i^+$, however, the fast curvature is only $3\mathrm{T}$: at the small temperatures of the formation regime that direction relaxes at a rate of order T, so on the full sheet the constant is of order T and not uniform as $\mathrm{T} \to 0$.

Part (ii) sets $\mathrm{T} = 0$ and starts with every residual at -1. Each selected-edge block has a zero mode, on which both residuals vanish, and rates $\varsigma_\pm = 2(1 + \varepsilon_{\mathrm{m}} + \varepsilon_{\breve{\omega}}) \pm 2\sqrt{(1-\varepsilon_{\mathrm{m}})^2 + \varepsilon_{\breve{\omega}}^2}$. A fast link ($\varepsilon_{\breve{\omega}} = 1$) gives $|r_i^m(t)| \leq e^{-(4-2\sqrt{2})t}$ for every ε_{m}, removing the model disagreement on the belief clock even with frozen model means; fast model means give $r_i^m = -e^{-(4+4\varepsilon_{\breve{\omega}})t}$. With slow means

and equally slow links ($\varepsilon_{\breve{\omega}} = \varepsilon_{\mathrm{m}} \le \frac{1}{2}$),

$$|r_i^b(t)| \le (1+\varepsilon_{\mathrm{m}})e^{-4t} + \varepsilon_{\mathrm{m}}, \qquad \big(1 - \tfrac{3}{2}\varepsilon_{\mathrm{m}}\big)e^{-6\varepsilon_{\mathrm{m}}t} - \tfrac{3}{2}\varepsilon_{\mathrm{m}} \le |r_i^m(t)| \le e^{-(6\varepsilon_{\mathrm{m}} - 2\varepsilon_{\mathrm{m}}^2)t} + \tfrac{3}{2}\varepsilon_{\mathrm{m}},$$

so for $\varepsilon_{\mathrm{m}} \le 1/25$ each matched pair has $|r_i^b| \le 2\varepsilon_{\mathrm{m}}$ and $|r_i^m| \ge \frac{1}{2}$ from $\frac{1}{4}\log(4/\varepsilon_{\mathrm{m}})$ to $1/(12\varepsilon_{\mathrm{m}})$: beliefs agree while models disagree, for a time of order $1/\varepsilon_{\mathrm{m}}$.

Part (iii) returns to the sampled process along any assignment path: a model residual of size at least one drops below $\frac{1}{2}$ before time t, on the event that residuals and configuration stay bounded, with probability at most $2\exp[-1/(192\mathrm{T}\bar{\varepsilon}t)]$, $\bar{\varepsilon} = \max(\varepsilon_{\mathrm{m}}, \varepsilon_{\breve{\omega}})$, for t up to order $1/\bar{\varepsilon}$. It is a bound on a joint event, not an unconditional persistence probability. The mechanism-level content, conjectural beyond the four-cycle, is that a link equilibrating on the belief clock removes any disagreement measured through it; the rates $4 - 2\sqrt{2}$ and $6\varepsilon_{\mathrm{m}}$, the window, and the constant 192 are four-cycle specifics.

8.4.5 Parents: the prepared parent and the refresh hypothesis

The last section is finite (MT Definition 8.8): fine states ξ, a parent map par, a positive prior and likelihoods with posterior p, and unobserved records with laws ψ_ξ. The kernels are

$$\mathsf{M}_{\xi\xi'} = \mathsf{a}_{\xi\xi'} \min\Big(1, \frac{p(\xi')}{p(\xi)}\Big) \quad (\xi \ne \xi'), \qquad \mathsf{C}_{\xi z} = \mathbf{1}_{\mathrm{par}(\xi)=z}, \qquad \mathsf{J}_{z\xi} = \frac{p(\xi)\mathbf{1}_{\mathrm{par}(\xi)=z}}{p_Z(z)},$$
$$\mathsf{D} = \mathsf{CJ}, \qquad \mathsf{T} = \mathsf{MD}, \qquad \mathcal{F}(q \otimes \psi'; o) = D_{\mathrm{KL}}(q\|p) + \mathbb{E}_q D_{\mathrm{KL}}(\psi'_\xi \| \psi_\xi) - \log \mathcal{Z}_o.$$

M is a Metropolis move with symmetric proposals; C reads the parent; J draws a fine state from the posterior given the parent; D refreshes within the parent; T is one sweep. The free energy shows that matching record conditionals is a specialization of the joint objective. MT Theorem 8.10 shows that the kernels preserve p, that the refresh is the exact block update given the parent, that any p-stationary kernel lowers $D_{\mathrm{KL}}(q\|p)$ by an explicit reverse-conditional divergence, that the sweep lowers it strictly unless q is in conditional equilibrium on each connected component, and that convergence is geometric on each component, with no uniform rate.

The prepared-parent theorem (MT Theorem 8.11) reads

$$\mathsf{JT} = \overline{\mathsf{T}}\mathsf{J}, \qquad (\nu\mathsf{J})\mathsf{T}^n = \nu\overline{\mathsf{T}}^n\mathsf{J}, \qquad \overline{\mathsf{T}} = \mathsf{JMC}.$$

From a fine law $\nu\mathsf{J}$ in conditional equilibrium within parents, one sweep returns a fine law in conditional equilibrium with parent law $\nu\overline{\mathsf{T}}$. The parent history is Markov with the p_Z-reversible kernel $\overline{\mathsf{T}}$, and from any start after the first refresh, since after a refresh the fine state's law given the whole parent history depends only on the current parent. This is standard in substance: a chain resampled from block-conditional equilibrium at every step is a function of its block chain. The manuscript's parallel with the continuous reduced conjugacy (fast variable at its conditional optimum given the slow one) is structural; no theorem links the two.

Everything rests on MT Hypothesis 8.9, exact within-parent resampling at every sweep, which the manuscript calls adiabatic elimination with an infinitely shorter timescale. The partial refresh $\mathsf{D}_\omega = (1-\omega)I + \omega\mathsf{CJ}$, $\omega < 1$, keeps normalization, reversibility, and the parent, and loses the Markov property (MT Remark 8.12). With $\mathrm{par}(1) = \mathrm{par}(2) = z_1$, $\mathrm{par}(3) = z_2$, uniform p, and one proposal between 2 and 3,

$$\mathbb{P}(Z_2 = z_2 \mid Z_1 = z_1, Z_0 = z_1) = \tfrac{1}{6} + \tfrac{\omega}{12}, \qquad \mathbb{P}(Z_2 = z_2 \mid Z_1 = z_1, Z_0 = z_2) = \tfrac{1}{2} - \tfrac{\omega}{4}.$$

State 2 is the border of z_1 with z_2. A parent just arrived from z_2 has its fine state at the border with probability $1 - \omega/2$, a parent already in z_1 with probability $\frac{1}{3} + \omega/6$; the parent's past says where in the fiber the fine state sits, and that predicts the next move. The dependence $(1-\omega)/3$ vanishes only at $\omega = 1$. The parent path law is still retained exactly as a function of the fine chain (MT Section 11.8.1), and the cost of forcing a first-order Markov parent is the conditional mutual information $D_N^{(1)}$ of MT Theorem 11.44, positive here at $N = 3$.

8.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Definition 8.2	The fast–slow split, rate ratio, and block mobilities are declared.	CHOICE	The free energy selects none; a rate ratio can change the orbit.
MT Proposition 8.1	Any positive-definite block mobility descends; in product families local descent is a block of collective descent.	SOLID	Standard in substance; correlated recognition breaks block diagonality.
MT Lemma 8.3	Envelope identity; the reduced slow flow is the natural gradient of the reduced functional.	SOLID	Standard (envelope theorem, Danskin); needs a C^1 branch of minimizers.
MT Proposition 8.4	On the graph of conditional minimizers the rate-separated flow has the reduced slow component; projection is a conjugacy.	SOLID	Kinematic: says nothing about approaching the graph.
MT Remark 8.5	Off the graph no semiconjugacy; the graph is not invariant; correlated metrics move the slow set and can break descent; congruence repairs both.	SOLID	Negative witnesses in two and three dimensions.

Result	What it says	Status	Scope and caveats
MT Proposition 8.6	On the four-cycle a belief-agreeing, model-disagreeing interval of order $1/\varepsilon_{\mathrm{m}}$ exists exactly when model means and links are both slow; held sheets meet the Tikhonov hypotheses.	TOY	Four agents, scalar Gaussian means, translation links, supplied rates; (ii) at zero temperature on a held matching.
MT Open Problem 8.7	Tracking of the reduced flow by the rate-separated or sampled flow.	OPEN	Posed for the sampled four-cycle class; learning outside.
MT Theorem 8.10	Kernels preserve the posterior; refresh is the exact block update; sweeps descend and converge geometrically on components.	SOLID	Standard in substance; no uniform rate.
MT Hypothesis 8.9	Every sweep resamples the fine state exactly given the parent.	CHOICE	Idealized infinite separation; substantive.
MT Theorem 8.11	From a prepared start the parent history is Markov with a reversible lumped kernel.	COND	Hypothesis: instantaneous refresh. Standard in substance.
MT Remark 8.12	Under partial refresh the parent is not Markov; the past dependence is $(1-\omega)/3$.	SOLID	Refutation by a three-state witness; constant specific to it.

Conditions: which ones matter

Conditions that can fail in the program's cases: the declared split, rate ratio, and mobilities; a C^1 branch of minimizers with fast curvature uniformly bounded below, which fails after source rows are optimized even when every source energy and frozen-weight curvature is positive (MT Proposition 6.8) and on the full four-cycle sheet, where the constant is $3\mathrm{T}$; a block-diagonal metric with f-independent slow block, which correlated recognition breaks; a product family for local descent; instantaneous refresh for the Markov parent; and, on the four-cycle, a held matching in (i) and (ii), zero temperature in (ii), bounded residuals and configuration in (iii), and no learning.

Hygiene: C^1 and C^2 regularity, solutions staying in the open domain, integrable log normalizers, finite state spaces with positive prior and likelihoods, and proposal row sums at most $\frac{1}{2}$, which secures positive holding.

8.6 Toy versus general

General and independent of N: block descent, the envelope identity and reduced flow, the kinematic agreement on $\mathcal{S}^\star$, the parent sweep's descent, and the Markov parent under instantaneous refresh. The negative witnesses (a slow velocity off the graph, the correlated-metric slow set, the three-

state partial refresh) are refutations with general force; their constants $7 - 4\sqrt{3}$ and $(1 - \omega)/3$ are specific. The four-cycle dichotomy is TOY: four agents, scalar Gaussian means with fixed variances, translation links ($\mathbb{R}$ acting on means, an abelian noncompact group), supplied rates, a held matching in (i) and (ii), zero temperature in (ii). These exclude the target of 512 or more agents with nonabelian links such as $\mathrm{GL}(K, \mathbb{R})$ or $\mathrm{SU}(N)$. The mechanism, that a link fast on the belief clock absorbs disagreement measured through it so that model incoherence with belief coherence needs slow links as well as slow models, is plausible on larger graphs and unproved there; the rates, the window, the bound $\varepsilon_{\mathrm{m}} \leq 1/25$, and the persistence constant are four-cycle specifics. The only place the open tracking problem is closed, the held sheets, is itself a toy, as the chapter says.

8.7 Open ends

MT Open Problem 8.7 asks when Tikhonov's and Fenichel's theorems apply to the rate-separated flow at c_*. Its three hypotheses (a C^2 objective with C^1 block metrics and an f-independent slow metric; a C^1 branch of minimizers with fast curvature at least c_1 relative to the fast metric on a compact set; a reduced orbit inside that set and a fast start in the basin) are verified only on held sheets. Source-row optimization can destroy the curvature bound, sampled dynamics with source jumps lie outside a deterministic theorem, and no explicit bound in ε on the slow orbit or its Fisher duration exists. The problem is posed in one class, the sampled four-cycle with slow model means and links at equal rates and no learning: settled positively by a bound on the distance between the sampled slow coordinates and the reduced flow of the assignment path over a bounded slow-time interval, explicit in ε_{m}, T, and the number of assignment changes, tending to zero, with probability tending to one when changes per unit slow time are bounded; negatively by rates and temperatures tending to zero along which that distance stays bounded below with probability bounded below. Learning and correlated belief–model metrics are outside the posed class; for the latter the chapter establishes only that the rate declaration must name its slow set. MT Open Problem 5.32 asks for convergence of the discrete alternating scheme, and MT Open Problem 11.38 asks how a parent's dynamics follow from the network it summarizes, which the prepared parent answers only under instantaneous refresh. The uniform rate of the parent sweep and persistence under sampling with jumps at the formation temperature are declined without open problems.

Does it deliver what it claims? NARROWER THAN IT SOUNDS

Claimed. The opening promises descent for every positive-definite block mobility, local in product families; a reduced slow flow conjugate to its lift on the graph of the conditional optimum, the exact route for information-discarding maps; a Markov parent under instantaneous refresh, failing under finite refresh; and, on the four-cycle, a model channel lagging the belief channel exactly when links are slow too.

Delivered. All of it, as stated. The general results are the kinematics of an adiabatic reduction (envelope theorem, reduced natural gradient, agreement on the graph of minimizers) and its discrete, infinitely separated case (lumpability), all standard in substance, plus clean witnesses

that correlated metrics move the slow set and finite separation creates memory. The four-cycle dichotomy is a complete toy result with explicit rates and a pathwise probability bound.
Gap. The dynamical content of a two-timescale theory, that the actual flow at small ε follows the reduced one, holds only on the four-cycle's held sheets, where the chapter verifies Tikhonov's hypotheses; it is not established on the full sheet (fast curvature of order the temperature), with source jumps, for correlated metrics, or with learning. The "exact route" is exact on the graph of minimizers and unproved as dynamics. The Markov parent rests on a named idealization, and parent dynamics under finite refresh is open. The split, rate ratio, and mobilities are declared, and the formation theorem does not use the belief–model split. The scope paragraph declines each of these; the title promises more than the theorems deliver.

Chapter 9

Coarse Channels

9.1 The question and the answer

MT Chapter 9 opens Part II of the main manuscript. It asks what a coarse description of the agent network at the base point c_* keeps of three laws: the normalized generative joint $\mathbb{P}(do, dy)$ of MT chapter 4, its selected posterior $\mathbf{\Pi}_o$, and a recognition law $\mathbb{Q}_o$. The answer is one identity. Take a coarse channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$, a Markov kernel that reads the latent coordinates but neither the observation nor the recognition law, and push all three laws through it. The evidence $p(o)$ is then the same at both scales, the pushed posterior is a posterior of the pushed joint, and the free energy splits exactly (MT Theorem 9.6):

$$\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_o) = \mathcal{F}_{\mathbb{P}^C}(\mathbb{Q}_o C) + \Delta(o), \qquad \Delta(o) \in [0, +\infty],$$

where $\Delta(o)$ is what recognition gets wrong, relative to the posterior, inside the coarse cells the channel no longer resolves. Applied to a finite block I of agents and a parent state space, the same identity produces a complete normalized parent triple, the pointwise parent datum (MT Corollary 9.10), on which chapters 10 to 13 build. Around this core the chapter collects kernel tools (composition, data processing with its equality case, Bayes recovery), proves that three operations physicists all call coarse-graining obey different theorems, treats symmetry, defines scale diagrams, and ends with labeled mixtures whose retained label carries Fisher information of order N^2 at a tie.

The verdict in advance: the core identity is the relative-entropy chain rule applied twice to the joint law of fine and coarse variables. It is standard in substance; the contribution is careful typing and a clean statement of what the identity does not give. Exact means an identity with an explicit nonnegative remainder. It does not mean closure of the coarse law in a model family, a selected channel, or parent dynamics, and the chapter claims none of these.

9.2 The picture

A coarse channel is a noisy detector attached to the latent coordinates: given a fine configuration y it emits a coarse value z with probability $C(y, dz)$, and a deterministic statistic such as a block sum is the noiseless case. Pushing a law through it is what a stochastic block-spin rule does to a Gibbs measure.

Physics analogue (exact)

One step of Kadanoff blocking. A block-spin weight $T(\sigma' \mid \sigma) \geq 0$ with $\sum_{\sigma'} T(\sigma' \mid \sigma) = 1$ maps e^{-H} to $e^{-H'(\sigma')} = \sum_{\sigma} T(\sigma' \mid \sigma) e^{-H(\sigma)}$, and this trace condition leaves the partition function invariant [13, 43]. A coarse channel is such a weight on the agents' latent coordinates, and evidence preservation (MT Corollary 9.7) is the trace condition with $p(o)$ in the role of Z. The correspondence is exact for one step, and everything Kadanoff and Wilson [82] add to that step is absent. There is no lattice and no length to rescale: the agents sit at one base point and blocks are vertex sets of a graph. The coarse law is not projected onto a finite set of couplings (an effective action first appears in chapter 11). No single rule is iterated, so there is no flow on a parameter space, no fixed point, and no exponent. The standing scope declines all of these (MT Section 1.4); the asymptotic apparatus is in Supplement B.

The split itself is simple. Recognition and posterior are two laws on the fine space; the channel coarsens both; their relative entropy splits into the relative entropy of the coarse laws plus the relative entropy of the two conditional laws of the fine configuration given each coarse value, averaged over coarse recognition. The common term $-\log p(o)$ carries this over to free energies. The coarse free energy is never the larger: the coarse description looks better only because it no longer sees the errors it discarded, which the manuscript calls bookkeeping, not learning.

A two-spin example (the companion's arithmetic) shows which errors survive. Let the posterior be the Ising pair $\mathbf{\Pi}(s_1, s_2) \propto e^{J s_1 s_2}$ and recognition the uniform product law, a mean-field guess, so that $D_{\mathrm{KL}}(\mathbb{Q} \| \mathbf{\Pi}) = \log\cosh J$. A channel keeping the bond $s_1 s_2$ finds both laws uniform inside each bond cell: $\Delta = 0$ and the coarse gap is the whole $\log\cosh J$. A channel keeping only s_1 finds both coarse laws equal to $(1/2, 1/2)$: the coarse gap is zero and $\Delta = \log\cosh J$, so the one-spin description is perfectly calibrated and the entire error of that guess sits in the discarded cell.

Physics analogue (exact)

Constrained (Landau) free energy. On finite spaces and for a deterministic statistic c, the coarse free energy of a coarse law q' is the minimum fine free energy over fine laws with image q' (MT Proposition 9.9, part (i)); for $q' = \delta_z$ it is $-\log[p(o)\mathbf{\Pi}_o(c = z)]$, the constrained free energy at fixed order parameter. For a noisy channel this fails (part (ii)): the coarse free energy of q' can lie strictly below every fine free energy compatible with q'.

Physicists call three different moves coarse-graining: integrating out (pushing a law through a kernel), restricting the configurations that enter an energy (fields constant on blocks), and compressing an operator onto a subspace (a multigrid coarse operator). MT Section 9.4 proves that they obey different theorems, so a proof about one transfers nothing to another.

Physics analogue (structural)

Feshbach projection. The Galerkin coarse propagator reproduces the fine dynamics on kept signals exactly when the operator never maps discarded directions into kept ones (MT Proposition 9.14), which in Feshbach language is the vanishing of the coupling that would

generate a self-energy. The block-operator algebra is the same; energy-dependent self-energies are not treated.

At the end of the chapter the retained coordinate is a label γ from a finite menu whose log weight grows like N. The weights are then a softmax of extensive arguments, like the weights of coexisting phases with free energies Nf_γ: a strict winner hides the label, a tie makes it a very sensitive function of the parameter, and a window of width $1/N$ interpolates.

Physics analogue (structural)

Finite-size rounding of a first-order transition. Borgs and Kotecký [11] prove for lattice models with finitely many coexisting phases that the transition is rounded over a window of width of order inverse volume; at coexistence the variance of the total conjugate extensive quantity is then of order volume squared. The retained-label theorems (MT Section 9.8) are the same softmax calculus with no lattice and no Gibbs measure; extensive growth of the log weights is assumed, not derived from an agent model. The manuscript names this correspondence itself.

9.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
coarse channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$	Markov kernel on latent coordinates, stochastic by default, reading neither o nor $\mathbb{Q}_o$; composites act on the right	block-spin weight $T(\sigma' \mid \sigma)$
selected posterior version $\mathbf{\Pi}_o$	a regular conditional law of y given o, fixed at every admitted o	conditional Gibbs state
extended free energy $\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_o)$	$-\log p(o) + D_{\mathrm{KL}}(\mathbb{Q}_o \Vert \mathbf{\Pi}_o)$ for every law; $+\infty$ off absolute continuity	Gibbs–Bogoliubov functional
bridge lift $\widehat{\mathbb{Q}}_o(dy, dz)$	joint law $\mathbb{Q}_o(dy)C(y, dz)$ of fine and coarse values	joint law of spins and block spins
discarded information $\Delta(o)$	expected relative entropy of recognition to posterior given the coarse value	information lost inside blocks
Bayes reverse kernel R_Q	reverse channel by Bayes' rule under Q; recovering a pair means restoring both laws	classical Petz map; sufficiency
parent state space $\mathsf{Z}_A = \mathsf{B}_A \times \mathsf{M}_A \times \mathbf{\Xi}_A \times \mathsf{H}_A$	parent belief samples, model presentations, evaluator inputs, retained marks of the candidate parent A of block I	block variables plus bookkeeping
evaluator ev_A	map from a parent model value to a normalized kernel generating (b_A, o, h_A)	coupling-to-Hamiltonian map
energy precomposition	substitute a restricted configuration into an energy, $\bar{E}(z) = E(\iota z)$	constrained configurations

Term or symbol	Plain meaning	Physics analogue
Galerkin data S, M, L, H_c	prolongation, reference form, operator, coarse propagator	multigrid coarse operator
scale diagram	functor from resolutions to kernels; composites along routes agree	semigroup property of blocking
finite labeled family	mixture whose component label stays in the sample	multiphase ensemble with phase label
log mass λ_γ, weight W_γ	log weight of label γ; softmax of the λ	$-\beta F_\gamma$; phase weight
extensive size N, density s_γ	index along which log masses grow linearly; $\lim \lambda_\gamma / N$	volume; free-energy density
strict winner, tie, window w	one maximal density; several; $\theta = \theta_0 + \mathsf{w}/N$	one phase; coexistence; rounding window
natural selector	data-dependent partition rule compatible with relabeling and gauge	choice of blocking scheme

Exact means an identity with a remainder, never a truncation-free flow; scale, resolution, and depth are indices with no length attached. C is always the channel, never a covariance. Δ is the discarded information here and a matrix (the cut excess) in chapter 10, and W is a label weight here and an edge form there. N is an extensive size that the manuscript allows, but does not require, to be the number of agents. In the house style, declared marks an input the theory does not select (the block I, the label A, the kernel C_A), which is substantive; selected version and admitted observation (finite positive evidence) are null-set bookkeeping, and vertical over the identity of c_* is automatic at one base point.

9.4 The formulas, explained

9.4.1 Channels, contraction, and recovery

A channel sends P to $(PK)(B) = \int K(x, B)P(dx)$, and $P(KL) = (PK)L$ (MT Proposition 9.2): stochastic matrices on standard Borel spaces. Every channel contracts relative entropy; when $D_{\mathrm{KL}}(P\|Q) < \infty$, equality holds exactly when dP/dQ depends on x only through the output (MT Theorem 9.3), and then the Bayes reverse kernel of Q also recovers P (MT Corollary 9.4). This is the information-loss theorem of Csiszár [20], Kullback and Leibler [48] and the commutative case of Petz [60]. Finiteness cannot be dropped: a three-point example has $D_{\mathrm{KL}}(PK\|QK) = D_{\mathrm{KL}}(P\|Q) = +\infty$ and no common recovery (MT Corollary 9.5). On the singular branch of the theory, where recognition is not absolutely continuous, identities hold as $+\infty = +\infty$ and say nothing.

9.4.2 The exact coarse free energy

The central display (MT (9.3)) is

$$\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_o) = \mathcal{F}_{\mathbb{P}^c}(\mathbb{Q}_o^c) + \Delta(o), \qquad \Delta(o) = \int_{\mathsf{Z}} D_{\mathrm{KL}}\big(\widehat{\mathbb{Q}}_o(dy \mid z) \| \widehat{\mathbf{\Pi}}_o(dy \mid z)\big)\mathbb{Q}_o^c(dz).$$

The first term on the right is computed entirely at the coarse scale, from the pushed joint (evidence $p(o)$, posterior $\mathbf{\Pi}_o C$) and the pushed recognition law $\mathbb{Q}_o^c = \mathbb{Q}_o C$. The remainder compares, for each coarse value z, what recognition and posterior say about the fine configuration given z, weighted by how often recognition produces z. If C is the identity, $\Delta = 0$. If C sends everything to one point, the coarse free energy is $-\log p(o)$ and $\Delta = D_{\mathrm{KL}}(\mathbb{Q}_o \| \mathbf{\Pi}_o)$: the whole gap is discarded. For a deterministic statistic, $\Delta = 0$ exactly when recognition has the posterior's shape inside every fiber $c^{-1}(z)$. For Gaussian laws and a linear projection $(z, w) \mapsto z$, Δ is the expected relative entropy of the two conditional Gaussians of w given z.

Is this more than the chain rule and data processing? In mathematical substance, no: the proof disintegrates the bridge lift over y, where the common conditional $C(y, \cdot)$ drops out, and over z, where the coarse term and Δ appear (MT Lemma 3.12). The theorem adds typing that later chapters use: the pushed posterior is a coarse posterior version at every declared observation, including exceptional ones, which matters because the theory evaluates at the one observed record; the identity holds in the extended reals without absolute continuity; with finite fine relative entropy, $\Delta(o) = 0$ exactly when one kernel recovers both recognition and posterior; and recovery of a whole family needs the hypothesis for every member at once (MT Remark 9.8 handles the support boundaries). For the ELBO (MT (9.4)),

$$\mathcal{L}^{\mathrm{ext}}_{\mathbb{P}^c}(\mathbb{Q}_o C) = \log p(o) - D_{\mathrm{KL}}(\mathbb{Q}_o C \| \mathbf{\Pi}_o C) \geq \log p(o) - D_{\mathrm{KL}}(\mathbb{Q}_o \| \mathbf{\Pi}_o) = \mathcal{L}^{\mathrm{ext}}_{\mathbb{P}}(\mathbb{Q}_o),$$

and MT Corollary 9.7 draws the right conclusion: the coarse ELBO is higher by $\Delta(o)$ at unchanged evidence, which is neither model improvement nor evidence ascent, and ELBOs of separately declared models compare tightness only at equal evidence.

On finite spaces, where MT Proposition 9.9 is stated, and for a deterministic statistic the coarse free energy is the constrained minimum of the fine one, attained by $q'(c(y))\mathbf{\Pi}_o(y \mid c(y))$. For a noisy channel it is not: with a uniform binary posterior, the symmetric channel with flip probability $1/4$, and $\mathbb{Q} = (1, 0)$, the only fine law with image $(3/4, 1/4)$ is $\mathbb{Q}$, and its free energy exceeds the coarse one by $2\log 2 - \frac{3}{4}\log 3 \approx 0.56$ (MT Proposition 9.9).

9.4.3 The pointwise parent datum

For a block I, a parent label A, and a parent kernel $C_A : \mathsf{Y}_I \rightsquigarrow \mathsf{Z}_A$,

$$\mathbb{P}_A(do, dz \mid X) = \int_{\mathsf{Y}_I} C_A(y, dz)\mathbb{P}_I(do, dy \mid X), \qquad \mathbf{\Pi}_{A,o,X} = \mathbf{\Pi}_{I,o,X} C_A, \qquad \mathbb{Q}_{A,o,X} = \mathbb{Q}_{I,o,X} C_A,$$

and $\mathcal{F}_I = \mathcal{F}_A + \Delta_A$ (MT Corollary 9.10). The corollary takes $\mathbb{Q}_{I,o,X} \ll \mathbf{\Pi}_{I,o,X}$ as an input, which is harmless because the exact coarse free energy identity holds without it. Does a parent law always exist? Yes, trivially: a pushforward exists for every declared block and kernel. Is the parent an agent with its own dynamics? No, and the statement says what is missing. The block, label, and kernel are inputs that nothing selects (MT Open Problem 9.19). The parent's model coordinate is generative only through an evaluator; disintegrating $\mathbb{P}_A$ always supplies one, so the parent always has a generative mechanism, but it is whatever $\mathbb{P}_A$ says, and an evaluator declared independently (say, a Gaussian agent with stated parameters) agrees with $\mathbb{P}_A$ only under a separate compatibility hypothesis that disintegration does not validate. The parent's belief and model marginals do not determine its law unless one is a point mass and the auxiliary factors are trivial. And the construction is static: no fine evolution induces a parent evolution (MT Open Problem 11.38, the defining open problem of the phase, chapter 11). How holonomy enters the parent is the subject of chapter 10.

9.4.4 Three operations

Energy precomposition, $\bar{E}(z) = E(\iota z)$ and $\bar{P}(dz) = e^{-\bar{E}(z)}\bar{\nu}(dz)/\bar{Z}$, needs a declared reference $\bar{\nu}$. On a countable space with $E = -\log p$ it is conditioning on $X \in \iota(\mathsf{Z})$, and conditioning can increase relative entropy: $(1, 1, 98)/100$ and $(1, 2, 97)/100$ are about 0.003 apart, their restrictions $(1/2, 1/2)$ and $(1/3, 2/3)$ about 0.059 (MT Proposition 9.12). So no kernel produces precomposed laws, and data processing, evidence preservation, and ELBO monotonicity do not transfer. On a continuous fiber the restricted set is typically null, as for block-constant configurations, and the answer depends on the reference: $\bar{E}(z) = z^2$ gives $\mathcal{N}(0, 1/2)$ against Lebesgue measure and $\mathcal{N}(0, 1/4)$ against $e^{-z^2}dz$.

Galerkin restriction (MT Definition 9.13) takes an injective prolongation S, a form $M \succ 0$, and an operator form L, with $H = M^{-1}L$, $S^+ = (S^\top MS)^{-1}S^\top M$, and $H_c = S^+HS$; the form $S^\top LS$ is the aggregation multigrid coarse operator [57]. The residual $S^+H - H_cS^+ = S^+H(I - SS^+)$ vanishes exactly when (MT (9.8)) H maps the M-orthogonal complement of $\operatorname{range} S$ into itself; for symmetric L this is invariance of $\operatorname{range} S$, and without symmetry the two conditions are independent (MT Proposition 9.14). As an illustration outside the manuscript, for block aggregation with $M = I$ and L a graph's adjacency matrix it is the equitable-partition condition: every vertex of a block has the same number of neighbors in each block. Restricting a recognition family is yet another operation: it keeps the joint and optimizes the same ELBO over a subset.

9.4.5 Symmetry and scale diagrams

An equivariant channel, $K(gx, B) = K(x, g^{-1}B)$, commutes with the action on laws (MT Proposition 9.15); at c_* gauge covariance is equivariance for ρ_b, ρ_m, and relabeling covariance is equivariance for permutations. The Bayes reverse kernel is defined only almost everywhere, and whether it can be chosen equivariantly everywhere is settled for finite spaces, for compact stabilizers by Haar averaging (MT Proposition 9.16), and for any stabilizer when the posterior has a weakly continuous version (MT Corollary 9.17).

A scale diagram (MT Definition 9.20) is a functor from resolutions to kernels, with right-acting composites $K_{\ell r} = K_{\ell k} K_{kr}$. On a chain this is automatic (MT Proposition 9.21). It has content on a poset that is not a chain, where two routes must agree, and for a blocking rule indexed by block factor, where it is the semigroup law $K_{b_1 b_2} = K_{b_1} K_{b_2}$. Block sums satisfy it; majority rule with ties broken by the first member does not, since $(0,0,1,0,1,0)$ goes to 1 by pairs then triples and to 0 by triples then pairs. Mutually inverse kernels between finite spaces are permutations, so no stochastic isomorphism links levels of different sizes. The Dobrushin contraction bound is not in this chapter: it is part of MT Theorem 11.5 (chapter 11), and the scale diagram here carries no contraction rate.

9.4.6 Labeled families and Fisher information

A finite labeled family (MT Definition 9.22) keeps the label in the sample, $P_\theta(\gamma, x) = W_\gamma(\theta) p_{\gamma,\theta}(x)$ with $W_\gamma \propto e^{\lambda_\gamma}$. With $\xi_\gamma = \nabla \lambda_\gamma$ and $\bar{\xi} = \sum_\gamma W_\gamma \xi_\gamma$, the score is $\xi_\gamma - \bar{\xi} + \mathsf{s}_{p_\gamma}(x)$ and the Fisher tensor is (MT (9.13))

$$g^F(P) = \mathrm{Cov}_W(\xi_\gamma) + \sum_\gamma W_\gamma g^F(p_\gamma),$$

label information plus weighted branch information (MT Proposition 9.23), the law of total covariance for the score. If $\lambda_\gamma = -\beta F_\gamma(\theta)$, then ξ_γ is the extensive quantity conjugate to θ in phase γ and the first term is its between-phase variance, the coexistence part of a susceptibility. Observed-data information is complete minus missing information, Louis's identity [51] averaged over records (MT Proposition 9.25); a channel acting within branches never touches the label information (MT Corollary 9.26); and along a word of channels the Fisher losses and the discarded relative entropies telescope (MT Proposition 9.27), dividing by an extensive size giving per-agent bookkeeping, not a thermodynamic density.

The asymptotics assume extensive growth, $\lambda_{\gamma,N} = O(N)$ in C^3 with $\lambda_{\gamma,N}/N \to \mathsf{s}_\gamma$. With the branch densities $N^{-1} g^F(p_{\gamma,N})$ converging in C^2, a strict winner makes the Fisher density per N converge to the winning branch's, with exponentially small errors (MT Theorem 9.28). At a tie where the weights converge, $W_{\gamma,N}(\theta_0) \to \bar{\mathsf{p}}_\gamma$, and the branch Fisher tensors are $O(N)$, $N^{-2} g^F(P_N)(\theta_0) \to \mathrm{Cov}_{\bar{\mathsf{p}}}(\mathsf{v}_\gamma)$ with $\mathsf{v}_\gamma = \nabla \mathsf{s}_\gamma(\theta_0)$, of rank at most $\min(d, |\mathrm{supp}\, \bar{\mathsf{p}}| - 1)$ (MT Proposition 9.29). In the window $\theta = \theta_0 + \mathsf{w}/N$, given center ratios c_γ of the tied weights, the label law tends to a softmax of $\mathsf{w} \cdot \mathsf{v}_\gamma + \log \mathsf{c}_\gamma$ and the Fisher tensor in w to (MT (9.23))

$$D^2_{\mathsf{w}} \log \sum_{\gamma \in S} \mathsf{c}_\gamma e^{\mathsf{w} \cdot \mathsf{v}_\gamma},$$

the Hessian of a finite log-partition function (MT Theorem 9.30); the center ratios are extra finite-N data. MT Proposition 9.31 shows, with one parameter-independent branch law throughout, that extensive growth alone gives none of the three conclusions (diverging density $N/4$ at a linear tie, nonconverging weights, sub-extensive scales, a rare tied label with diverging density, uniform convergence without convergence of second derivatives), and MT Remark 9.32 identifies the N^2 scale as information about the label: in the binary examples the unlabeled configuration carries none.

What is N, and is this a thermodynamic limit? N indexes a sequence of finite labeled families whose log masses are assumed to grow linearly; the number of agents is offered as an example. The label set is fixed while N grows. The theorems are the finite-mixture calculus of phase coexistence, not a thermodynamic limit of the agent network: there is no infinite-volume measure, and nothing derives extensive growth or a fixed label set from an agent model (Supplement B instantiates the three regimes for its lattice law with $N = L^2$ cells, deriving the branch inputs while still assuming a growth condition on the raw label weights, Corollary B9.7; that lattice carries no agent sections). The natural label sets of an N-agent network, such as its partitions, grow with N; the theorems apply to a fixed finite menu of coarse alternatives whose log weights are supplied as extensive.

9.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Definition 9.1	A coarse channel is a kernel on latent coordinates reading neither recognition nor observation, applied in common to the three laws.	CHOICE	Fitted coarse joints, different channels per law, and observation coarsening are outside it.
MT Propositions 9.2 and 9.15	Kernels compose and push laws functorially; equivariant kernels commute with the action.	SOLID	Standard (Kleisli category of the Giry monad [34]).
MT Theorem 9.3 and corollaries 9.4 and 9.5	Data processing with its equality case; pairwise Bayes recovery; nothing follows from $+\infty = +\infty$.	SOLID	Standard in substance (Kullback, Csiszár, commutative Petz). Needs finite divergence.
MT Theorem 9.6	Evidence and posterior version preserved; $\mathcal{F} = \mathcal{F}^c + \Delta$ in the extended reals; $\Delta = 0$ iff the discarded conditionals agree, and with finite divergence iff one kernel recovers both.	SOLID	Standard in substance (chain rule); typing new. One common channel, observation untouched.
MT Corollary 9.7	Coarse ELBO exceeds fine ELBO by Δ at equal evidence.	SOLID	The rise is discarded information, not model improvement.
MT Proposition 9.9	Constrained-minimum reading holds for deterministic statistics and fails for noisy channels.	SOLID	Part (ii) is a two-state counterexample.
MT Corollary 9.10	Every declared block and parent kernel give a normalized parent triple with the chain rule; evaluator by disintegration; marginals do not fix the law.	SOLID	Static; block, label, kernel are inputs; a declared evaluator needs a compatibility hypothesis.

Result	What it says	Status	Scope and caveats
MT Proposition 9.12	Precomposition is conditioning, can raise relative entropy, is reference dependent on null sets.	SOLID	Negative: pushforward theorems do not transfer.
MT Proposition 9.14	Galerkin propagation is exact iff the discarded complement is invariant.	SOLID	Standard linear algebra.
MT Proposition 9.16 and corollary 9.17	Equivariant Bayes reverse kernels for finite spaces, compact stabilizers, continuous posteriors.	SOLID	General noncompact case open.
MT Proposition 9.21	Functoriality empty on chains, contentful on other posets and rule semigroups; tie-broken majority fails.	SOLID	A resolution index; no rescaling, no flow.
MT Propositions 9.23, 9.25 and 9.27	Label-plus-branch Fisher decomposition, missing information, telescopes.	SOLID	Standard in substance (total covariance; Louis).
MT Theorems 9.28 and 9.30 and proposition 9.29	Strict winner hides the label; a tie gives N^2 information; the $1/N$ window gives a log-sum-exp Hessian.	COND	Extensive growth, branch convergence, center ratios supplied, not derived; label set fixed as N grows.
MT Proposition 9.31	Each conclusion fails without its hypotheses.	SOLID	Negative.
MT Open Problems 9.18 and 9.19	Equivariant versions for noncompact stabilizers; a natural partition selector.	OPEN	See section 9.6.

Conditions: which ones matter

Conditions that matter. The channel must be one kernel for all three laws, blind to the observation and to recognition; different channels per law, fitted coarse joints, and observation coarsening (even invertible) change the evidence event or the comparison target and are outside the theorem. The difference form and the recovery characterization need finite fine relative entropy, which fails on the singular branch. The constrained-minimum reading needs a deterministic statistic and is stated for finite spaces. Fisher contraction needs a parameter-independent channel, since a parameter-dependent one can create information (MT Corollary 7.42). The label asymptotics need extensive growth, convergence of the tied weights, C^2 convergence of branch densities in the strict and window cases (only $O(N)$ branch information at the tie), center ratios in the window, and a fixed label set. Equivariant reverse kernels need a compact stabilizer or a continuous posterior. Hygiene: standard Borel spaces, measurable kernels, the choice of versions on null sets, fixed-support strata, and C^3 log masses.

The chapter has no toy content: its small examples (three-point infinite equality, the two-state noisy channel, three-point conditioning, six-agent majority, binary tie families) are counterexamples, which establish negative statements at full generality.

9.6 Open ends

MT Open Problem 9.18 asks for everywhere-defined equivariant Bayes reverse kernels under a noncompact stabilizer when the posterior has no continuous version, jointly measurable across orbits; it is settled by an equivariant measurable selection theorem or a counterexample in the stated class. An elementary remark outside the manuscript bounds its reach for the program's Gaussian models: for a faithful representation the stabilizer of a nondegenerate Gaussian law is compact, so the Haar-averaged version of MT Proposition 9.16 applies. The open case concerns laws with noncompact stabilizers, such as degenerate ones or nonfaithful actions.

MT Open Problem 9.19 is the substantive one. It asks for a data-dependent partition, membership law, or structural channel that is natural under relabeling (relabeling the input relabels the unlabeled output) and reads gauge data only through invariant scores. No analyzed cost selects one, and the reason is monotonicity. For fixed recognition the coarse free energy never increases along a composition of channels, so minimizing it favors the channel to a point, which discards everything and returns $-\log p(o)$; minimizing Δ favors the identity; an unpenalized sum of within-block divergences is minimized by singletons (MT Remark 10.20); the Gaussian recognition costs of Supplement G select trivial endpoints or need an external coefficient, a price per block much like a chemical potential; and common recovery of a whole experiment needs simultaneous hypotheses. The problem is settled by a relabeling-natural, gauge-compatible functional that is not monotone under refinement and has a nondegenerate optimizer, or by an impossibility theorem in a stated class. In Kadanoff–Wilson renormalization the blocking is supplied by lattice geometry and never selected; here no geometry supplies it. Parent dynamics, MT Open Problem 11.38, is posed in MT chapter 11 and discussed in chapter 11.

Does it deliver what it claims? DELIVERED

The chapter says that one coarse channel applied to the joint, the posterior, and a recognition law preserves the evidence and the posterior version and splits the free energy into the coarse free energy plus the discarded conditional information; that this specializes to a candidate parent; that pushforward, precomposition, and Galerkin restriction are different operations; and that a retained label carries information of order N^2 at a tie. Each is proved as stated. The central theorem is standard in substance (the relative-entropy chain rule and commutative Petz sufficiency, carefully typed), and the chapter is explicit that exactness buys no closure, no selected channel or block, and no parent dynamics. Two wordings outside the statements promise more than the statements give. The chapter opening places the N^2 information at a tie among N agents, while the theorems concern a fixed finite label set with supplied extensive log masses. The introduction (MT Section 1.3) says this chapter separates Markov

channels, energy restriction, and parameter aggregation; the chapter separates pushforward, precomposition, and Galerkin restriction, and never treats parameter aggregation.

Chapter 10

Gauge-Retained Network State and Fixed Sectors

10.1 The question and the answer

Grouping agents into a block does not average away their gauge data: the links Θ_e^x are independent group elements on the edge copies of the interaction multigraph, one family per channel $x \in \{b, m\}$, and closed walks carry holonomy. MT Chapter 10 asks what a coarse description of a block must keep of these data and which part of the block's state survives exact aggregation. The answers come in six pieces. A rooted spanning tree turns the block's gauge data into one root-framed datum (member coordinates carried to the root, the based holonomy, dressed boundary edges) that transforms under a single root gauge transformation (MT Proposition 10.5). The configurations that survive exact aggregation form the kernel of the connection Laplacian, which with positive-definite weights is isomorphic to the holonomy-fixed vectors at the root (MT Proposition 10.17). The members' marginal laws admit one parent law exactly when, carried to the root, they coincide in a holonomy-invariant law (MT Theorem 10.9); compact holonomy gives a closed form, and for Gaussian parents noncompact holonomy generically allows no parent at all (MT Corollary 10.11). Cut links compress exactly into rectangular endpoint maps, and into a single invertible link with new self terms exactly when an explicit matrix is invertible, with one balanced choice that has no law-level counterpart and no continuous extension to singular cuts. The frame of a meta-agent is invisible to its members and cannot be computed from their frames by any symmetric rule that does not depend on a reference point (equivalently, any symmetric left-equivariant rule) (MT Proposition 10.27). A declared noisy record at one base point distinguishes loop holonomies (MT Theorem 10.28).

These results feed chapters 11 to 14; Supplement G realizes them for Gaussian families. The scope paragraph fences them: the fixed-sector rank is an admissibility criterion, not a cost, keeps singletons admissible, and does not require beliefs to agree; the marginal-law score takes its block and weights as inputs, selects no partition, supplies no kernel, and does not recover the dependence structure of a joint law; the operational record recovers no link values.

10.2 The picture

At one base point the agent network is a lattice gauge system over a zero-dimensional base: sites are agents, links $\Theta_e^x \in G$ live on edge copies, and a passive gauge transformation acts by $\Theta_e^x \mapsto g_i \Theta_e^x g_j^{-1}$ for $e : j \to i$. Because each agent also carries a frame, the links read relative to the agents' own frames (the per-edge twists) are invariant, and the complete invariant is the tuple of twists up to one simultaneous conjugation, which contains the loop holonomies up to conjugation (MT Proposition 2.12); structurally this is the composite link of a gauge-Higgs model with a group-valued Higgs field. Blocking means choosing a root, carrying every member's data to it along a spanning tree, and keeping the loops the tree does not use.

Physics analogue (exact)

Maximal-tree gauge and Wilson loops. Gauging every tree link to the identity, uniquely up to one element per block, turns each non-tree link into the holonomy of the loop it closes (MT Proposition 10.5). This is the maximal-tree gauge of lattice gauge theory [17], which the manuscript cites, and the retained holonomy homomorphism carries the block's Wilson loops and more: the based holonomies up to one root conjugation.

Physics analogue (exact)

Zero modes of the covariant lattice Laplacian. The connection Laplacian is the covariant kinetic energy of a matter field: each edge penalizes the mismatch between a member's vector and its neighbor's vector carried across the link. Its zero modes are the covariantly constant configurations, which exist with a given root value exactly when every loop returns that value to itself, as a charged field on a ring has a constant mode only for trivial flux. The kernel is the degree-zero sheaf cohomology of the block [37], and for orthogonal links and scalar weights the operator is the vector diffusion of Singer and Wu [66] (MT Proposition 10.17).

To give a block one parent law, carry each member's marginal law to the root. The result depends on the walk only through a holonomy action, so an honest parent must be holonomy invariant, and for a compact holonomy group the best invariant parent in forward relative entropy is the transported mixture averaged over the group: a twirl, with the block's represented holonomy in the role of the symmetry. A planar rotation loop forces a zero parent mean and an isotropic covariance; a loop that stretches volume admits no Gaussian parent at all.

Parallel links between two blocks become one effective link only at a price. Their quadratic energy completes to a single link, the weight-averaged twist, plus a self-energy measuring how much the twists disagree. Twists I and $-I$ with equal weights average to zero and leave the blocks coupled only through self terms: the two transports interfere destructively. Requiring the effective link to balance its self terms across the cut makes it unique; for orthogonal twists and scalar weights it is the orthogonal polar factor of the mean twist.

Physics analogue (structural)

Link smearing with projection to the group. Lattice practice replaces a link by a sum of parallel transports and projects the sum back to the group, often by polar decomposition. The balanced link in the orthogonal case is such a projection (MT Proposition 10.23, part (iii)). The projection jumps where the sum is singular, and MT Proposition 10.26 proves that no continuous rule avoids this.

The frame of a meta-agent is pure gauge for its members, and it cannot be computed as a symmetric, reference-independent average of their frames.

Physics analogue (exact)

The phase of the mean of the roots of unity. The phases $1, \omega, \ldots, \omega^{n-1}$ with $\omega = e^{2\pi i/n}$ are permuted by multiplication by ω, so a symmetric rotation-covariant average would have to be fixed by ω, which is impossible. The frame no-go (MT Proposition 10.27, part (ii)) is this argument for any group with an element of order n.

Finally, the operational record reads, with Gaussian noise, the displacement $(\mathrm{Hol} - I)z$ of a probe agent's state carried around a loop, as an interferometer reads the holonomy seen by a test particle. The analogy is structural, and the record kernel is declared to have this form rather than derived from the agents' dynamics.

10.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
levelwise fiber state $\mathfrak{X}_\ell$	all fiber data at depth ℓ: group, frame torsor, associated fibers, multigraph, links, holonomy groups, cross maps	configuration space of a lattice gauge and matter theory
scale intertwiner	homomorphism $\kappa_\ell : G_\ell \to G_{\ell+1}$ with compatible fiber maps	embedding of gauge groups under blocking
cross map $\Phi, \widetilde{\Phi}$	declared map between belief and model bundles	none
tree transport $\Theta^x_{I \leftarrow i}$	link product along the tree path from i to the root	parallel transporter
based holonomy group $\mathrm{Hol}^x_I(r)$; $\mathrm{Hol}^x(p_e)$	all loop products at the root; the loop closed by one non-tree edge	Wilson-loop data; independent loop
block-dressed edge $\widehat{\Theta}^x_e$	boundary link with tree transports attached at both ends	link between block roots
holonomy-stabilized law	law invariant under every holonomy element	invariant (singlet) state
averaging maps $\mathsf{C}_x, \mathsf{S}_x$	weighted mean of transported features and its right inverse	gauge-covariant block averaging

Term or symbol	Plain meaning	Physics analogue
connection Laplacian L_I; fixed fiber $\ker L_I$, rank f_I	covariant quadratic mismatch energy; its zero modes and their number	covariant lattice Laplacian; zero modes
full, partial, no fixed sector	$f_I = K$, $0 < f_I < K$, $f_I = 0$	flat, partly flat, fully twisted
marginal-law score $\mathfrak{D}_I^x$	least weighted forward relative entropy from transported marginals to an invariant parent	none
Haar-symmetrized mixture $\bar{M}$	mixture of transported marginals averaged over the holonomy group	twirled state
exact common recovery	one fixed reverse kernel restores every law of an experiment	sufficient statistic
cut blocks $\Lambda_{II}, \Lambda_{IJ}, \Lambda_{JJ}$; excess Δ	quadratic coupling of two block roots; self-energy left after completing the square	frustration of parallel links
anchored, no-new-anchor, balanced	one link and weight plus new self terms; without self terms; with self terms matched across the link	effective link plus mass terms
endpoint maps S_e^Q, S_e^K	factorized edge energy through an edge space	sheaf restriction maps
probe i_0, record r	agent whose state is carried around a loop; noisy readout	test particle; interferometer

Holonomy always means the product of independent link variables around a closed walk of the agent multigraph, neither the holonomy of a base connection nor a topological property of the bundle. Fixed means holonomy-fixed; sector means a subspace or its rank, not a superselection sector. Δ is now a positive semidefinite matrix, and W_e are positive semidefinite forms on the fiber. Represented means seen through ρ; a nonfaithful representation can hide principal holonomy. Balanced names one symmetry condition, not an optimum. K is the fiber dimension, and $\mathrm{GL}(K)$ is real unless stated.

10.4 The formulas, explained

10.4.1 Depth maps and the rooted datum

A fiber state (MT Definition 10.1) collects the depth-ℓ group, frame torsor, associated fibers, multigraph, links, holonomy groups, and cross maps. A map to the next depth is a homomorphism κ_ℓ, a compatible torsor map (unique once one frame is sent to one frame), and fiber maps $q_{\ell,x}$; the induced map of associated fibers exists exactly when $q_{\ell,x} \circ \widehat{\rho}_{\ell,x}(g) = \widehat{\rho}_{\ell+1,x}(\kappa_\ell(g)) \circ q_{\ell,x}$ (MT Proposition 10.2), and defect pairs have gauge-invariant equality loci (MT Proposition 10.3). This is standard associated-bundle functoriality.

For a block I with root r_I and a spanning tree, the tree transport, the holonomy of a non-tree edge $e : j \to i$, and the dressed edge of a boundary edge from block J are

$$\Theta^x_{I\leftarrow i} = \Theta^x_{p_{I,i}}, \qquad \mathrm{Hol}^x(p_e) = \Theta^x_{I\leftarrow i}\Theta^x_e(\Theta^x_{I\leftarrow j})^{-1}, \qquad \widehat{\Theta}^x_e = \Theta^x_{I\leftarrow i}\Theta^x_e(\Theta^x_{J\leftarrow j})^{-1},$$

and a gauge transformation sends them to $g_{r_I}\Theta^x_{I\leftarrow i}g_i^{-1}$, $g_{r_I}\mathrm{Hol}^x(p_e)g_{r_I}^{-1}$, and $g_{r_I}\widehat{\Theta}^x_e g_{r_J}^{-1}$ (MT Proposition 10.5): after transport only the roots' gauge remains. The holonomy homomorphism does not depend on the tree; a tree change multiplies transports on the left by holonomy elements. A tree block has trivial holonomy and pure-gauge internal data; for abelian G each holonomy is gauge invariant; graph-flat links (MT Proposition 2.13) have trivial holonomy. The manuscript keeps the raw root-framed datum rather than a gauge quotient, since quotienting each holonomy separately loses its orientation relative to the root features and boundary legs, and an orbit space would need a quotient theorem it does not supply. A parent carries holonomy either by full groupoid covariance of all fine laws, versions, kernel, and evaluator, or by placing the rooted datum in its retained marks H_A with no invariance claim (MT Proposition 10.6); neither selects the block.

The natural linear coarse feature is the weighted mean of transported features, with its right inverse,

$$(\mathsf{C}_x z)_I = \sum_{i\in I} w_{Ii}\rho^{\mathbb{W}}_x(\Theta^x_{I\leftarrow i})z_i, \qquad (\mathsf{S}_x\hat{z})_i = \rho^{\mathbb{W}}_x(\Theta^x_{I\leftarrow i})^{-1}\hat{z}_I.$$

It is gauge covariant but tree dependent (MT Proposition 10.7): for two vertices joined by edge copies with $\mathrm{SO}(2)$ links I and R_ϑ, equal weights, and equal features e_1, the two trees give averaged features of norms 1 and $|\cos(\vartheta/2)|$, both invariant under root gauge transformations. Averaging is tree independent on holonomy-fixed features and, for trivial represented holonomy, on all features. The tree is a gauge fixing for the retained datum but part of the model for the averaged feature, as gauge-covariant smearing of matter fields depends on the paths unless the field is covariantly constant.

10.4.2 The holonomy-conditioned marginal law

Carry each marginal to the root, $\bar{q}^x_i = \widehat{\rho}_x(\Theta^x_{p_i})q^x_i$, and score the block by (MT (10.17))

$$\mathfrak{D}^x_I = \inf_{Q\in\mathscr{Q}^x_{I,\mathrm{fix}}(r)} \sum_{i\in I} w_i D_{\mathrm{KL}}(\bar{q}^x_i\|Q), \qquad \mathscr{Q}^x_{I,\mathrm{fix}}(r) = \{Q \in \mathscr{M}^x_r : \widehat{\rho}_x(\mathsf{k})Q = Q \text{ for all } \mathsf{k} \in \mathrm{Hol}^x_I(r)\},$$

with $\inf\varnothing = +\infty$ and $\mathscr{M}^x_r$ a declared class of candidate parents. MT Theorem 10.9 proves that the score depends on neither walks, root, nor frames, and that an attained zero means all transported marginals equal one invariant law, which then defines a holonomy-stabilized parallel family of member laws; an unattained zero means only approximability. For compact represented holonomy with Haar probability $d\mathsf{k}$, put $\bar{P} = \sum_i w_i\bar{q}^x_i$ and $\bar{M}(B) = \int \bar{P}(\mathsf{k}^{-1}B)d\mathsf{k}$; every invariant Q then satisfies

$$\sum_{i\in I} w_i D_{\mathrm{KL}}(\bar{q}^x_i\|Q) = \sum_{i\in I} w_i D_{\mathrm{KL}}(\bar{q}^x_i\|\bar{M}) + D_{\mathrm{KL}}(\bar{M}\|Q),$$

so the minimizer is $\bar{M}$ for unrestricted parents and its moment-matched Gaussian for Gaussian ones (MT Proposition 10.10). That the forward-relative-entropy barycenter of laws is their mixture is standard; the symmetrization and the gauge typing are the chapter's.

On the running three-agent loop with $\mathrm{SO}(2)$ links, loop rotation ϕ with $\sin\phi \neq 0$, and transported marginals $\mathcal{N}(m_i, \sigma^2 I_2)$, the only invariant Gaussians are $\mathcal{N}(0, sI_2)$ and (MT Remark 10.12)

$$\mathfrak{D}_I^b = \log\Big(1 + \frac{1}{2\sigma^2}\sum_i w_i |m_i|^2\Big), \qquad s_\star = \sigma^2 + \tfrac{1}{2}\sum_i w_i |m_i|^2 :$$

the parent forgets direction and absorbs the members' means into its variance. Frame-aligned links give $\frac{1}{2}\log\det(I_2 + \mathrm{Cov}_w(m)/\sigma^2)$, never larger, and a link of determinant 2 gives $+\infty$. The last case is general (MT Corollary 10.11): with Gaussian parents and linear holonomy on $\mathbb{R}^K$, an invariant parent exists iff the represented holonomy group is conjugate into $\mathrm{O}(K)$; one loop holonomy with an eigenvalue off the unit circle, or a nondiagonalizable one, gives $+\infty$ for all marginals; and for a block containing a loop with independent $\mathrm{GL}(K)$ links, $|\det| = 1$ on that loop is Lebesgue-null, so the score is $+\infty$ almost everywhere and distinguishes configurations only after the links are restricted to a group conjugate into $\mathrm{O}(K)$ or a parent family with invariant members is declared.

Two fences. A zero score is not common recovery: it reads only marginals, and joint laws with equal marginals cannot both be recovered through a channel that reads only marginals (MT Remark 10.14). A cross map commuting with the links carries invariant belief parents to invariant model parents (MT Corollary 10.15) without identifying the bundles.

10.4.3 Structural fixed sectors

The connection Laplacian of a block is (MT (10.21))

$$z^\top L_I z = \sum_{e:j\to i} (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j),$$

with represented links Θ_e and edge forms $W_e \succeq 0$. How does its kernel relate to $\mathrm{Fix}(\mathrm{Hol}_I)$? With positive-definite weights, evaluation at the root is an isomorphism from $\ker L_I$ onto the fixed space of the represented holonomy on each component, so f_I is its dimension, and $f_I = K$ exactly when the links are graph flat on the block (MT Proposition 10.17). With merely semidefinite weights the kernel is set by the edgewise constraints $W_e^{1/2}(z_i - \Theta_e z_j) = 0$, and holonomy can hide in weight-null directions: a triangle with loop product $2I$, which fixes nothing, still has a two-dimensional kernel.

Aggregation onto fixed sections is Galerkin compression: with $\iota_{\mathcal{P}}$ block diagonal and spanning each $\ker L_I$, the coarse form $\iota_{\mathcal{P}}^\top \Lambda \iota_{\mathcal{P}}$ sees no internal edges and is positive definite exactly when no fixed section is a fine zero mode (MT Proposition 10.18); nested sections compose (MT Proposition 10.19). This structural rank is independent of the marginal-law score (MT Remark 10.20): a two-vertex tree has trivial holonomy and arbitrarily distant transported marginals, while a triangle with nontrivial $\mathrm{SO}(3)$ holonomy has $f_I = 1 < 3$ and zero divergence for isotropic Gaussians. Pairwise transported

divergence also depends on the walk unless the laws are stabilized: with $\mathsf{k} = \mathrm{diag}(1,-1,-1)$ and two copies of $\mathcal{N}(e_2, I_3)$, two walks give 0 and 2.

10.4.4 Cut links

For blocks I, J in root coordinates with dressed edges $\widehat{\Theta}_e$ and weights $W_e \succ 0$ on the cut copies, put $\Lambda_{II} = \sum_e W_e$, $\Lambda_{IJ} = \sum_e W_e \widehat{\Theta}_e$, and $\Lambda_{JJ} = \sum_e \widehat{\Theta}_e^\top W_e \widehat{\Theta}_e$. Completing the square (MT Lemma 10.22, MT (10.26)),

$$\mathcal{E}_{\mathrm{cut}} = (z_I - \bar{\Theta} z_J)^\top \Lambda_{II} (z_I - \bar{\Theta} z_J) + z_J^\top \Delta z_J, \qquad \Delta = \sum_e (\widehat{\Theta}_e - \bar{\Theta})^\top W_e (\widehat{\Theta}_e - \bar{\Theta}) \succeq 0,$$

with $\bar{\Theta} = \Lambda_{II}^{-1} \Lambda_{IJ}$ the weight-averaged twist. The excess is the weighted spread of the twists, zero exactly when all dressed edges agree. $\bar{\Theta}$ transforms like a dressed edge but can be singular (I and $-I$ average to 0): a mean of group elements need not be one. One invertible link with new self terms at both ends (an anchored presentation) exists exactly when $\det \Lambda_{IJ} \neq 0$, positive for a GL^+ link; the admissible weights form the Loewner interval $\Lambda_{IJ}\Lambda_{JJ}^{-1}\Lambda_{IJ}^\top \preceq W \preceq \Lambda_{II}$, and the presentation is unique exactly when $\Delta = 0$, the only case without new self terms. This is Schur-complement algebra.

Requiring the self term at J, carried along the link, to equal the one at I, equivalently $\Theta^\top \Lambda_{II} \Theta = \Lambda_{JJ}$, selects one presentation (MT Proposition 10.23): its weight is the matrix geometric mean [8] of the interval endpoints,

$$W_{\mathrm{bal}} = W_{\mathrm{lo}}^{1/2} \big(W_{\mathrm{lo}}^{-1/2} W_{\mathrm{hi}} W_{\mathrm{lo}}^{-1/2} \big)^{1/2} W_{\mathrm{lo}}^{1/2}, \qquad \Theta_{\mathrm{bal}} = W_{\mathrm{bal}}^{-1} \Lambda_{IJ},$$

gauge covariant, the same read from either end, and in the scalar-weight orthogonal case the polar factor of $\bar{\Theta}$. Two negative results bound it. At the level of Gaussian laws, with $\sum_e w_e D_{\mathrm{KL}}(q_I \| (\widehat{\Theta}_e)_\# q_J)$ in place of the quadratic energy, a cut with excess has no presentation as one divergence to an image of q_J plus self terms reading one law each whose mean part is the balanced link, for any law class at J closed under covariance rescaling (MT Proposition 10.24); the self term at I would have to carry $\Lambda_{II} - W_{\mathrm{bal}}$, which depends on the covariance at J. For twists I and $2I$ the excess is $\frac{w_1 w_2}{2(4w_1 + w_2)} \Sigma_J^{-1}$. And MT Proposition 10.26, despite its label, is an established no-go: two identity-weighted $\mathrm{SO}(2)$ twists rotating by $\pm \arccos t$ give $\Lambda_{IJ} = 2tI_2$ and $\Theta_{\mathrm{bal}} = \mathrm{sign}(t) I_2$, which jumps at $t = 0$.

What survives exactly is the rectangular closure (MT Theorem 10.25): factor $W_e = (S_e^Q)^\top S_e^Q$, put $S_e^K = S_e^Q \Theta_e$, and the edge energy $\|S_e^Q z_i - S_e^K z_j\|^2$ compresses exactly under block injections, associatively, with parallel copies merged into one edge space of dimension $\mathrm{rank} \sum_e \delta_e^\top \delta_e \leq \dim \mathbb{V}_I + \dim \mathbb{V}_J$. This is the cellular-sheaf Laplacian category of Hansen and Ghrist [37], and the price of exactness is that the coarse edge is a pair of rectangular maps, not an element of the structure group.

10.4.5 The frame sector

What each member sees of a coarse datum (u_I, q_I) is the coordinate of $[u_I, q_I]$ in its own frame, $\mathcal{N}(U_i\bar{m}, U_i\bar{S}U_i^\top)$ for the Gaussian fiber of $\mathrm{GL}^+(K)$ (MT Proposition 10.27, part (i)). Data indistinguishable to every member form one free orbit $\{(u_I \cdot g, \widehat{\rho}_x(g)^{-1}q_I)\}$, of dimension K^2 for $\mathrm{GL}^+(K)$: the parent frame is pure gauge for the members. Part (ii): if G contains $v \neq e$ with $v^n = e$, no $F : G^n \to G$ is both left equivariant, $F(gg_1, \ldots, gg_n) = gF(g_1, \ldots, g_n)$, and symmetric, because left multiplication by v permutes $(e, v, \ldots, v^{n-1})$. $\mathrm{GL}^+(K)$ with $K \geq 2$ has such elements for every $n \geq 2$; on $\mathbb{R}^d$ the arithmetic mean works. A rule $u_I = \sigma \cdot F(a_1, \ldots, a_n)$ is independent of the reference σ exactly when F is left equivariant, so no symmetric rule that is independent of the reference point computes a meta-agent's frame from its members'; a symmetric rule that fixes a reference point exists. The theory does not need one, since the parent frame is invisible; the rooted datum uses a root, a choice that is equivariant but not permutation symmetric. The no-go concerns maps on all of G^n and does not exclude rules near consensus: as a standard fact outside the manuscript, Riemannian centers of mass for a left-invariant metric [45] give symmetric equivariant averages of sufficiently concentrated configurations. The obstruction is global.

10.4.6 The operational record

Fix a probe i_0, a closed walk at it, and labels ω with prior masses ψ_ω, each prescribing link values on the walk and hence a represented loop holonomy $\mathrm{Hol}_\omega^{\mathrm{pr}}$ on the probe's belief and model coordinates. The record joint is

$$P^{\mathrm{rec}}(\omega, dy, do, dr) = \psi_\omega \mathbb{P}(dy, do)\mathcal{N}\big(r; (\mathrm{Hol}_\omega^{\mathrm{pr}} - I)z_{i_0}(y), \Sigma_{\mathrm{r}}\big)dr.$$

MT Theorem 10.28 proves that it is normalized and that given ω and o the record is $(\mathrm{Hol}_\omega^{\mathrm{pr}} - I)Z$ plus independent noise, so two labels share a record law exactly when their images of Z agree in law; that a trivial and an invertible $\mathrm{Hol} - I$ give different laws unless the probe's posterior state is the point mass at zero, with $r^\top \Sigma_{\mathrm{r}}^{-1} r$ having mean $\dim \mathbb{V}_{\mathrm{pr}}$ in the first case and strictly more in the second; and that the posterior over ω and the law of that statistic are gauge invariant.

Can an experiment at one base point detect network-loop holonomy? At the level of laws, yes, if the experiment is modeled by a record kernel reading the loop transport of the probe's state: the theorem shows such a record is a legitimate, gauge-covariant part of the joint whose law separates holonomies unless the probe is pinned at zero. It does not show that any record kernel of the program's interaction model has this form. The latent law is independent of ω, so here the links do not influence the agents' states. Holonomy in the kernel of ρ, or holonomy fixing the support of the probe's state, is invisible. No link value is recovered, and no sample-size or power statement is made.

10.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Propositions 10.2 and 10.3	Depth maps exist iff fiber maps intertwine; defect equality loci are gauge invariant.	SOLID	Standard associated-bundle functoriality.
MT Definition 10.4	A block keeps the raw root-framed datum, not a gauge quotient.	CHOICE	Stated after MT Proposition 10.5; a design decision, not a necessity theorem.
MT Propositions 10.5 and 10.6	Gauge laws of transports, holonomy, dressed edges; tree links gauge to the identity; two ways a parent carries holonomy.	SOLID	Standard in substance (maximal-tree gauge [17]); neither alternative selects the block.
MT Proposition 10.7	Linear averaging of transported features is covariant but tree dependent.	SOLID	Tree independent on holonomy-fixed features and, for trivial represented holonomy, on all features.
MT Theorem 10.9	Marginal-law score is walk, root, and frame independent; attained zero iff one invariant law.	SOLID	Marginals only; no kernel, no joint recovery.
MT Proposition 10.10	Compact holonomy: minimizer is the Haar-symmetrized mixture or its moment-matched Gaussian.	COND	Compact represented holonomy: holds for $\mathrm{SU}(N)$ links, fails for $\mathrm{GL}(K)$ links.
MT Corollary 10.11	Gaussian parents exist iff holonomy is conjugate into $\mathrm{O}(K)$; generic $\mathrm{GL}(K)$ loops give infinite score.	SOLID	Negative for the noncompact target.
MT Remark 10.12	Three-agent $\mathrm{SO}(2)$ loop: explicit score.	TOY	Illustrates the compact closed form.
MT Propositions 10.17 to 10.19	Fixed fiber is $\ker L_I$, the holonomy-fixed space for positive-definite weights; compression and nesting exact.	SOLID	Standard in substance (sheaf cohomology). Semidefinite weights can hide holonomy.
MT Remark 10.20	Structural rank and marginal-law occupation are independent; singletons minimize unpenalized divergence sums.	SOLID	Witnesses for negative statements.
MT Lemma 10.22 and proposition 10.23	Completed square and excess; one-link presentations iff $\det \Lambda_{IJ} \neq 0$, unique iff no excess; unique balanced one.	SOLID	Schur complements and geometric means; balance is a chosen condition.
MT Propositions 10.24 and 10.26	No balanced law-level presentation with excess; no continuous extension to singular cuts.	SOLID	Negative; the second despite its label prefix.

Result	What it says	Status	Scope and caveats
MT Theorem 10.25	Exact compression of cuts into rectangular endpoint maps.	SOLID	The coarse edge is not a group element.
MT Proposition 10.27	Coarse frame is a free orbit invisible to members; no symmetric left-equivariant $G^n \to G$ under order-n torsion.	SOLID	Excludes global rules only.
MT Theorem 10.28	A declared noisy record of the loop displacement distinguishes holonomies unless the probe is pinned at zero.	SOLID	Record declared; latent law ignores the label; no link values or power.

Conditions: which ones matter

Conditions that matter. Compactness of the represented holonomy decides whether the marginal-law parent has a closed form and, for Gaussian parents, whether it exists; it fails for generic $\mathrm{GL}(K)$ links. The parent family matters: the Gaussian family forces conjugacy into $\mathrm{O}(K)$. Positive-definite versus semidefinite edge weights decides whether the fixed fiber is the holonomy-fixed space. A nonfaithful representation hides holonomy from the fixed sector and the record. One-link cut compression needs $\det \Lambda_{IJ} \neq 0$. Composing averaging maps across depths needs nested trees and multiplied weights. The record theorem needs its declared kernel and a label-independent latent law; the covariance branch of the parent alternatives needs covariance of every law, version, kernel, and evaluator. Attainment of the infimum matters too: without it a zero marginal-law score gives only approximability by invariant parents, not one invariant law. Hygiene: bimeasurable actions, metrizable compact groups, Polish fibers, and finite second moments.

10.6 Toy versus general, and the noncompact target

The one result proved on a small instance and not as a counterexample is the three-agent $\mathrm{SO}(2)$ loop of MT Remark 10.12. Its mechanism, that nontrivial compact holonomy forces the parent mean into the fixed space and inflates the parent covariance, is general by the compact closed form; its formulas are special to plane rotations. The chapter's other small configurations are witnesses for general negative statements.

For the program's target of 512 or more agents with links in $\mathrm{GL}(K,\mathbb{R})$, $\mathrm{GL}(K,\mathbb{C})$, or $\mathrm{SU}(N)$, the statements sort by whether they use compactness. The rooted datum, tree dependence, the kernel-holonomy isomorphism, fixed-section compression, the cut calculus, the frame sector, and the operational record use only group structure and linear algebra, and hold for all three groups with complex links realified. The torsion hypothesis of the frame no-go holds for all three ($e^{2\pi i/n} I \in \mathrm{GL}(K,\mathbb{C})$ and $\mathrm{diag}(e^{2\pi i/n}, e^{-2\pi i/n}, 1, \ldots, 1) \in \mathrm{SU}(N)$ for $N \geq 2$), a direct check of the stated hypothesis. The marginal-law parent is where compactness decides. For $\mathrm{SU}(N)$ links the holonomy lies in a compact group, the closed form applies (a relatively compact holonomy group

has the fixed laws of its closure), and Gaussian constituents give a finite score. For $\mathrm{GL}(K,\mathbb{R})$ links with Gaussian parents the score is infinite for almost every configuration on every block containing a loop; under the default complete interaction graph every block of three or more agents contains a triangle, so for generic independent links a finite Gaussian score is available only on blocks whose internal multigraph is a forest, unless links or parent family are restricted. For $\mathrm{GL}(K,\mathbb{C})$ the conjugacy criterion applies configuration by configuration, but the almost-everywhere statement is proved for real $\mathrm{GL}(K)$ only. Whether link dynamics drive holonomies into compact conjugacy classes is not addressed. By an elementary remark outside the manuscript, in the defining representation (realified for $\mathrm{SU}(N)$) a generic real matrix and a generic element of $\mathrm{SU}(N)$ on $\mathbb{R}^{2N}$ have no eigenvalue one, so with positive-definite weights a block containing a loop with generic links has $f_I = 0$ and is retained only through the rooted datum. This is specific to that representation: MT Definition 10.16 allows any linear representation, and one with a trivial summand, such as the adjoint representation of $\mathrm{SU}(N)$, keeps $f_I > 0$ at every link configuration. The balanced cut link lies in $\mathrm{GL}(\mathbb{V})$, orthogonal in the scalar-weight orthogonal case; no statement keeps it in a smaller group containing the twists.

10.7 Open ends

The chapter poses no open problem of its own; its one open-problem label names an established negative proposition. It feeds the open questions of chapter 9: the natural selector (MT Open Problem 9.19), since the fixed-sector rank keeps singletons admissible and the marginal-law score takes the block as input; equivariant reverse kernels beyond compact stabilizers (MT Open Problem 9.18); and parent dynamics (MT Open Problem 11.38). Two gaps are named without being posed: an orbit space of rooted data would need a quotient theorem, and no statement connects the declared record to the record kernels of the interaction model.

Does it deliver what it claims? DELIVERED

The chapter says it determines what a coarse description of a block retains of its gauge data and which part of its state survives exact aggregation, and its opening lists the rooted datum, the kernel-holonomy fixed sector, the holonomy-conditioned parent law, the cut compressions with their balanced choice and two obstructions, the frame no-go, and a record sensitive to loop holonomy. Each is proved at the stated generality. Much is standard in substance: maximal-tree gauge fixing, the kernel of a sheaf Laplacian, the mixture as forward-relative-entropy barycenter, Schur complements, the matrix geometric mean. The solid new content is mostly negative: no law-level balanced presentation, no continuous extension of balanced links, no symmetric reference-independent frame average, no Gaussian parent under noncompact holonomy. Three limits of reach remain. What a block must retain is answered by a choice (the raw root-framed datum), not a necessity theorem. The record distinguishes holonomies because it is declared to read them. For the $\mathrm{GL}(K,\mathbb{R})$ target, the holonomy-conditioned Gaussian parent exists, for almost every link configuration, only on blocks without internal

loops; the chapter states this itself, and its positive parent results reach that target only under a restriction of the links or of the parent family.

Chapter 11

Exact Agent-Network Coarse-Graining

11.1 The question and the answer

The technical chapter (MT Chapter 11) applies one coarse channel, a Markov kernel from fine to coarse configurations (MT Definition 9.1), to the interacting agent network at the base point c_* and asks what the step does to the network's energy function and couplings, and what must be carried along so that nothing is lost. The coarse effective action is minus the logarithm of a conditional partition function: the fine Boltzmann weight averaged over the fine configurations compatible with each coarse configuration (MT Theorem 11.3). Written in interaction coordinates, one component for every set of coupled agents, the step is an exact nonlinear map from fine couplings to coarse couplings, and the derivative of any composite of such steps along an exact trajectory has norm at most $3^{|V_n|} - 1$, a constant fixed by the number of target agents and independent of the number of steps (MT Proposition 11.19). What must be kept is collected in a table (MT Table 11.1): induced multi-agent couplings, the joint law of fine and coarse variables, the joint law of attention events rather than attention rows, the gauge marks of each block, and the path law with its memory.

Has this resolved the renormalization of the agent network at one base point? Not in the sense a physicist gives the word, and the chapter does not say that it has. Exact and proved are the effective action as minus the log density of the pushed evidence measure relative to the pushed reference, for every channel; the exact step on couplings, together with an identity for the residual of any declared truncation; the finite-step multipliers (the linearized exact flow) and their depth-independent bound; and the coarse path law with an exact memory recurrence, filter, and finite-memory cost. Absent are four things a physicist would expect. There is no closed truncation: apart from sectors produced by independence, symmetry, or couplings already carried by the coarse variables, no proper subspace of couplings is shown to be mapped into itself, and no truncation residual is estimated (MT Open Problem 11.25). There is no flow on one fixed set of couplings: across a step that reduces the number of configurations the coupling spaces have different dimensions and admit no identification (MT Proposition 11.20), and the main manuscript constructs no rescaling, fixed point, or beta function (MT Section 1.4). There is no thermodynamic limit: every statement concerns a finite network, and infinite networks are deferred to Supplement B, where the limit questions are open (Open Problem B6.19). And there is no parent dynamics: the parent exists at every instant as an exact pushforward (MT Corollary 9.10), but whether its evolution closes on itself is MT Open Problem 11.38, the defining open problem of the current phase.

The chapter uses the exact coarse free energy of chapter 9 (MT Theorem 9.6) and the root and holonomy marks of chapter 10, and it feeds chapter 12 and the formation witnesses of chapter 13, whose parents are exact marked parents in the present sense.

11.2 The picture

Take N agents at c_* with states y_i (belief samples, model presentations, labels) and a joint law equal to a reference probability times a record likelihood. Read the reference ϱ as the a priori measure, $L_o = e^{-H_o}$ as a Boltzmann weight at unit temperature, $\mathsf{m}_o = e^{-H_o}\varrho$ as the Gibbs weight, and its mass M_o as the partition function; in Bayesian terms M_o is the evidence and the normalized $\bar{\mathsf{m}}_o$ the posterior. A Kadanoff block map is a deterministic channel sending each block to one block variable; a noisy block map is a stochastic channel; both are admitted, and random or soft blocks enter through the structural channel of chapter 12. The exact coarse theory is the pushforward of every law. The chapter says what that pushforward looks like as an effective Hamiltonian with couplings, and what a reduced description cannot discard.

Physics analogue (exact)

The conditional-partition formula is the position-space renormalization formula with a weight function, in the block-spin tradition of Kadanoff [43] reviewed by Wilson and Kogut [79]: $e^{-H'(s')} = \sum_s P(s' \mid s) e^{-H(s)}$ up to normalization. The a priori measure is explicit, so fiber-size (entropy) terms such as those of a majority rule are carried by the pushed reference. Lengths and fields are not rescaled and nothing is truncated; the identification is exact at the level of the formula and brings no fixed-point theory with it.

Physics analogue (exact)

The two-agent example of MT Remark 11.14 is decimation: summing over spin 2 of a pair with fields ξ_1, ξ_2 and coupling ξ_{12} gives the coarse field $\xi_1' = \xi_1 + \operatorname{artanh}(\tanh \xi_2 \tanh \xi_{12})$, so the pair coupling survives only as a field and only through the polarization $\tanh \xi_2$ of the eliminated spin. With three neighbors the same mechanism produces a three-spin coupling (MT Proposition 11.26), as decimation of the two-dimensional Ising model produces four-spin couplings. This is why exact coarse-graining leaves the pairwise world.

11.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
depth ℓ, V_ℓ	Coarse-graining level and its finite agent set	Blocking step; not time
C_ℓ, C	Coarse channel	Block-spin weight function $P(s' \mid s)$

Term or symbol	Plain meaning	Physics analogue
ϱ, m_o, M_o	Reference probability; evidence measure $e^{-H_o}\varrho$; its mass	A priori measure; Gibbs weight; partition function
H_o, H_o^c	Action (minus log density relative to the reference); coarse action	Hamiltonian at unit temperature; block Hamiltonian
$C^\dagger$, U	Reverse (Bayes) kernel of fine given coarse; its conditional expectation	Conditional Gibbs measure at fixed block spins
$\mathcal{A}(\psi)$, U^ψ	Exact change of the coarse action under a bounded change ψ of the fine action; tilted reverse expectation	Change of the effective Hamiltonian; perturbed conditional ensemble
$\lVert\cdot\rVert_{\mathrm{osc}}$, δ	Half the range of a function; largest total-variation distance between reverse conditionals	Sup norm of an energy modulo constants; Dobrushin coefficient
ν_ℓ, $\mathsf{P}_{\ell,B}$	Product reference law; projector onto the pure B-body part	Noninteracting a priori measure; projection onto k-body terms
$\mathcal{G}_\ell$, $\mathsf{g} = (\mathsf{g}_B)$	One bounded component per nonempty set of agents, normed by $\sum_B \lVert\mathsf{g}_B\rVert_\infty$	All k-body couplings of a finite system
asm, hd	Couplings to action; action to couplings	Writing and reading off a Hamiltonian
$\mathcal{T}_\ell$	Exact interaction step	Exact coupling recursion, untruncated, unrescaled
D_ℓ, $D_{n\leftarrow\ell}$	Derivative of one step on an exact orbit; ordered product	Linearized RG matrix along a trajectory
$\mathsf{P}_\ell^{\mathrm{ret}}$, $r^{\mathcal{G}}$	Declared retained projection; its residual	Truncation of coupling space; truncation error
hyperedge	Component on three or more agents	Multi-spin coupling
posterior bridge, bridge kernels	Joint posterior law of fine and coarse variables; its pair conditionals	Joint law of spins and block spins
η, α, β	Event law over edge copies; receiver occupancy; attention row	Who attends to whom; its receiver marginal; the conditional
$D_N^{(k)}$, $\mathrm{h}_k^{\mathrm{ex}}$	Cost of a memory-k path law; excess conditional entropy	Truncation cost of a k-th order Markov model; excess entropy

Some glyphs mislead. M_o is an evidence, not a magnetization, and $\mathcal{G}_\ell$ is a coupling space, not a group. Three objects share a D: the multipliers D_ℓ, the edge divergences D_e, and the memory defects $D_N^{(k)}$. H is an action early in the chapter and an entropy in the memory theorems. The dagger in $C^\dagger$ is Bayes reversal (the L^2 adjoint of the lift), not complex conjugation. Here z is the

coarse sample of one arrow, whereas in chapter 12 Z_ℓ is the coordinate block at depth ℓ. "Exact" means without approximation, not solvable in closed form; "closure" means that the kept state suffices for the next exact step; a "multiplier" is a derivative and a "cocycle" an ordered product of derivatives along an orbit, not a cohomology class. The main manuscript avoids the words relevant, fixed point, and beta function by design.

11.4 The formulas, explained

11.4.1 The effective action is a conditional partition function

For one arrow $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$, with $C^\dagger_\varrho(z, dy)$ the law of the fine state given the coarse state under the reference joint law $\varrho(dy)C(y, dz)$,

$$\exp[-H^c_o(z)] = \int \exp[-H_o(y)]C^\dagger_\varrho(z, dy), \qquad M_o = \int \exp[-H^c_o(z)]\varrho^c(dz), \qquad \varrho^c = \varrho C$$

(MT Theorem 11.3). The coarse Boltzmann weight at z is the a priori average of the fine weight over the configurations the channel can send to z, and the partition function is the same at both scales. For the identity channel $H^c_o = H_o$; for a channel to a point the action becomes the number $-\log M_o$; for a deterministic block map on a finite space with uniform reference, $e^{-H^c_o(z)}$ is the average of e^{-H_o} over the fiber of z, the Kadanoff block Hamiltonian with the entropy $\log|\chi^{-1}(z)|$ moved into the reference. The theorem holds for any channel between standard Borel spaces and is standard in substance (the density of a pushforward is a conditional expectation). It says nothing about the form of H^c_o: on a finite network it always exists but may couple every coarse agent to every other, while in infinite volume the analogous object can fail to exist as a quasilocal Hamiltonian [72], which no finite-network statement reaches.

11.4.2 Perturbing the fine action: cumulants and contraction

Perturb the fine evidence measure by a bounded ψ. With $C^\dagger$ the reverse kernel of the normalized joint law,

$$\mathcal{A}(\psi)(z) = -\log \int e^{-\psi(y)}C^\dagger(z, dy), \qquad D\mathcal{A}(\psi)[h] = \mathsf{U}^\psi h, \qquad D^2\mathcal{A}(\psi)[h, k] = -\operatorname{Cov}_{C^\dagger_\psi(z,\cdot)}(h, k)$$

(MT Theorem 11.4). $\mathcal{A}$ is the conditional cumulant generating function of ψ, $\langle\psi\rangle_z - \frac{1}{2}\langle\psi;\psi\rangle_z + \cdots$, the cumulant expansion of any integrating-out calculation. It is real analytic on all bounded perturbations, order preserving, passes constants through, and never expands the sup norm (MT (11.11)). Its linear part U is a conditional expectation: it contracts every L^p norm, its L^2 loss is the expected conditional variance, and it contracts oscillation by the Dobrushin coefficient δ of the reverse kernel [24] (MT Theorem 11.5). For Kadanoff blocks the last bound is empty: under a deterministic block map with a nondegenerate coarse law the reverse conditionals at distinct coarse states live on disjoint fibers, so $\delta = 1$ and the bound says only that nothing expands; strict contraction needs a noisy channel. Every norm here is a norm of the whole action. A uniform perturbation of size h per

agent has sup norm of order Nh, so these statements control the total action, not couplings per agent.

11.4.3 Interaction coordinates

Under a product reference $\nu_\ell = \bigotimes_i \nu_{\ell i}$ equivalent to the normalized law (MT Definition 11.6), let $\mathbb{E}_{\ell,B}$ average out the agents outside B. The Hoeffding projectors and the assembly of an action from couplings are

$$\mathsf{P}_{\ell,B} = \sum_{B' \subseteq B} (-1)^{|B|-|B'|} \mathbb{E}_{\ell,B'}, \qquad \psi_{\mathsf{g}} = \sum_{\varnothing \neq B \subseteq V_\ell} \mathsf{g}_B$$

(MT Definition 11.9). $\mathsf{P}_{\ell,B} f$ is the part of f that depends on all of y_B jointly and has zero average in each single agent of B: the cluster, or ANOVA, expansion of the Hamiltonian into k-body terms. For ± 1 spins with uniform reference the B-component is $J_B \prod_{i \in B} y_i$. Gibbs-measure theory also uses the vacuum normalization relative to a reference configuration [33]; the chapter has it too (MT (11.40)) but uses only the Hoeffding components in the step. MT Theorem 11.10 proves that couplings and actions modulo constants are isomorphic, with $\|\mathrm{asm}_\ell\| \leq 1$ and $\|\mathrm{hd}_\ell\| \leq 3^{|V_\ell|} - 1$, sharp: for independent biased ± 1 coordinates the product $\prod_i Y_i$, of oscillation one, has components whose norms sum to nearly $3^n - 1$. The size of an action and the total size of its couplings are equivalent only up to a factor exponential in the number of agents. The decomposition is gauge covariant when the frame change acts agent by agent (MT Corollary 11.11), which is the case for the program's vertex gauge; preserving the product reference alone is not enough.

Two premises limit the scope, and both matter for the program. The bounded tier (MT Remark 11.7) admits finite and categorical agents and continuous couplings of compact group-valued states, so $\mathrm{SU}(N)$-valued states are inside; it excludes every polynomial coupling on $\mathbb{R}^K$, since $y_i^\top \mathsf{N} y_j$ is unbounded. The Gaussian realization with $\mathrm{GL}(K,\mathbb{R})$ or $\mathrm{GL}(K,\mathbb{C})$ links is therefore outside this machinery; its exact marginalization (Kron reduction, a Schur complement) is in Chapter G7. And product equivalence at both depths, automatic for strictly positive laws on finite products, is not preserved by every channel: a channel cloning one binary agent into two yields a diagonal law that no product law matches (MT Proposition 11.8).

11.4.4 The exact step and the native growth bound

The exact interaction step is $\mathcal{T}_\ell = \mathrm{hd}_{\ell+1} \overline{\mathcal{A}}_\ell \, \mathrm{asm}_\ell$ (MT Definition 11.12): assemble the couplings into an action, integrate out exactly modulo constants, read off the coarse couplings. The couplings perturb a base law whose own image is carried separately, so $\mathcal{T}_\ell(0) = 0$ for every model and channel (MT Lemma 11.13). Along an exact orbit the multipliers are $D_\ell = \mathrm{hd}_{\ell+1} \overline{\mathsf{U}_\ell^{\psi_{\mathsf{g}_\ell}}} \, \mathrm{asm}_\ell$, every interior pair $\mathrm{asm}_k \, \mathrm{hd}_k$ cancels, and

$$D_{n \leftarrow \ell} = \mathrm{hd}_n \, \overline{\mathsf{U}_{n-1}^{\psi_{\mathsf{g}_{n-1}}}} \cdots \overline{\mathsf{U}_\ell^{\psi_{\mathsf{g}_\ell}}} \, \mathrm{asm}_\ell, \qquad \|D_{n \leftarrow \ell}\|_{\mathcal{G}_\ell \to \mathcal{G}_n} \leq 3^{|V_n|} - 1$$

(MT Proposition 11.19). Every tilted conditional expectation has oscillation norm at most one, assembly has norm at most one, and only the final extraction can exceed one.

What is bounded is the operator norm of the linearized composite step from all depth-ℓ couplings to all depth-n couplings, each measured by the sum of the sup norms of its components: a perturbation of total size one in the fine couplings changes the total size of the depth-n couplings by at most $3^{|V_n|} - 1$ to first order, however many steps intervene and however large the fine network was. It is not a bound per coupling or per agent. As a number it is useful only near the top of a blocking: with pair blocks 512 agents admit nine steps, and the constant is 2, 8, 80, and 6560 for targets of one, two, four, and eight blocks, but about 10^{122} for the first step's target of 256 agents. The real content is independent of N: in the action norm the linearized exact flow is a product of contractions, and all apparent amplification is one change of norm at the target. Supplement B turns this into the statement that the exact step has no relevant direction under tempered comparisons (Corollary B5.18); relevance requires an extensive normalization declared from outside.

Physics analogue (structural)

In Wilson's renormalization a uniform field $h \sum_i s_i$ is relevant because the per-site field grows as $b^{y_h} h$ under blocking by b in d dimensions, while the whole-action size hN becomes $b^{y_h - d} hN$, which does not grow since $y_h \leq d$. Relevance is a statement about couplings per site. The native bound is the exact finite-network form of the other half: with no rescaling and no per-site normalization, nothing grows.

11.4.5 Residuals, invariant sectors, and the finite-network obstruction

A declared truncation is a gauge-intertwining idempotent $\mathsf{P}^{\mathrm{ret}}$ on couplings, for example onto single-agent and pair components (MT Definition 11.21), with residual $r^{\mathcal{G}}_{\ell+1} = (I - \mathsf{P}^{\mathrm{ret}}_{\ell+1})\mathcal{T}_\ell(\mathrm{g}_\ell)$. MT Proposition 11.22 proves that the coupling and action residuals vanish together, agree in norm up to $3^{|V_{\ell+1}|} - 1$, and vanish on the retained range exactly when that range is mapped into its successor. It is an identity, not an estimate. The proved invariant pairs are generated by structure, not dynamics (MT Proposition 11.15, MT Remark 11.24): independent blocks under a block-product channel generate no inter-block couplings; a finite symmetry preserves its fixed sector, such as the even sector under a global spin flip; and couplings already carried by the coarse variables of a deterministic channel pass through unchanged. Anything beyond is MT Open Problem 11.25.

For finite agent spaces the coupling space has dimension $\prod_i |\mathrm{supp}\, \nu_{\ell i}| - 1$, so $2^N - 1$ for N binary agents (MT Proposition 11.20). A step that reduces the number of configurations reduces the dimension (six binary agents in three binary blocks go from 63 to 7), so no identification of the two coupling spaces exists and no single space carries a flow; if every step divides the vertex count by at least $b > 1$, the depth is at most $\log_b |V_\ell|$. This is not a small-network artifact. It holds at $N = 512$ and at $N = 6 \times 10^{23}$ alike: a finite system can be blocked only logarithmically often, and its coupling space shrinks with each blocking. Physics removes the obstruction by passing to infinite volume and identifying couplings of the same shape across scales by translation invariance; neither ingredient is in the main manuscript, and Supplement B records that finite networks do not carry the asymptotic classification (Proposition B5.15). The dimension count needs finite agent spaces; termination holds for all.

11.4.6 Hyperedges and bridge kernels

Marginalizing a variable multiplies the factors that touch it and leaves one factor on the union of its neighbors, so iterated elimination produces factors on elimination cliques (MT Proposition 11.26); this is variable elimination with fill-in. Eliminating the center y_0 of a star of three ± 1 neighbors, with field ξ_0 and couplings ξ_r, leaves a coefficient $2\,\mathrm{sech}^2(\xi_0)\tanh(\xi_0)\xi_1\xi_2\xi_3 + O(\xi^5)$ on $y_1y_2y_3$: nonzero for small nonzero couplings when the eliminated spin carries a field, and identically zero at $\xi_0 = 0$, where spin-flip symmetry kills odd terms. Pairwise closure fails in general; "generic" is shown within the star family, not over all networks, although the mechanism is that of decimation.

The posterior bridge is the joint posterior law of fine and coarse variables, and the bridge kernels are its pair conditionals between a fine agent and a meta-agent, Bayes adjoints of each other (MT Definition 11.27, MT Proposition 11.28). The meta-to-agent kernel does not reconstruct the fine posterior: when the likelihood couples blocks, a fine agent's conditional law depends on neighboring block variables too, just as the configuration inside a block is not fixed by its own block spin. For a deterministic block map, lifting and conditioning returns a coarse observable unchanged; for a randomized channel it does not.

11.4.7 Meta-attention

An attention row is the conditional law of a source given a receiver, and a conditional law cannot be coarse-grained without the law of what it conditions on, as lumping a transition matrix needs occupation weights. The recursive state is the joint event law η over oriented edge copies:

$$\eta^c_{IJ}(z) = \mathbb{E}_{\mathbb{B}_o}\Big[\sum_{e:j\to i,\ i\in I,\ j\in J} \eta_e(Y)\ \Big|\ Z = z\Big], \qquad \alpha^c_I = \sum_J \eta^c_{IJ}, \qquad \beta^c_{IJ} = \eta^c_{IJ}/\alpha^c_I$$

(MT Proposition 11.29). The next depth pushes η^c, not β^c, and events inside a block become meta-self events, so a meta-agent attends to itself whenever its members attend to each other.

Where the receiver weights come from is a modeling choice with consequences. In the single-edge model (MT Definition 6.5) each record carries one joint receiver–source label, and the receiver law within a block is evidence weighted, $\alpha'_i \propto a_{i|I}\widetilde{c}_i\mathcal{Z}_i$, where $\mathcal{Z}_i = \sum_j \pi_{ij}e^{-D_{ij}/\tau_i}$ is the partition function of receiver i's row and $\widetilde{c}_i$ the likelihood ratio of its record under the event; the block row is the α'-weighted average of member rows (MT Proposition 11.30). In the per-receiver model of chapter 6, every receiver has its own label, and a block row needs a structural selector that draws one receiver with fixed weights independently of everything; its posterior equals its prior, and the block row collapses to the prior-weighted average (MT Proposition 11.31). The two agree when $\widetilde{c}_i\mathcal{Z}_i$ is constant on the block and otherwise differ ($3/4$ against $1/2$ in the chapter's two-receiver example); the collapse is the loss of evidence weighting. Keeping all receivers' labels is exact, and collapsing them to one row is a further channel whose cost enters MT (9.3).

Physics analogue (exact)

Grouping the sources of one row gives the Gibbs row of the coarse prior $\pi^c_J = \sum_{j\in J}\pi_j$ and coarse energy $D^c_J = -\tau\log[(\pi^c_J)^{-1}\sum_{j\in J}\pi_j e^{-D_j/\tau}]$ (MT Remark 11.32): the free energy of a macrostate whose microstates have degeneracies π_j. In the single-edge model, receivers are merged with weights proportional to their own partition functions, as subsystems are when their ensembles are merged.

The event-weighted mean of transported link representations between blocks maps the root frame of J to that of I covariantly, but it need not lie in the group (the mean of rotations by $\pm\pi/2$ is zero) or be invertible, and it is only a first moment of the joint law of marks and features that exact closure keeps (MT Proposition 11.33). It is the blocked link of lattice gauge theory, which leaves the group and is usually projected back; here nothing is projected.

11.4.8 Path laws and memory

Applying the channel at every time of a path is again a coarse channel, so the free-energy chain rule holds for path laws; the ordering index is an inference index, not physical time (MT Section 11.8). The coarse path law is exact but generally not Markov. MT Theorem 11.35 is the exact condition of Kemeny and Snell [46, Thm. 6.3.2], on standard Borel spaces, for Markov coarse dynamics from every initial law: equal jump probabilities into each coarse set from all fine states with the same coarse value. MT Remark 11.36 shows a chain whose coarse process is Markov from one start without lumpability, and a deterministic three-cycle with labels $(0,0,1)$ whose stationary coarse process has memory. For a linear recursion $\xi_{n+1} = \mathsf{T}\xi_n$ with resolved coordinate $w_n = \mathsf{C}\xi_n$, lift J ($\mathsf{CJ} = I$), and $\mathsf{Q} = I - \mathsf{JC}$,

$$w_{n+1} = \mathsf{CTJ}w_n + \sum_{k=0}^{n-1}\mathsf{CTQ}(\mathsf{QTQ})^{n-1-k}\mathsf{QTJ}w_k + \mathsf{CTQ}(\mathsf{QTQ})^n\mathsf{Q}\xi_0$$

exactly (MT Theorem 11.39): a Markovian term, memory kernels, and the unresolved initial state. This is the discrete, linear form of the projection-operator method of Nakajima [54] and Zwanzig [86], the last term playing the role of the noise.

For a hidden finite chain observed through labels, the exact filter is the forward algorithm, and its terminal posterior determines every prediction but may take infinitely many values (MT Theorem 11.42), so a finite-state Markov parent does not follow even in principle. The memory-k approximation whose rows are the true conditionals given the last k symbols is the unique forward-KL optimum, with cost

$$D_N^{(k)} = D_{\mathrm{KL}}\big(\mathbb{P}_N^{\mathrm{path}}\|\widehat{\mathbb{P}}_N^{(k)}\big) = \sum_{t=0}^{N-1} I\big(Z_t; Z_{0:t-k-1} \mid Z_{\max(0,t-k):t-1}\big)$$

(MT Theorem 11.44): what the discarded past still says about the present given the kept past. The reverse divergence is not controlled (the memory-one projection of the three-cycle gives the impossible word 000 probability $1/6$), and using the projection as a prior changes a recognition law's

free energy by an unsigned amount. For a stationary start, MT Theorem 11.45 gives $D_N^{(k)}/N \to \mathrm{h}_k^{\mathrm{ex}}$ with $\sum_j \mathrm{h}_j^{\mathrm{ex}} \le H(Y_0)$ and $\mathrm{h}_k^{\mathrm{ex}} \le H(Y_0)/(k+1)$, where N is now the path length: the excess entropy [19] is bounded by the entropy of the hidden state. For the program this budget is extensive, up to $512 \log 2 \approx 355$ nats for 512 binary agents, and the theorem is stated for finite hidden spaces only. The prepared parent of chapter 8 is Markov when the fine state inside each parent is refreshed to its exact conditional law at every step (MT Theorem 8.11), which is adiabatic elimination by hypothesis (MT Hypothesis 8.9); with partial refresh the parent history has memory (MT Remark 8.12), whose first-order cost is $D_N^{(1)}$.

11.4.9 The retained-state table and what "exact" means

MT Table 11.1 lists each retained object with its exact operation, the statement that derives it, and its admission hypotheses, and the chapter says it is a catalog, not a completeness theorem. Its first row keeps the complete normalized generative, posterior, and recognition laws, and its last the path law with its memory. Closure is therefore achieved by keeping everything; the other rows name parts of those objects that a smaller description must not discard, several with witnesses that they cannot be dropped in general (hyperedges, receiver laws, memory, the marked kernel rather than its mean, and in chapter 12 the graph datum). "Exact" means exact as $\sum_y e^{-H(y)}$ is exact, a lossless change of description, not exact as one-dimensional Ising decimation is, a finite set of couplings with a closed recursion; no reduced retained set is shown to close. Blocks and channel are inputs, and must be: at the uniform law on seven outcomes, no rule that sees only the law and its Fisher metric and treats all outcomes alike can select three two-dimensional blocks (MT Proposition 11.46).

11.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Theorem 11.3	Coarse action is minus the log of a conditional partition function; evidence preserved	SOLID	Any channel; standard in substance
MT Theorem 11.4, MT Theorem 11.5	Analytic, nonexpansive action increment; cumulant derivatives; L^p and Dobrushin contraction	SOLID	Whole-action norms; $\delta = 1$ for deterministic block maps
MT Theorem 11.10, MT Corollary 11.11	Couplings and actions isomorphic; extraction constant $3^{\lvert V\rvert} - 1$, sharp; gauge covariant	SOLID	ANOVA standard; sharp constant specific
MT Remark 11.7, MT Proposition 11.8	Polynomial couplings on $\mathbb{R}^K$ excluded; product equivalence can fail	SOLID	Negative; Gaussian $\mathrm{GL}(K)$ couplings are out of scope
MT Lemma 11.13, MT Proposition 11.23	Exact step fixes the origin, has the tilted derivative, is gauge covariant	SOLID	Bounded tier; covariance of a.e. classes

Result	What it says	Status	Scope and caveats
MT Proposition 11.19	Multipliers telescope; norm at most $3^{\lvert V_n\rvert} - 1$ at every depth	SOLID	Bounded tier; vacuous as a number for large targets
MT Proposition 11.20	No identification of coupling spaces across a reducing step; blocking terminates	SOLID	Negative; every finite network, not only small ones
MT Proposition 11.22, MT Remark 11.24	Residual identity; independence, symmetry, and pullback sectors are invariant	SOLID	No residual estimate; trivial pairs only
MT Remark 11.14	Decimating one binary agent	TOY	Two agents; textbook mechanism
MT Proposition 11.26	Pairwise closure fails (star), not universally (parity)	SOLID	Negative by counterexample; "generic" within the star family
MT Proposition 11.28	Bridge kernels adjoint; no reconstruction of the fine posterior	SOLID	Standard in substance
MT Proposition 11.29, MT Proposition 11.31	Push event laws; evidence-weighted versus prior-weighted receivers	SOLID	Inside two declared label models
MT Theorem 11.35, MT Theorem 11.39, MT Theorem 11.42	Lumpability; discrete Nakajima–Zwanzig; exact filter	SOLID	Standard in substance
MT Theorem 11.44, MT Theorem 11.45	Memory-k cost; excess entropy at most hidden entropy	SOLID	Finite spaces; stationary start; budget extensive in N
MT Table 11.1, read as closure	Exact closure keeps the listed objects	VAGUE	Rows proved or declared; sufficiency trivial; no minimality claim
MT Proposition 11.46	Law and Fisher metric cannot select blocks	SOLID	Negative
MT Definition 11.6, MT Definition 11.21	Channel, blocks, reference, retained projection are inputs	CHOICE	Declared, not selected
MT Open Problem 11.25, MT Open Problem 11.38	Closed truncation; closed parent dynamics	OPEN	See section 11.6

The only toy positive is the two-agent decimation; the other small instances are counterexamples, hence solid negative results. The mechanisms they display (fielded eliminated agents generate multi-agent couplings, hidden coordinates produce memory, evidence weighting differs from prior weighting) plausibly survive at large N but are proved only where stated. The general results hold

at every finite N; their only N-dependence is in $3^{|V_n|}-1$ and $H(Y_0)$, the quantities a large-N theory would have to replace with intensive ones.

Conditions: which ones matter

The conditions that can fail in the program's models are these. The bounded tier admits finite, categorical, and compact group-valued states ($\mathrm{SU}(N)$ included) and excludes the quadratic couplings of $\mathbb{R}^K$-valued agents, so the coupling results say nothing about the Gaussian $\mathrm{GL}(K,\mathbb{R})$ or $\mathrm{GL}(K,\mathbb{C})$ realization. Product equivalence at both depths fails for channels that impose hard constraints among coarse agents. The channel and blocks are declared and must not read the recognition law. Everything is finite-network, with no limit taken. The memory results need finite hidden spaces and, for the budget, a stationary start. The rest is hygiene (standard Borel spaces, measurable reverse kernels, almost-everywhere versions, integrability), with one item touching noncompact groups: pointwise equivariant conditional versions are settled only for finite and compact stabilizers and weakly continuous posteriors (MT Open Problem 9.18), so for $\mathrm{GL}(K)$ covariance holds for almost-everywhere classes.

11.6 Open ends

MT Open Problem 11.25 asks for a closed truncation. It is posed for finite agent spaces with full-support product references, at least two blocks, and deterministic block-product channels whose block maps read every member of their block (majority or block-sum maps, not decimation), for product base laws and for all base laws. It is settled by a proper invariant pair, invariant for every base law in an open set, whose range is not inside the sum of the pullback sector and any grouping sector, or by a proof that generically every proper invariant pair lies in such a sum. The exclusion of decimation sets aside the classical exactly closed recursion of the one-dimensional Ising chain [35].

MT Open Problem 11.38 is the defining problem. The exact parent exists at every instant as the image of the fine law, so parents do evolve, but only as images of the fine evolution; the question is whether that evolution is a dynamics of its own. In a differentiable case the obstruction is the semiconjugacy defect

$$\delta_t = \partial_t \mathsf{F}_t + D\mathsf{F}_t \mathscr{V}_t - \overline{\mathscr{V}}_t \circ \mathsf{F}_t$$

between the fine field pushed through the moving coarse map and a candidate parent field; for Markov evolutions the exact condition is semigroup intertwining. Existing statements answer the unrestricted question both ways (lumpable kernels lump, a parent carrying the full filter reproduces every path law, the three-cycle admits no first-order parent), so it is posed for a class: switching diffusions at c_* of the four-cycle type of chapter 13 (finitely many source assignments, a quadratic energy on each assignment's sheet, overdamped diffusion, a rate-one clock redrawing the assignment), with a parent that keeps the assignment and a projection of the configuration but hides coordinates that the receiver rows of a non-negligible assignment read. Candidate parents are time-homogeneous Markov processes of the retained variables, possibly with a regular finite-dimensional filter (MT Definition 11.37), started from the exact image. It asks whether every member admits

such a parent whose path law agrees in total variation as the regime parameter $\epsilon \to 0$, on fixed windows up to the first expected switch and on whole histories with diverging numbers of switches; a member for which every such parent stays a bounded distance away settles it negatively. In physics terms: does adiabatic elimination with finitely many auxiliary variables close the retained dynamics when hidden coordinates drive the switching? The one positive instance is the four-cycle's inherited parent on one episode window (MT Proposition 13.22), a toy case. The limit is $\epsilon \to 0$ at any finite size, not $N \to \infty$.

Three gaps are not posed as open problems of the main manuscript: no truncation residual is estimated; the unbounded tier on $\mathbb{R}^K$ exists only in the Gaussian case of Supplement G; and a thermodynamic limit of the exact step belongs to Supplement B, where fixed-depth and diagonal limits and universality are open (Open Problem B6.19).

Does it deliver what it claims? NARROWER THAN IT SOUNDS

The chapter says it determines what one exact step does to the agent network at c_* and which state must be kept so that the step closes, with an effective action, residual, finite-step multipliers, and native growth bound; the introduction calls this exact coarse-graining of the agent network. The proofs deliver those objects: the effective action for every channel, its analytic nonexpansive calculus, an exact step on couplings with an exact residual identity, a depth-independent derivative bound, exact memory identities and costs, and witnesses that hyperedges, receiver laws, marked kernels, and memory cannot be dropped in general. The delivery is narrower than the wording in four respects. Closure is achieved by keeping the complete laws; no smaller closed set and no truncation with a bounded error are proved. The coupling machinery holds only in the bounded tier, which excludes the program's Gaussian $\mathrm{GL}(K)$ couplings. Everything is finite-network, so there is no flow on fixed couplings (proved impossible across reducing steps) and no thermodynamic limit. And parent dynamics is open. The scope paragraph says most of this; "which state must be kept so that the step closes" is where a physicist will read more than is there. Much of the content is standard in substance; specific are the typing at c_*, the sharp constant and telescoping bound, the obstruction, the attention-event bookkeeping, and gauge covariance throughout.

Chapter 12

The Multiscale Graph VFE and Closing the Tower

12.1 The question and the answer

The technical chapter (MT Chapter 12) asks how to place everything exact coarse-graining produces (adaptive attention event laws and links, random partitions, and the influence of parents on their children) into one variational free energy across depths, and how a tower of depths ends. Its answer is one normalized generative law and one recognition law over the whole tower, with free energy

$$\mathcal{F}^{\mathrm{ext}}_{\mathrm{tower}}[\mathbb{Q}_\lambda; X_*, o] = -\log p_\theta(o \mid X_*) + D_{\mathrm{KL}}\big(\mathbb{Q}_\lambda(\cdot \mid o, X_*) \| \mathbf{\Pi}_{\theta,o,X_*}\big)$$

(MT Definition 12.1); everything else in the chapter is exact bookkeeping of this object or is labeled as something else. For a tower built bottom-up by coarse-graining channels, both laws factor top-down automatically, each depth conditioned on the complete next-depth state (coordinates and graph datum) and the partition mark (MT Lemma 12.5); dropping the graph datum costs a sum of mutual informations (MT Proposition 12.6); and the free energy splits into observation, top-prior, partition, graph, and parent–child relative entropies (MT Theorem 12.7). Partitions may be random, from a finite catalog under two inequivalent normalizations (MT Proposition 12.13). The tower cannot close on itself: a finite acyclic model has a parentless latent with a declared prior (MT Proposition 12.20). What remains of the idea that the agents constitute their own top lives at three loci (the inference loop, learning of the top prior, and delayed feedback across episodes), each an evaluation of fixed normalized joints, and delayed feedback with a declared update rule is itself the free energy of one lifted joint (MT Proposition 12.25). Whether descent of this free energy produces a persistent hierarchy is open (MT Open Problem 12.27).

Do parents act back on their children in a proved sense? In four senses of different strength. In a declared top-down hierarchy the downward kernel reads the parent coordinates, so the parent enters every child's prior by construction, a modeling choice; parent and child optimization are then coupled exactly when a mixed derivative is nonzero, which the chapter proves only as a criterion in a binary example (a kernel odds ratio different from one), declining any family-wide claim without a proof in that family (MT Proposition 12.19). In an adaptive tower the downward kernel is the Bayes reversal of the bottom-up coarse-graining, and the chapter declines to read it causally. Through the inference loop, each child's effective prior at a mean-field fixed point is formed from the parent's

factor, itself formed from all children (MT Remark 12.21), a standard fact. Across episodes, the parent inferred at one episode sets the children's prior at the next inside one declared joint (MT Proposition 12.24). Not proved is a parent with its own dynamics acting back over time: that is MT Open Problem 11.38 and the third result required by MT Open Problem 12.27.

12.2 The picture

Picture depths $\ell = 0, \dots, L$: agents at depth 0, meta-agents at depth 1, a declared top at depth L. The state at each depth includes, besides coordinates, the graph datum: who attends to whom in both channels, the links on every edge copy, and the root and holonomy marks of each block. Going up, a structural channel takes the complete depth-ℓ state and produces a partition mark R_ℓ and the depth-$(\ell+1)$ state. Read downward, the same joint law is a Markov chain in decreasing depth whose transitions are the conditional laws of finer given coarser states. The free energy of the whole stack is one Gibbs variational functional, and its per-depth decomposition is the chain rule.

Physics analogue (exact)

Bayes reversal of an adaptive tower is time reversal of a Markov chain with depth as time: the chain $Y_0 \to (R_0, Y_1) \to (R_1, Y_2) \to \cdots$ read backward is Markov, and its downward transition is the reverse block-spin kernel of chapter 11. No downward causation follows, as time reversal of a diffusion does not make the future cause the past.

Physics analogue (exact)

The two normalizations of a partition catalog are the two ways to combine Gibbs measures on several sectors: a mixture of separately normalized measures with weights fixed by hand (hierarchical), or one Gibbs measure on the disjoint union, in which each sector weighs its a priori weight times its partition function, $\mathsf{a}_\gamma e^{-F_\gamma}$ (global). The first fixes the probabilities of the phases; the second lets their free energies take part.

Physics analogue (exact)

The coarse links of MT Definition 12.17 are maximal-tree gauge fixing inside each block [17]: tree links go to the identity, the remaining internal links carry the block's Wilson loops as self-loops of the coarse vertex, and boundary links connect block roots. Unlike smearing of gauge links, nothing is averaged or projected, and every parallel copy is kept.

12.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Z_ℓ	Full coordinates at depth ℓ (beliefs and models)	Block variables
$\mathfrak{N}_\ell = (\eta_\ell, \Theta^{b,\ell}, \Theta^{m,\ell}, \mathfrak{h}_\ell)$	Graph datum: event laws, links, root and holonomy marks	Coarse lattice with gauge field and bond occupations
$Y_\ell = (Z_\ell, \mathfrak{N}_\ell)$, $\mathbf{Y}$	Complete state at depth ℓ; all depths together	Block configuration; multiscale configuration
R_ℓ, γ	Structural mark (realized partition or membership pattern)	Random blocking pattern
$C^{\mathrm{struct}}_{\ell+1\leftarrow\ell}$	One kernel from Y_ℓ to $(R_\ell, Y_{\ell+1})$ applied to all three laws	Configuration-dependent block transformation
adaptive tower	Tower generated bottom-up by structural channels	Iterated blocking
$K^\ell_\downarrow$	Law of depth-ℓ coordinates given the next depth	Prolongation; reverse block-spin kernel
$\mathcal{F}^{\mathrm{att},x}_{i,\ell}$	Row free energy: mean edge divergence plus τ times relative entropy to the prior	Helmholtz free energy of a receiver's source choice
$\mathcal{J}_{\mathrm{comp}}$	Energies appended without a normalizer	Lyapunov function, not a free energy
$\mathfrak{G}$, a_γ, $\mathcal{Z}_\gamma$	Partition catalog; label weights; branch partition functions	Sectors, their weights and partition functions
$\mathfrak{m}_\ell$, $\mathfrak{M}_\ell$	Soft membership of a vertex; joint membership of an edge's endpoints	Fuzzy block assignment
declared root, $\mathbb{P}_L$	Top prior supplied by no other latent	Boundary condition at the top
episode n, θ_n, $\mathcal{U}_n$	Learning episode, parameter, declared update	Sweep of a learning loop
latent route; parameter route	One joint across episodes; separate joints linked by updates	Hidden Markov model over episodes; learning by updates
lifted joint, $\widetilde{o}_n$	One joint with parameters and record copies latent; latent record copy	State augmentation in filtering
TC, ϵ_{e}, T	Total correlation across episodes; aggregate update step; descent temperature	Multi-information; learning rate; temperature

Here Y_ℓ includes the graph datum and Z_ℓ is the coordinate block at depth ℓ, unlike the coarse sample z of chapter 11. A learning episode (index n) is not a coherence episode of formation (chapter 13). The fraktur $\mathfrak{m}_\ell$ is a membership kernel, unrelated to the evidence measure m_ℓ of chapter 11, and R_ℓ is a partition mark, not a rotation. "Top-down" names the direction of a generative arrow and, in an adaptive tower, carries no causal meaning. "Closing the tower" means ending it with a declared root. "Participation" is interpretive vocabulary, not a mathematical property.

12.4 The formulas, explained

12.4.1 One free energy and the structural channel

The tower free energy is the evidence identity of chapter 5 on a latent space that contains every depth; with $L = 0$ it is the population free energy of one network. The commitment, and the reason it is a choice rather than a theorem, is that one normalized generative law covers all depths: independently chosen per-depth energies added together do not define a free energy, and the chapter calls such a sum a composite potential.

At depth ℓ the closed scale datum holds the full coordinates and the graph datum on a finite multigraph, the support of the source prior at depth 0 and the multigraph induced by coarse links deeper down (MT Definition 12.2). A structural step is one recognition-independent channel $C^{\mathrm{struct}}_{\ell+1\leftarrow\ell} : \mathsf{Y}_\ell \rightsquigarrow \mathsf{R}_\ell \times \mathsf{Y}_{\ell+1}$, applied alike to the generative marginal, the posterior, and the recognition law, whose laws of R_ℓ may therefore differ (MT Definition 12.3). Kadanoff blocks are a constant mark with a deterministic block-product map. Random blocks are a mark drawn by the channel, possibly from the fine configuration, which makes the step adaptive. What the blocking may not read is a law: choosing blocks by inspecting the posterior or recognition law is excluded, since that is not one channel applied to three laws. Soft blocks are membership kernels $\mathfrak{m}_\ell(I \mid i, R_\ell)$, a hard partition being the deterministic case, with a joint endpoint kernel $\mathfrak{M}_\ell$ for correlated assignment of an edge's two ends (MT Definition 12.14). Under independent draws the coarse event law is

$$\eta^{x,\ell+1}_{IJ}(y,\gamma) = \sum_{i\neq j} \eta^{x,\ell}_{ij}(y)\mathfrak{m}_\ell(I \mid i,\gamma)\mathfrak{m}_\ell(J \mid j,\gamma) + \delta_{IJ}\sum_i \eta^{x,\ell}_{ii}(y)\mathfrak{m}_\ell(I \mid i,\gamma),$$

the loop term using one draw for both ends of a self-edge (MT Proposition 12.15). The random $\eta_{\ell+1}$ is part of the state, not its mean, and unit-weight membership in several parents replicates mass rather than splitting it. Nested assignments compose by Chapman–Kolmogorov, and a multi-depth channel is exact only if the state carries every mark the next kernel reads (MT Definition 12.16).

12.4.2 Adaptive reversal and the cost of dropping the graph datum

The standing hypothesis MT Hypothesis 12.4 factors the generative law from the top,

$$\begin{aligned}\mathbb{P}_\theta(do, d\mathbf{Y} \mid X_*) = \mathbb{P}^{\mathrm{obs}}_\theta(do \mid Z_0, X_*)\mathbb{P}_L(dY_L \mid X_*)\\ \times \prod_{\ell=0}^{L-1} \mathbb{P}^\ell_R(dR_\ell \mid Y_{\ell+1})\mathbb{P}^\ell_{\mathfrak{N}}(d\mathfrak{M}_\ell \mid Y_{\ell+1}, R_\ell)K^\ell_\downarrow(dZ_\ell \mid Y_{\ell+1}, R_\ell, \mathfrak{N}_\ell),\end{aligned}$$

with X_* suppressed in the depth factors, and recognition likewise. Each depth reads the complete next depth and nothing deeper: a Markov property in scale, not mean field, since dependence within and between adjacent depths is arbitrary. For a tower generated bottom-up by structural channels it is automatic (MT Lemma 12.5): deeper variables are produced from $Y_{\ell+1}$ by kernels that see nothing finer, and successive disintegration gives the top-down factors, which are reversal (bridge) kernels rather than independent choices. When the parent is a function of the child and the

mark, the downward kernel is supported on compatible child configurations, the constrained sum of a Kadanoff block read backward.

If the depth-ℓ factors read only $(Z_{\ell+1}, R_\ell)$, the reduced product differs from the tower law by

$$D_{\mathrm{KL}}\big(\mathbb{P}^{\mathrm{tow}}\|\mathbb{P}^{\mathrm{red}}\big) = \sum_{\ell=0}^{L-1} I_{\mathbb{P}^{\mathrm{tow}}}\big(Y_\ell; \mathfrak{N}_{\ell+1} \mid Z_{\ell+1}, R_\ell\big)$$

(MT Proposition 12.6): what the coarse graph datum says about the fine state beyond the coarse coordinates and the partition. The chapter's four-vertex witness, where fine event laws with equal coarse coordinates push to a boundary event or a meta-self event, costs $\log 2$. The identity is the chain rule; its content is that coarse attention structure is part of the coarse state.

12.4.3 The hierarchical decomposition and the row free energies

Under the depth-Markov factorization, with common domination and integrability,

$$\begin{aligned}\mathcal{F}_{\mathrm{tower}} = & -\mathbb{E}_{\mathbb{Q}_\lambda} \log p_\theta^{\mathrm{obs}}(o \mid Z_0, X_*) + D_{\mathrm{KL}}(\mathbb{Q}_L\|\mathbb{P}_L) \\ & + \sum_{\ell=0}^{L-1} \mathbb{E}_{\mathbb{Q}_\lambda}\big[D_{\mathrm{KL}}(\mathbb{Q}_R^\ell\|\mathbb{P}_R^\ell) + D_{\mathrm{KL}}(\mathbb{Q}_{\mathfrak{N}}^\ell\|\mathbb{P}_{\mathfrak{N}}^\ell) + D_{\mathrm{KL}}(\mathbb{Q}_Z^\ell\|K_\downarrow^\ell)\big]\end{aligned}$$

(MT Theorem 12.7): the data misfit at the finest depth, the top's recognition law against its declared prior, and at each depth the costs of the partition mark, the graph datum, and the coordinates against their parent-conditioned priors, averaged under recognition. With one depth it is misfit plus prior relative entropy. It is the measure-level chain rule, standard in substance; its value is that every candidate hierarchy has exact accounting.

With transported divergences $D_e^{x,\ell}$, copy priors $\pi_i^{x,\ell}$, and temperatures $\tau_{i,\ell}^x$, the row free energy $\mathcal{F}_{i,\ell}^{\mathrm{att},x} = \sum_e \beta_e^{x,\ell} D_e^{x,\ell} + \tau_{i,\ell}^x D_{\mathrm{KL}}(\beta_i^{x,\ell}\|\pi_i^{x,\ell})$ (MT Definition 12.8) is uniquely minimized by the Gibbs row $\beta_e^* \propto \pi_e e^{-D_e/\tau}$ with minimum $-\tau \log \mathcal{Z}$, infinite-energy edges included (MT Proposition 12.9): the Gibbs variational principle for $U - \tau S$ with the prior as degeneracy. With $\tau \neq 1$ this energy-unit form is not an independently weighted sector of one evidence bound. Appending such row energies, partition terms, cross-scale terms, and a link regularizer to a fixed population free energy gives the composite potential of MT Definition 12.10, a descent potential with no evidence interpretation; the same terms from a normalized Gibbs prior inside the joint belong to the tower free energy instead.

12.4.4 Random partitions and the two normalizations

A finite catalog $\mathfrak{G}$ gives each label γ a carrier, a reference ν_γ, a raw factor $\widetilde{p}_\gamma$, a weight a_γ, and a likelihood, with branch partition function $\mathcal{Z}_\gamma = \sum_y \nu_\gamma(y)\widetilde{p}_\gamma(y)$ (MT Definition 12.12):

$$\mathbb{P}_{\mathrm{h}}(\gamma, y, o) = \bar{\mathsf{a}}_\gamma p_\gamma(y) L_\gamma(o \mid y), \qquad \mathbb{P}_{\mathrm{f}}(\gamma, y, o) = \frac{\mathsf{a}_\gamma \nu_\gamma(y)\widetilde{p}_\gamma(y) L_\gamma(o \mid y)}{\sum_\delta \mathsf{a}_\delta \mathcal{Z}_\delta}.$$

Both are normalized with the same within-branch laws, but the label weight is $\bar{a}_\gamma$ hierarchically and proportional to $a_\gamma \mathcal{Z}_\gamma$ globally; the joints coincide exactly when all $\mathcal{Z}_\gamma$ agree, rescaling one branch's raw factor changes only the global joint, and the two free energies differ only in normalizers (MT Proposition 12.13). A catalog creates no edge. The partition term of MT Definition 12.11 enters the free energy only when its Gibbs factor is part of the generative law, and no canonical partition selector follows from the free energy (MT Open Problem 9.19).

12.4.5 Coarse links

Under a hard partition with one root and spanning tree per block, every fine edge copy survives between the blocks of its endpoints, with the dressed link $\Theta_e^{x,\ell+1} = \Theta_{I\leftarrow i}^{x,\ell}\Theta_e^{x,\ell}(\Theta_{J\leftarrow j}^{x,\ell})^{-1}$ (MT Definition 12.17). By MT Proposition 12.18, no link content is lost: internal tree copies have identity links, internal non-tree copies carry the based holonomy of their fundamental loops, closed-walk products reproduce the block holonomy, and coarse divergences and rows are gauge invariant. A loop divergence vanishes on a non-tree copy exactly when the parent law is fixed by that holonomy. Once holonomy is nontrivial the trees are part of the coarse channel, not a gauge choice, although the divergences do not depend on them for holonomy-stabilized parent laws. Overlapping membership is excluded, since an edge would need a root transport into every block containing an endpoint.

12.4.6 How a parent enters its children's prior

In a declared top-down hierarchy whose downward kernel factors over parents and reads the parent coordinates, the cross-scale term of the free energy is

$$\mathcal{F}^\ell_{\mathrm{cross}} = \mathbb{E}_{\mathbb{Q}_\lambda} D_{\mathrm{KL}}\big(\mathbb{Q}^\ell_Z(\cdot \mid Y_{\ell+1}, R_\ell, \mathfrak{N}_\ell, o, X_*) \| K^\ell_\downarrow(\cdot \mid Y_{\ell+1}, R_\ell, \mathfrak{N}_\ell, X_*)\big)$$

(MT Proposition 12.19), and the parent can also couple to child marks through the partition and graph factors. A kernel that ignores the parent contributes no parent–child cross-derivative through this factor, and normalization alone implies no coupling. When the kernel reads the parent and the recognition parameterization transmits the dependence, the mixed derivative can be nonzero, so optimization is coupled both ways although the generative arrow points down. For a binary parent and child with product recognition $q_b(1) = u$, $q_y(1) = v$, the mixed derivative is $-\log[K_\downarrow(1 \mid 1)K_\downarrow(0 \mid 0)/(K_\downarrow(0 \mid 1)K_\downarrow(1 \mid 0))]$, zero exactly at unit odds ratio. That criterion in a two-variable example is the whole of the proved bidirectional coupling.

12.4.7 Closing the tower and the three loci

In a finite directed acyclic model where every record has a latent parent, a topological order starts at a parentless latent whose factor depends only on parameters and structural data (MT Proposition 12.20). The proof is one line of graph theory and the result an established negative: within one finite acyclic joint the top cannot be constituted by the agents it organizes. Same-time reciprocal structure among latents is admissible only as a globally normalized undirected model with its partition function (MT Section 4.2, MT Proposition 4.6); an energy appended without its normalizer is not an evidence bound.

The first locus is the inference loop (MT Remark 12.21): mean-field coordinate updates of parent and child read each other [9, Sec. 10.1.1], and at a fixed point a child's effective prior is formed from the population's beliefs through a latent inside the joint. This is mean-field self-consistency, as when each spin of a mean-field magnet sees a field set by its neighbors, whose fields include its own; it is a schedule against one fixed joint, not participation in time. The second is learning the top prior (MT Proposition 12.22): with independent episodes sharing a top-prior parameter, the optimum over an unrestricted family is the aggregate recognition law, the variational form of empirical Bayes [63] and the aggregate-posterior prior of Hoffman and Johnson [38] and Tomczak and Welling [70].

The third is delayed feedback. In the latent route (MT Definition 12.23) one joint spans episodes, and the parent read out of episode n parameterizes episode $n+1$, so

$$\mathbb{P}^{\mathrm{ep}}\big(\mathbf{Y}^{(n+1)} \in \cdot \mid o_{0:n}\big) = \int \mathbb{P}^{(n+1\leftarrow n)}(\cdot \mid z, X)\mathbb{P}^{\mathrm{ep}}\big(Z^{(n)}_{\mathrm{par}} \in dz \mid o_{0:n}\big)$$

(MT Proposition 12.24): the children's next prior is the filtered parent pushed through the transition. The joint is acyclic, so it still needs a declared root at episode 0. In the parameter route each episode has its own joint and the parameter is updated by a declared map of the selected recognition coupling. MT Proposition 12.25 shows that such a route is one joint: make the parameters and latent record copies $\widetilde{o}_n$ latent states with Dirac transitions and evaluate the update at the copy, not at the recognition law being scored. The lifted joint's evidence is the product of the episode evidences, its posterior the product of episode posteriors, and

$$\mathcal{F}^{\mathrm{lift}}[\mathbb{Q}] = \sum_{n=0}^{n_1}\Big[-\log p_{\theta_n,V}(o_n \mid X) + D_{\mathrm{KL}}\big(\mathbb{Q}^y_n \| \mathbf{\Pi}_{\theta_n,V,o_n,X}\big)\Big] + \mathrm{TC}(\mathbb{Q}^y),$$

the sum of the episode bounds plus a cross-episode total correlation that vanishes at product laws. In physics terms this is state augmentation: a learning loop becomes filtering once the parameters are latent. The point is typing: no generative factor reads the recognition law being scored, so the typing prohibition of chapter 4 (MT Hypothesis 4.3) holds.

12.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Definition 12.1	One generative and one recognition law over the whole tower	CHOICE	The modeling commitment; the rest is accounting
MT Definition 12.3, MT Definition 12.14	One structural channel for three laws; blocking reads states, not laws; soft membership	CHOICE	Covers Kadanoff, random, adaptive, and soft blocks
MT Lemma 12.5	Adaptive towers factor top-down; downward factors are reversal kernels	SOLID	Standard in substance

Result	What it says	Status	Scope and caveats
MT Proposition 12.6	Dropping the graph datum costs a sum of mutual informations	SOLID	Chain rule; $\log 2$ witness illustrative
MT Theorem 12.7	Five-term decomposition of the tower free energy	SOLID	Standard in substance; depth-Markov form
MT Proposition 12.9	Gibbs rows minimize row free energies	SOLID	Gibbs variational principle; $\tau \neq 1$ not a sector of one bound
MT Definition 12.10	Appended energies give a descent potential	CHOICE	No evidence interpretation, as stated
MT Proposition 12.13	The two normalizations differ by branch partition functions	SOLID	Finite catalog and carriers
MT Proposition 12.15, MT Definition 12.16	Event pushforward under soft membership; composition	SOLID	Product form needs independent draws
MT Proposition 12.18	Coarse links keep all holonomy; trees belong to the channel	SOLID	Hard partitions only
MT Proposition 12.19, general part	No parent dependence, no cross-derivative; normalization gives no coupling	SOLID	Declared top-down hierarchy
MT Proposition 12.19, criterion	Bidirectional coupling iff odds ratio $\neq 1$	TOY	Binary parent and child; no family theorem
MT Proposition 12.20	A finite acyclic tower has a declared root	SOLID	Negative; every record needs a latent parent
MT Remark 12.21	Parent and child updates read each other	SOLID	Standard
MT Proposition 12.22	Optimal top prior is the aggregate recognition law	SOLID	Empirical Bayes in substance
MT Proposition 12.24, MT Proposition 12.25	Delayed feedback is the free energy of one joint	SOLID	Declared measurable selection and update
MT Section 12.7.2, read as participation	The loci realize what a cycle was reaching for	VAGUE	Interpretation; each identity proved
MT Open Problem 12.27, MT Open Problem 12.26	Persistent hierarchy; feedback by labels alone	OPEN	See section 12.6

Toy content is confined to the binary odds-ratio criterion and the four-vertex $\log 2$ witness; only the first stands in for a general claim, which the chapter explicitly does not make.

Conditions: which ones matter

Four conditions matter. The depth-Markov form is automatic for adaptive towers but must be built into a declared hierarchy. The structural channel must not read a law, which excludes posterior-dependent blocking. Coarse links are constructed for hard partitions only. And no generative factor may read the recognition law being scored, which is why the lifted joint evaluates its update at a latent record copy. A fifth concerns reading, not truth: energy-unit row free energies with $\tau \neq 1$ and the composite potential are not evidence bounds. The rest is hygiene: standard Borel spaces and regular conditional versions, common domination and integrability, extended-real conventions, joint measurability of selector and update, and discrete records for the displayed episode decomposition.

12.6 Open ends

MT Open Problem 12.27 asks for the hierarchy the program is after: a normalized partition law and parent–child kernels at c_* such that descent of the tower free energy produces persistent nested blocks, with pushforwards and link lifts that compose, a controlled closure residual, and non-flat holonomy kept rather than forced flat. What holds is the exact accounting, the fast–slow typing of chapter 8, nested formation under a supplied hierarchy of interaction strengths (MT Theorem 13.17), and one endogenous level on the four-cycle (MT Proposition 13.22). Unproved is that descent produces a second level that depends on the first. The problem is posed for models with a partition catalog inside both laws, recognition coordinates descending by reversible diffusion at temperature T with rate-one mark redraws, and T-independent data, as $\mathrm{T} \to 0$. A positive answer needs three results: a two-level lineage formed with probability bounded below whose second level is recursive in the manuscript's strict sense (it depends on the first by a margin over counterfactual handoffs and is not a supplied-scale sequence) and is a coherence episode (held by a restoring interaction against thermal noise, on horizons short compared with its stationary recurrence or with probability exceeding its stationary value by a margin); an intrinsic partition selector; and a separated slow manifold on which coarser variables feed back consistently into finer kernels. A negative answer is a proof that no second level produced by descent in this class meets these conditions. The recursion and coherence rules are decoded in chapter 13. The limit is zero temperature on networks of any finite size, not $N \to \infty$.

MT Open Problem 12.26 asks whether delayed feedback can be carried by the labels alone. In a label-sufficient latent route the episode model is that of a parameter route with the aggregate update $\theta_{n+1} = (1-\epsilon_{\mathrm{e}})\theta_n + \epsilon_{\mathrm{e}}\mathbf{\Pi}^{\mathrm{lab}}_{\theta_n,V,o_n,X}$, and the parent read-out sees only the label and a preference coordinate. Does some such route's filtered label law follow the aggregate update to $o(\epsilon_{\mathrm{e}})$ per episode, uniformly over histories and horizons of order $1/\epsilon_{\mathrm{e}}$, or does every route miss by order ϵ_{e}? The records are restricted either to include an uninformative value while moving the label posterior by an amount bounded below on average, or to have generic likelihood vectors (no

nonzero polynomial vanishes on their cone). In physics terms: can a stochastic-approximation learning rule be represented, to first order in its step, as filtering in a fixed latent model whose memory is a label and one preference coordinate? No natural partition selector is known either (MT Open Problem 9.19), and parent dynamics (section 11.6) is what would make the static and episodic senses of parents acting back dynamical.

Does it deliver what it claims? DELIVERED

The chapter says it places adaptive event laws and links, random partitions, and the influence of parents on their children inside one variational free energy, factors that free energy for adaptive towers, and closes the tower with a declared root, with participation at three loci. It does exactly that: one Gibbs variational functional for the tower, automatic top-down factorization, an identity for the five-term decomposition, an exact cost for dropping the graph datum, the two partition normalizations distinguished exactly, coarse links that keep all holonomy, a proof that a declared root is unavoidable, and a proof that delayed feedback with a declared update is the free energy of one joint. Two remarks keep the verdict honest. What is delivered is accounting and typing, largely standard in substance (chain rule, Markov-chain reversal, Gibbs variational principle, mean-field coordinate ascent, empirical Bayes, state augmentation); nothing is proved about what descent of this free energy does, and the hierarchy it would have to produce is MT Open Problem 12.27. And "the influence of parents on their children" is delivered as a typed possibility: in a declared hierarchy the parent enters the children's prior by construction, bidirectional coupling is proved only as a binary criterion, and in adaptive towers the downward kernel is a Bayes reversal the chapter declines to read causally. "Closing the tower" means closing it by declaration; the chapter proves that a finite acyclic tower cannot close itself.

Chapter 13

Formation Witnesses

This chapter of the companion mirrors MT Chapter 13, with the proofs in MT Appendix C. It is where the program's central word, formation, receives theorems, and where the theorems most need to be read against their hypotheses: the headline result forms collectives out of a hierarchy the model is given, not one it finds. The technical chapter's scope paragraph declines each stronger reading; the opening paragraph of the introduction summarizes the chapter more strongly than its theorems support.

13.1 The question and the answer

Earlier chapters establish that an exact parent exists: coarse-graining the agent network by a Markov channel gives a normalized parent law that preserves the evidence and splits the free energy exactly (MT Theorem 9.6; see chapter 11). They also establish that this parent is in general not an autonomous Markov process (MT Theorem 11.39), and how a parent's own dynamics is fixed by the network it summarizes is the defining open problem of the phase (MT Open Problem 11.38). The formation chapter asks something more concrete. In one fully normalized experiment at the base point, can coherent groups of agents form one after another while each parent is started from the complete law of the level below, keeps every loop coordinate and every attention label, and learns its own copy of a model parameter from its own law? Three statements answer, and they must be read together.

MT Theorem 13.17 says yes for a chain of 2^{L-1} triangles at every finite depth $L \geq 2$: the number of coherent components passes through $3 \cdot 2^{L-1} \to 2^{L-1} \to \cdots \to 1$ with probability tending to one, and every error made by replacing a level with its parent vanishes, separately for the free energy, the learning signal, the attention history, and the continuous path. The succession is put in by hand. The interaction strengths are supplied as $\mathrm{J}_\ell = \epsilon^{-\mathrm{Q}_\ell}$ with a separation doubly exponential in the depth, the initial state is a prepared displacement of a common mean, the learned parameter is pinned by a calibration record, and the attention record is saturated. The chapter says so: formation in this family is hierarchical relaxation of the prepared mean, and attention and learning are perturbations the proof controls, not causes (MT Remark 13.18). What the theorem establishes is that exact marked parents can be carried through such a relaxation without their errors accumulating, at every finite depth, with constants that depend on the depth in an uncontrolled way.

MT Proposition 13.22 removes the supplied hierarchy. On a homogeneous cycle of four agents, where an edge carries energy only while one of its endpoints attends to it, a pair forms by a feedback between attention and belief relaxation, with probability at least $3e^{-1/4}/512 \approx 4.6 \times 10^{-3}$ uniformly in the small parameter, and an exact parent follows the episode for one unit of time. The pairing is seeded by the random initial attention assignment, the law of the process stays symmetric, there is one level only, and nothing is learned. MT Proposition 13.24 shows where this mechanism stops: in the whole gated family on any finite graph, the coherent blocks of the stationary law are, up to an error vanishing with the temperature, the components of a minimum edge cover, a forest of stars, and every further merge costs a factor of the temperature.

The chapter therefore proves that exact parents can track a supplied hierarchy, that one endogenous level can form on a small graph, and that in the family containing that mechanism the stationary law stops at one level. It does not prove that collectives form through the agents' coupled inference in more than one level, the program aim of MT Section 1.1; that is MT Open Problem 13.25, and whether learning or loop holonomy can decide which components form is MT Open Problem 13.20. The seeded contrast formation from every Gaussian initial law, the pairing-selection theorem, and the parent-activation continuation results belong to the interpretation chapter of the technical manuscript and are treated in chapter 14.

13.2 The picture

The triangle chain: springs of very different stiffness. Give each agent a belief mean $\bar{k}_i$ and a model mean $\bar{m}_i$, and each oriented edge $e : j \to i$ a translation ω_e, the link, which shifts the sender's frame into the receiver's. The mismatch on an edge is $r_e = \bar{k}_i - \bar{k}_j - \omega_e$, and the energy is a sum of squared mismatches weighted by spring constants. Inside each triangle the springs, and the spring tying each agent's belief to its own model, have strength J_1. The bridge between triangles j and $j+1$ has strength $\mathrm{J}_{\ell(j)}$ with $\ell(j) = 2 + v_2(j)$, where $2^{v_2(j)}$ is the largest power of two dividing j; with four triangles ($L = 3$, twelve agents) the three bridges have levels two, three, and two. The bridges thus sit on the levels of a binary tree, like the couplings of Dyson's hierarchical model [27], a resemblance that is structural only. The strengths are separated by enormous factors, $\mathrm{J}_1 \gg \mathrm{J}_2 \gg \cdots \gg \mathrm{J}_L = 1$.

The system starts displaced, every mismatch of order A, with thermal fluctuations of size at most $\sqrt{\mathrm{T}} = \epsilon^3 \mathrm{A}$, and relaxes by overdamped Langevin dynamics. The stiffest springs relax first, on times of order $1/\mathrm{J}_1$ up to a logarithm, and the triangles become internally coherent; on times of order $1/\mathrm{J}_2$ the level-two bridges relax and triangles pair up; and so on, until on times of order one the chain is a single component. In the twelve-agent example the counts are $12 \to 4 \to 2 \to 1$. An edge is coherent when its conditional root-mean-square mismatch is below $\eta_L \mathrm{A}$, a small fixed fraction of the initial displacement, so formation here means that a group's internal relative displacement has decayed below a resolution set by the preparation. The mean dynamics is linear and, relative to the displacement, deterministic up to corrections of order ϵ^3, and the grouping is fixed entirely by which

springs are strong. A parent is the law with the already relaxed distortions integrated out; it keeps the loops, the gauge directions, and the complete attention assignment.

Physics analogue (exact)

Linear response after a field is switched off, with adiabatic elimination of the fast modes. The reference preparation of MT Definition 13.13, a Gaussian with mean $\mathrm{A}\mathbf{v}_0$ and covariance $\mathrm{T}H^{-1}$ on the non-gauge coordinates, is exactly the Gibbs state of $\frac{1}{2}\mathbf{x}^\top H\mathbf{x} - \mathbf{h}^\top\mathbf{x}$ in the static field $\mathbf{h} = \mathrm{A}H\mathbf{v}_0$, and the chain's dynamics is the Ornstein–Uhlenbeck relaxation after the field is switched off at time zero. Integrating out relaxed modes is exact Gaussian integration, with the Schur complement H_ℓ as retained stiffness. The identification is a theorem for the Gaussian core, where the record is off and the parameter frozen and the mean follows $\bar{\mathbf{x}}(t) = e^{-H_{\ell'}t}\bar{\mathbf{x}}(0)$ (MT Proposition 13.15); in the full theorem the record and learning add perturbations that the proof bounds.

Why the regime is so extreme. The reason, informally, is that two requirements pull against each other. Coherence must be many thermal widths wide, so $\mathrm{J}_\ell\mathrm{A}^2/\mathrm{T}$ must diverge. But a parent integrates out fast modes while slower modes are still moving, and the fast modes lag behind the position the slow modes dictate. The parent's error is that lag measured in thermal units, and at low temperature a small lag is many thermal widths. In the Gaussian core the loss is exactly of order $(\mathrm{A}^2/\mathrm{T})(\mathrm{J}_{\ell+1}^4/\mathrm{J}_\ell^3)$ (MT (13.37)); keeping it small while A^2/T is large forces each level's stiffness to beat the next by a large power, and iterating over L levels gives a separation doubly exponential in L. Separately, the regime makes the network's stored energy, of order $\mathrm{A}^2\mathrm{J}_1$, negligible against the unit-scale calibration term. That is what makes the amplitude tiny, and it is what makes learning blind to the network.

The four-cycle: attention feedback and a frustrated loop. Four agents sit on a ring, each receiver attending one of its two neighbors. The prepared state puts every mean at zero and a unit translation on every link, so every edge has mismatch -1 and the links carry a circulation of four. If the assignment is a perfect matching, only the two matched edges carry energy: they contract, while the unattended edges absorb the circulation and stretch. Each receiver's partner edge soon becomes much cheaper than its exterior edge, and at $\mathrm{T} = \epsilon^8$ the attention posterior then keeps the partner. Attention caused the contraction and the contraction justifies the attention. Nothing in the energies prefers one pair; the initial assignment, drawn uniformly, decides, and the law stays invariant under the symmetries of the ring at every time.

Physics analogue (structural)

A frustrated plaquette, with symmetry broken along single trajectories. The flux around the loop forbids all four bonds from being satisfied at once, so the strain must sit somewhere, and attention decides where. The broken symmetry is of the finite-volume kind: each sample path picks a pair and holds it over the episode while the ensemble law stays symmetric, as a finite Ising magnet below its ordering temperature has a symmetric Gibbs state although single

trajectories sit in one well. No thermodynamic limit is taken; law-level selection is the subject of chapter 14.

Why the gated mechanism stops at one level. On a fixed attention assignment, a sheet, the energy stiffens exactly the attended residuals; every other direction, including any rigid translation of one block relative to another, is held only by the reference prior, with stiffness of order T in energy units. All sheets share the minimum energy zero, so a sheet's stationary weight is its phase-space volume, of order $\mathrm{T}^{\mathrm{k}_S/2}$ with k_S the number of stiff directions. The law prefers the sheets with the fewest stiff directions, and each attended edge joining two blocks adds two of them and costs a factor T. Blocks float relative to each other and nothing pulls them together.

Physics analogue (structural)

Order by disorder. Among degenerate ground states, harmonic fluctuations favor those with the most soft modes, because each stiff mode contributes a factor proportional to $\sqrt{\mathrm{T}}$ to the partition function; this is how fluctuations lift a ground-state degeneracy in frustrated magnets. Here the degenerate states are the attention sheets and the selection favors fragmentation. The counting is the same; the degenerate set is a finite family of label sectors rather than a continuous manifold.

13.3 Objects and notation

The technical chapter uses a local notation table (MT Table 13.1). The decoder covers what a first reading needs.

Term or symbol	Plain meaning	Physics analogue
Configuration $\mathbf{x} = (\bar{k}, \bar{m}, \omega)$	Belief means, model means, link translations; agent laws are Gaussian with fixed variance, so only means move	Lattice fields: two matter fields on sites, links on bonds
Link ω_e, group $(\mathbb{R}, +)$	Independent edge variable shifting the sender's frame into the receiver's	Noncompact abelian gauge link
Residual r_e, cost $\frac{1}{2} r_e^2$	Edge mismatch; the cost is a relative entropy between unit Gaussians	Covariant difference; bond energy
Loop $h_q = \mathfrak{h}_q/\sqrt{3}$	Normalized total translation $\mathfrak{h}_q$ around a triangle	Plaquette flux
$\mathcal{U}, \mathcal{H}, \mathcal{K}$	Normal (gauge-invariant, flux-free) distortions; loops; pure-gauge directions	Massive modes; flux sector; gauge orbit
Stiffness H; strengths $\mathrm{J}_\ell = \epsilon^{-\mathrm{Q}_\ell}$	Quadratic form of the energy; supplied spring constants, one per level	Dynamical matrix; coupling hierarchy
A, $\mathbf{v}_0$, margin m_L	Size and direction of the initial displacement; unrelaxed bridges stay at least $\mathrm{m}_L \mathrm{A}/2$ from zero	Applied field switched off at time zero

Term or symbol	Plain meaning	Physics analogue
Calibration record $o_{\mathrm{cal}} = 2$	Precise observation of the learned parameter itself	External clamp
Learned scalar ϑ	Global multiplier of the stiffness, ϑH, learned by natural gradient	Slowly ramped coupling
Source, assignment S, sheet	Which neighbor each receiver attends; one fixed assignment and its slice of state space	Label field; sector
Source record E_i, ϕ_S, B_0	Bounded observation of the attended edge's mismatch, total effect at most $B_0 = 10^{-4}$	Weak bounded potential
$\mathbf{\Pi}$, $q^\star$, q_t	Physical posterior; best law of the restricted family; evolving recognition law	Gibbs state; variational optimum; time-dependent density
Marked channel $C_{\ell\leftarrow\ell'}$, parent	Keep retained coordinates and every label, integrate out the rest	Exact decimation
Schur complement H_ℓ	Stiffness of retained coordinates after minimizing over discarded ones	Tree-level effective stiffness
Discarded loss $\Delta_{\ell':\ell}$	Mean relative entropy of the discarded coordinates' actual law to their equilibrium law, given the retained ones	Adiabatic-elimination error
Window I_ℓ, activation, k_L	Interval on which level ℓ is coherent; start of the next parent; how many relaxation times to wait	Timing of decimations
Coherent edge, component	Conditional root-mean-square residual at most $\eta_L \mathrm{A}$; cluster of such edges	Cluster at a resolution

Some traps. T is the temperature, not the frame comparison T_{ij}. J_ℓ is a spring constant while calligraphic $\mathcal{J}$ is the set of assignments, and a source is the neighbor a receiver attends, not a source term in a field theory. H is a stiffness matrix, not a Hamiltonian or a holonomy. The root is the set of gauge directions, not the root of a tree. A parent is a marginal law, not an agent: the scope paragraph declines that parents are coarse agents with their own belief and model fibers. Learning is the deterministic drift of one scalar. Coherence is measured against the preparation amplitude A, not the thermal width. Complete, exact, marked, and retained are emphasis on objects that keep all their information; declared marks a modeling choice, as in the declared parent kinetics. NOT-CLAIMED marks a declined claim, and declining is not refuting.

13.4 The formulas, explained

13.4.1 The carrier

The chain of MT Definition 13.1 has $N = 3 \cdot 2^{L-1}$ agents and configuration space $\mathbb{R}^{10 \cdot 2^{L-1}-1}$. By MT Lemma 13.2,

$$\mathbf{x}^\top H \mathbf{x} = \mathrm{J}_1 \sum_i \mathfrak{d}_i(\mathbf{x})^2 + \sum_e \mathrm{J}_{\ell(e)} r_e(\mathbf{x})^2, \qquad |r_q|^2 = |O_\triangle r_q|^2 + h_q^2.$$

The first sum ties each belief to its own model through the discrepancy $\mathfrak{d}_i = \bar{k}_i - \bar{m}_i$, with the strongest spring; the second weights each edge mismatch by its level. A passive frame change $\bar{k}_i \mapsto \bar{k}_i + g_i$, $\bar{m}_i \mapsto \bar{m}_i + g_i$, $\omega_e \mapsto \omega_e + g_i - g_j$ leaves every r_e and $\mathfrak{d}_i$ unchanged, the translation-group case of the frame invariance proved in general (MT Corollary 4.10). The second identity splits a triangle's squared mismatches into two gauge-invariant contrasts and the flux h_q^2: the vertex means telescope around the triangle, so the mismatches sum to $-\mathfrak{h}_q$ whatever the means are, and a triangle with flux cannot be fully satisfied. Space splits orthogonally into $\mathcal{U} \oplus \mathcal{H} \oplus \mathcal{K}$, and H vanishes exactly on $\mathcal{K}$.

Physics analogue (exact)

The quadratic form is the action of a Gaussian, noncompact lattice abelian gauge-Higgs model in Stückelberg form, in the lattice formulation of Wilson [81] but with no plaquette term: flux costs energy only through the matter coupling, and the model channel is a second matter field tied on site to the first. The group is $(\mathbb{R}, +)$, abelian and one-dimensional, so nothing about nonabelian or compact groups can be read off.

13.4.2 One normalized experiment and its three laws

MT Definition 13.3 makes the carrier a probability model. Each agent has a model sample $m_i \sim \mathcal{N}(\bar{m}_i, \sigma_{\mathrm{ev}}^2)$ and, given it, a belief sample $k_i \sim \mathcal{N}(m_i, 1 - \sigma_{\mathrm{ev}}^2)$. The recognition family fixes the coupling of the two samples to its optimal form, at the price $\mathfrak{d}_i^2/(2(1-\sigma_{\mathrm{ev}}^2))$ per agent (MT Proposition 13.4), summing to $\frac{1}{2}\mathbf{x}^\top R_{\mathrm{ev}} \mathbf{x}$. The configuration prior is Gaussian with precision $\Omega_0(\vartheta)/\mathrm{T}$, where

$$\Omega_0(\vartheta) = \vartheta H + P_{\mathcal{K}} - \mathrm{T} R_{\mathrm{ev}}.$$

The first term is the physics, a stiffness multiplied by the learned ϑ; the second normalizes the gauge directions; the third cancels the restriction price so that the optimizer has a temperature-independent energy. The last two are bookkeeping.

Three observations complete the joint. A root observation gives the gauge coordinates an independent Gaussian law, a form of gauge fixing. A calibration record $o_{\mathrm{cal}} = 2$ from $\mathcal{N}(\vartheta, \mathrm{T})$ observes the learned parameter itself; it is what the parameter will learn. And each receiver has a binary record reading the edge it attends,

$$\Pr(E_i = 1 \mid \mathbf{x}, S_i = j) = \tfrac{1}{2} \exp\Big[-\delta \chi\Big(\sqrt{\mathrm{J}_1/\mathrm{T}}\, r_{ij}(\mathbf{x})\Big)\Big], \qquad \chi(r) = 1 - e^{-r^2/2}, \qquad \delta = 10^{-4}/N.$$

The argument of χ is the mismatch in units of the thermal width $\sqrt{\mathrm{T}/\mathrm{J}_1}$, and χ saturates at one. Summed over receivers, the configuration-dependent part ϕ_S of the record's negative log-likelihood lies in $[0, 10^{-4}]$, so attention differs from its prior by a factor of at most $e^{\pm 10^{-4}}$ (MT (13.47)): attention is blind to the configuration by construction. The attention prior is uniform except at one receiver i_0, whose prior is $\mathrm{sig}(4(\vartheta - 3/2))$; the learned parameter controls one attention weight, through the prior only.

Three laws coexist (MT Definition 13.5; MT Proposition 13.6): the physical posterior $\mathbf{\Pi}$, the restricted optimizer $q^\star$, and the evolving recognition law q_t. The scaled free energy is $\mathcal{F}_\mathrm{T}(q, \vartheta) = \mathrm{T} D_\mathrm{KL}(q \| q^\star_\vartheta) + \mathrm{C}_\mathrm{T}(\vartheta)$, and the optimizer differs from the posterior by at most $2e^{B_0} N \mathrm{T}^2/((1-\sigma_\mathrm{ev}^2)\mathrm{J}_1)$ in free energy (MT (13.16)), negligible here. The theorem is proved for both $\mathbf{\Pi}$ and $q^\star$.

13.4.3 Marked parents and the discarded loss

The parent at level ℓ comes from the deterministic channel $C_{\ell \leftarrow \ell'}((y, S), \cdot) = \delta_{(P_\ell y, S)}$, which projects onto the retained space W_ℓ (unrelaxed normal directions, all loops, all gauge directions), keeps the whole assignment, and leaves the observations alone (MT Definition 13.7). The parent law is a marginal, with sheet potential

$$V_{\ell,S}(y, \vartheta) = \tfrac{1}{2}\vartheta y^\top H_\ell y + \mathrm{C}^\mathrm{G}_\ell(\vartheta) + \mathrm{T}\psi_{\ell,S}(y, \vartheta), \qquad 0 \le \psi_{\ell,S} \le B_0,$$

by MT Proposition 13.8: the Schur-complement stiffness, the Gaussian determinant of the integrated modes (their entropy, which depends on ϑ), and the record averaged over the discarded coordinates, still below 10^{-4}. The chain rule of relative entropy gives what a parent loses,

$$\mathcal{F}^{\ell'}_\mathrm{T}(q^{\ell'}, \vartheta) - \mathcal{F}^{\ell}_\mathrm{T}\big((C_{\ell \leftarrow \ell'})_\# q^{\ell'}, \vartheta\big) = \mathrm{T}\Delta_{\ell':\ell}(q^{\ell'}, \vartheta),$$

where $\Delta_{\ell':\ell}$ is the mean relative entropy of the discarded coordinates' actual conditional law to the optimizer's conditional. It vanishes exactly when the discarded coordinates are in conditional equilibrium, the condition for exact adiabatic elimination. These identities are standard in substance.

The parent is not a coarse agent. At the top level the retained space has dimension $4 \cdot 2^{L-1}$: all N gauge directions and all $N/3$ loops, with every normal direction integrated out (MT (13.5)), alongside the full assignment with 2^N values. For $L = 2$ the carriers go $\mathbb{R}^{19} \to \mathbb{R}^9 \to \mathbb{R}^8$ with 64 assignments throughout, and the hidden conditionals are not Gaussian once the record is on (MT Remark 13.19). The top parent is the zero-mode and flux sector of the Gaussian theory together with the complete label field.

13.4.4 Kinetics and learning

The microscopic kinetics (MT Definition 13.9) are overdamped Langevin dynamics on each sheet with a rate-one clock that redraws the whole assignment from the attention posterior,

$$\mathscr{L}^0_\vartheta \mathsf{f}(\mathbf{x}, S) = \mathrm{T}\Delta_\mathbf{x}\mathsf{f} - \nabla_\mathbf{x} U_{\vartheta,S} \cdot \nabla_\mathbf{x}\mathsf{f} + \sum_{S'} \beta_{S'}(\mathbf{x}, \vartheta)\big[\mathsf{f}(\mathbf{x}, S') - \mathsf{f}(\mathbf{x}, S)\big],$$

a diffusion at temperature T with unit mobility plus a heat-bath update of the labels. The parent generator has the same form with the parent potential and inherited jump rows, conditional averages of the fine rows, which are reversible and compose across levels (MT Proposition 13.10). The parent kinetics is declared, not derived: the time evolution of a true marginal carries memory (MT Theorem 11.39; compare Zwanzig [86]), and the chapter checks its Markov closure afterwards through conclusions (c) and (e).

Each level learns its own copy of ϑ by natural gradient,

$$\dot{\vartheta}_\ell = -\epsilon \mathsf{m}(\vartheta_\ell)^{-1} g_\ell(q_t^\ell, \vartheta_\ell), \qquad \mathsf{M}(\vartheta) = \frac{1}{\mathrm{T}} + \frac{1}{2}\operatorname{Tr}\big[(\Omega_0(\vartheta)^{-1}H)^2\big] + 16\pi_{i_0,+}(1-\pi_{i_0,+}),$$

with $\mathsf{m} = \mathrm{T}\mathsf{M}$ (MT Definition 13.11; MT Proposition 13.12). The metric is dominated by the calibration term, so $\mathsf{m} \approx 1$, and the covector g_ℓ (MT (13.27)) is $\vartheta - 2$ plus the stored elastic energy $\frac{1}{2}\mathbb{E} y^\top H_\ell y$, at most of order $\mathrm{A}^2\mathrm{J}_1$, plus entropic and attention terms of order T. Hence $\vartheta(t) \approx 2 - e^{-\epsilon t}$. Along the coupled dynamics the free energy decreases through three nonnegative dissipation terms, the Fisher information of the diffusion, the symmetrized entropy production of the label jumps, and the natural-gradient term $\epsilon \mathsf{m}^{-1} g_\ell^2$ (MT (13.28)): an H-theorem, standard in substance.

13.4.5 The regime

The regime of MT Definition 13.13 is

$$\mathrm{Q}_L = 0, \quad \mathrm{Q}_\ell = 2\mathrm{Q}_{\ell+1} + 20, \quad \mathrm{J}_\ell = \epsilon^{-\mathrm{Q}_\ell}, \quad \mathrm{A}^2 = \epsilon^{3\mathrm{Q}_1+20}, \quad \mathrm{T} = \mathrm{A}^2\epsilon^6, \quad t_* = \frac{2}{\epsilon}.$$

Then $\mathrm{Q}_1 = 20(2^{L-1}-1)$ and $\mathrm{T} = \epsilon^{60\cdot 2^{L-1}-34}$ (MT Remark 13.14). Since $N = 3\cdot 2^{L-1}$, this is $\mathrm{T} = \epsilon^{20N-34}$: the temperature exponent is linear in the number of agents, and the stiffness range is $\epsilon^{-20(N/3-1)}$. The last two columns of the table are arithmetic on the class of schedules in which the sharpness problem is posed (section 13.4.7): the least admissible Q_1 there, rounded, and its temperature.

L	N	Q_1	A	T	least Q_1, base $4/3$	its T
2	6	20	ϵ^{40}	ϵ^{86}	14	ϵ^{68}
3	12	60	ϵ^{100}	ϵ^{206}	25.3	ϵ^{102}
4	24	140	ϵ^{220}	ϵ^{446}	40.4	ϵ^{147}
8	384	2540	ϵ^{3820}	ϵ^{7646}	171	ϵ^{539}
9	768	5100	ϵ^{7660}	ϵ^{15326}	235	ϵ^{730}

The ratios the proof actually uses are the stiffness range $\epsilon^{-\mathrm{Q}_1}$, the noise-to-signal ratio $\mathrm{T}/\mathrm{A}^2 = \epsilon^6$, the learning mobility ϵ, and the stored energy relative to calibration, $\mathrm{A}^2\mathrm{J}_1 = \epsilon^{2\mathrm{Q}_1+20}$. The window of level ℓ is $I_\ell = [t_\ell, 2t_\ell]$ with $t_\ell = k_L \log(1/\epsilon)/\mathrm{J}_\ell$; the last closes at order $k_L \log(1/\epsilon)$, long before the horizon $2/\epsilon$ on which the parameter moves. The entrance constant must exceed thresholds proportional to Q_1 (MT (13.38)), and ϵ_L is not computed. The regime is a formal asymptotic regime at each fixed depth, with no limit in which N grows at fixed ϵ.

13.4.6 The theorem, conclusion by conclusion

MT Theorem 13.17 assumes the experiment and kinetics at a fixed depth (H1), the regime (H2), an initial law in the preparation neighborhood (H3), and scheduled activations (H4). The preparation neighborhood consists of smooth transports of the Gaussian reference that move each sample by less than $\epsilon\sqrt{\mathrm{T}}$ in the energy norm (MT (13.31)): a thermalized cloud around the displaced mean, slightly deformed, possibly depending on ϵ and L. The reader's phrase "coherence is linear hierarchical relaxation of the prepared mean" has a precise meaning here. Up to perturbations the proof bounds, the mean obeys $\dot{\bar{\mathbf{x}}} = -\vartheta H\bar{\mathbf{x}}$ from $\mathrm{A}\mathbf{v}_0$. In the flag coordinates of MT (C.2), layer ℓ has stiffness of order J_ℓ and couples to the slower layers only through strengths of order $\mathrm{J}_{\ell+1}$, so the solution decays level by level. On I_ℓ the layers up to ℓ have decayed below $\eta_L\mathrm{A}$, while the higher bridges have moved by at most of order $\mathrm{J}_{\ell+1}t_\ell\mathrm{A} \to 0$ and keep their prepared margin.

Conclusion (a) gives unique non-exploding solutions for every process, and learned parameters in $[1, 21/10]$ that stay within $\mathrm{C}_L\epsilon^{2\mathrm{Q}_1+20}$ of the calibration curve $2 - e^{-\epsilon t}$ (MT (13.39)): learning is relaxation to the calibrated value, the same at every level because no level can see the network.

Conclusion (b) is the formation statement. Except on an event of probability at most $\mathrm{C}_L(1 + t_*)\exp(-\mathrm{c}_L/\epsilon^6)$, the fine process and its parents show the counts $3 \cdot 2^{L-1} \to 2^{L-1} \to \cdots \to 1$ (MT (13.41)): on I_ℓ every edge of level at most ℓ is coherent, every belief agrees with its model to within $\eta_L\mathrm{A}$, and every higher bridge stays a fixed multiple of A from zero; the final component persists to the horizon, and the loops stay random and retained. The exponent $\epsilon^{-6} = \mathrm{A}^2/\mathrm{T}$ is the signal-to-noise ratio. By MT Remark 13.18 the coherence proof uses the record only through $0 \le \phi_S \le B_0$ and the parameter only through $1 \le \vartheta \le 21/10$, so it gives the same components with the record off and the parameter frozen at one. Attention and learning play no role in (b).

Conclusion (c) bounds the free energy discarded at each activation, $\Delta_{\ell-1:\ell} \le \mathrm{C}_L\epsilon^{14}$ early and $\mathrm{C}_L\epsilon^2$ at the last level, and the difference of consecutive learning signals by $\mathrm{C}_L\mathrm{T}\epsilon^7$ and $\mathrm{C}_L\mathrm{T}\epsilon$ (MT (13.42)): the parents are faithful in free energy and in learning gradient. The exponent 14 is $\mathrm{A}^2\mathrm{J}_{\ell+1}^2/(\mathrm{T}\mathrm{J}_\ell) = \epsilon^{-6}\epsilon^{20}$, the general Fisher-budget estimate; the Gaussian core shows the true early loss is much smaller.

Conclusion (d) shows the model is not static: from every activation to the horizon the distinguished receiver's mass on source $+$ rises by more than one half, and the first hidden conditionals change by relative entropy more than $1/200$ (MT (13.43)). Both are driven by ϑ moving toward two, through the prior $\mathrm{sig}(4(\vartheta-3/2))$ and the stiffness ϑH, and both happen after formation, since $\vartheta \le 1+\mathrm{C}_L\epsilon\log(1/\epsilon)$ until the last window closes. This is an engineered stress test showing that the lineage tracks kernels that really move, not a mechanism of formation.

Conclusion (e) says that after each activation the retained coordinates of the fine process and of the parent stay within $\mathrm{C}_L\mathrm{A}\epsilon^2$ of one deterministic path except with probability $\mathrm{C}_L(1 + t_*)e^{-\mathrm{c}_L/\epsilon^2}$ (MT (13.44)), and that their attention histories can be coupled to disagree with probability at most $\mathrm{C}_L\epsilon^6$ (MT (13.45)). The resolution $\mathrm{A}\epsilon^2 = \sqrt{\mathrm{T}}/\epsilon$ lies between the thermal width and the displacement. This small-noise statement is weaker than closeness of path laws, which the chapter declines along

with exact lumpability: the parent reproduces the fine trajectory at a resolution coarser than the fluctuations it carries.

MT Remark 13.18 fixes the reading of the whole theorem. The record is saturated, since at a residual of order A its argument is of order $\epsilon^{-(3+\mathrm{Q}_1/2)}$, so it cannot tell a coherent edge from an incoherent one at the coherence resolution; learning is calibration; and the loops have root-mean-square size of order $\sqrt{\mathrm{T}/\mathrm{J}_1}$, so holonomy is thermal noise. The chapter's own summary says the reductions succeed in a regime "in which nothing adaptive decides the outcome".

13.4.7 The Gaussian core and whether the doubling is needed

MT Proposition 13.15 computes the loss with the record off and the parameter frozen. Part (a) gives $\Delta = \frac{1}{2\mathrm{T}}\mathfrak{v}^\top H_{zz}^{-1}\mathfrak{v}$, with $\mathfrak{v}$ the force on the discarded coordinates: the lag of the fast modes behind their slaved position, weighted by compliance, in thermal units. Part (b) gives, uniformly on each early window,

$$\Delta_{\ell-1:\ell}(t) = \frac{\mathrm{A}^2}{\mathrm{T}}\frac{\mathrm{J}_{\ell+1}^4}{\mathrm{J}_\ell^3}\big[\mathfrak{y}_\ell(\mathbf{v}_0) + o(1)\big], \qquad \mathfrak{y}_\ell(\mathbf{v}_0) > 0.$$

Informally, the next level moves at rate $\mathrm{J}_{\ell+1}$ and pulls through couplings of order $\mathrm{J}_{\ell+1}$, the current level answers with compliance $1/\mathrm{J}_\ell$, so the lag force is of order $\mathrm{A}\mathrm{J}_{\ell+1}^2/\mathrm{J}_\ell$ and its compliance-weighted square gives the display; the general estimate behind (c) is larger by $(\mathrm{J}_\ell/\mathrm{J}_{\ell+1})^2$. Part (c) turns this into a criterion: with $\mathrm{A}^2/\mathrm{T} = \epsilon^{-\mathrm{Q}_\mathrm{T}}$ and a large enough entrance constant, the core loss vanishes on every early window exactly when $3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} > \mathrm{Q}_\mathrm{T}$ for every $\ell < L$, and is $O(\epsilon^{\mathrm{Q}_\Delta})$ exactly when $3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} \geq \mathrm{Q}_\mathrm{T} + \mathrm{Q}_\Delta$, forcing $\mathrm{Q}_1 \geq (\mathrm{Q}_\mathrm{T} + \mathrm{Q}_\Delta)[(4/3)^{L-1} - 1]$.

Is the doubling required? Not shown, and the chapter says so. It enters the proof through the full Fisher budget of MT Lemma C.7 and the history estimate of MT Lemma C.11. What is proved is that in the Gaussian core, for this preparation and schedule, a separation growing like $(4/3)^L$ in the exponent is necessary. Whether the full family works under the base-$4/3$ schedules $3\mathrm{Q}_\ell \geq 4\mathrm{Q}_{\ell+1} + 20$, $\mathrm{Q}_\ell \geq 14$, with depth-independent constants, is MT Open Problem 13.16. Either way the separation diverges with the number of agents: since $L - 1 = \log_2(N/3)$, $(4/3)^{L-1} = (N/3)^{\log_2(4/3)} \approx (N/3)^{0.415}$, so the exponent of the stiffness range grows linearly in N under the doubling and like a power of N under the best schedule the core allows. That is the quantitative sense in which this family has no large-N limit.

13.4.8 The four-cycle witness

MT Definition 13.21 puts four agents on a ring with edge cost and record

$$\mathcal{D}_{e_i}(\mathbf{x}) = \tfrac{1}{2}\big[(r_i^b)^2 + (r_i^m)^2\big], \qquad \Pr(E_i = 1 \mid \mathbf{x}, S) = \tfrac{1}{2}\exp\big[-\mathcal{D}_{e(i,S_i)}(\mathbf{x})/\mathrm{T}\big],$$

the cost adding belief- and model-channel relative entropies and the record reading only the attended edge, with weight $1/\mathrm{T}$. An attended edge is stiff against the thermal scale; an unattended edge carries no energy at all. This is the gating. The prior is uniform, nothing is learned, every coordinate has unit mobility, $\mathrm{T} = \epsilon^8$ with $\epsilon \leq 1/100$, the formation time is $t_\epsilon = 2\log(1/\epsilon)$, and the

horizon is $t_\epsilon + 1$. The preparation is an equal mixture of narrow Gaussians around $\pm\mathbf{x}^\circ$, where $\mathbf{x}^\circ$ has all means zero and all links one, with a uniform assignment, so it names no pair. On the path that holds a matching from $\mathbf{x}^\circ$, $r_{\mathrm{int}} = -e^{-(8+\mathrm{T})t}$ and $r_{\mathrm{ext}} = e^{-\mathrm{T}t}[-\frac{3}{2} + \frac{1}{2}e^{-8t}]$ (MT (C.95)): matched mismatches decay at rate eight, exterior ones go to about $-3/2$ and then decay at the thermal rate. The circulation, equal to minus the sum of the belief mismatches, drops from four to about three and then relaxes only at rate T; it is what stretches the unattended edges.

Part (a) of the proposition gives the episode: with probability at least $p_0 = 3e^{-1/4}/512$ the assignment is one perfect matching throughout; at time zero each matched pair has belief relative entropy above $1/4$, so the contrast is entered during the episode; after t_ϵ the matched relative entropies are at most ϵ^4; and after time $1/4$ every exterior belief relative entropy exceeds $1/2$. The pair is held with restoring rates $8 + \mathrm{T}$ and $4 + 5\mathrm{T}$, and the kinetics started from the stationary law realize the event with probability at most $4e^{-1/(4\mathrm{T})} < p_0/2$. Part (b) is the parent channel, discarding the three fast normal directions of the attended edges and keeping the assignment, the loop, and one fast coordinate. Part (c) bounds the discarded conditional relative entropy by $\bar{\eta}_\epsilon$ (MT (13.52)). Part (d) puts the parent activated at t_ϵ within total variation $2p_\epsilon/(1-p_\epsilon) + 4\sqrt{8/3}e^{-1/\mathrm{T}}$ of the native retained path law conditioned on the same prefix, assignment path included. The constants are explicit at every $\epsilon \le 1/100$; at $\epsilon = 1/100$ they give a path-law error of about 1.5 percent and a discarded loss of about 0.045 nats, both vanishing as $\epsilon \to 0$. Unlike the chain, the comparison here is in total variation of path laws, because on a held matching the hidden coordinates are exactly independent Ornstein–Uhlenbeck modes.

Ingredient	Role	Origin
Attention-contraction feedback	Attended edges contract, contracted edges win attention, the matching persists	endogenous
Energy scale	All edges have the same cost structure	homogeneous
Which pair forms	Initial assignment, drawn uniformly; the law stays symmetric	random seed
Circulated preparation	A flux around the ring that survives on the episode's time scale and stretches the unattended edges	supplied
Source gating	Energy only while attended, with weight $1/\mathrm{T}$	supplied
Temperature ϵ^8	Makes attention an arg-min of edge costs	supplied
Euclidean kinetic metric	Beliefs, models, and links all move at rate one; models are not slow	supplied
Duration-one episode; no learning	The pair is proved held for one time unit; parameters fixed	supplied

One observation places the witness. The four-cycle belongs to the gated family, so by MT Proposition 13.24 its stationary law already sits on the two perfect-matching sheets, up to a fraction of order T of its mass, with the matched pairs as its coherent blocks. The paired block partition the episode reaches is the typical partition of equilibrium; what distinguishes the episode from equilibrium coherence is its entrance clause (the pairs are incoherent at time zero) and the seeded choice of matching. Physically, the witness is relaxation from a prepared symmetric incoherent start into one of two paired equilibrium sectors, the choice made by a sampled label and held by

attention feedback. That is a genuine coupled-inference mechanism at one level, not a collective the equilibrium law would not already contain.

13.4.9 The gated family and depth one

MT Definition 13.23 places the four-cycle's laws, records, and kinetics on any finite simple graph without isolated vertices, with a product prior independent of T. On a sheet the energy is $V_S = \frac{1}{2}\mathbf{x}^\top H_S \mathbf{x}$, where H_S sums over receivers the residual pair of the attended edge, and both laws are Gaussian on each sheet with precision $H_S/\mathrm{T} + \Omega$ (MT (13.55)). By MT Proposition 13.24, (a) every sheet has a stiff band of order one on the k_S attended directions and a soft band of order T elsewhere, and the prior enters only sheet weights, never stiffness; (b) the configuration marginal is a Gibbs law within $\mathrm{T}\sum_i \log(1/\min_j \pi_{ij})$ of the energy $V_{\min} = \sum_i \min_j \mathcal{D}_{e(i,j)}$, in which each receiver pays only for its cheapest edge and which is flat under rigid translation of any block of a covering edge set; (c) the counting is

$$\operatorname{rank} \mathbf{a}_F = |F| + N - \mathrm{cc}(F), \qquad \mathrm{k}_S \geq 2(N - \mathfrak{n}), \qquad q_\Omega\{S : \mathrm{cc}(\mathrm{att}(S)) \leq \mathfrak{n} - m\} \leq \mathrm{K}_m \mathrm{T}^m,$$

with equality in the middle exactly for minimum edge covers, the spanning forests of $\mathfrak{n}$ stars, and sheet weights proportional to $\mathrm{T}^{\mathrm{k}_S/2}$ up to controlled factors (MT (13.57)); (d) at resolution a the stationary block partition differs from a minimum-edge-cover partition with probability at most

$$\mathrm{K}_1 \mathrm{T} + \frac{|E|\|\Omega\|}{2\mathrm{s}_*^2}\mathrm{a}^2 + 4|E| e^{-\mathrm{a}^2/(4\mathrm{T})},$$

the mass of merged sheets, the chance that an unheld edge between blocks falls inside the resolution, and the chance that a held edge fluctuates outside it. On an even ring or an even complete graph the blocks are the pairs of a perfect matching, and on a star graph the block is the star: the stationary law fragments into the smallest blocks the graph allows.

Is this large-N, and is it static? It holds on every finite graph, so it applies verbatim at 512 or more agents and is not a small-network result. But it concerns $\mathrm{T} \to 0$ at a fixed graph with graph-dependent constants. K_1 contains the factor $(1 + \|\Omega\|/\mathrm{h}_*)^{N-\mathfrak{n}}$, exponential in N, times a ratio whose numerator runs over every merged assignment (MT (13.58)), and the second term grows with the number of edges. Nothing keeps the bound small as N grows at fixed temperature, so the limit of many agents at fixed temperature, where the number of ways to merge competes with the cost of each merge, is not addressed. The statement is also purely static: it has nothing on relaxation times, metastability, or how long sampled paths can hold a larger block. Since the dynamic question is posed only outside the family (MT Open Problem 13.25), a transient second coalescence inside the gated family is neither excluded nor posed.

13.4.10 The authoring rules R13 to R16 in physics terms

Four authoring rules govern what the technical documents may call learning-driven selection, recursion, a coherence episode, and a settled open problem.

R13, the Curie rule, is Curie's principle applied to learning. If model, records, parameterization, and kinetics share a symmetry, a claim that learning selects a structure the symmetry does not fix must name the symmetry-breaking datum it amplifies and compare against a frozen-learning run, or else declare the selecting mechanism, for example a rule acting on sampled paths, and then the claim concerns sampled paths. A unique, equivariant law-level dynamics from a symmetric start cannot produce an asymmetric law, the finite-volume fact that a finite system's Gibbs state is symmetric. The rule is substantive: it is why the four-cycle result claims only a sampled-path pairing.

R14 defines recursion by a counterfactual. One declares a handoff time, a first-stage event with its blocks, a second-stage event whose block is a union of first-stage blocks, a statistic such as the second stage's probability, and a class of handoff states that includes, with probability bounded below, states from histories without first-stage formation. The second stage is recursive when it has probability bounded below and its statistic after actual first-stage formation beats its value after a counterfactual handoff by a margin. If it happens with probability tending to one from every handoff in the class, or its proof uses only properties that counterfactual handoffs share, the sequence is a supplied-scale sequence: the hierarchy is in the Hamiltonian, as in any system whose separated couplings order each on its own scale whether or not the previous scale has ordered. The physical distinction is between a hierarchy of couplings and a hierarchy of products, whose stage-two units are made of stage-one units and would not form otherwise. The definition is substantive. The chapter does not run the test on the triangle chain, but each level's coherence follows from that level's stiffness acting on the prepared displacement, so in R14's terms the chain is a supplied-scale sequence; this is the companion's reading.

R15 defines a coherence episode. The restoring clause requires a named interaction whose rate times the squared resolution, divided by the temperature, diverges on the sheets the kinetics select: by equipartition a mode of stiffness k has width $\sqrt{\mathrm{T}/k}$, so the block sits many thermal widths inside its resolution, held by energy rather than luck. The horizon clause requires horizons short compared with the time equilibrium takes to revisit the event, or a probability exceeding the equilibrium value by a margin. Restored coherence that fails the horizon clause, for instance because equilibrium has it with probability tending to one, is restored equilibrium coherence, not formation; coherence failing the restoring clause is coincidental. The recurrence branch is how a physicist says a broken-symmetry state is real, since tunneling times diverge; the margin branch admits the transients of relaxation from any prepared, atypical start. The rule thus excludes thermal accidents and equilibrium ordering from the word formation but not preparation-driven relaxation. The four-cycle meets both clauses, the horizon clause through its entrance requirement. In the chain the intermediate windows meet the margin branch, but the final single component is what the restricted optimizer looks like, every residual being thermal under it, so by R15's own terms the chain's last stage is restored equilibrium coherence. The chapter does not evaluate R15 for the chain; this is the companion's reading.

R16 requires each open problem to be posed in a stated class with two nontrivial settling branches, neither met by an existing statement or a trivial construction. It is bookkeeping for well-posedness,

and it is why the open problems below are long and cannot be closed by pointing at the witnesses already proved.

13.5 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
MT Lemma 13.2	Gauge-invariant residuals; triangle energy splits into contrasts and flux; orthogonal split into normal, loop, gauge directions	SOLID	Standard in substance
MT Proposition 13.6	Normalized joint, quadratic restriction cost, free energy as relative entropy to the optimizer, gap $O(N\mathrm{T}^2/\mathrm{J}_1)$; with MT proposition 13.4	SOLID	Standard in substance
MT Proposition 13.8	Parents are exact marginals with Schur stiffness and bounded tilt; free energy drops by exactly $\mathrm{T}\Delta$; with MT proposition 13.10 and MT proposition 13.12, reversible inherited rows and an H-theorem	SOLID	Standard in substance: Gaussian integration, chain rule
MT Definition 13.9	Langevin plus heat-bath label jumps; declared parent kinetics; own-law learning (MT definition 13.11)	CHOICE	The parent kinetics is a Markov closure
MT Definition 13.13	Doubling recursion, amplitude, temperature ϵ^{20N-34}, horizon, prepared mean, calibration	CHOICE	Where the outcome is put in
MT Theorem 13.17(b)	Component counts $3 \cdot 2^{L-1} \to \cdots \to 1$ on the windows	TOY	Triangle chain, $(\mathbb{R}, +)$, scalar fibers, supplied hierarchy; linear relaxation; constants depend on L
MT Theorem 13.17(a), (d)	Learning follows calibration; attention mass and hidden kernels move after formation	TOY	Movement engineered; no effect on formation
MT Theorem 13.17(c), (e)	Discarded loss and score error vanish; histories couple; paths concentrate	TOY	Resolution $\sqrt{\mathrm{T}}/\epsilon$, not total variation of path laws
MT Remark 13.18	Record saturated, learning calibration-dominated, loops thermal; coherence unchanged without record and learning	SOLID	Negative content fixing the theorem's reading
MT Proposition 13.15	Exact loss; two-sided asymptotic; base-$4/3$ separation criterion	TOY	Sharp, for the Gaussian core of the chain only

Result	What it says	Status	Scope and caveats
MT Proposition 13.22	A pair forms by attention feedback with probability at least 4.6×10^{-3}; exact parent over one time unit	TOY	Four agents; gating, seed, circulation, $\mathrm{T} = \epsilon^8$, no learning; symmetric law
MT Proposition 13.24	Stationary blocks are minimum-edge-cover components; each merge costs T	SOLID	No-go inside the gated family only; stationary; $\mathrm{T} \to 0$ at fixed graph
MT Open Problem 13.16	Is the doubling needed; depth-uniform constants	OPEN	Chain family, base-$4/3$ schedules
MT Open Problem 13.20	Learning or holonomy changes, in law, which block forms	OPEN	Four-cycle with learned rows
MT Open Problem 13.25	A recursive (R14) second coalescence that is an episode (R15)	OPEN	Outside the gated family

Conditions: which ones matter

Conditions that decide the chain's outcome, several chosen precisely to switch off the feedback the program is interested in: the supplied strengths with the doubling, the prepared displacement with a strict margin, the tie $\mathrm{T} = \mathrm{A}^2\epsilon^6$, the calibration record, the saturated record, and source palettes confined to each triangle. Conditions that exclude the program's targets: the group $(\mathbb{R}, +)$ with scalar fibers, unit mobility (models as fast as beliefs), and the uncomputed ϵ_L with depth-dependent constants. For the four-cycle the conditions that matter are the supplied rows of the table in section 13.4.8; for the gated family they are exclusive single-label attention, a scalar translation group, and temperature-independent energies and prior ratios. Hygiene: the root observation and gauge weight, the cancellation term $-\mathrm{T}R_{\mathrm{ev}}$, the preparation neighborhood of smooth transports, positivity and non-explosion of densities, the choice of entrance constant and tolerance above their thresholds, the Gaussian reference, and $\mathrm{T} \leq 1$.

13.6 Toy versus general

None of the three results has hypotheses that include the program's target, 512 or more agents with nonabelian, noncompact groups. For the triangle chain the mechanism-level content concerns coarse-graining, not formation: exact marked parents can be carried through a hierarchical relaxation with separately vanishing errors, provided the stiffness separation beats the adiabatic lag in thermal units; that this tension appears in every construction of the type is plausible but unproved. The doubling, the exponents 20, 6, and 14, the constants 10^{-4}, $[1, 21/10]$, $1/200$, and $57/100$, the sigmoid prior, and the chain topology are specifics, and the theorem, though valid for arbitrarily many agents, is not a large-N result. For the four-cycle the mechanism-level content is the attention-contraction feedback, with an order-one flux deciding which edges stretch; it is local and could operate in large graphs, but whether it forms many pairs at once, how neighboring pairs interact, and whether the law picks them are not addressed, and the floor $3e^{-1/4}/512$, the rates 8

and 4, and the one-unit horizon are specifics. For the gated family the mechanism-level content, that exclusive gating puts the label into a sheet's weight and never its stiffness, is proved on every finite graph; it is the chapter's one result that is not about a small network, and it does not reach non-exclusive or multi-head attention, temperature-dependent couplings, interactions that stiffen the directions between blocks, nonabelian links, dynamics, or many agents at fixed temperature.

On the reader's own intuitions the chapter is neutral. Formation in the chain is continuous: the dynamics is smooth relaxation, and the discrete counts come from the resolution $\eta_L \mathrm{A}$ and scale separation, not from a transition. No phase transition, condensation, or thermodynamic limit appears; every result is at fixed finite N with T or ϵ tending to zero. The gated no-go shows fragmentation at fixed N and low temperature in one family, and rules condensation neither in nor out elsewhere or as $N \to \infty$ at fixed temperature.

13.7 Open ends

The scope paragraph's declined claims map onto three open problems. Order-one or nonabelian holonomy, a model-channel record, slow per-agent models, and a record informative at the coherence scale are absent from the chain and reappear as the ingredient list of MT Open Problem 13.20: a family in which learning or holonomy changes, in law, which components form needs at least one of a record standardized at the coherence scale, for example through $\chi(r/(\eta_L \mathrm{A}))$; model means and links both slow relative to beliefs, since a fast link or fast model erases the model disagreement on the belief clock (MT Proposition 8.6; see chapter 8); loops of order A or nonabelian links; or interacting model-channel records. The problem is posed on the four-cycle with learned attention rows, as a total-variation difference, bounded below, between the conditional laws of the first formed block in the learned and frozen arms, or between initial means differing only in their loop coordinate, or a proof that these differences vanish for almost every mean. In a slower-mobility regime the learned and frozen first-block laws coincide in the limit, so learning selects nothing there (MT Proposition C.17), and a parent activated after hidden thermalization has separately vanishing errors on the seeded sub-event (MT Proposition C.25); the original coupled mobility and the loop comparison remain open (chapter 14).

Depth-uniform constants and a non-doubling regime are MT Open Problem 13.16; a positive answer would still leave a stiffness range growing like a power of N. Arbitrary families and initial laws, mixed-level source support, permanent membership, and control as the depth tends to infinity are folded into MT Open Problem 13.25: a second endogenous coalescence, with learned parameters and from non-matching assignments, in a source-gated model with temperature-independent data outside the gated family, recursive in the sense of R14, held in the sense of R15, and followed by parents with separately vanishing errors, or a proof that none exists in the class. A dynamic second coalescence inside the gated family falls between this problem and the depth-one proposition. Intrinsic selection of the contrasts is MT Open Problem 9.19, a persistent multilevel hierarchy formed by descent is MT Open Problem 12.27, and behind all of them stands MT Open Problem 11.38: the

chapter's parents are declared Markov closures checked against the fine process in two special families, which does not settle how a parent's dynamics follows from the network it summarizes.

Does it deliver what it claims? NARROWER THAN IT SOUNDS

What the chapter says it does. Its opening asks whether coherent components can form in succession in one normalized experiment while every parent is activated from the complete preceding law, keeps every loop and label, and learns its own parameter, and says two witnesses answer yes under explicit restrictions. The opening paragraph of the introduction describes a finite-depth family that forms successive meta-agents under free-energy descent with learning, and the program aim is that collectives form through the agents' coupled inference. The introduction's reading map, by contrast, describes the chapter accurately.

What the statements give. For the chain the question as posed is answered yes (MT Theorem 13.17): exact marginal parents that keep loops and labels and learn from their own laws track the fine process with separately vanishing errors at every finite depth. The succession itself is linear relaxation of a supplied displacement through a supplied stiffness hierarchy; attention is within 10^{-4} of its prior, learning follows calibration and moves after formation, loops are thermal, and the coherence proof works with attention and learning switched off (MT Remark 13.18). The meta-agents are marginal laws, not coarse agents; the temperature is ϵ^{20N-34} and the constants depend on the depth without control. The four-cycle witness (MT Proposition 13.22) and the depth-one no-go (MT Proposition 13.24) are delivered as stated.

The gap. Nothing shows a collective forming because of coupled inference beyond one level. The one endogenous witness is a sampled-path event on four agents with a symmetric law and no learning, whose end state equilibrium already favors, and the opening's "two witnesses answer yes" fits the chain but not the four-cycle, which has neither succession nor learning. Recursive endogenous formation (MT Open Problem 13.25), law-level selection by learning or holonomy (MT Open Problem 13.20), and the depth-uniform control that any statement about hundreds of agents needs (MT Open Problem 13.16) are open. The technical chapter's scope paragraph is candid about all of this; it is the introduction's opening summary and the program aim that promise more than these theorems deliver.

Chapter 14

Interpretation and Its Limits

This chapter covers MT Chapter 14 together with the proof appendices that carry its four-cycle results, from MT Appendix C.9 to MT Appendix C.13. The technical chapter is short, but it is where the program's interpretive claims meet its most delicate formation theorems, and eight of the propositions treated here are stated, not only proved, in those appendices.

14.1 The question and the answer

The technical chapter asks what the preceding chapters commit to, and which readings of them an observation or a proof could count against. Its answer has two layers that should be kept apart. The first is interpretive and proves almost nothing: it sorts readings into declared rules of reading (definitions), commitments the mathematics uses (standing hypotheses), and unsettled questions (open problems). Its verdicts are that the architecture is local-first rather than background-free, that the frame-change invariants are compatible with several structural-realist positions and select none, that environment messages create no agents, and that nothing proved before the formation chapter distinguishes the program from an ordinary hierarchical model with learned priors, whereas formation by the inference could. The second layer is mathematical and substantial: a line of results on the homogeneous four-cycle of MT Definition 13.21 asking whether learning, rather than a random seed or a supplied hierarchy of strengths, decides which collective forms.

The answer of the second layer fits in one paragraph. On the four-cycle, formation in the technical sense, meaning that a block enters coherence with a probability above its stationary value, happens from every Gaussian initial law with a probability bounded below, but that probability is supplied by the preparation's random initial assignment and source learning plays no role in it (MT Proposition 14.4). Law-level selection does happen when learning is slow: at a small fixed temperature, as the learning mobility tends to zero, the learned source prior picks one of the two pairings of the cycle for almost every Gaussian initial mean, while the arm with a frozen prior ends symmetric (MT Theorem 14.5, a theorem despite its label). That selection amplifies an asymmetry already present in the initial law, it appears only at late times, and at those times the selected pair's coherence is a property of equilibrium, not a formation event (MT Proposition C.16); on the early horizon where formation happens, slow learning leaves the law of the first formed block unchanged (MT Proposition C.17). Early formation without law-level selection, late law-level selection without formation: the two halves do not meet, and making them meet is the open problem MT Open Problem 13.20. The remaining results concern the parent. The fixed parent channel of the four-cycle can be activated

with separately vanishing conditional and dynamical errors, both after the late selection and on the early seeded episode, provided the activation time waits for the discarded modes to thermalize; activation at the moment the coherence criterion is first met provably fails.

In the chain of the theory this chapter follows the formation chapter (chapter 13), whose finite-depth theorem forms collectives through a supplied hierarchy of interaction strengths and whose endogenous four-cycle witness forms one pairing from a symmetric preparation without learning. The present chapter asks whether learning can do the choosing. Table 14.1 maps the statements by time horizon and observable, the organizing principle the technical text uses but never tabulates.

Table 14.1: The four-cycle continuation line. "Early" is the original episode horizon, of order $\log(1/\epsilon)$ at temperature $\mathrm{T} = \epsilon^8$; "late" is the selection window $t \sim \epsilon_{\mathrm{L}}^{-1} \log(1/\epsilon_{\mathrm{L}})$ at fixed or slowly vanishing temperature.

Statement	Horizon	What is compared or bounded	Role of learning
MT Proposition 14.4	early	probability of a contrast (entrance) event	idle
MT Theorem 14.5	late, fixed T	one-time assignment law, learned versus frozen	amplifies the initial law
MT Corollary 14.6	late, $\mathrm{T} \to 0$ jointly	same, along implicit mobility thresholds	same
MT Proposition C.15	late	distance of the law from its instantaneous equilibrium	adiabatic following only
MT Proposition C.16	late window $\mathrm{T}^{-1/4}$	path law and coherent partition	fixes the label only
MT Proposition C.17	early	conditional law of the first formed block	none if $\epsilon_{\mathrm{L}} = o(\mathrm{T}^{1/4})$
MT Proposition C.18	late activation	discarded conditional information	parent inherits rows
MT Proposition C.20	late window $\mathrm{T}^{-1/4}$	retained path law, parent versus fine	parent learns from its own law
MT Proposition C.25	early activation	discarded information, both directions	idle for $\epsilon_{\mathrm{L}} \in [0, 1]$
MT Proposition C.26	first completion	discarded information diverges	idle
MT Proposition C.27	windows $\ll \mathrm{T}^{-1}$	retained path law, parent versus fine	idle

14.2 The picture

14.2.1 Scaffolding, idle posits, and what survives a change of frame

Each agent carries two law points at the base point, a belief law and a model law, which the cross maps do not identify (MT (14.1)). The base, the principal bundle, and the measurable spaces are coordination scaffolding: they make the agents' interfaces comparable, but no inference law constructs them and no law is placed on them. That is the meaning of local-first and not background-free, and a physicist will recognize lattice gauge theory, where the lattice and the group are fixed

background and only site fields and links fluctuate. The verdict describes the present architecture; it would change if edges became structural coordinates updated by inference (MT Open Problem 6.9) or links were recovered from population records. Because the scaffolding is not an inferential target, its lack of a law decides neither a Neo-Kantian reading (frames as revisable constitutive conditions of meaning) nor a noumenal one (the bundle as structure reached only through representations).

The chapter's one new definition is a parsimony rule: a posit is idle when changing it alters the law of no record and of no statistic of a record in the model, and the idle-wheel criterion (MT Definition 14.1) removes idle posits. It plays the role gauge redundancy plays in physics, but it is a declared rule of reading, close in spirit to constructive empiricism without being derived from it, and it is relative to the records a model declares. What survives it is structural data,

$$\mathcal{I}_{\mathrm{struct}} = \Big\{\text{ELBO and relative-entropy values},\ [\mathrm{Hol}^{b}(p)]_{\mathrm{conj}},\ [\mathrm{Hol}^{m}(p)]_{\mathrm{conj}},\ \big[(\mathring{\Theta}^{b}_{e}, \mathring{\Theta}^{m}_{e})_{e}\big]\Big\}$$

(MT (14.3)): scalars invariant because the kernels are covariant (MT Proposition 4.9), loop classes in both channels, invariant because a passive re-expression conjugates a loop product, and the per-edge twists (each link expressed in its endpoints' own frames) modulo one simultaneous conjugation, which is finer than the loop classes because it keeps how the twists of different edges sit relative to one another. For abelian links, as on the four-cycle, the loop class is the loop sum $\sum_i \omega_i$. On a graph-flat family the loop classes are trivial (MT Proposition 2.13) and give a noumenal reading nothing to do; a nontrivial loop class is a referent inside the interaction complex, which a declared record at c_* can read (MT Theorem 10.28). The technical text finds these invariants compatible with epistemic, moderate, and ontic structural realism and gives no argument from theory change, since they are invariants of frame change inside one theory, whereas Worrall's structure [83] survives a change of theory.

Physics analogue (exact)

The loop classes $[\mathrm{Hol}^{x}(p)]_{\mathrm{conj}}$ are the gauge-invariant content of a closed loop in lattice gauge theory, the data of a Wilson loop before the trace. On the interaction graph this is an identity of structure: links are group elements on edges, a passive re-expression acts as $\Theta_e \mapsto g_i \Theta_e g_j^{-1}$, and only conjugacy classes of loop products are invariant (MT Proposition 2.12). Nothing carries over to a continuum connection: graph-link holonomy is neither base-connection holonomy nor bundle topology, and the technical text keeps these and gauge redundancy apart.

14.2.2 Participation and why it does not yet distinguish the program

Participation, in the technical sense, is reciprocal inferential dependence through records inside one joint law, with no live posterior feeding a same-step generative factor (MT Hypothesis 4.3). It occurs in the same-time inference loop (block coordinate ascent, as in any connected hierarchical model), in parameter learning (whose optimal prior over an unrestricted family is the aggregate recognition law, MT (12.34)), and in delayed feedback through a declared kernel (still the free energy of one joint, MT Proposition 12.25). The verdict is blunt: none of these separates the program from a

hierarchical model with learned priors, and the resemblance to Wheeler's participatory proposal [78] is neither derived from it nor evidence for it. Formation could separate them: collectives constituted by the coupled inference, held by a restoring interaction, and dependent on the preceding state. Hence the formation theorems in this chapter.

14.2.3 The four-cycle: a ring of agents choosing whom to listen to

Four agents sit on a ring. Each has a scalar Gaussian belief (mean $\bar{k}_i$, unit variance) and a scalar Gaussian model (mean $\bar{m}_i$, variance $1/2$); each edge carries a translation link ω_i in the abelian group $(\mathbb{R}, +)$. The cost of an edge is the transported relative entropy of its residuals. Each agent listens to exactly one neighbor, and an edge carries energy only when an endpoint listens to it (source gating). Of the sixteen listening assignments, the two perfect matchings, which split the ring into two mutually attending pairs, are the candidate collectives. The kinetics is Langevin diffusion at temperature T on the energy sheet of the current assignment, interrupted by a rate-one clock at which every agent redraws whom it listens to from its posterior. Everything is symmetric under the rotations and reflections of the ring.

Seeded formation. From a Gaussian configuration and a uniformly random assignment, with small probability the assignment is already a matching and the configuration lies near a seed point where every edge is equally mismatched; the attended residuals then relax, the unattended ones grow, and two pairs form. The probability is that of a lucky initial condition, independent of the small parameter and unchanged if the prior learns (MT Proposition 14.4).

Selection by slow learning. Now let the listening prior learn slowly from the population's recognition law. Its rest point is the recognition marginal, so learning has the form of the expectation-maximization update of mixture weights, run in continuous time; averaged over the fast kinetics, the listening probabilities climb the logarithm of an evidence function in the Fisher metric of the simplex. At low temperature that evidence is largest at the two pure matchings and has a saddle at the uniform prior, unstable exactly in the direction that favors one matching over the other. An asymmetric Gaussian initial law leaves, during the early transient, an asymmetric imprint on the rows of the order of the mobility; the saddle amplifies it, and after a time of order $\epsilon_{\mathrm{L}}^{-1} \log(1/\epsilon_{\mathrm{L}})$ the prior sits at one matching while the recognition law tracks the corresponding equilibrium. The frozen arm has nothing to amplify and relaxes to its symmetric equilibrium (MT Theorem 14.5).

Physics analogue (structural)

The theorem has the dynamical signature of mean-field symmetry breaking. The learned four-cycle is a nonlinear Markov process, whose generator depends on its own law through the rows, as in the McKean–Vlasov limit of the Curie–Weiss model; the rest point of learning is a Curie–Weiss-type self-consistency (the listening marginal computed under the rows equals the rows), and below the theorem's temperature threshold its symmetric solution is unstable along one direction. The limits fail to commute in the familiar way: at fixed time the rows move by at most $\epsilon_{\mathrm{L}} t/2$ (part (a) of MT Proposition 14.4), so $\epsilon_{\mathrm{L}} \to 0$ first and then $t \to \infty$ gives the symmetric frozen answer, while $t \propto \epsilon_{\mathrm{L}}^{-1} \log(1/\epsilon_{\mathrm{L}})$ gives a selected matching. The imprint, of the

order of the mobility, acts as a vanishing field whose effect survives because its amplification time diverges.
It is not spontaneous symmetry breaking in the strict sense. There is no thermodynamic limit, and the frozen kinetics has a unique invariant law (the proof uses the Harris theorem for exactly this). Nothing is spontaneous either: the rows learn from the law, so they are deterministic and carry no fluctuations that could choose a branch, and the branch is fixed by a definite datum, the sign of an analytic function of the initial mean. A symmetric initial law keeps the rows uniform forever (part (c) of MT Proposition 14.4), as a mean-field magnetization started at zero stays at zero. The accurate description is explicit symmetry breaking by the initial data, amplified by a dynamical instability; the technical text's word is "amplifies".

Physics analogue (structural)

Why matchings win at low temperature: every listening sheet has the same minimum energy, zero, at the same configuration, so the sixteen sheets are degenerate. They differ in how many directions they stiffen: a matching stiffens four residual directions, a path six, an oriented cycle seven (the residuals around a loop obey one linear relation). Each extra stiff direction costs a factor of order $\sqrt{\mathrm{T}}$ in thermal phase-space volume, so the evidence peaks at the matchings. Selection among degenerate ground states by the volume of thermal fluctuations is the mechanism of order by disorder; the technical documents do not use that term.

Late persistence, early null, and the parent. Once the prior sits at a matching, the selected pairs stay coherent, but the stationary law at those rows realizes the same persistence with probability tending to one, and the frozen arm shows persistent pairs too, split half and half between the two pairings (MT Proposition C.16). Late, learning changes which pairing, not whether there is pairing. Early, a slow prior has barely moved, and the law of the first formed block is the same with or without learning (MT Proposition C.17). The parent of a pairing is obtained by a fixed channel that keeps the assignment, the loop coordinate $\sum_i \omega_i$, the slow directions, and one fast mode (the sum of the two pairs' average residuals), and discards three fast modes (the difference of the pairs' average residuals and each pair's belief-model residual difference). The parent is exact as a summary when the discarded modes, given the kept ones, are thermal, and exact as a dynamics when its own evolution reproduces the kept trajectories. Both hold after the late selection and, on the early seeded episode, at deterministic activation times after the discarded modes have relaxed to the thermal scale; the first moment the coherence criterion is met is too early by at least a factor of two in logarithmic time, and there the discarded information diverges (MT Proposition C.26).

Physics analogue (exact)

The discarded conditional relative entropy is the error term of integrating out fast variables. The chain rule $D_{\mathrm{KL}}(q\|\mu) = D_{\mathrm{KL}}(C_{\#}q\|C_{\#}\mu) + \mathcal{H}_{\mathrm{opt}}(q)$ (MT (C.133)) is an identity, so the fine divergence is the parent's divergence plus a remainder that vanishes exactly when the fast modes are in conditional equilibrium given the slow ones: adiabatic elimination written as an

identity. The dynamical statements are the structural analogue of a valid reduced Langevin equation, proved only on stated windows. Those windows lie on a plateau between the thermalization time of the discarded modes, $\log(1/\epsilon) = \frac{1}{8}\log(1/\mathrm{T})$, and the slow time T^{-1} on which weakly restored directions equilibrate and the pairing can change; a quasi-stationary state between fast relaxation and slow escape is what a physicist calls metastability.

14.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Idle posit	A posit changing the law of no record or record statistic (MT Definition 14.1).	Gauge redundancy; unobservable absolute frame.
$[\mathrm{Hol}^x(p)]_{\mathrm{conj}}$	Conjugacy class of the product of links around closed walk p in channel x.	Wilson loop before the trace.
Frame, law, mean agreement	Three inequivalent notions of agents agreeing (MT Proposition 2.18).	Pure gauge; equal states up to transport; equal first moments.
$\mathbf{x} = (\bar{k}, \bar{m}, \omega)$	Four-cycle configuration: belief means, model means, link translations; twelve coordinates.	Site fields and abelian link fields.
r_i^b, r_i^m, $\mathcal{D}_{e_i}$	Belief and model residuals on edge i and the edge cost, a transported relative entropy.	Covariant difference and its energy.
S, σ_i	Listening assignment; $\sigma_i = +1$ when i listens to $i+1$.	Discrete label field.
$S^{(0)}$, $S^{(1)}$, matching	The two assignments that split the ring into two mutually attending pairs.	The two dimer coverings of a plaquette.
π_{ij}, $p_i = \pi_{i,i+1}$	Learned source prior (listening probabilities), one row per receiver.	Mean-field order parameter.
ϵ_{L}, T	Learning mobility; temperature ($\mathrm{T} = \epsilon^8$ in the original regime).	Slow relaxation rate; temperature.
$q_t^{\epsilon_{\mathrm{L}}}$, q_t^0	Recognition law of the kinetics, learned arm and frozen arm (rows fixed at $1/2$).	Ensemble with and without feedback.
$q^\star$, $\mu_{\mathrm{T},p}$	Restricted optimizer $\propto \pi_S e^{-U_S/\mathrm{T}}$, the invariant law at fixed rows p.	Gibbs state at fixed couplings.
ν_{T}^M	Gaussian law on the sheet of matching M, assignment held at M.	Pure phase.
Contrast formation	A block internally coherent and externally distinct, entered after time zero.	Nucleation of a bound pair.
$\mathrm{j}_{\mathrm{T}}(\mathrm{m}_0)$	Transient imprint: integrated early asymmetry of the frozen law along the matching direction.	Integrated response to an initial perturbation.

Term or symbol	Plain meaning	Physics analogue
Restored equilibrium coherence	Coherence with a restoring interaction that is typical at stationarity; not formation.	Equilibrium order.
C, W_M, Z_M	Parent channel; kept subspace (nine dimensions); discarded subspace (three fast modes).	Blocking map; kept and integrated-out variables.
$\mathcal{H}_{\mathrm{opt}}$, $\mathcal{H}_{\mathrm{phys}}$	Mean discarded conditional relative entropy against the optimizer or the physical posterior.	Error of adiabatic elimination.
Retained-path TV	Total variation between the parent's path law and the fine kept-coordinate path law.	Validity of a reduced dynamics.
θ, $\mathrm{t_c}$	Deterministic activation time; first qualifying completion time.	Fixed measurement time; first-passage time.

Several words carry nonstandard senses. "Episode" means a coherence episode in the sense of rule R15, not a learning episode. "Formation" is always a threshold event at declared resolutions (relative entropies below ϵ^4 inside, above $1/2$ across the boundary, with an entrance clause), and "selection" always a property of the law, never of a sample path. "Actual" means the law the kinetics really produces, not a Gaussian or stationary substitute. "Physical" refers to the generative posterior, "optimizing" to the restricted optimizer. The regime parameter appears as ϵ in the appendices, ξ in the chapter text, and $\mathrm{T}^{1/8}$ in MT Proposition C.17; the original coupled regime sets the mobility ϵ_{L} equal to it. The label `conj:phil-contrast-formation` survives from when the statement was a conjecture; the printed heading says Theorem, and it is proved.

14.4 The formulas, explained

14.4.1 Energy, invariant law, and learning rule

On edge e_i the residuals are $r_i^b = \bar{k}_{i+1} - \bar{k}_i - \omega_i$ and $r_i^m = \bar{m}_{i+1} - \bar{m}_i - \omega_i$, and

$$\mathcal{D}_{e_i}(\mathbf{x}) = \tfrac{1}{2}\big[(r_i^b)^2 + (r_i^m)^2\big] = D_{\mathrm{KL}}\big(q_{i+1}^b \| (\Theta_{e_i})_{\#} q_i^b\big) + \sigma_{\mathrm{ev}}^2 D_{\mathrm{KL}}\big(q_{i+1}^m \| (\Theta_{e_i})_{\#} q_i^m\big)$$

(MT (13.50)): the relative entropy between one agent's law and its neighbor's law carried across the link. The sheet energy $V_S = \sum_i \mathcal{D}_{e(i,S_i)}$ counts an edge twice if both ends attend to it and not at all if neither does. The restricted optimizer, which is the invariant law of the kinetics at a fixed prior, is

$$q^\star(\mathbf{x}, S) \propto \pi_S e^{-U_S(\mathbf{x})/\mathrm{T}}, \qquad U_S = V_S + \tfrac{\mathrm{T}}{2}|\mathbf{x}|^2 + \tfrac{\mathrm{T}}{2}\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}.$$

The first term is the interaction; the second comes from the unit Gaussian prior on configurations and gives every direction a weak restoring rate of order T; the third comes from the model evaluator and penalizes belief-model discrepancy, and the physical posterior omits it. At zero temperature

every sheet has a flat zero-energy valley; at finite temperature each sheet is a Gaussian whose weight is its phase-space volume.

The learning rule of the source prior is

$$\dot{\pi}_{ij} = \epsilon_{\mathrm{L}}\big(q_t(S_i = j) - \pi_{ij}\big)$$

(MT (14.4)): the rows relax at rate ϵ_{L} toward the current recognition marginal. It is the natural-gradient flow of the free energy in the prior row under the Fisher metric of the simplex, with the recognition label marginal as rest point, as for the aggregate prior. The right side uses the law q_t, not a sample, so the rows are deterministic; at $\epsilon_{\mathrm{L}} = 0$ they stay at $1/2$, the frozen arm against which every selection claim is measured. Part (a) of MT Proposition 14.4 gives well-posedness and $|\pi_{ij}(t) - 1/2| \le \epsilon_{\mathrm{L}} t/2$, so nothing learned can matter on horizons short compared with $1/\epsilon_{\mathrm{L}}$.

14.4.2 Where the probability of formation comes from

Part (b) of MT Proposition 14.4 bounds the probability of contrast formation below by

$$\frac{e^{-1/4}}{16} \sum_{\pm} \mathcal{N}(\mathsf{m}_0, I_{12})\big\{|\mathbf{x} \mp \mathbf{x}^{\circ}| < \varrho_{\circ}\big\}, \qquad \mathbf{x}^{\circ} = (0, 0, \mathbf{1}_4), \quad \varrho_{\circ} = \tfrac{1}{100},$$

uniformly in the regime parameter. The factor $1/16$ is the probability that the uniform initial assignment is a given matching; $e^{-1/4}$ is the probability that the redraw clock is silent until time $1/4$, long enough for a cost gap to open; the Gaussian mass is the probability of starting within $1/100$ of a seed point, where all means vanish and every link translation is ± 1, so every edge is equally mismatched. On this sub-event the mechanism of MT Proposition 13.22 runs: attended residuals decay at rates of order one, unattended ones grow toward $3/2$ in magnitude, and posterior redraws keep the matching except with exponentially small probability. The bound holds equally with frozen rows, and the technical text draws the right conclusion: a claim that learning creates a contrast must compare the arms at the level of the law. Two calibrations: the explicit estimate $e^{-1/4}\varrho_{\circ}^{12}/737280$ times a Gaussian factor is about 10^{-31} at $\mathsf{m}_0 = 0$, so its only role is to be positive and ϵ-independent; and the seeded sub-event is genuine formation under R15, with a restoring interaction and a probability exceeding its value from the invariant law, at most $4e^{-1/(4\mathrm{T})}$, by a margin bounded below.

14.4.3 The row flow, its saddle, and the imprint

Averaging the learning rule over the invariant law at fixed rows gives the reduced field

$$\mathscr{V}_i(p) = \bar{\beta}_i(p) - p_i = p_i(1 - p_i)\partial_{p_i} \log \mathsf{Z}(p), \qquad \mathsf{Z}(p) = \sum_{\sigma} \pi_{\sigma}(p)\mathsf{w}_{\sigma}, \qquad \mathsf{w}_{\sigma} = \det(\mathsf{A}_{\sigma}/\mathrm{T})^{-1/2}$$

(MT (C.96)). Here $\bar{\beta}_i(p)$ is the stationary probability that receiver i listens to $i + 1$, w_{σ} is the Gaussian volume of sheet σ, and Z is the evidence of the prior. The factor $p_i(1 - p_i)$ is the inverse Fisher metric of a Bernoulli row, so the reduced flow is the replicator equation with fitness $\partial_{p_i} \log \mathsf{Z}$,

natural-gradient ascent of the log evidence. With the matching weight normalized to one, paths have relative weights $\mathsf{r}_E, \mathsf{r}_N \approx \sqrt{40/7}\mathrm{T}$ and oriented cycles $\mathsf{r}_C \approx \sqrt{5/3}\mathrm{T}^{3/2}$, the count of stiff directions made exact. At the uniform prior the linearized field has one positive eigenvalue $\alpha \geq 7/5$ along $\mathbf{v} = (1, -1, 1, -1)$, toward one matching and away from the other, and negative eigenvalues elsewhere (MT (C.114)); both matching vertices are sinks. The threshold T_0 is a temperature below which the sheet ratios are at most $1/8$: a sufficient condition of the proof, not a located critical temperature, and nothing is proved above it.

The seed that pushes the rows off the saddle is the transient imprint

$$\mathsf{j}_{\mathrm{T}}(\mathsf{m}_0) = \mathbf{v} \cdot \mathsf{I}(\mathsf{m}_0), \qquad \mathsf{I}_i(\mathsf{m}_0) = \int_0^\infty \left[q_t^0(\sigma_i = +1) - \tfrac{1}{2}\right] dt$$

(MT (C.100)), the time-integrated early asymmetry of the frozen law. It is an entire function of the initial mean and changes sign under rotation of the ring by one site, so it vanishes at every rotation-invariant mean, the seed points included. The proof shows it is not identically zero, so its zero set is Lebesgue-null: that is the excluded set of means. After the transient the rows sit at $1/2 + \epsilon_{\mathrm{L}}\mathsf{I}(\mathsf{m}_0) + o(\epsilon_{\mathrm{L}})$; the component along $\mathbf{v}$ grows like $\epsilon_{\mathrm{L}} e^{\alpha\epsilon_{\mathrm{L}} t}$ and reaches order one at $t \approx \alpha^{-1}\epsilon_{\mathrm{L}}^{-1}\log(1/\epsilon_{\mathrm{L}})$, hence the window $t_{\mathrm{L}} = \mathrm{C}_{\mathrm{T}}\epsilon_{\mathrm{L}}^{-1}\log(1/\epsilon_{\mathrm{L}})$ with $\mathrm{C}_{\mathrm{T}} > 1/\alpha$. The theorem states: there is $\mathrm{T}_0 > 0$ such that for every $\mathrm{T} \in (0, \mathrm{T}_0]$ and Lebesgue-almost every m_0 one matching M satisfies

$$\lim_{\epsilon_{\mathrm{L}} \to 0} \sup_{t_{\mathrm{L}} \leq t \leq 2t_{\mathrm{L}}} q_t^{\epsilon_{\mathrm{L}}}(S \neq M) = 0,$$

while for every $\mathrm{T} > 0$ and every m_0 the frozen difference $q_t^0(S = S^{(0)}) - q_t^0(S = S^{(1)})$ tends to zero as $t \to \infty$. The order of limits is fixed temperature, then mobility to zero, with the window scaled by the mobility; the conclusion concerns the one-time law of the assignment on a moving window, not histories, events, or $t \to \infty$ at fixed mobility.

14.4.4 What the corollary and the tracking bound add

MT Corollary 14.6 lets the temperature vanish too. Along any $\mathrm{T}_n \downarrow 0$ with tolerances $\eta_n \to 0$ there are mobility thresholds $d_n \downarrow 0$, the same for almost every mean, below which the selection error on the window (now with $\mathrm{C}_{\mathrm{T}} = 2$ for all temperatures) is at most η_n, the learned law is η_n-close in relative entropy to its instantaneous equilibrium, and the frozen law is η_n-close to its symmetric equilibrium, so the frozen matching probabilities tend to $1/2$. It adds a single joint path $(\mathrm{T}(\epsilon_{\mathrm{L}}), \epsilon_{\mathrm{L}}) \to 0$ and an assignment-law contrast tending to $1/2$ in total variation. It adds no rate: the thresholds exist by a measure-theoretic argument and have no values, they can be taken below T_n^a for any $a > 7/2$ only as a cap, and the matching may change with n, because nothing controls the imprint's sign as the temperature vanishes.

MT Proposition C.15 makes the adiabatic half explicit. For $\mathrm{T} \leq 1/100$ the frozen entropy production dominates $\mathrm{c}_*\mathrm{T}^{5/2}/[4 + 3\log(1/\mathrm{T}) + 8\log(1/p_{\min})]$ times the relative entropy, and

$$\mathrm{TV}(q_t, \mu_{p(t)}) \leq e^{-\ell t/2}\sqrt{\frac{13(12 + |\mathsf{m}_0|^2)}{4\mathrm{T}}} + \frac{2\epsilon_{\mathrm{L}}\mathrm{D}(\mathrm{T}, \epsilon_{\mathrm{L}})}{\mathrm{c}_*\mathrm{T}^{7/2}}, \qquad \ell = \frac{\mathrm{c}_*\mathrm{T}^{5/2}}{\mathrm{D}(\mathrm{T}, \epsilon_{\mathrm{L}})}$$

(MT (C.121)), with D logarithmic. The first term is memory of the initial condition, erased at the mixing rate ℓ; the second is the lag of a law chasing an equilibrium that moves at a rate of order $\epsilon_{\mathrm{L}}/\mathrm{T}$, divided by the mixing rate, which is where the exponent $7/2$ comes from. For $\epsilon_{\mathrm{L}} = \mathrm{T}^{a}$, $a > 7/2$, tracking holds on the window: an explicit adiabatic condition, not a selection result, since whether the equilibrium has moved to a matching depends on the imprint, for which no lower bound as $\mathrm{T} \to 0$ is available.

14.4.5 Persistence, and why it is equilibrium

MT Proposition C.16 assumes the learned law is δ-close to ν_{T}^{M} and the frozen law δ-close to its symmetric equilibrium. Over a window h the learned path law is then within $(\mathsf{h}+2)\delta$ of a stationary Ornstein–Uhlenbeck path held at M, the frozen one within $\delta + 16\mathrm{T} + 31\mathrm{T}\mathsf{h}$ of an equal mixture of the two held-matching paths, and coherence at resolution a fails with probability at most

$$(\mathsf{h}+2)\delta + \mathsf{b}(\mathrm{T}, \mathsf{a}, \mathsf{h}), \qquad \mathsf{b} = 8\sqrt{2/\pi}\,\mathsf{a} + (8 + 3400\mathsf{h})\mathrm{T}/\mathsf{a}^{2}$$

(MT (C.124)). The first term of b is the chance that an exterior residual, of variance of order one, is accidentally below a; the second is the chance that thermal noise pushes an internal residual above a during the window. The choice $\mathsf{a} = \mathrm{T}^{1/4}$, $\mathsf{h} = \mathrm{T}^{-1/4}$ puts the resolution far above the thermal scale (restoring ratio $4\mathsf{a}^{2}/\mathrm{T} \to \infty$) and the window far below the hopping time T^{-1}; the error is then of order $3400\mathrm{T}^{1/4}$, small only far below $\mathrm{T} = 10^{-14}$. The last clause decides the classification: with rows near the matching vertex, the stationary law at those frozen rows realizes the same persistence event with probability at least $1 - 4(\mathsf{h}+2)\delta - \mathsf{b}$. The event is typical at equilibrium, fails the horizon clause of R15, and is restored equilibrium coherence. The coherent-partition laws of the two arms differ by nearly $1/2$ because the frozen arm splits its equilibrium pairs evenly (MT (C.125)).

14.4.6 The early first-block null

MT Proposition C.17 keeps the original contrast detector on the horizon $H_{\mathrm{T}} = 2\mathrm{T}^{-1/8}$ and takes the block that completes first. Coupled with common noise, clock, and initial sample, the arms differ only when a redraw differs, and a row displaced by $\epsilon_{\mathrm{L}} t/2$ changes a redraw probability by at most $\epsilon_{\mathrm{L}} t$. The conditional first-block laws differ by at most $\mathsf{d}_{\mathrm{T}}/\mathsf{c}(\mathsf{m}_0)$, with $\mathsf{d}_{\mathrm{T}} = \min\{1, 8\epsilon_{\mathrm{L}}\mathrm{T}^{-1/4}\}$ and c the seeded bound above (at most of order 10^{-30}), so they agree when $\epsilon_{\mathrm{L}} = o(\mathrm{T}^{1/4})$, at extraordinarily small temperatures; at the original mobility $\mathrm{T}^{1/8}$ the bound is vacuous.

14.4.7 Conditioning, and what a vanishing discarded term means

The late parent starts from the full recognition law conditioned on a completed coherence prefix E on $[t_{\mathrm{L}}, t_{\mathrm{L}}+1]$, pushed through the channel. MT Proposition C.18 bounds the conditioned relative entropy and the optimizing discarded term by

$$\kappa_E = \frac{\eta + h_b(\rho)}{1-\rho}, \qquad h_b(r) = -r\log r - (1-r)\log(1-r),$$

and the physical discarded term by $\kappa_E + \mathrm{T}$, where η bounds the unconditioned relative entropy to equilibrium and ρ the failure probability of the prefix. The binary entropy is the price of conditioning, small only for a typical event. MT Remark C.19 shows the price is real: conditioning a thermal Gaussian on the sign of one discarded coordinate costs $\log 2$ at every temperature, and conditioning on the whole realized history pins the discarded coordinates and makes the relative entropy infinite. A parent is activated from a law given a high-probability event, never from one history. With $\eta, \delta \le \mathrm{T}$ and $\epsilon_{\mathrm{L}} \le \mathrm{T}^2$, $\kappa_E = O(\mathrm{T}^{1/4}\log(1/\mathrm{T}))$.

MT Proposition C.20 compares paths. The parent diffuses on the kept subspace and, at each ring of its own clock, reinserts the discarded coordinates as fresh thermal draws, redraws the assignment from the posterior at the reconstructed configuration, and projects back; its rows learn from its own law. Its path law, including the assignment history and hence the loop, is within

$$2\rho + (\mathsf{h}+3)\delta + \frac{4\mathsf{h}d}{1-4d}, \qquad d = \delta + \epsilon_{\mathrm{L}}\mathsf{h},$$

of the conditioned fine kept-coordinate path law (MT (C.137)), which is $O(\mathrm{T}^{1/4})$ for $\mathsf{h} = \mathrm{T}^{-1/4}$: the cost of conditioning, the distance of both processes from the held-matching stationary path, and the probability that the parent leaves the matching.

14.4.8 The activation clock

On the early seeded episode, at $\mathrm{T} = \epsilon^8$ and any $\epsilon_{\mathrm{L}} \in [0,1]$, MT Proposition C.25 bounds the discarded information at a deterministic activation time θ from both sides. Asymptotically,

$$\mathcal{H}_{\mathrm{opt}}(q^\circ_\theta) \le \tfrac{9}{2}\varrho_\circ^2 e^{-8(\theta-\log(1/\epsilon))} + o(1), \qquad \mathcal{H}_{\mathrm{opt}}(q^\circ_\theta) \gtrsim \frac{\varrho_\circ^2}{7}\mathsf{c}_{\mathsf{m}} e^{8(\log(1/\epsilon)-\theta)} - \log 2 \quad (\theta \le \log(1/\epsilon)),$$

with q°_θ the law conditioned on the seeded prefix and $\mathsf{c}_{\mathsf{m}} > 0$ depending only on the initial mean. The physics is one line. The discarded discrepancy coordinates start at offsets of order $\varrho_\circ$ and decay at rate about 4, while their thermal width is of order $\sqrt{\mathrm{T}} = \epsilon^4$; a displaced Gaussian has relative entropy of order squared offset over variance, $\varrho_\circ^2 e^{-8\theta}/\mathrm{T} = \varrho_\circ^2 e^{-8(\theta-\log(1/\epsilon))}$. So the discarded information is large before the thermalization time $\log(1/\epsilon)$ and small after it. The upper bound needs $\theta - \log(1/\epsilon) \to \infty$ and $\epsilon^4\theta \to 0$, the lower bound diverges when $\log(1/\epsilon) - \theta \to \infty$, and the window $\theta = \log(1/\epsilon) + O(1)$ is undecided. The original witness of MT Proposition 13.22 activates at $2\log(1/\epsilon)$, safely on the good side.

MT Proposition C.26 treats the clock a physicist might pick first: the first time any block meets the contrast criterion. That criterion asks for residuals below about ϵ^2, far coarser than the thermal scale ϵ^4. Seeded residuals fall below ϵ^2 by about $\frac{1}{2}\log(1/\epsilon)$, and near $\frac{1}{4}\log(1/\epsilon)$ when a pair's initial discrepancy is small, so the first completion occurs by $\theta_\circ = \frac{1}{2}\log(1/\epsilon) + 1$. There the discarded information is at least of order $\mathrm{T}^{-5/8}$, driven by seeds whose first pair completes near $\frac{1}{4}\log(1/\epsilon)$ while the other matched pair, whose discrepancy the channel also discards, still carries an imprint of order $\varrho_\circ\epsilon$. The divergence persists when the law is conditioned on the whole contrast event. This

is an established negative result: no parent of this type activated at the first qualifying completion has a vanishing conditional error, in either arm and at any mobility in $[0, 1]$.

MT Proposition C.27 bounds the path error of a parent activated at a good deterministic time by

$$24\exp\Big[-\frac{1}{76800\mathrm{Th}}\Big] + 4\big(1+\sqrt{8/3}\big)\Lambda_{\mathrm{L}}(\theta+\mathsf{h})\mathsf{h}e^{-1/\mathrm{T}}, \qquad \Lambda_{\mathrm{L}}(t) = 2e^{\epsilon_{\mathrm{L}}t} - 1,$$

uniformly in the mean and in $\epsilon_{\mathrm{L}} \in [0, 1]$, with Λ_{L} bounding how far learned rows can tilt a redraw. The first term is the probability that thermal diffusion over the window moves the configuration a distance $1/20$ off its deterministic path, negligible when $\mathrm{Th} \to 0$; the second is the Arrhenius probability of a redraw across a cost gap larger than one. Windows up to $\mathrm{T}^{-1+\mathrm{q}}$, $\mathrm{q} \in (0, 1)$, are covered, and nothing is concluded at order T^{-1}. The late proof cannot be reused here because the early activation law is at total-variation distance essentially one from the stationary kept-coordinate Gaussian (MT Remark C.28), so the early proof follows the energy gap along the conditioned path.

14.5 What is proved, and how solid it is

The first scorecard covers the interpretive layer.

Result	What it says	Status	Scope and caveats
MT Definition 14.1	A posit that changes the law of no record or record statistic is removed.	CHOICE	Declared rule of reading, not derived from constructive empiricism.
Local-first verdict, MT (14.1)	Agents' law points are primary; base, bundle, spaces are scaffolding with no law.	CHOICE	Describes the present architecture; changes if edges become inferred coordinates.
MT (14.3)	Evidence values, loop classes, and per-edge twists up to conjugation survive frame change.	SOLID	Invariance proved in the cited statements; reading it as "structure" is interpretation.
Realist and Neo-Kantian readings	Compatible with the invariants; none selected.	VAGUE	The chapter declines to select and names what support would need.
MT Hypothesis 14.2	All data reach an agent as clamped states or node messages.	CHOICE	Free for kernels by MT Theorem 5.17; removes no conditioning.
MT Hypothesis 14.3	Boundary environment nodes are called agents.	CHOICE	Naming only; adds no state, action, blanket, or free energy.

Result	What it says	Status	Scope and caveats
Participatory reading	Participation via inference loop, learning, delayed feedback; formation could distinguish the program.	VAGUE	Negative half rests on proved identifications; positive half is programmatic.
MT Open Problem 14.7	Identify a coarse-graining object with a physical law.	OPEN	Needs a named target system and held-out evidence; none supplied.
MT Open Problem 14.8	An observation at c_* that could count against the program.	OPEN	Only model-internal separations are proved.

The second covers the continuation line. Every theorem-type statement is proved only on the four-cycle, with abelian translation links on scalar Gaussian means, so all are TOY; the two general items are marked separately.

Result	What it says	Status	Scope and caveats
MT Proposition 14.4	Contrast formation from every Gaussian initial law with probability bounded below; same with frozen rows; symmetric laws keep rows uniform.	TOY	Probability from the matching seed; learning idle; floor about 10^{-31} at $\mathrm{m}_0 = 0$.
MT Theorem 14.5	Fixed small T, $\epsilon_{\mathrm{L}} \to 0$: learned assignment law concentrates on one matching on $[t_{\mathrm{L}}, 2t_{\mathrm{L}}]$ for a.e. mean; frozen arm symmetric.	TOY	Amplifies the initial-law asymmetry; one-time law; no rate in T.
MT Corollary 14.6	Joint limit $\mathrm{T}, \epsilon_{\mathrm{L}} \to 0$ along implicit thresholds; frozen matching probabilities tend to $1/2$.	TOY	Thresholds existential; matching may vary with T_n.
MT Proposition C.15	Entropy-production rate of order $\mathrm{T}^{5/2}$; tracking for $\epsilon_{\mathrm{L}} = \mathrm{T}^a$, $a > 7/2$.	TOY	Tracking only; no power-law selection.
MT Proposition C.16	Selected pair coherent at resolution $\mathrm{T}^{1/4}$ for time $\mathrm{T}^{-1/4}$; partition contrast near $1/2$.	TOY	Restored equilibrium coherence; no entrance claim.
MT Proposition C.17	For $\epsilon_{\mathrm{L}} = o(\mathrm{T}^{1/4})$ learned and frozen first-block laws agree in the limit.	TOY	Silent at the original mobility $\mathrm{T}^{1/8}$.
MT Proposition C.18	Late activation: relative entropy and discarded information $O(\mathrm{T}^{1/4} \log(1/\mathrm{T}))$.	TOY	Averaged over a declared prefix, not per history.
MT Proposition C.20	Parent path law within $O(\mathrm{T}^{1/4})$ of the conditioned fine kept path over $\mathrm{T}^{-1/4}$.	TOY	Finite window; own-law learning; no recursion.

Result	What it says	Status	Scope and caveats
MT Proposition C.25	Discarded information vanishes if $\theta - \log(1/\epsilon) \to \infty$, diverges if $\log(1/\epsilon) - \theta \to \infty$.	TOY	Seeded sub-event; window $\log(1/\epsilon) + O(1)$ open.
MT Proposition C.26	At first qualifying completion, discarded information at least of order $\mathrm{T}^{-5/8}$, also given the whole contrast event.	TOY	Established negative result on the four-cycle.
MT Proposition C.27	Early parent path error vanishes when $\mathrm{Th} \to 0$, uniformly in mean and mobility.	TOY	Windows shorter than T^{-1}.
MT Lemma C.22	Discarded information under conditioning, mixing, and extra variables; a Donsker–Varadhan lower bound.	SOLID	General; standard in substance (chain rule).
MT Remark C.19	Conditioning on a probability-$1/2$ event costs $\log 2$; on a realized history, infinity.	SOLID	General counterexample shown on a Gaussian.

Conditions: which ones matter

What matters is what makes the four-cycle special and what fixes the comparison. The graph has exactly two perfect matchings; links are abelian translations on scalar means, so loop holonomy is the number $\sum_i \omega_i$ and plays no role in any selection; edges carry energy only when attended. The temperature is small and fixed ($\mathrm{T} \le \mathrm{T}_0$ or $1/100$ late, $\mathrm{T} \le 10^{-16}$ for the early null, $\mathrm{T} = \epsilon^8$ with $\epsilon \le 1/100$ for the early parent), and the bounds are asymptotic, some with large constants. The rows learn from the law, not from samples, which makes them deterministic and excludes fluctuation-driven selection: a modeling commitment, not hygiene. The initial law is Gaussian with full support, and "almost every mean" excludes the imprint's zero set, which contains every rotation-invariant mean. The kinetics are a declared realization along which the free energy does not increase, not selected by it; the parent channel is fixed by hand. The late results need $\eta, \delta \le \mathrm{T}$ and $\epsilon_{\mathrm{L}} \le \mathrm{T}^2$; the early ones condition on the seeded sub-event, not the whole contrast event. Hygiene to take on trust: standard Borel spaces, measurability of the first-block mark, well-posedness, finite moments, density envelopes, and form-domain extensions of functional inequalities. Observational closure costs nothing for kernels; the boundary-agent hypothesis is a naming decision.

14.6 Toy versus general

14.6.1 Mechanism and small-network specifics

Every formation statement here lives on four agents; the program's target is 512 or more agents with nonabelian or non-compact groups such as $\mathrm{GL}(K, \mathbb{R})$ or $\mathrm{SU}(N)$. The four-cycle grounds and validates; it does not set direction, and none of its results transfers without proof. Table 14.5

separates plausible mechanism-level content, the companion's reading and conjectural beyond four agents, from four-agent specifics.

Table 14.5: Mechanism-level content versus four-cycle specifics.

Result	Mechanism-level content (conjectural beyond four agents)	Four-cycle specifics
Seeded formation	A favorable initial condition relaxes into a coherent block; the probability is the measure of such conditions.	Seed points $\pm\mathbf{x}^\circ$, floor near 10^{-31}, factors $1/16$, $e^{-1/4}$.
Symmetric laws	A unique equivariant law-level dynamics cannot break a symmetry of its initial data; elementary and general, stated only for the four-cycle.	The form $\mathrm{m}_0 = (\mathrm{c}_1\mathbf{1}_4, \mathrm{c}_2\mathbf{1}_4, 0)$.
Pairing selection	Slow law-level learning is natural-gradient ascent of the log evidence; low temperature favors sheets with fewest stiff directions; a transient imprint decides the branch after time $\epsilon_{\mathrm{L}}^{-1}\log(1/\epsilon_{\mathrm{L}})$.	Two matchings, one unstable direction, ratios $\sqrt{40/7}\mathrm{T}$, $\sqrt{5/3}\mathrm{T}^{3/2}$, $\alpha \geq 7/5$, the imprint computation.
Tracking	Mobility must be small against the frozen mixing rate.	Rate $\mathrm{T}^{5/2}$, exponent $7/2$.
Persistence	A structure selected in the prior is coherent in equilibrium on windows short against the hopping time.	Resolution $\mathrm{T}^{1/4}$, window $\mathrm{T}^{-1/4}$, flux $31\mathrm{T}$.
Early null	Learning slower than formation cannot bias the first block.	Threshold $o(\mathrm{T}^{1/4})$, horizon $2\mathrm{T}^{-1/8}$.
Parent closure	Exact to the extent discarded modes are in conditional equilibrium; activation must follow their thermalization; a detector coarser than the thermal scale fires too early.	Time $\log(1/\epsilon)$, divergence $\mathrm{T}^{-5/8}$, windows $\mathrm{T}^{-1/4}$, $\mathrm{T}^{-1+\mathrm{q}}$.

Some hypotheses exclude the target outright. Scalar fibers mean $K = 1$, and abelian links mean holonomy cannot act, so whether holonomy selects components is not touched; it is one branch of MT Open Problem 13.20. The stationary preference for the fewest stiff directions does generalize within the gated family: minimum edge covers dominate the stationary law on any finite graph (MT Proposition 13.24). But nothing is proved about which of the combinatorially many covers slow learning selects, or whether hopping times between covers grow with N, which is where genuine metastability and a large-N transition could enter; the four-cycle rules none of this in or out. Recursion is out of reach too: eligible blocks have two or three agents, so the pairs cannot coalesce within the event class, and recursive formation in the sense of R14 is posed outside the gated family (MT Open Problem 13.25).

14.6.2 What the four-cycle says about waiting out the transient

The reader's intuition is that meta-agent formation is soft and continuous: wait out the transient and keep the slow-to-relax mode. The four-cycle gives this a precise form at $N = 4$ in three respects and qualifies it in two.

The activation-clock results are the sharpest support. A parent on the seeded episode is exact, as a summary and as a dynamics, when activated at a deterministic time after the discarded

modes have thermalized, and provably not at the first moment a threshold criterion is met (MT Propositions C.25 and C.26). "Wait out the transient" is exactly $\theta - \log(1/\epsilon) \to \infty$, and the negative result says that a first-passage clock driven by a detector coarser than the thermal scale is the wrong clock. The waiting is bounded on the other side too: valid activation times and windows lie below the slow time T^{-1}, so the meta-agent is a quasi-stationary description on a plateau, not a permanent object. Second, what survives the transient late is equilibrium structure: the selected pair is coherent because the equilibrium at the learned prior is paired. Third, the channel does keep the slow directions, relaxing at rates of order T, with the loop and the assignment, and every discarded coordinate is fast.

The first qualification is that the channel was not chosen by a slow-mode criterion: it is supplied, it keeps one fast mode as well, and no statement derives it from relaxation rates. The second cuts deeper. Waiting out the transient selects nothing by itself. In a symmetric system with law-level learning, which pairing forms is a property of sample paths, and at the level of the law nothing is selected unless a datum breaks the symmetry (part (c) of MT Proposition 14.4, rule R13). Soft continuous formation and law-level selection of a culture are different claims; on the four-cycle the first holds and the second needs an asymmetric initial law. Whether large networks add a genuine source of selection, such as branch choice by fluctuations under learning from finite samples, or a transition with diverging hopping times, nothing here addresses.

14.7 Open ends

Physical-law identification (MT Open Problem 14.7) asks for a named target system whose data the model does not generate, a record kernel separating two coarse-graining objects that differ in a gauge-invariant datum, and held-out evidence, robust across blocking schemes, that a baseline margin is or is not met. Single-point testability (MT Open Problem 14.8) asks for a test at c_*, fixed before the data, whose failure would falsify nontrivial holonomy on a walk or formation of a block, or a proof that no record statistic distinguishes the alternatives. What exists is model-internal (the holonomy record of MT Theorem 10.28, the learned-versus-frozen selection); no target system, noise model, estimator, or baseline is supplied.

The central formation open end is MT Open Problem 13.20: at the original coupled mobility $\epsilon_{\mathrm{L}} = \mathrm{T}^{1/8}$, a difference bounded below between the learned and frozen conditional laws of the first formed block, or between two initial means differing only in their loop coordinate, with parents of vanishing errors; or a proof that these differences vanish for almost every mean. The late selection and the early null bracket this regime without entering it. Also undecided are the activation window $\theta = \log(1/\epsilon) + O(1)$, the discarded information given the whole contrast event at a deterministic time, windows of order T^{-1} or longer, and a power-law selection regime, which needs a quantitative lower bound on the imprint. Beyond the four-cycle, repeated endogenous formation (MT Open Problem 13.25), dynamic closure of the pointwise parent (MT Open Problem 11.38), edge creation (MT Open Problem 6.9), and a natural partition selector (MT Open Problem 9.19) remain open; the last two mark where the local-first verdict would change.

Does it deliver what it claims? NARROWER THAN IT SOUNDS

The technical chapter says its readings are definitions, hypotheses, and open problems, and that it proves a proposition and a theorem that sharpen the participatory reading: formation from every Gaussian initial law with learning idle, and law-level pairing selection by slow source learning against a symmetric frozen arm. Every proved statement does exactly what it says, the scope paragraph is accurate, and the interpretive verdicts (local-first, no realist position selected, no agents created by naming) correctly describe the formalism. For the claim that matters to the program, the result is narrower than it sounds. What is selected is one of two pairings of a four-agent ring with abelian links, in the one-time assignment law, by amplifying an asymmetry of the initial law, at fixed small temperature with mobility tending to zero. The selected pair's later coherence is equilibrium coherence, not formation; the early formation that passes the formation test is seeded and unaffected by slow learning; parent closure is proved as averaged, finite-window statements on declared prefixes, and the first-completion clock is refuted. The chapter itself concedes that assignment selection does not establish the recursive participatory claim. The gap is the meeting point of formation and selection, MT Open Problem 13.20, with recursion, large networks, nonabelian holonomy, and contact with any observed system all open.

Chapter 15

Related Work

This chapter covers MT Chapter 15, a positioning chapter with one small established remark, and adds what it does not tabulate: which of the theory's main results are standard mathematics in new typing and which are new.

15.1 The question and the answer

The technical chapter asks where the construction sits among existing work and what it owes. The neighbors flow a distribution along a scale parameter (Bayesian renormalization), coarse-grain graphs whose edges carry scalars (network renormalization, graph coarsening), constrain learned maps over a base manifold (equivariant networks), or share, omit, or fit a group-level model (multi-agent active inference). The present theory computes the parent law from the fine joint by an exact Markov channel at fixed evidence (MT Theorem 9.6, MT Corollary 9.10), with group-valued independent links at one point. The central comparison, with Bayesian renormalization, rests on two statements: evidence bounds of different joints cannot be compared unless their evidences are related (MT Proposition 5.11), and an evidence-preserving channel applied to both the posterior and the recognition law orders the bounds by data processing (MT Corollary 9.7). The scope paragraph states that no other construction is identified with this one and that every qualification of another work is the technical text's reading.

15.2 The picture

Table 15.1 maps the neighbors; the grade refers to the relation between the neighbor's central object and the theory's corresponding object.

The pattern is that the linear algebra is almost entirely standard and the probabilistic and gauge typing around it is the theory's own. Represented-link quadratic energies are sheaf Laplacian energies with rectangular restriction maps, whose kernels are global sections; the coarse operator is that of algebraic multigrid; loops are handled by maximal-tree gauge fixing. The theory places these operators inside normalized joint laws with exact evidence accounting and two channels that may carry inequivalent representations, and proves formation theorems there. Its relation to renormalization in the physicist's sense is weaker than the vocabulary suggests: the main

Table 15.1: Neighbors of the construction.

Neighbor	Its object	Relation here	Grade
Bayesian renormalization	Posterior flow on parameter space along a Fisher information scale; stiff and sloppy directions.	Different construction; composition needs a common target (Open Problem GB.1).	loose
Information-geometric renormalization	Channels from microscopic to observable, a metric on Hamiltonians, monotone correlation functions.	Only the classical Fisher contraction for one kernel (MT Theorem 7.8).	structural
Network (Laplacian) renormalization	Heat kernel $e^{-tL}/\operatorname{Tr} e^{-tL}$, spectral entropy, specific-heat peaks.	Spectra gauge invariant, node affinities not (MT Remark 15.1).	structural
Algebraic multigrid	Galerkin coarse operator $P^\top AP$ with aggregation prolongation.	Same operator, inside normalized laws (MT Proposition 9.14).	exact
Matrix-weighted consensus	Laplacian with matrix edge weights; consensus subspace.	The Gaussian interaction family (Definition G3.1).	exact
Cellular sheaves	Sheaf Laplacian; H^0 as its kernel; Kron obstruction.	Represented-link energies are sheaf energies (MT Proposition 10.17).	exact
Connection Laplacians	Orthogonal transports on a graph; convergence to a manifold operator.	Special case: orthogonal links, scalar weights.	exact
Maximal-tree gauge fixing	Tree links set to the identity; non-tree links carry loop holonomies.	The rooted datum (MT Proposition 10.5).	exact
Equivariant networks	Weight sharing over a global group or local frames on a base manifold.	Links are independent data at one point, not a learned layer.	loose
Multi-agent active inference	Shared models, collective motion, group-level blankets.	Parents computed from the fine joint.	loose
Path-space averaging	Relative-entropy bounds for effective diffusions.	Not growing-history, switching-source comparisons.	structural

manuscript's renormalization is an exact finite-step channel with no rescaling, fixed points, or flow of couplings, and the asymptotic apparatus lives in Supplement B.

Physics analogue (exact)

Kron reduction of a Gaussian network is integrating out Gaussian sites: the Schur complement of the precision is the precision of the marginal. The Kron obstruction of Hansen and Ghrist [37] says that for vector stalks an eliminated site can induce couplings no pairwise restriction maps express, much as decimation generates couplings outside the original family. Supplement G classifies when matrix-weighted Kron reduction stays in its family (Proposition G7.6, Theorem G7.9); the scalar case is closed (Lemma G7.7). Galerkin closure (MT Theorem 10.25) avoids the obstruction by restricting the form to a subspace: block-spin averaging rather than decimation.

Physics analogue (exact)

MT Remark 15.1 is the lattice-gauge fact that a propagator between different sites is not gauge invariant. Under vertex reframings $P = \mathrm{diag}(P_i)$ the generator $R^{-1}L$ is conjugated, so spectrum, heat trace, and spectral entropies are invariant, while off-diagonal heat-kernel blocks transform as $P_i^{-1}(\cdot)P_j$, like $\langle \phi_i \phi_j^\dagger \rangle$ without a connecting Wilson line. A partition selector built on node affinities therefore needs invariant block statistics.

15.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Information scale	In Bayesian renormalization, a scale on parameter space set by Fisher distinguishability.	A renormalization scale, by its authors' analogy.
Stiff and sloppy directions	Parameter directions well and poorly constrained by data.	Relevant and irrelevant directions, loosely.
$D_\theta^{\mathcal{D}}$	Pushed inverse Fisher metric, the diffusion tensor of the data-space flow.	Diffusion constant of a Fokker–Planck equation.
$\chi(t)$	Spectral specific heat $-d\mathcal{S}/d\log t$; peaks mark resolution scales.	Specific heat.
Sheaf, stalk, restriction map	A vector space per vertex and edge, maps from vertex to edge spaces.	Matter field and parallel transport on a lattice.
H^0	Global sections: vertex data consistent across every edge.	Covariantly constant configurations.
Kron reduction	Eliminating vertices by a Schur complement.	Decimation of Gaussian sites.
Galerkin form $P^\top AP$, aggregation	Restricting a form to a subspace of piecewise-constant block fields.	Kadanoff block-spin averaging.
Maximal-tree gauge	Gauge in which the links of a spanning tree are the identity.	Axial or tree gauge.
Evidence-preserving channel	Coarse channel that leaves the observation marginal unchanged.	Integrating out at fixed partition function.

"Renormalization" in the main manuscript means exact finite-step coarse-graining of the latent law; Wilsonian words appear only when describing other authors' work. "Channel" means a classical Markov kernel, not the quantum channel of Bény and Osborne [6]. "Gauge" means passive re-expression of frames at one point with independent links, not a connection on a base manifold, which is why the comparison with gauge-equivariant networks [14] is loose.

15.4 The formulas, explained

The display of Bayesian renormalization the technical text engages is the pushed inverse Fisher metric

$$D^{\mathfrak{O}}_{\theta} = \partial\mathfrak{O}(\theta) g^{F}(\theta)^{-1} \partial\mathfrak{O}(\theta)^{\top}$$

(MT (15.1)), with $\mathfrak{O}$ the map from parameters to data. In the late-data regime the posterior is Gaussian with covariance proportional to $(g^F)^{-1}$; the delta method pushes it to data space with covariance proportional to $D^{\mathfrak{O}}_{\theta}$, which Berman et al. [7] read as the diffusion tensor of a flow on data. The technical text's reading is that this is the leading local approximation of the pushforward, exact for affine $\mathfrak{O}$, and that continuing it beyond the late-data regime is a renormalization construction, not a theorem of Bayesian asymptotics.

The comparison turns on MT Proposition 5.11: for joints indexed by recognition laws Q and Q',

$$\mathcal{L}(Q'; X) - \mathcal{L}(Q; X) = \mathsf{e}(Q') - \mathsf{e}(Q) - \big[D_{\mathrm{KL}}(Q' \| P_{\theta,Q'}(\cdot \mid o, X)) - D_{\mathrm{KL}}(Q \| P_{\theta,Q}(\cdot \mid o, X))\big],$$

with e the log evidence of each joint. A change of bound mixes a change of evidence with a change of tightness, so the optimized bounds of two independently declared joints, such as a model at one scale and a model whose prior is another scale's posterior, are not ordered. An evidence-preserving channel fixes e, and data processing (MT Lemma 3.12) then orders the divergences. In physics terms, variational free energies of different Hamiltonians cannot be compared unless their partition functions are related, and integrating out keeps the partition function fixed. Hence the typing prohibition does not refute Bayesian renormalization, but inserting a posterior as the generative top factor at another scale declares a new model, comparable only through a supplied comparison map.

For network renormalization the key object is the spectral specific heat $\chi(t) = -d\mathcal{S}/d\log t$ of Villegas et al. [74] (MT (15.2)). Spectral quantities of the reframed generator survive frame changes, but a scale diagnostic selects no partition (MT Open Problem 9.19), and the node affinities one would use to build blocks are not invariant. For algebraic multigrid the coarse operator $P^{\top}AP$ with P piecewise constant on disjoint aggregates [57] is the theory's Galerkin form. The technical documents add a criterion for exact coarse message passing, invariance of the M-orthogonal complement of the coarse range under the fine operator (MT Proposition 9.14), and the placement of the operator inside normalized laws with gauge covariance and exact evidence accounting.

15.5 What is proved, and what is standard

The chapter proves one remark; the rest is attribution. The scorecard adds the statements it leans on and the companion's reading of which main results are standard in substance, which is information, not criticism: it tells the reader what a result can and cannot do.

Result	What it says	Status	Scope and caveats
MT Remark 15.1	Spectra and heat traces of the reframed generator are invariant; off-diagonal blocks are not.	SOLID	Elementary similarity; standard in substance.
MT Proposition 5.11	Differences of bounds of different joints mix evidence and tightness.	SOLID	Algebra of the evidence identity; the point is the typing.
MT Corollary 9.7	A recognition-independent channel preserves evidence and orders bounds.	SOLID	Standard in substance (data processing).
MT Corollary 5.7	One evidence bound whose gap is the relative entropy to the posterior, in the extended reals.	SOLID	Standard in substance (Gibbs variational principle); new in its measure-level, many-agent typing.
MT Theorem 9.6	Fine free energy equals coarse free energy plus discarded conditional information.	SOLID	Standard in substance (chain rule).
MT Theorem 11.3	Coarse likelihood as a conditional partition function; composition along a tower.	SOLID	Standard in substance (integrating out), with care about versions.
MT Theorem 7.8	Fisher information contracts under a parameter-independent kernel.	SOLID	Classical, as the technical text says.
MT Proposition 10.17	The kernel of the represented-link energy is the holonomy-fixed sector.	SOLID	Standard in substance (sheaf H^0, cited).
MT Proposition 10.5	Tree links gauge to the identity; non-tree links carry loop holonomies.	SOLID	Standard (maximal-tree gauge fixing, cited).
Proposition G3.2	Energy and kernel criterion for matrix-weighted Laplacians.	SOLID	Standard in substance (consensus literature, cited).
MT Theorem 10.28	A declared record's law separates loop-holonomy classes.	SOLID	New as a construction; elementary proof.
Open Problem GB.1	Relate Bayesian renormalization to Gaussian aggregation through a common monotone.	OPEN	Needs a common state space and a transported monotone.

Conditions: which ones matter

The comparisons are claims about other authors' work; their accuracy depends on the cited sources, which the companion has not rechecked, and the scope paragraph marks each qualification as the technical text's reading. The Bayesian-renormalization contrast rests only on the evidence identity, not on how the Gaussian pushforward is read. The sheaf and connection-Laplacian identifications hold for quadratic energies with represented links, not for

the non-Gaussian laws of the general theory, where only the kernel and the holonomy-fixed sector are claimed.

15.6 Standard versus new

Standard in substance, and valuable to know as such, are the evidence-bound identity and its gap (the Gibbs variational principle), the local-global potential identity (coordinate ascent in mean-field variational inference, MT Theorem 5.23), attention as the posterior of a latent source label (mixture-model responsibilities, MT Proposition 6.2), the exact coarse free energy (the chain rule), the effective action (integrating out), Fisher contraction, the kernel of represented-link energies (sheaf cohomology in degree zero), maximal-tree gauge fixing, the Galerkin operator of algebraic multigrid, matrix-weighted Laplacians, and the Kron obstruction for vector stalks. For each, the technical documents supply careful typing: measure-level statements in the extended reals, two channels with independent links, and a separation of Markov channels, energy restriction, and parameter aggregation. The typing is the contribution, and it is where the documents' length comes from.

New in content are the formation results: the finite-depth theorem under a supplied hierarchy (MT Theorem 13.17), the endogenous four-cycle witness (MT Proposition 13.22), the depth-one result for the gated family (MT Proposition 13.24), and the selection and parent line of chapter 14. Their tools are standard (the Harris theorem, logarithmic Sobolev inequalities, couplings, saddle escape) and their statements new, within the scopes chapters 13 and 14 describe. New as constructions, with elementary proofs, are the operational holonomy record and the use of the no-distinguished-target identity against posterior-flow constructions. Supplement G's classification of Kron closure is set against cited prior art for the obstruction and the scalar case; its assessment belongs to the companion's Supplement G chapters.

15.7 Open ends

The bridge between Bayesian renormalization and Gaussian aggregation (Open Problem GB.1) needs a common state space, a map between the Bayesian scale parameter and a block ratio, and a functional proved monotone along both constructions, or a Gaussian hierarchy in a stated class with no such functional. A combined network scheme, sketched but not posed as a numbered problem, would infer a partition from gauge-invariant diagnostics of edge-event laws through a structural channel with a named law and push the complete law exactly; its partition part is MT Open Problem 9.19. The growing-history, switching-source path comparison that averaging theory does not supply is part of MT Open Problem 11.38.

Does it deliver what it claims? DELIVERED

The technical chapter says it positions the construction among existing work without identifying it with any of them, with the Bayesian-renormalization comparison resting on the no-distinguished-target identity and on evidence-preserving channels. It does that: each comparison is anchored to a statement of the theory, the Bayesian-renormalization contrast follows correctly from the evidence identity, and the one proved remark is correct and elementary. Two limits. The chapter does not say, result by result, which of the theory's statements are standard mathematics in new typing; the companion's reading is that most of the linear algebra and the variational identities are, and the formation results are new. And two neighbors one would expect here, matrix-weighted consensus and maximal-tree gauge fixing, are treated in Supplement G and the network-state chapter rather than in this chapter. The attributions to other authors are the technical text's reading and have not been rechecked by the companion.

Part II

The Supplements

Chapter 16

Supplement G, Part 1: The Gaussian Realization and Its Geometry

Supplement G asks what the single-point theory becomes when every belief and model law is multivariate Gaussian (Chapter G0). This chapter covers its introduction and its first five chapters; aggregation, Kron reduction, nonflat links, the operator arrow, and reciprocal pairs are in chapter 17. Three facts frame what follows. The supplement is a realization, not the ambient theory: it asserts nothing for a non-Gaussian belief fiber beyond the exponential-family statements of G1, and the main manuscript uses none of its results. The Gaussian lives on real fibers, $\mathbb{R}^K$ for beliefs and $\mathbb{R}^{d_m}$ for model presentations with $K \neq d_m$ allowed, and the structure group reaches them only through its represented image $\Gamma_b = \rho_b(G) \subseteq \mathrm{GL}(K)$; some statements hold for every Γ_b, and a named set is proved under the specialization $\Gamma_b = \mathrm{GL}^+(K)$ (Hypothesis G2.1). And the belief precision analyzed throughout is that of the fast belief at frozen model latents, $p(k \mid m, o, X)$, not the joint belief–model precision.

Most of the mathematics is finite-dimensional linear algebra and Gaussian integration, much of it textbook in substance. What is specific to the program is the typing (which objects are laws, which are operators, which are artifacts of a chart or frame) and the analysis of when a directed linear-Gaussian network yields the precision a physicist would write first, a matrix-weighted graph Laplacian plus local mass terms. Only Proposition G1.3 is tagged STANDARD; every other statement in G1 to G5 is ESTABLISHED with a proof, and none rests on computation. In one line each: G1 shows that boundary points and operator fixed points carry no law; G2 gives the information form, the forest precision, and one Schur lemma behind every later marginal; G3 shows that the Laplacian-plus-self-term shape is a declared restriction and says when it holds, exactly in one chart and at law level on forests and pairs; G4 prices recognition restrictions and finds that none selects a partition; G5 separates law-level invariants from chart and frame artifacts.

16.1 Exponential families and the operator layer (G1)

16.1.1 The question and the answer

Later chapters apply algebraic maps (aggregation, rescaling, Schur complementation) to matrices in a cone, and a matrix in the cone need not be the precision of any normalizable law. Chapter G1 asks what happens when a map of natural parameters leaves the natural domain N. On its boundary

nothing is left: log partition, mean parameter, Fisher metric, relative entropies, and evidence bound diverge or are undefined (Proposition G1.7); a map can keep every iterate inside while its only fixed point is on the boundary (Proposition G1.8); and pinning, quotienting, and keeping a mass coordinate are three different repairs (Section G1.3). Inside, relative entropy is a Bregman divergence (Proposition G1.15) and affine mean constraints have a unique information projection (Proposition G1.16).

16.1.2 The picture

Physics analogue (exact)

An exponential family is a Gibbs ensemble: the statistic T lists observables, the natural parameter λ their conjugate fields, $\mathsf{A}(\lambda)$ is a Massieu function, $\nabla\mathsf{A}$ the thermodynamic averages, and $\nabla^2\mathsf{A} = \mathrm{Cov}(\mathsf{T})$ the fluctuation–response relation, which is also the Fisher metric (Proposition G1.3). For the Gaussian, $\mathsf{T} = (y, yy^\top)$ and $\lambda = (\mathsf{h}, -\frac{1}{2}\Lambda)$, a source and a coupling matrix. The information projection is the family-restricted form of minimum relative entropy [41].

The boundary results are an infrared statement. For the scalar Gaussian with $\mathsf{T}(z) = -z^2/2$, λ is the stiffness and $\mathsf{N} = (0, \infty)$; the map $\lambda \mapsto \lambda/2$ keeps every iterate normalizable, but its limit $\lambda = 0$ is the flat weight on the line ((G1.8)), inside the operator cone $[0, \infty)$ and outside the probability layer.

Physics analogue (structural)

This is the massless limit of a free field, whose zero mode has no normalizable measure. Physicists fix a boundary value, remove the zero mode, or keep a small mass as a running coupling; G1's pinning (Definition G1.9), quotienting (Definition G1.11), and mass sector (Definition G1.13) are these three repairs, and they give three different objects. A pseudoinverse and pseudodeterminant in the full-space density formula compute the quotient normalizer, on a smaller space with another reference measure, while the full-space integral stays infinite (Proposition G1.12).

16.1.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Canonical family $(\nu, \mathsf{T}, \lambda, \mathsf{A}, \mathsf{N})$	Laws $\propto e^{\langle\lambda,\mathsf{T}\rangle}$ relative to ν; N is where $\mathsf{A} < \infty$	Gibbs ensemble
Regular, minimal	N open; no affine redundancy, so parameters determine laws	No redundant coordinates
Operator cone $\mathcal{W}$; probability layer $\mathcal{W} \cap \mathsf{N}$	Parameters preserved by the maps; the part that denotes laws	Couplings with unphysical points; normalizable couplings

Term or symbol	Plain meaning	Physics analogue
Bregman divergence; information projection	Error of the linear extrapolation of A; closest member under affine mean constraints	Availability; maximum entropy

Traps: the sans-serif A is the log partition, the italic A_i of later chapters are self terms; the bold $\boldsymbol{\mu}$ is a mean parameter, for the Gaussian the pair (μ, M_2) of mean and second moment (Definition G2.3); "operator" means a matrix acted on by an algebraic map, nothing quantum.

16.1.4 The formulas, explained

Inside the domain,

$$D_{\mathrm{KL}}(Q_{\lambda'}\|Q_\lambda) = \mathsf{A}(\lambda) - \mathsf{A}(\lambda') - \langle \nabla\mathsf{A}(\lambda'), \lambda - \lambda' \rangle$$

(Proposition G1.15): the difference of log partitions minus the first-order change predicted from λ'. Convexity makes it nonnegative; for nearby parameters it is the Fisher quadratic, and at finite separation it is not, a difference the supplement insists on. For a closed affine set of mean parameters meeting the interior, the projection $Q_{\widehat{\lambda}}$ is unique and every constraint-satisfying member obeys

$$D_{\mathrm{KL}}(Q_{\lambda'}\|Q_\lambda) = D_{\mathrm{KL}}(Q_{\lambda'}\|Q_{\widehat{\lambda}}) + D_{\mathrm{KL}}(Q_{\widehat{\lambda}}\|Q_\lambda),$$

as a squared distance splits under orthogonal projection (Proposition G1.16); the Bernoulli examples of Remark G1.17 show that interior feasibility, closedness, and convexity are each needed. The scope paragraph is a real warning: no coarse-graining statement follows from exponential-family structure alone, since an operator arrow needs, among other things, a declared rescaling, and a law-level arrow needs finite normalizers at every scale.

16.2 The Gaussian realization at the base point (G2)

16.2.1 The question and the answer

Chapter G2 asks what the directed law becomes when every kernel is linear Gaussian, and what the precision blocks mean. The precision is at once the coupling matrix and the Fisher metric of the mean (Proposition G2.4); a zero block is conditional independence, not zero covariance (Proposition G2.5); the marginal precision of a set of directions is a Schur complement below the restricted precision, with equality exactly under Λ-orthogonality (Lemma G2.6). A forest assembles into an explicit sparse precision (Proposition G2.16), every kernel is frame covariant (Proposition G2.17), and the learning step has a closed form that fails to exist in two named cases (Proposition G2.18).

16.2.2 The picture

Physics analogue (exact)

In $q \propto \exp(\mathsf{h}^\top y - \frac{1}{2} y^\top \Lambda y)$, Λ is the quadratic part of a lattice action (the inverse propagator), h an external source, and $\Sigma = \Lambda^{-1}$ the propagator. A zero off-diagonal block means no direct bond while the propagator still connects the sites through paths ((G2.15)). The Schur complement is integrating out, exact at tree level for a Gaussian. The inequality $\Lambda_{bb}^{-1} \preceq (\Lambda^{-1})_{bb}$ (Corollary G2.7) is the Le Chatelier–Braun inequality: a block fluctuates less with the rest clamped than with the rest free to relax, which is why a factorized recognition law, whose block covariances are the clamped ones, underestimates every marginal fluctuation.

The directed model (Definition G2.13) is a state-space model on a tree: each nonroot belief is its parent's belief carried by the link Θ_{ij}, multiplied by a gain F_i, shifted by a bridge term $\mathsf{B}_i m_i$ and an offset, plus noise of precision Λ_i, with linear-Gaussian observations. The product $\Omega_i = F_i \Theta_{i\,\mathrm{par}(i)}$ is the transition. At frozen model latents the belief sector is the forest $k_i \mid k_{\mathrm{par}(i)} \sim \mathcal{N}(\Omega_i k_{\mathrm{par}(i)}, \Lambda_i^{-1})$, and model latents, offsets, and the model part of observations move only h: model presentations act on beliefs as an external source, never as a coupling. This is the Born–Oppenheimer move of the main manuscript's fast–slow split (MT Section 8.2), cleaner than usual because the fast precision does not depend on the frozen variables at all.

16.2.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
(h, Λ); $\mu = \Lambda^{-1}\mathsf{h}$, $\Sigma = \Lambda^{-1}$	Information vector, precision; mean, covariance	Source, couplings; propagator
Four charts	Information (h, Λ), natural $(\mathsf{h}, -\frac{1}{2}\Lambda)$, expectation (μ, M_2), moment (μ, Σ)	Choices of thermodynamic variables
F_i, Θ_{ij}, Ω_i	Gain, represented link, transition	Transfer matrix times transporter
Restricted versus marginal precision	$E^\top \Lambda E$ versus $(E^\top \Lambda^{-1} E)^{-1}$	Clamped versus relaxed stiffness
M-coordinate	Maximizing expected complete log likelihood over generative parameters	Learning (EM's M-step)

Traps: one subscript on Λ is a kernel's innovation precision, two a block of the assembled matrix (the W_i of G3 follows the same rule); symmetric-matrix coordinates scale off-diagonal entries by $\sqrt{2}$; the moment chart is not Legendre dual to the natural chart, the expectation chart is; "pullback" means restriction along an injective parameter map.

16.2.4 The formulas, explained

The log normalizer $\mathsf{A}(\mathsf{h}, \Lambda) = \frac{1}{2}\mathsf{h}^\top \Lambda^{-1}\mathsf{h} - \frac{1}{2}\log\det\Lambda + \frac{n}{2}\log 2\pi$ is a source energy mediated by the propagator, a fluctuation determinant, and a measure constant; the Fisher information in μ at fixed Σ is Λ (Proposition G2.4), so the stiffness of the energy along a direction is the distinguishability of mean shifts along it. The conditional mean $\mathbb{E}[Y_b \mid Y_{-b}] = \mu_b - \Lambda_{bb}^{-1}\sum_{c\neq b}\Lambda_{bc}(y_c - \mu_c)$, with conditional covariance Λ_{bb}^{-1}, is the local equilibrium of block b in its neighbors' field, and as an update the coordinate-ascent step of G4. With an agreement coupling $\Lambda_{bc} = -W_{bc} \preceq 0$, a neighbor's deviation pulls the block toward it in the Λ_{bb} inner product but not necessarily in the Euclidean one (Proposition G2.5, part (iii)): agreement is relative to the receiver's own metric, the only one available under general linear frames.

The forest precision has the blocks

$$\Lambda_{i\,\mathrm{par}(i)} = -\Lambda_i\Omega_i, \qquad \Lambda_{ii} = \Lambda_i + \sum_{i'\in\mathrm{ch}(i)} \Omega_{i'}^\top \Lambda_{i'}\Omega_{i'},$$

and zero between nonadjacent agents (Proposition G2.16): each factor puts its noise stiffness on the child, a bond $-\Lambda_i\Omega_i$ to the parent, and a back-reaction on the parent. The bond need not be symmetric, which is the seed of G3. The belief–model cross block is $K \times d_m$, so no family of square agent blocks hosts the joint precision when $K \neq d_m$. Frame covariance (Proposition G2.17) is lattice-gauge covariance of a matter action, with gains transforming by conjugation, covariances by congruence, and observation maps absorbing the inverse frame; it holds for every Γ_b, and for $\Gamma_b \subseteq \mathrm{O}(K)$ ordinary eigenvalues are also invariant (Proposition G2.2), which fails under $\mathrm{GL}^+(K)$.

The learning step (Proposition G2.18) is least squares plus residual covariance, $\mathbb{E}\log\mathcal{N}(t; Fw, \Sigma) \leq -\frac{1}{2}\log\det(2\pi e\Sigma_*)$, attained when $\Sigma_* \succ 0$; an exact regression in some direction makes the likelihood unbounded, the variance-collapse degeneracy of maximum likelihood. The named cases are a record factor at one observed value and a link learned inside the invertible group, whose supremum can sit at a singular map outside the group: for $K = 1$ with independent centered k_i, k_j, the optimized term $-\frac{1}{2}\log(2\pi e(\mathbb{E}k_i^2 + \mathsf{t}^2\mathbb{E}k_j^2))$ increases as $\mathsf{t} \to 0$ and is never attained. Noncompactness of the link group can make learning ill posed; the scalar witness says nothing about frequency at large K. The communication formulas (Proposition G2.19, Proposition G2.20) are Gaussian conditioning and the Gaussian channel; the record's mutual information is gauge invariant and at rank one takes every value in $(0, \infty)$.

16.3 The interaction family (G3)

16.3.1 The question and the answer

The coarse-graining results of the second half act on one shape of precision, the interaction family $\mathcal{I}(V)$ (Definition G3.1):

$$\Lambda_{ii} = A_i + \sum_{j\neq i} W_{ij}, \qquad \Lambda_{ij} = -W_{ij} \quad (i \neq j), \qquad A_i \succeq 0, \quad W_{ij} = W_{ji} \succeq 0.$$

Chapter G3 asks when the belief precision of G2 has this shape. It is a restriction, not a consequence of linear-Gaussian directedness (Proposition G3.3, Proposition G3.7). With orientation-preserving frame changes, positive-determinant transitions on a forest can be gauged away and a pair can be brought into the family exactly when $(-1)^K \det \Lambda_{12} > 0$ (Proposition G3.8). A reversing gain stays outside in every chart reached by orientation-preserving frames, and a cycle with the wrong determinant sign, or for odd K the three-agent v-structure, in every chart reached by any frames (Proposition G3.9). In the link-trivializing chart two computable conditions decide membership (Proposition G3.10), and there the family is adopted as a standing hypothesis (Hypothesis G3.13).

16.3.2 The picture

A member's quadratic form is

$$y^\top \Lambda y = \sum_i y_i^\top A_i y_i + \sum_{i<j} (y_i - y_j)^\top W_{ij} (y_i - y_j)$$

(Proposition G3.2): K-component site variables, matrix springs W_{ij} penalizing disagreement, and pinning or mass terms A_i. Consensus configurations cost only $v^\top(\sum_i A_i)v$; they are the zero modes of the Laplacian part L, lifted by self terms alone, and at $A = 0$ they are the non-normalizable zero modes of G1. A singular bond leaves disagreement free along its kernel, so $\ker L$ can exceed the consensus subspace on a connected graph. With definite bonds under flat comparison, one definite self term per connected component makes $\Lambda \succ 0$ (Proposition G3.15).

Physics analogue (exact)

For $K = 1$, $W_{ij} = J$ on lattice bonds and $A_i = m^2$, a member is the massive lattice Laplacian $-\Delta + m^2$ of the lattice Gaussian free field; for general K it is the matrix-weighted Laplacian of the consensus literature [71]. With nontrivial links the agreement energy $\sum (k_i - \Theta_{ij} k_j)^\top W (k_i - \Theta_{ij} k_j)$ is the covariant lattice Laplacian of a matter field in a background gauge field [16]: the connection Laplacian of Singer and Wu [66] for orthogonal links, a sheaf Laplacian [37] for general linear ones. The program's links are independent edge variables, as lattice links are, so this is exact at the level of the quadratic form. G3's family is what the covariant Laplacian becomes in the pure-gauge sector: when every loop product is the identity the links are a coboundary $\Theta_{ij} = U_i U_j^{-1}$ (MT Proposition 2.13), and $z_i = U_i^{-1} k_i$ turns every residual into $U_i(z_i - z_j)$ (Hypothesis G3.4), giving an ordinary matrix-weighted Laplacian with weights

$U_i^\top W U_i$. With a nontrivial Wilson loop no gauge does this; that case is excluded here and treated in chapter 17.

16.3.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
$\mathcal{I}(V)$, A_i, W_{ij}, L	Interaction family; self term; edge weight; Laplacian part	Massive matrix-weighted Laplacian; mass; bond
Link-trivializing chart	Coordinates $z_i = U_i^{-1} k_i$ under flat comparison, unique up to one right factor per component	Gauge setting every link to one
Block reframing $\mathbf{P} = \mathrm{diag}(\mathbf{P}_i)$	Frame change agent by agent, $\Lambda \mapsto \mathbf{P}^\top \Lambda \mathbf{P}$	Site-dependent passive gauge transformation
Gain absorption; orientation criterion	Frames turning every transition into the identity; $(-1)^K \det \Lambda_{12} > 0$	Maximal-tree gauge; ferromagnetic bond sign ($K = 1$)
Chart-independent witness; v-structure	Outside the family in every admissible chart; a child with two parents	Frustrated plaquette; explaining away

Traps: Ω_i is a transition, not a holonomy or frequency; $W_i := \Lambda_i \Omega_i$ with one subscript is the candidate weight of the edge to the parent, W_{ij} an edge weight; "flat" means trivial loop products of links at the one base point, with no base manifold in sight; an "admissible" chart is one reached by block reframings in the group a statement names.

16.3.4 The formulas, explained

Why the shape is not automatic is visible in one block. A receiving factor contributes the bond $-\Lambda_i \Omega_i$, and membership needs it to be $-W$ with W symmetric positive semidefinite, which nothing in a linear-Gaussian model forces. Even with links removed the gain remains: $\widetilde{\Lambda}_i = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ with the quarter-turn $\widetilde{\Omega}_i = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ gives the bond $\left(\begin{smallmatrix} 1 & -2 \\ 3 & -1 \end{smallmatrix}\right)$, and $F_i = -I$ gives a symmetric bond of the wrong sign (Proposition G3.7). In a fixed chart every off-diagonal block must be symmetric, $\binom{N}{2} K(K-1)/2$ linear conditions, so the family is Lebesgue-null once $K, N \geq 2$ (Proposition G3.3).

Thinness concerns one chart; the law-level question is whether some frame change puts the precision into the family, and the answer is gauge-theoretic. A reframing $k_i = \mathbf{P}_i z_i$ sends Ω_i to $\mathbf{P}_i^{-1} \Omega_i \mathbf{P}_{\mathrm{par}(i)}$, the transformation law of a link (Proposition G3.8, part (o)): gain times link behaves as one link, and no gauge transformation can tell a gain from a link. On a tree every link configuration is pure gauge, so frames $\mathbf{P}_r = I$ at roots and $\mathbf{P}_i = \Omega_i \mathbf{P}_{\mathrm{par}(i)}$ turn every residual into $z_i - z_{\mathrm{par}(i)}$, leaving pure agreement bonds $W = \mathbf{P}_i^\top \Lambda_i \mathbf{P}_i$ and self terms only at roots: the maximal-tree gauge applied to transitions. The frames must lie in the frame group, which under $\mathrm{GL}^+(K)$ requires $\det \Omega_i > 0$; for a forest with invertible transitions this is also necessary (part (iii)). For a pair the only

obstruction is the sign

$$(-1)^K \det \Lambda_{12} > 0$$

(part (ii)). Informally: a member with invertible bond has $\Lambda_{12} = -W$, $W \succ 0$, so $(-1)^K \det \Lambda_{12} = \det W > 0$, and a reframing multiplies $\det \Lambda_{12}$ by $\det \mathbf{P}_1 \det \mathbf{P}_2 > 0$. The sign is a $\mathbb{Z}_2$ gauge invariant of the edge. Admitting orientation-reversing frames, $\Gamma_b = \mathrm{GL}(K)$, removes it (part (iv)).

Physics analogue (exact)

For $K = 1$ this is the gauge structure of Ising spin glasses [52]. The family is the set of precisions with ferromagnetic couplings $\Lambda_{ij} \leq 0$ and nonnegative self terms; positive rescalings cannot flip a bond sign (the orientation criterion); $k_i \mapsto -k_i$ is the Mattis gauge transformation, which makes any sign pattern on a tree ferromagnetic; and on a loop the product of bond signs is invariant (frustration). Proposition G3.9(ii) is the $K \times K$ version: the sign of the product of block determinants around a cycle of $n_\circ$ agents is invariant under every $\mathrm{GL}(K)$ reframing, and membership forces it to be $(-1)^{Kn_\circ}$. Part (iii) reads explaining away as frustration: two parents of one child are coupled through $\Lambda_{jj'} = \Omega_j^\top \Lambda_i \Omega_{j'}$, and the triangle has the wrong determinant sign for every odd K. The correspondence is exact at the level of these signs and says nothing beyond them.

In the link-trivializing chart membership is decided exactly (Proposition G3.10): each W_i must be symmetric positive semidefinite, and each self term

$$A_i = \Lambda_i - W_i + \sum_{i' \in \mathrm{ch}(i)} \left(W_{i'} \Lambda_{i'}^{-1} W_{i'} - W_{i'} \right)$$

must be positive semidefinite (no $-W_i$ at a root), with $A_i + O_i \succeq 0$ once observation precisions are added. $\Lambda_i - W_i$ is what a factor leaves on the child after paying for its bond, and $W_{i'} \Lambda_{i'}^{-1} W_{i'} - W_{i'}$ what each child's factor leaves on the parent. For a scalar transition ω these are $\Lambda(1-\omega)$ and $\Lambda\omega(\omega - 1)$: a contracting gain leaves a negative self term on the parent, an expanding one on the child, and only $\omega = 1$ is safe factor by factor (Proposition G3.11, Corollary G3.12). Globally a root prior or observation can pay for a negative contribution, so factorwise membership is strictly stronger than the hypothesis (Remark G3.14). A physicist can read the condition as a ban on net tachyonic self terms.

Two consequences need emphasis. First, the adopted hypothesis is stronger than law-level membership. It requires membership in the link-trivializing chart, and the scope paragraph says nothing selects another chart. The prior of part (iii) of Proposition G3.10, with transition $\frac{1}{2}$, is outside the family in that chart and, by gain absorption, inside it in its gain-absorbing chart. The choice matters downstream: aggregation identifies agents by equal coordinates in the chosen chart, so the chart decides whether agreement means link-transported equality $k_i = \Theta_{ij} k_j$ or equality with the prediction $k_i = \Omega_i k_{\mathrm{par}(i)}$. That is a modeling decision, and the supplement makes it explicitly. Second, membership is not invariant under general block reframings, which preserve inertia but not the symmetry of off-diagonal blocks (Remark G3.6); frame-independent statements need the invariants of G5.

16.4 Recognition restrictions (G4)

16.4.1 The question and the answer

Fix a Gaussian model with posterior $p^\star = \mathcal{N}(\mu, \Lambda^{-1})$, $\Lambda \succ 0$. By the restriction principle (MT Proposition 5.9), shrinking the recognition family lowers the optimized bound by the rise in minimal relative entropy to the posterior. Chapter G4 computes that loss for covariance factorization (the determinant gap, Theorem G4.3), for mean restrictions (Proposition G4.8), and for identifying agents ($+\infty$, Proposition G4.12), and shows that the gap selects no structure (Proposition G4.16).

16.4.2 The picture

For a Gaussian candidate,

$$D_{\mathrm{KL}}(Q\|p^\star) = \tfrac{1}{2}\left[\mathrm{Tr}(\Lambda\Sigma_Q) + (\mu_Q-\mu)^\top\Lambda(\mu_Q-\mu) - n - \log\det\Lambda - \log\det\Sigma_Q\right]$$

(Proposition G4.1). The mean enters only the quadratic term and the covariance only the trace and log determinant, so mean and covariance restrictions are separate problems whose costs add (Corollary G4.10).

Physics analogue (exact)

The bound's gap is the slack in the Gibbs–Bogoliubov–Feynman inequality. For a Gaussian field with a block-factorized trial law, the optimal means are exact and the optimal block covariances are the inverse on-block stiffnesses Λ_{bb}^{-1} (Theorem G4.2): Weiss mean-field theory, which drops the feedback of fluctuations through the environment, with the gap as its free-energy error [9, 75]. The mean-restriction cost $\frac{1}{2}\mu_\perp^\top(B_\perp^\top\Lambda^{-1}B_\perp)^{-1}\mu_\perp$ is the free-energy cost of constraining a Gaussian field's mean: half the squared displacement weighted by the inverse relaxed susceptibility. The bare stiffness $B_\perp^\top\Lambda B_\perp$ overstates it unless the constrained directions are Λ-orthogonal to the rest ((G4.23)).

For two blocks the gap is their mutual information, $-\frac{1}{2}\sum_a \log(1-\check{\rho}_a^2)$ in the canonical correlations (Proposition G4.4); for more blocks it is the dual total correlation $h(Y) - \sum_b h(Y_b \mid Y_{-b})$ of Han [36] (Corollary G4.5), built from the conditional entropies that Λ_{bb} encodes, not the total correlation built from marginal ones (for three unit scalars with pairwise correlation $\frac{1}{2}$, $\frac{1}{2}\log\frac{27}{16}$ against $\frac{1}{2}\log 2$). The gap is observation free because a data-independent joint precision makes conditioning on o move only the posterior mean (Corollary G4.7); the mean cost is quadratic in the data (Corollary G4.9).

Identification of agents is the case that matters for coarse-graining. Let $\mathrm{range}\, S$ be the cluster-constant configurations of a hard aggregation, of dimension $r = n_c K < n$. A recognition law making all agents of a cluster literally equal lives on that subspace, to which the posterior, having a Lebesgue density, gives probability zero. Absolute continuity fails (clause (H3) of MT Hypothesis 5.3), and the relative entropy is $+\infty$. Thickening by a variance ε in each identified direction gives

$$\mathbb{D}_\varepsilon = \mathbb{D}_{\mathrm{tie}} + \frac{n-r}{2}\log\frac{1}{\varepsilon} - \frac{n-r}{2} + \frac{1}{2}\log\det(B_\perp^\top\Lambda^{-1}B_\perp) + \frac{\varepsilon}{2}\mathrm{Tr}(B_\perp^\top\Lambda B_\perp)$$

(Proposition G4.12), one $\frac{1}{2}\log(1/\varepsilon)$ per frozen continuous coordinate, as a delta-function constraint costs entropy logarithmic in the resolution. Identification is therefore a generative construction, the fine law conditioned on zero within-cluster disagreement (Proposition G7.4); at finite cost a recognition law can only tie its means, the mean-tie family with cost $\mathbb{D}_{\mathrm{tie}}$ (Corollary G4.11). Restricting the energy to $\operatorname{range} S$ and renormalizing, the "hard trace," changes the evidence (Proposition G4.13); with a data-independent joint precision its posterior is the image of the fine one under $P_\Lambda = (S^\top \Lambda S)^{-1} S^\top \Lambda$, an evidence-preserving coarse channel (MT Corollary 9.7).

16.4.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Recognition restriction; partition $\mathcal{P}$	Shrinking the candidate family; blocks with block-diagonal candidate covariance	Trial ensemble; mean-field clusters
Cost, gap	Loss of optimized bound, the rise of minimal relative entropy	Variational free-energy error
Mean tie; identification	Candidate means cluster constant; candidate supported on $\operatorname{range} S$	Constrained order parameter; delta-function constraint
Hard trace; P_Λ	Energy restricted to $\operatorname{range} S$; the map carrying the fine posterior to it	Constrained partition function; Galerkin projection

Traps: "restriction" restricts the variational family, not a tensor; "structure" is the partition used by the recognition law; the orthonormal basis B is unrelated to the bridge B_i; "understates the covariance" is said of the optimal factorized law, never of the model.

16.4.4 The formulas, explained

The determinant gap,

$$\log p_\theta(o \mid X) - \mathcal{L}^\star_{\mathcal{P}} = \frac{1}{2}\left[\sum_{b\in\mathcal{P}} \log\det \Lambda_{bb} - \log\det \Lambda\right] \geq 0,$$

compares the product of on-block stiffness determinants with the full determinant; by Fischer's inequality it vanishes exactly when every cross block does. When every block is a union of whole agents, a block reframing shifts both terms by the same Jacobian, so the gap is invariant under every reframing, whatever the frame group (Proposition G5.9, part (iii)); a partition that splits an agent is not.

Does any analyzed cost select a partition scale? No. The gap grows under refinement by an explicit nonnegative amount (Proposition G4.14) and vanishes at one block, so minimizing it selects the unrestricted family (Proposition G4.16). Non-nested partitions are ordered by no structural

feature: for $\Lambda = \left(\begin{smallmatrix} 1 & a & 0 \\ a & 1 & b \\ 0 & b & 1 \end{smallmatrix}\right)$ the partitions $\{\{1,2\},\{3\}\}$ and $\{\{1\},\{2,3\}\}$ have equal block sizes and swap order when $|a|$ and $|b|$ are exchanged (Proposition G4.15). The mean-tie cost vanishes at the finest partition and identification is infinite for every nontrivial cluster. Each cost is extremized at an endpoint, the Gaussian instance of MT Open Problem 5.14 and MT Open Problem 9.19. One functional the chapter writes down is not examined in this role: the combined cost $\mathbb{D}_{\mathrm{tie}} + \mathbb{D}_{\mathrm{fact}}$ of Corollary G4.10 (mean-tie cost plus cluster factorization gap, both displayed in Corollary G4.11), with both restrictions indexed by the same clusters, is the exact loss of one admissible family, and its terms move in opposite directions under merging (the first cannot decrease because merging shrinks the allowed means; the second cannot increase). The no-selection statements concern each cost separately; whether the sum has a nondegenerate minimizer is not analyzed.

16.5 Fisher geometry and gauge invariants (G5)

16.5.1 The question and the answer

Chapter G5 asks which geometric statements about the Gaussian family concern laws and which concern a chart or frame. "The Fisher mean block is the precision" holds in the moment chart but not, for coordinate directions, in the expectation chart (Proposition G5.1, Proposition G5.2). Under frame changes, the generalized spectrum of the pencil (L, Λ) is invariant when regular (Proposition G5.5), while ordinary eigenvalues and single subspace determinants are frame artifacts once the frame group contains the positive scalars (Proposition G5.9), and under $\Gamma_b \supseteq \mathrm{GL}^+(K)$ a coupling matrix keeps only its rank (Proposition G5.10). Standard: the moment-chart metric of the multivariate normal [67] with its affine isometries [61], exponential-chart duality [4], congruence invariance and Sylvester's law [39]. New: the typing of which quantities survive the represented frame group, and a scoped statement of when a graph Laplacian is a Fisher tensor.

16.5.2 The picture

The two mean blocks differ because $\partial/\partial\mu$ at fixed Σ and at fixed M_2 are different tangent vectors: the second must shrink the covariance to compensate ($\Sigma = M_2 - \mu\mu^\top$), and that extra motion has Fisher length, governed by the unbounded Mahalanobis norm $\mu^\top\Lambda\mu$. The quotient metric, a Schur complement, is exactly Λ and sits below the restricted one, the clamped-versus-relaxed ordering of G2. As with $C_p \neq C_V$, a derivative depends on what is held fixed; an identity mixing mean and covariance sectors must name its chart and say whether it restricts or quotients.

Under a general linear frame group there is no metric on the fiber, so the eigenvalues of one quadratic form are chart data: rescaling all frames by $\mathrm{c} > 0$ multiplies every eigenvalue of Λ by c^2, and a threshold on absolute eigenvalues can accept or reject any direction by choice of c (Proposition G5.9, part (i)). Only relative quantities and law functionals survive.

Physics analogue (exact)

Normal-mode frequencies are generalized eigenvalues of the potential form relative to the kinetic form and do not depend on the generalized coordinates, whereas eigenvalues of the stiffness matrix alone do. The pencil $Lx = \mathrm{d}\Lambda x$ is this construction with the Laplacian part as potential and the full precision as kinetic form; both transform by the same congruence, so the d_k are invariant. Each $\mathrm{d}_k \in [0, 1]$ is the fraction of precision along its direction that bonds supply rather than self terms: consensus sits at 0, directions no self term touches at 1 (Proposition G5.7).

16.5.3 Objects and notation

Term or symbol	Plain meaning	Physics analogue
Restriction versus quotient metric	Metric on coordinate directions versus orthogonal to a fiber	Clamped versus relaxed response
Block reframing $\mathbf{P}$	New-to-old frame change, $\Lambda \mapsto \mathbf{P}^\top \Lambda \mathbf{P}$, $\Sigma \mapsto \mathbf{P}^{-1}\Sigma\mathbf{P}^{-\top}$	Site-dependent gauge transformation
Pencil (L, Λ), regular	Pair of forms, $Lx = \mathrm{d}\Lambda x$; determinant not identically zero	Normal-mode problem; no shared zero mode
Agent-aligned law functional; residual fiber label	Gap, mutual information, relative entropy of unions of agent blocks; congruence class of one semidefinite matrix	Gauge-invariant observable; inertia (Sylvester)

Traps: d is a generalized eigenvalue, not a dimension; L is the Laplacian part, not the bound $\mathcal{L}$; a "regular" pencil and a "regular" exponential family are unrelated; the mass a below is a coefficient on a reference precision.

16.5.4 The formulas, explained

In the moment chart,

$$g^F\big((\dot\mu_1, \dot\Sigma_1), (\dot\mu_2, \dot\Sigma_2)\big) = \dot\mu_1^\top \Lambda \dot\mu_2 + \tfrac{1}{2}\operatorname{Tr}(\Lambda\dot\Sigma_1\Lambda\dot\Sigma_2),$$

and every affine map $(\mu, \Sigma) \mapsto (Q\mu + \mathrm{v}, Q\Sigma Q^\top)$ is an isometry; block reframings act this way, so the Fisher geometry of the recognition family is gauge invariant for any frame group. In the expectation chart $g_{[\mu\mu]} = (1 + \mu^\top\Lambda\mu)\Lambda + \Lambda\mu\mu^\top\Lambda$, with eigenvalues relative to Λ of $1 + \mu^\top\Lambda\mu$ ($n - 1$ times) and $1 + 2\mu^\top\Lambda\mu$. The link between Laplacian and Fisher geometry is narrow and exact:

$$D_{\mathrm{KL}}\big(\mathcal{N}(\mu_i, W_{ij}^{-1}) \| \mathcal{N}(\mu_j, W_{ij}^{-1})\big) = \tfrac{1}{2}(\mu_i - \mu_j)^\top W_{ij}(\mu_i - \mu_j),$$

summing over edges with definite weights to $\frac{1}{2}\mu^\top L\mu$ (Proposition G5.3). Outside Gaussian laws at that fixed covariance, or a frozen base law to second order, the chapter states that a sample-space

Laplacian is neither a recognition Fisher tensor nor a natural-gradient operator, a scope limit on any use of a graph Laplacian as a preconditioner.

Pencil invariance is $\det(\mathbf{P}^\top L\mathbf{P} - \mathrm{d}\mathbf{P}^\top \Lambda \mathbf{P}) = \det(\mathbf{P})^2 \det(L - \mathrm{d}\Lambda)$. The pencil is regular exactly when $\ker L \cap \ker A = \{0\}$ (Proposition G5.6); at the massless boundary $A = 0$ it is (L, L), irregular, and says nothing about the graph until a pin, quotient, or mass is declared. With a mass, $\Lambda = L + \mathrm{a}\Lambda_{\mathrm{ref}}$ gives $\mathrm{d} = \mathrm{d}_{\mathrm{ref}}/(\mathrm{d}_{\mathrm{ref}} + \mathrm{a})$ (Proposition G5.8), roughly the massive-propagator ratio $p^2/(p^2 + \mathrm{a})$ on a lattice; the small-eigenvalue counting exponent survives, and the limits $\mathrm{a} \to 0$ and $\mathrm{d}_{\mathrm{ref}} \to 0$ do not commute ((G5.18)). A subspace log determinant picks up the Jacobian $\log\det(B'^\top \mathbf{P}^\top \mathbf{P} B')$ ((G5.19)), which orthogonal reframings kill and general linear ones do not; the gap of G4 cancels it for agent-aligned blocks, while an agent-splitting gap moves from $\frac{1}{2}\log\frac{4}{3}$ to $\frac{1}{2}\log 4$ in the $N = 1$, $K = 2$ witness. Under $\Gamma_b \supseteq \mathrm{GL}^+(K)$ every semidefinite matrix of rank r is congruent to $\mathrm{diag}(I_r, 0)$, so a common reframing leaves $K + 1$ classes and no continuous modulus; under $\mathrm{SO}(K)$ or $\mathrm{O}(K)$ a definite matrix keeps its K eigenvalues (Proposition G2.2). With a noncompact frame group the strength of a coupling matrix by itself is gauge; with a compact one it is physical.

One caution the wording does not make explicit. The pencil is formed after splitting $\Lambda = L + A$ in the chart where the family is imposed (Definition G5.4), the link-trivializing chart of Hypothesis G3.13. Its spectrum is data of the law together with that split, hence with its links, not of the law alone: a precision in the family in two inequivalent charts can have two pencils. For $N = 2$, $K = 1$, $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$ splits with $W = 1$, $A = (1, 1)$ and has spectrum $\{0, \frac{2}{3}\}$, while its reframing by $\mathrm{diag}(1, 2)$, $\left(\begin{smallmatrix} 2 & -2 \\ -2 & 8 \end{smallmatrix}\right)$, lies in the family with $W = 2$, $A = (0, 6)$ and spectrum $\{0, 1\}$. This does not contradict Proposition G5.5, which concerns one congruence of one pair; it limits Remark G3.6, which lists pencil eigenvalues among frame-independent membership criteria, since forming the pencil presupposes a chart in which membership already holds.

16.6 What is proved, and how solid it is

Every row is proved in the supplement or cited there to a checked source; there is no NUM statement and no TOY positive result in this part. COND rows rest on a frame-group or graph hypothesis that fails in cases the program cares about.

Result	What it says	Status	Scope and caveats
G1. Exponential families			
Proposition G1.7	No law, Fisher metric, or bound at a boundary point; an operator map can converge there (Proposition G1.8)	SOLID	Negative; scalar witness suffices
Section G1.3	Pin, quotient, and mass sector are three distinct repairs	CHOICE	Their two propositions are SOLID
Proposition G1.16	Cumulants and Fisher metric; relative entropy is Bregman; unique affine projection	SOLID	Standard in substance; Proposition G1.3 is STANDARD

Result	What it says	Status	Scope and caveats
G2. Gaussian realization			
Hypothesis G2.1	Represented frame group, specialized to $\mathrm{GL}^+(K)$ where named	CHOICE	Not met by $\mathrm{SU}(N)$ or a realified $\mathrm{GL}(K,\mathbb{C})$
Definition G2.13	Directed linear-Gaussian forest	CHOICE	General DAGs get one sentence
Lemma G2.6	Zero block is conditional independence; Schur complement below restricted precision	SOLID	Standard in substance
Proposition G2.17	Every kernel frame covariant for every Γ_b; explicit forest blocks (Proposition G2.16)	SOLID	Belief sector at frozen model latents
Proposition G2.18	Closed-form learning step; no maximizer if the residual covariance is singular	SOLID	Link witness is scalar: possibility, not frequency
G3. Interaction family			
Hypothesis G3.13	The family, adopted in link-trivializing coordinates	CHOICE	The chart fixes what agreement means
Hypothesis G3.4	Belief links form a graph coboundary	CHOICE	Automatic on forests; fails for generic links on cycles
Proposition G3.3	Null set in a fixed chart; not reframing invariant	SOLID	Chart-level
Proposition G3.8	On forests, $\mathrm{GL}^+(K)$ frames absorb positive-determinant transitions; pair criterion $(-1)^K \det \Lambda_{12} > 0$; $\mathrm{GL}(K)$ frames reach every invertible pair	COND	Forest or pair; stated frame group; nothing sufficient for cycles or smaller groups
Proposition G3.9	Reversing gain, cycle determinant sign, odd-K v-structure exclude membership	SOLID	Negative; (ii), (iii) for $\mathrm{GL}(K)$; (iii) on three agents
Proposition G3.10	Exact membership test; prior balance sufficient, not necessary; factorwise stronger	SOLID	Link-trivializing chart
G4. Recognition restrictions			
Theorem G4.3	Exact mean, clamped covariances; loss is the gap, the dual total correlation (Corollary G4.5)	SOLID	Standard in substance
Corollary G4.7	Gap independent of the observation; mean cost carries the data (Corollary G4.9)	SOLID	Needs data-independent joint precision
Proposition G4.12	Identification costs $+\infty$; regularized, $\frac{n-r}{2}\log\frac{1}{\varepsilon}$	SOLID	Negative; absolute continuity

Result	What it says	Status	Scope and caveats
Proposition G4.16	Gap monotone on chains, unordered off them, minimized by one block	SOLID	Negative; combined cost not examined
G5. Fisher geometry and invariants			
Proposition G5.2	Mean block is Λ in the moment chart, larger in the expectation chart; quotient Λ	SOLID	Standard in substance [67]
Proposition G5.3	Edge relative entropy at covariance W_{ij}^{-1} is the Laplacian quadratic	SOLID	Gaussian, fixed covariance only
Proposition G5.5	Regular pencil spectrum invariant; in $[0, 1]$	SOLID	Invariant of law plus link-fixed split
Proposition G5.9	Absolute spectra, subspace determinants frame dependent; agent gaps invariant	SOLID	(i) needs positive scalars
Proposition G5.10	A semidefinite coupling matrix is labeled by rank alone	COND	Needs $\Gamma_b \supseteq \mathrm{GL}^+(K)$; false for compact groups
Open			
Open Problem G1.14	Mass-coordinate scaling; also Open Problem G1.18, Open Problem GB.2	OPEN	Section 16.8

Conditions: which ones matter

Conditions that can fail where the program lives: flat comparison (Hypothesis G3.4), without which the interaction hypothesis is not formulated; membership in the link-trivializing chart (Hypothesis G3.13); the $\mathrm{GL}^+(K)$ specialization (Hypothesis G2.1), needed by absorption and the rank-only label; frozen model latents; absolute continuity (MT Hypothesis 5.3, clause H3), which identification violates; a nondegenerate target and regular pencil, which fail at the massless boundary $A = 0$; a data-independent joint precision; and a nonsingular residual covariance in the learning step. Hygiene: standard Borel spaces, sigma-finite references, measurability, regularity and minimality (automatic for the Gaussian), definite receiving covariances (Hypothesis G2.14), finite second moments, and orthonormal bases used as a computational device.

16.7 Toy versus general, and what this part implies for $\mathrm{GL}(K)$ networks at large N

Nothing in G1 to G5 is TOY. Every theorem holds for arbitrary finite N and K (the two-agent orientation criterion has a forest version at every N), and every small instance is a counterexample to a universal statement, hence a solid negative result that shows a claim fails, not how often it fails at $N = 512$. Thinness holds at every N, and the cycle-sign condition applies to any cycle with

invertible blocks inside any network. Nothing here is a large-N statement either: no thermodynamic limit, per-agent density, or scaling law.

The interaction family at scale. Two conditions stand between a 512-agent Gaussian precision with cycles and the family. Flat comparison requires every loop product of the belief links to be the identity, automatic on a forest but violated by generic independent $\mathrm{GL}(K)$ links on a graph with many cycles; then the hypothesis behind aggregation and the operator arrow is not formulated, and the nonflat case is in chapter 17. And the law-level criterion is complete only for forests and pairs; for precisions whose graph has cycles, including those created by co-parents, only the necessary sign condition is proved.

Frame groups. For $\mathrm{GL}(K,\mathbb{R})$, $\Gamma_b = \mathrm{GL}(K)$ contains $\mathrm{GL}^+(K)$ and the positive scalars, so every statement needing either holds. Part (iv) of Proposition G3.8 reaches every invertible pair, so the orientation obstruction disappears there, and the forest construction of part (i) uses positive determinants only to keep its frames in $\mathrm{GL}^+(K)$; the cycle-sign and odd-K v-structure obstructions, stated for $\mathrm{GL}(K)$ frames, remain. The gauge-invariant data of a Gaussian network under $\mathrm{GL}(K,\mathbb{R})$ are agent-aligned law functionals, pencil spectra relative to the link-fixed split, cycle determinant signs, and holonomy conjugacy classes (MT Proposition 2.12); absolute eigenvalues and single subspace determinants are gauge artifacts. The supplement treats real Gaussians only. A complex group enters through a real representation, for instance the realification on $\mathbb{R}^{2K}$, whose image lies in $\mathrm{GL}^+(2K)$ and contains the positive scalars; statements proved for every Γ_b hold, the rank-only label and the GL^+ sufficiency constructions do not apply as stated because their frames need not be complex linear, and obstructions proved for larger groups still apply. For $\mathrm{SU}(N)$ and other compact groups, $\Gamma_b \subseteq \mathrm{O}(K)$ after choosing an invariant inner product: spectra are invariant, coupling matrices keep continuous eigenvalue moduli, and no absorption result is stated.

Where compactness and orthogonality enter. No theorem here assumes compactness. It is excluded by the $\mathrm{GL}^+(K)$ specialization, brings extra invariants when present, and its absence is what the scalar link witness exploits (informally, on a compact group a continuous objective attains its maximum, a standard fact the supplement does not state). A Euclidean structure enters as a computational device (orthonormal bases), through orthogonal reframings that preserve spectra and subspace determinants, through the averaging pushforward $y \mapsto S^\dagger y$ of Corollary G2.10, whose equality condition uses the Euclidean complement and is chart-level under $\mathrm{GL}(K)$, and through the finite part of the regularized identification cost, whose divergent coefficient $(n-r)/2$ is frame free.

16.8 Open ends

The supplement records three open problems here: the law of a retained mass coordinate under a scale map, settled by an explicit induced law with a proof that its orbit stays in the natural domain or reaches the boundary (Open Problem G1.14); whether the affine information projection commutes with gauge actions or coarse parameter maps, settled by an intertwining theorem or a

counterexample (Open Problem G1.18); and whether the Bayesian-renormalization information scale, built on the Fisher metric of generative parameters, transfers to Λ read as the Fisher metric of a recognition mean, settled by making the coarse mean a generative parameter with nondegenerate Fisher metric and transferring the construction, or by a family where that metric is singular (Open Problem GB.2). G5 declines any contraction claim for varying precision (Open Problem G9.4), and G4 is a Gaussian instance of MT Open Problem 5.14 and MT Open Problem 9.19. Four gaps are visible but not recorded as open problems: law-level membership for precisions whose graph has cycles; the relation of the family to a covariant Laplacian with nontrivial holonomy; the combined mean-tie and factorization cost as a selector; and complex Gaussians and complex or unitary frame groups beyond a real representation. These describe what the documents do not contain; they are not proposals.

16.9 Verdict

Does it deliver what it claims? DELIVERED

What the part says it does. Explicit Gaussian forms of the theory's objects: exponential-family boundary and projection results; the information form, Schur lemma, and frame-covariant kernels of the directed law; an exact account of when the belief precision is a matrix-weighted Laplacian plus self terms, with that shape adopted as a hypothesis rather than derived; restriction costs, with the statement that neither the factorization gap nor the mean cost selects a structure; and a separation of Fisher and spectral quantities into invariants and chart or frame artifacts.

What the statements give. All of it, at arbitrary finite N and K, with proofs. Much is textbook in substance and presented as such. The program-specific content, the membership analysis of G3 and the invariance typing of G5, is elementary linear algebra done carefully; for $K = 1$ its sign conditions are the Mattis-gauge and frustration conditions of spin glasses.

The limits a reader must carry. The interaction family is the pure-gauge sector, and the chart in which it is imposed, not the law, decides what agreement means downstream. Law-level membership is settled on forests and pairs and only bounded by a necessary sign condition on cycles, where a 512-agent network lives. The pencil spectrum is data of a law with its link-fixed split, so the remark offering it as a frame-independent membership criterion claims more than the propositions give. The combined recognition cost is not examined as a selector. Absorption and the rank-only label need $\mathrm{GL}^+(K)$ frames; under $\mathrm{GL}(K, \mathbb{R})$ they hold and absorption is easier, compact groups keep more invariants and lose absorption, and complex Gaussians are not treated. None of this is a defect in a proof, and the part states most of it itself.

Chapter 17

Supplement G, Part 2: Gaussian Coarse-Graining

This chapter covers the second half of Supplement G: score tangents and the Hermite spectrum (Chapter G6), aggregation and Kron reduction (Chapter G7), nonflat links, relaxation, and quotient volume (Chapter G8), the operator arrow (Chapter G9), reciprocal pairs, anchored stars, and loop holonomy (Chapter G10), the decision metatheory (Appendix GA), and the supplement's open problems (Appendix GB.2). Everything acts on the Gaussian interaction family explained with the first half in chapter 16.

In short, this half of Supplement G carries out one finite step of exact coarse-graining for Gaussian agent networks and does so completely: which step maps are exact, when the natural parameter families close under them, where they provably do not, and which further declarations (a rescaling, a reference measure, a repair of a singular boundary) a flow would need. It proves no flow, fixed point, or thermodynamic limit, and says so. Most positive statements are standard mathematics in the theory's typing (Galerkin operators of algebraic multigrid, Gaussian marginalization, Mehler's identity, block Gauss–Seidel, Tarski's decision method); the content without a textbook counterpart among the supplement's citations is the Kron closure classification, the bookkeeping of holonomy under coarse-graining, and a set of sharp negative results.

Three threads run through the part. Aggregation freezes each cluster into a rigid body while Kron reduction integrates hidden agents out; both are exact, and they close on very different families. Curvature becomes mass: a link that cannot be gauged away inside a block, around a folded loop, or across a strict cut reappears after coarse-graining as a positive-semidefinite self term. And closure is neither selection nor a flow: a step between spaces of different dimension has no fixed point until an identification and a rescaling are declared.

17.1 Score Tangents and the Hermite Spectrum (G6)

17.1.1 The question, the answer, and the picture

At one law there are three candidate spaces of directions of motion: the tangent space of a finite family (for a Gaussian, mean and covariance moves), the Hilbert space $L^2_0(\nu)$ of all centered square-integrable scores, and the bounded perturbations $e^{-t\varphi}$ of the main manuscript's nonlinear action calculus (MT Theorem 11.4). The chapter asks how they nest at a Gaussian and what

a channel on b independent copies does to scores. It proves an exact information budget for every parameter-independent block channel (Theorem G6.4), the exact nesting (Remark G6.6), which Hermite directions admit exponential tilts (Proposition G6.7), and the spectrum $b^{1-k/2}$ of the linearized normalized-sum map at the standard Gaussian (Theorem G6.8). Its point is a warning: every exact arrow contracts Fisher information, so the eigenvalue $b^{1/2} > 1$ is created by the extensive replication factor $\sqrt{b}$, a declared normalization, and not by coarse-graining. Nothing later in Supplement G or in the main manuscript uses these results; Supplement B builds mode exponents on the spectrum and poses nonlinear attraction at the Gaussian block law as Open Problem B5.22.

Physics analogue (exact)

Copy a perturbed law onto b sites and push through $\bar{\xi} = b^{-1/2}\sum_i \xi_i$. At $\mathcal{N}(0,1)$ the block law is again $\mathcal{N}(0,1)$, and the score operator $\mathscr{L}_b$ is the derivative of this central-limit map at its fixed point. Its eigenvalues $b^{1-k/2}$ are the scaling of cumulants under normalized summation: in Wilsonian language the mean direction is relevant, the variance direction marginal, and every $k \geq 3$ irrelevant. The supplement identifies its $b = 2$ case with the linearization of Jona-Lasinio [42] and avoids the Wilsonian words, whose conventions belong to Supplement B.

The supplement adds the factorization $\mathscr{L}_b = \mathsf{U}_b\mathscr{I}_b$. The replication lift $\mathscr{I}_b h = \sum_i h(x_i)$ has norm exactly $\sqrt{b}$, since b independent sites carry b times the Fisher information of one, and it is not a physical arrow: no parameter-independent kernel maps $\mathcal{N}(x,1)$ to $\mathcal{N}(x\mathbf{1}_b, I_b)$ (Proposition G6.3). The conditional expectation U_b only loses information (MT Theorem 11.5). A uniform field couples to an extensive block magnetization; after dividing by $\sqrt{b}$ the mean direction looks amplified, but blocking learned nothing new.

17.1.2 Objects and formulas

Term or symbol	Plain meaning	Physics analogue
Score h; $L_0^2(\nu)$	Velocity $\partial_t \log p_t$ of a path with L^2-differentiable square-root density; all centered square-integrable functions	Tangent vector; $\delta \ln p$
$\mathscr{I}_b$; $\mathscr{L}_b$	Replication onto b independent sites; the block map linearized on tangents	Extensive source; linearized RG map
Bounded action quotient	Perturbations $e^{-t\varphi}$, φ bounded, modulo constants	Bounded perturbation of a Hamiltonian

Traps: h is a score, not the information vector `h` of later chapters; a block here is independent copies of one scalar law, not interacting agents; the label prefixes `rg` and `grg` are historical, and no Wilsonian flow is built anywhere in Supplement G. For a channel from b copies to an output z, the

pushed score is $\mathscr{L}_b h = \mathbb{E}[\sum_i h(x_i) \mid z]$ and

$$b\|h\|^2 - \|\mathscr{L}_b h\|^2 = \int \mathrm{Var}\Big(\sum_{i=1}^{b} h(x_i) \;\Big|\; z\Big)\nu^{(b),c}(dz) \geq 0$$

(Theorem G6.4): Fisher information of b copies, minus what survives, equals the expected conditional variance of the block score, the Rao–Blackwell form of information loss. Equality is local sufficiency at one parameter, not recovery of the experiment. For the normalized sum,

$$\mathscr{L}_b \widehat{\mathrm{He}}_k = b^{1-k/2}\widehat{\mathrm{He}}_k \quad (k \geq 1), \qquad \sigma(\mathscr{L}_b) = \{b^{1-k/2} : k \geq 1\} \cup \{0\},$$

with $\widehat{\mathrm{He}}_k = \mathrm{He}_k/\sqrt{k!}$, from Mehler's identity $\mathbb{E}[\mathrm{He}_k(\xi_i) \mid \bar{\xi}] = b^{-k/2}\mathrm{He}_k(\bar{\xi})$ times the replication factor b (Theorem G6.8). The operator is compact, injective, with norm $\sqrt{b}$ and dense nonclosed range: no component is destroyed, but undoing a step multiplies degree k by $b^{k/2-1}$, so reconstructing fine structure is ill-posed even here.

At a nondegenerate Gaussian on $\mathbb{R}^n$, the family tangent (polynomials of degree at most two, dimension $n + n(n+1)/2$) and the bounded actions (dense and proper in L^2_0, Proposition G6.5) meet only at zero, since nonzero polynomials are unbounded (Remark G6.6): Gaussian mean and covariance moves lie outside the main manuscript's analytic action calculus. And a direction can be a legitimate tangent with no exponential tilt: a Gaussian tilted by $e^{-t\mathrm{He}_3}$, a cubic potential, is normalizable for no $t \neq 0$, and by $e^{-t\mathrm{He}_4}$ only for the stable sign, exactly the familiar ϕ^3 and wrong-sign ϕ^4 instabilities (Proposition G6.7). For the program, the spectral theorem concerns a noninteracting block of scalar copies and says nothing about aggregating coupled K-dimensional agents.

17.2 Aggregation and Kron Reduction (G7)

17.2.1 The question, the answer, and the picture

Which coarse maps of the interaction family are exact, and when does the family close under them? In link-trivialized coordinates the family is a network of matrix springs and pins,

$$\mathsf{E}(z) = \frac{1}{2}\sum_i z_i^\top A_i z_i + \frac{1}{2}\sum_{i<j}(z_i - z_j)^\top W_{ij}(z_i - z_j), \qquad A_i \succeq 0, \quad W_{ij} = W_{ji} \succeq 0$$

(Definition G3.1): the self term or anchor A_i pins agent i to the origin of its fiber and W_{ij} is a matrix spring on the disagreement. Membership is declared (Hypothesis G3.13); directed models do not produce it in general, and for $K, N \geq 2$ it is a null set of precisions (Proposition G3.3).

Hard aggregation, which identifies each cluster with one coarse coordinate, always closes: self terms and cut weights add and internal edges drop out (Proposition G7.3); the coarse law is the fine law conditioned on zero within-cluster disagreement (Proposition G7.4) and a precision-dependent linear pushforward (Proposition G7.5). Kron reduction, which integrates hidden agents out, is exact

as a map of laws (Proposition G7.10) but leaves the family in general (Proposition G7.6). Its closure regimes are classified in Table G7.1, summarized by the G7 rows of section 17.7; the headline is that on a fixed flat graph with independent positive-definite weights and $K \geq 2$, closure for every assignment holds exactly when every hidden component touches at most two retained terminals (Theorem G7.14).

Aggregation makes each cluster a rigid body: internal springs are never stretched, member pins add, and cut springs add in parallel. Kron reduction lets hidden agents move and integrates them out, so retained agents feel effective springs softened by hidden fluctuations. One restricts the energy, the other marginalizes the law, and the marginal precision is dominated by the restricted one (Corollary G2.8).

Physics analogue (exact)

$S^\top \Lambda S$, with S the Boolean membership matrix tensored with I_K, is the Galerkin coarse operator of aggregation algebraic multigrid [50, 57]: the coarse energy is the fine energy on rigid-block configurations. Kron reduction is integrating out Gaussian fields; for scalar weights it is the star–mesh reduction of an electrical network [26, Lemma II.1], and its offset increment, led by $\frac{1}{2}\log\det M$, is the exact free energy of the eliminated modes.

Physics analogue (structural)

Aggregation is a Kadanoff block-spin map with a particular block spin: the coarse law is the law of $P_\Lambda y = (S^\top \Lambda S)^{-1} S^\top \Lambda y$, the rigid-block configuration closest to y in the energy norm (Proposition G7.5(i)), not the plain block average. Since this block spin depends on the couplings, aggregation is one parameter-independent channel only at fixed precision; no single kernel realizes it across the interior of the interaction cone (part (v)). Conditioning on the null event of zero disagreement also depends on the defining statistic, the Borel–Kolmogorov phenomenon behind the Jacobian of a delta-function constraint (Proposition G7.4); the linear constraint is part of the definition.

17.2.2 Objects and formulas

Term or symbol	Plain meaning	Physics analogue
Galerkin law; P_Λ	Coarse law of rigid clusters; energy-weighted block value	Block spin, coupling-dependent
Hidden, retained, terminal, ground	Eliminated agents; kept agents; kept agents adjacent to a hidden component; a kept zero vertex carrying each pin as an edge	Interior; boundary; Dirichlet ground
Cone $\mathcal{W}_H$	All $H\,\mathrm{diag}(d)H^\top$, $d \geq 0$, for one $H \in \mathrm{GL}(K)$	Couplings with common normal modes
Positive two-node presentation	Sum of positive-semidefinite terms on at most two nodes each	Nonnegative bond energies

Traps: "trace" in "graph-exponential trace closure" and "hard trace" means restricting the energy to identified configurations, the opposite of tracing out; "universal closure" quantifies over weight assignments on one graph; the pivot M is unrelated to the M of sections 17.4 and 17.5.

Aggregation. It gives $A_I = \sum_{i\in I} A_i$, $W_{IJ} = \sum_{i\in I, j\in J} W_{ij}$, $\mathsf{h}_I = \sum_{i\in I} \mathsf{h}_i$, and annihilates internal weights (Proposition G7.3). Internal edges vanish because the pair energy depends only on $z_i - z_j$: a translation-invariant pair statistic has constant diagonal, so node parameters simply add (Theorem G7.2, which closes energies, not normalized laws). With $B_\perp$ an orthonormal basis of the complement of $\operatorname{range} S$,

$$p\big(dy \;\big|\; B_\perp^\top z = 0\big) = \mathcal{N}\big((S^\top \Lambda S)^{-1} S^\top \mathsf{h}, (S^\top \Lambda S)^{-1}\big),$$

the event being $z_i = z_j$ inside each cluster. At fixed precision the pushed Fisher metric of the mean is $\Lambda S \Lambda_{\mathrm{c}}^{-1} S^\top \Lambda \preceq \Lambda$, with equality on rigid-block directions (Proposition G7.5(ii)): aggregation loses exactly the information in internal deformations, at every finite depth. When the precision moves, so does the channel (Open Problem G9.4).

Kron reduction. With hidden block $M = \Lambda_{\mathrm{hh}}$,

$$\Lambda^{\mathrm{eff}} = \Lambda_{\mathrm{rr}} - \Lambda_{\mathrm{rh}} M^{-1} \Lambda_{\mathrm{hr}}, \qquad \mathfrak{c}^{\mathrm{eff}} = \mathfrak{c} + \tfrac{1}{2}\log\det M - \tfrac{n_{\mathrm{h}}}{2}\log 2\pi - \tfrac{1}{2}\mathsf{h}_{\mathrm{h}}^\top M^{-1}\mathsf{h}_{\mathrm{h}}$$

(Proposition G7.10): direct stiffness minus softening through hidden agents, and an offset that adds the free energy of the eliminated modes, so raw mass and evidence are unchanged. Eliminating one agent k adds $W_{ik}M^{-1}W_{kj}$ between each pair of its retained neighbors. For $K = 1$ this is $w_{ik}w_{kj}/m \geq 0$, and scalar networks always close (Lemma G7.7). For matrices it need not be symmetric: $W_{13} = \operatorname{diag}(1,2)$, $W_{23} = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and unit pins manufacture $\frac{1}{19}\left(\begin{smallmatrix} 9 & 3 \\ 4 & 14 \end{smallmatrix}\right)$, which is no spring (Proposition G7.6). For a relay with hidden pin Q_0 and legs W, W' the output is symmetric exactly when $WQ_0^{-1}W' = W'Q_0^{-1}W$, the legs commuting after whitening by the pin; at a fixed positive-definite hidden pin the families passing every such test are exactly the subfamilies of one cone $\mathcal{W}_H$, $HH^\top = Q_0$ (Theorem G7.11), and such a cone closes on every graph (Theorem G7.9). In the frame H the network is K decoupled scalar networks.

The graph criterion and large N. With arbitrary independent positive-definite weights, treat each pin as an edge to ground. A hidden component with one terminal contributes nothing; with two it is one effective matrix spring (along a path compliances add, $W^{-1} = \sum_j W_j^{-1}$); with three or more it is a matrix star–mesh transform, nonsymmetric for some assignment. So for $K \geq 2$ universal closure holds iff every hidden component has at most two terminals, ground counted (Theorem G7.14). With a proper prior on every hidden agent, ground is always a terminal: closure iff each hidden component touches at most one retained agent, and otherwise the failing assignments contain an open dense set of full measure (Corollary G7.15).

This is the operative statement for integrating out agents at large N. With priors on all agents, $K \geq 2$, and independent matrix weights, integrating out a relay between two retained agents generically

leaves the positive matrix-spring family, while a hidden cluster hanging from one retained agent only stiffens its pin. Decimation, which keeps some agents and eliminates those between them, is made of relays. The coarse law stays an exact Gaussian; what fails is its presentation by positive pairwise springs. Degree-two hidden agents with zero pin close in the marked nonflat class; with pins they close with a generated pin and an induced link the law does not determine (Proposition G7.23). Branching is worse: a proper four-agent Gaussian has a three-node marginal that is no sum of positive terms on at most two nodes, even with new invertible links and redistributed pins (Proposition G7.16). The action is still quadratic (Möbius order two), so the obstruction is positivity of the pairwise split, not a three-body term; existence is proved, not frequency.

Series elimination with links. For degree-two hidden agents with zero pin (Theorem G7.20),

$$\Theta_{\mathrm{path}} = \Theta_1 \cdots \Theta_m, \qquad W_{\mathrm{path}}^{-1} = \sum_{i=1}^{m} \Theta_{\le i-1} W_i^{-1} \Theta_{\le i-1}^{\top}, \qquad Q_{\mathrm{loop}} = (I - \mathrm{Hol})^{\top} W_{\mathrm{path}} (I - \mathrm{Hol}).$$

Links multiply and compliances add after transport, resistors in series made matrix-valued and gauge covariant; a returning path leaves a self term fixed by its holonomy. The law sees only Q_{loop}: $(2I, W)$ and $(3I, W/4)$ give the same quadratic, so a folded holonomy is structural data the law does not determine.

17.3 Nonflat Links, Relaxation, and Quotient Volume (G8)

17.3.1 The question, the answer, and the picture

A twisted internal link leaves a positive-semidefinite self term $(I - \Theta_e)^{\top} W_e (I - \Theta_e)$ on its cluster, which only a singular weight can hide (Proposition G8.1). A Gaussian mean relaxes onto the fixed sector exponentially, by a flow that is exactly a Fisher natural gradient (Proposition G8.2). The Gaussian holonomy-constrained parent is the moment match (Proposition G8.3), Haar-projected under compact holonomy (Corollary G8.4); no parameter-independent kernel realizes the barycenter of two normals for all parameters (Proposition G8.5). Hard membership carries a Jacobian that changes raw normalizers only (Proposition G8.6). No partition is claimed to be selected by volume or transported divergence.

Force two agents joined by a twisted spring to be one agent. The spring wants $z_i = \Theta_e z_j$, identification imposes $z_i = z_j = y$, and every y with $(I - \Theta_e)y \neq 0$ stretches it: a quadratic in y alone, a mass term for the block. The edge has become a self-loop with holonomy Θ_e. The same bookkeeping gives the folded-loop term of section 17.2 and the generated self terms of section 17.5: frustration inside a block becomes mass after blocking.

Physics analogue (structural)

For $K = 2$, $W_e = wI$, and Θ_e a rotation by ϕ, the residue is $4w \sin^2(\phi/2) I$, the energy $|1 - e^{i\phi}|^2$ of hopping back to one site with phase ϕ; a flux through a ring lifts the uniform mode of a

hopping Laplacian in the same way. Here the flux is the declared link and the lifted mode is the rigid motion of the block.

17.3.2 Objects and formulas

Term or symbol	Plain meaning	Physics analogue
Weighted invisibility	$W_e^{1/2}(I-\Theta_e)=0$: the twist acts only where the weight is blind	Twist in an uncoupled direction
Forward-KL barycenter	Gaussian minimizing $\sum_i w_i D_{\mathrm{KL}}(q_i\|q)$	Moment matching; mean-field parent
$\mathsf{G}_{\mathcal{P}}$, $\mathrm{Jac}_{\mathcal{P}}$	Metric on cluster coordinates modulo translation; its volume factor	Collective-coordinate Jacobian

Internal residue. The residue vanishes exactly when $W_e^{1/2}(I-\Theta_e)=0$ (Proposition G8.1, from Lemma G3.5); positive-definite weights force $\Theta_e = I$. A common quotient on which every edge is positive definite exists exactly when all weights share one kernel preserved by every link, and only then is group-valued holonomy defined on the effective support. In part (v) each edge of a triangle is invisible and the loop product is $2I$, which fixes no vector, yet the zero-energy sections are all constant sections: with singular weights the kernel of the connection energy, not the loop product, governs the zero modes.

Relaxation. For constant $\mathsf{G} \succ 0$, $L \succeq 0$, the flow $\mathsf{G}\dot{z} = -Lz$ converges to the G-orthogonal projection $\Pi_0 z_0$ onto $\ker L$,

$$\|z(t)-\Pi_0 z_0\|_{\mathsf{G}} \le e^{-r_+ t}\|z_0-\Pi_0 z_0\|_{\mathsf{G}}, \qquad r_+ = \lambda_{\min}^{+}\big(\mathsf{G}^{-1/2} L \mathsf{G}^{-1/2}\big)$$

(Proposition G8.2): overdamped relaxation with conserved zero modes and a spectral gap, which for $\mathsf{G} = I$ is consensus dynamics. Since $D_{\mathrm{KL}}(\mathcal{N}(z,\mathsf{G}^{-1})\|\mathcal{N}(w,\mathsf{G}^{-1})) = \frac{1}{2}(z-w)^{\top}\mathsf{G}(z-w)$, it is exactly the natural gradient on that family; elsewhere the Fisher metric matches relative entropy only to second order. G must transform with the frame; $\mathsf{G} = I$ is equivariant only under orthogonal reframings.

Gaussian parents. Over nondegenerate Gaussians, $\sum_i w_i D_{\mathrm{KL}}(\mathcal{N}(\mu_i,\Sigma_i)\|\mathcal{N}(\mu,\Sigma))$ is minimized by

$$\bar{\mu} = \sum_i w_i \mu_i, \qquad \bar{\Sigma} = \sum_i w_i\big[\Sigma_i + (\mu_i-\bar{\mu})(\mu_i-\bar{\mu})^{\top}\big], \qquad \mathcal{D}_I^{\star} = \tfrac{1}{2}\Big[\log\det\bar{\Sigma} - \sum_i w_i \log\det\Sigma_i\Big]$$

(Proposition G8.3): within-constituent plus between-mean spread, and a score equal to the parent's entropy minus the mean constituent entropy, since $\frac{1}{2}\log\det(2\pi e\Sigma)$ is Gaussian entropy. Under a holonomy group with compact closure the parent is the moment match of Haar-averaged moments

(Corollary G8.4); for $\{\pm I_2\}$ and the single law $\mathcal{N}(u, I_2)$ it is $\mathcal{N}(0, I_2 + uu^\top)$, so the constraint bites without disagreement. Compactness matters: a loop holonomy with an eigenvalue off the unit circle fixes no nondegenerate Gaussian, and for independent $\mathrm{GL}(K)$ links $|\det| = 1$ on a loop is a null condition (MT Corollary 10.11). For the program's $\mathrm{GL}(K, \mathbb{R})$ links the constrained Gaussian parent is empty off a null set; a compact group such as $\mathrm{SU}(N)$ in its real representation meets the hypothesis. Nor is the barycenter a channel: a kernel sending $\mathcal{N}(\mu, \Sigma_0)$ to $\mathcal{N}(\mathsf{H}\mu + \mathsf{b}, \Sigma_1)$ for all μ must add noise $\Sigma_1 - \mathsf{H}\Sigma_0\mathsf{H}^\top$ (Proposition G8.5), and the equal-width barycenter of two normals needs noise $\varsigma^2/2$, which depends on the width. Merging two agents into one of the same width is a meta-agent map to which data-processing inequalities do not apply.

Quotient volume. Modulo translation, hard membership induces $\mathsf{G}_\mathcal{P} = \mathrm{diag}(n_I) - \mathbf{n}\mathbf{n}^\top/N$, with $\mathrm{pdet}\,\mathsf{G}_\mathcal{P} = |\mathcal{P}| \prod_I n_I/N$ and $\mathrm{Jac}_\mathcal{P} = (\mathrm{pdet}\,\mathsf{G}_\mathcal{P})^{K/2}$ (Proposition G8.6): the kinetic metric of rigid clusters with the center of mass removed (for two clusters, twice the reduced mass). The Jacobian cancels from normalized laws and shifts raw log normalizers, so partition scores built on raw normalizers depend on a volume convention; selection stays open (MT Open Problem 9.19).

17.4 The Operator Arrow (G9)

17.4.1 The question, the answer, and the picture

On operator parameters, aggregation is a positive linear semigroup action: self terms and cut weights add, internal edges vanish, nested aggregations compose (Proposition G9.2). On the interior of the interaction cone a step rescaled by a declared $\zeta_\ell > 0$ is a normalized Gaussian law step, realized by one parameter-independent kernel on a fixed-precision fiber exactly when $\zeta_\ell \le 1$ (Proposition G9.3). The only positive rules additive in scalar node masses are biadditive, $A_i = x_i A$, $W_{ij} = x_i x_j M$, closed under every hard partition (Proposition G9.5). Nothing is asserted about infinite hierarchies, limiting rays, or iterates.

Two facts keep aggregation from being a flow. Each step changes the number of agents, so steps compose maps between different spaces, and a fixed point needs a hierarchical identification of successive parameter spaces, which the supplement does not supply. And the bare map multiplies stiffness by the block size: a rigid block of b agents stretches b pins, and for equal cluster sizes

$$b\lambda_{\min}(\Lambda) \le \lambda_{\min}(S^\top \Lambda S), \qquad \lambda_{\max}(S^\top \Lambda S) \le b\lambda_{\max}(\Lambda)$$

(Proposition G9.2(iv)). With $b \ge 2$ iterates run to infinite stiffness, so no nontrivial fixed point can appear without a compensating rescaling. The step $\Lambda_{\ell+1} = \zeta_\ell^{-1} S_\ell^\top \Lambda_\ell S_\ell$ (Definition G9.1) types that rescaling, which plays the part, structurally, of the field-rescaling step of a Kadanoff–Wilson transformation [43, 80]; with no length to rescale at one base point, only it remains. The supplement fixes no ζ_ℓ and defers fixed rays and attraction to Supplement B.

Physics analogue (exact)

The biadditive boundary member $A = 0$ is a massless Gaussian field on a finite graph: consensus configurations are zero modes (Proposition G5.7), no normalized full-space law exists, and the pencil (L, Λ) is irregular (Proposition G5.6). The named repairs are the infrared regularizations of a zero mode: keep a mass, pin a site, or quotient the zero mode. They define different problems, and a pseudoinverse in the Lebesgue density is none of them.

17.4.2 Objects and formulas

Term or symbol	Plain meaning	Physics analogue
ζ_ℓ; information-vector rule	Declared rescaling of the coarse precision; how the source is carried	Field rescaling; source rescaling
Interior, boundary	$\Lambda \succ 0$; a singular member such as $A = 0$	Massive; massless
Declared repair	Pin, quotient, or retained mass	Infrared regulator

The step as a law. With $\mathsf{h}_{\ell+1} = S_\ell^\top \mathsf{h}_\ell$, the step sends $\mathcal{N}(\mu, \Lambda_\ell^{-1})$ to $\mathcal{N}(\zeta_\ell P_{\Lambda_\ell}\mu, \zeta_\ell \Lambda_{\mathrm{c}}^{-1})$, $\Lambda_{\mathrm{c}} = S_\ell^\top \Lambda_\ell S_\ell$ (Proposition G9.3). At $\zeta_\ell = 1$ it is the Galerkin pushforward; for $\zeta_\ell < 1$ it adds noise, via $y \mapsto \mathcal{N}(\zeta_\ell P_{\Lambda_\ell} y, \zeta_\ell(1-\zeta_\ell)\Lambda_{\mathrm{c}}^{-1})$; for $\zeta_\ell > 1$ no kernel exists on that fiber, because the coarse mean family would carry more Fisher information than the fine one. The rule $\mathsf{h}_{\ell+1} = \zeta_\ell^{-1/2} S_\ell^\top \mathsf{h}_\ell$ makes every ζ_ℓ a change of units. How the source is rescaled with the precision decides whether the step is a channel.

The biadditive family. Under every hard partition,

$$A_i = x_i A, \quad W_{ij} = x_i x_j M \qquad \longmapsto \qquad A_I = x_I A, \quad W_{IJ} = x_I x_J M, \qquad x_I = \sum_{i \in I} x_i,$$

and conversely a positive-semidefinite rule additive in the masses must be xA and xyM, because positivity makes each scalar projection monotone and monotone additive functions are linear (Proposition G9.5). The family is an all-to-all coupling with a rank-one scalar profile, a matrix mean-field coupling proportional to a product of masses. What selects it is this additivity hypothesis with positivity; neither is attraction. Translation invariance works one level earlier: difference-only pair energies make internal edges vanish and cut weights plain sums (Theorem G7.2), which is why additivity in masses is the natural equation. The supplement does not derive the biadditive form from translation invariance alone, and it says that closure is not selection. Under $\Gamma_b \supseteq \mathrm{GL}^+(K)$, which $\mathrm{GL}(K, \mathbb{R})$ satisfies, the factor M carries only its rank (Proposition G5.10); orthogonal frames would leave K eigenvalues.

Physics analogue (structural)

In the multiscale network renormalization of Garuccio et al. [32], two blocks are linked when any pair of members is, a nonlinear rule for probabilities, and the invariant linking probability is $1-e^{-\delta x_i x_j}$ with additive x_i: the product of masses appears in the additive variable $-\log(1-p_{ij})$. Gaussian weights add under aggregation, so the equation is linear from the start and $x_i x_j M$ is its only positive solution. The parallel is in form only.

17.5 Reciprocal Pairs, Anchored Stars, and Loop Holonomy (G10)

17.5.1 The question, the answer, and the picture

A tower in which each level's prior comes from the level above has a top with nothing above it; a tempting closure lets two agents supply each other's priors (the general question is MT Section 12.7). For two latents joined by edge copies in opposite directions, the kernel of the precision is the transported fixed space of the loop holonomy (Proposition G10.1), and the determinant factors through $\det(I - \mathrm{Hol}(p))$ (Proposition G10.8). The flat unanchored fold is therefore singular for all covariances (Corollary G10.2); an anchor, or a holonomy without eigenvalue one, restores a law whose normalizer depends on the links (Proposition G10.3). The standard replacement, an apex with a proper prior, has a mean-field schedule converging at the largest squared canonical correlation (Theorem G10.5). A strict coarse-graining with flat clusters and bare-link cuts forces trivial holonomy everywhere (Proposition G10.9), a conflict the default family does not have. Only one flat unanchored potential is singular; globally normalized cyclic Gaussians are legitimate.

Physics analogue (exact)

Two masses joined by springs and tied to nothing have a free center of mass and a non-normalizable Boltzmann weight. The flat fold is exactly this: each copy charges only disagreement after transport, so transport-consistent configurations cost nothing and the precision has a K-dimensional kernel. Mutual priors carry only relative information; an anchor is a spring to the wall.

Holonomy is the other cure. The conditions $y_1 = \Theta_e y_2$ and $y_2 = \Theta_{e'} y_1$ agree only on the fixed space of $\mathrm{Hol}(p) = \Theta_{e'}\Theta_e$, so without eigenvalue one only the origin costs nothing; structurally, a flux through a ring likewise removes the zero mode and makes the free energy depend on the flux, here through $\det(I - \mathrm{Hol})$. The supplement is explicit about the price: the fiber's origin now does a prior's work, and nontrivial holonomy needs at least one link declared independently of the frames, since frame comparisons telescope to the identity around every loop (MT Proposition 2.13). A declared prior is replaced by a declared link, not removed.

17.5.2 Objects and formulas

Term or symbol	Plain meaning	Physics analogue
Edge copy, reversal	One oriented interaction; parallel copies are distinct; a reversal carries the inverse link	One bond of a multigraph
Coboundary form	Links built from frame comparisons, telescoping around loops	Pure gauge
Strict no-new-anchor family	Presentations with each cut one bare link and no new self term	Coarse-graining that keeps pure gauge

Kernel, determinant, normalizer. With $\delta_e = (I, -\Theta_e)$, $\delta_{e'} = (-\Theta_{e'}, I)$, and $\Sigma_e, \Sigma_{e'} \succ 0$,

$$\Lambda = \delta_e^\top \Sigma_e^{-1} \delta_e + \delta_{e'}^\top \Sigma_{e'}^{-1} \delta_{e'}, \qquad \ker \Lambda \cong \ker(\mathrm{Hol}(p) - I), \qquad \det \Lambda = \frac{\det(I - \mathrm{Hol}(p))^2}{\det \Sigma_e \det \Sigma_{e'}}.$$

The kernel ignores the covariances, links enter the determinant only through the holonomy, and the link-dependent log normalizer $-\log|\det(I - \mathrm{Hol})|$ diverges on a singular locus larger than the flat point ($\mathrm{Hol} = \mathrm{diag}(1, 2)$ is singular). Under $\mathrm{GL}^+(K)$ the pair is proper for generic holonomy; under $\mathrm{SO}(K)$ with K odd every rotation has an axis and the unanchored pair is always singular (Corollary G10.7). For $K = 1$, links $\mathsf{g}, \mathsf{g}^{-1}$, and anchor a_0, $\det(\Lambda + a_0 I) = a_0^2 + a_0(\mathsf{g} + \mathsf{g}^{-1})^2$ (Proposition G10.3): varying a declared link changes the normalizer, so the global term cannot be treated as a per-agent constant while links are learned. This is a change of model, not of coordinates, and it fixes no sign for the full objective.

The anchored star. For an apex $y_0 \sim \mathcal{N}(0, A_0^{-1})$ with $y_i \mid y_0 \sim \mathcal{N}(\Theta_i y_0, \Sigma_i)$, precisions add, $\Lambda_{00} = A_0 + \sum_i \Theta_i^\top \Sigma_i^{-1} \Theta_i$, and on transport-consistent configurations the energy is exactly $v^\top A_0 v$ (Proposition G10.4): the prior supplies exactly the coercivity the fold lacks. Updating constituents from the apex, then the apex, converges to the exact mean,

$$\|\mu_0^{(t+1)} - \mu_0^\star\|_{\Lambda_{00}} \le \varrho_\star \|\mu_0^{(t)} - \mu_0^\star\|_{\Lambda_{00}}, \qquad \varrho_\star = \lambda_{\max}\big(\Lambda_{00}^{-1/2} B_\star \Lambda_{00}^{-1/2}\big) < 1$$

(Theorem G10.5), with $B_\star$ the part of the apex precision routed through the constituents and $\varrho_\star$ the largest squared canonical correlation between apex and constituents. The recursion is block Gauss–Seidel, whose rate also governs the deterministic-scan Gibbs sampler [64]. Without observations $\varrho_\star = 1 - \lambda_{\min}(\Lambda_{00}^{-1/2} A_0 \Lambda_{00}^{-1/2})$; read at large N with fixed A_0, Λ_{00} grows and $\varrho_\star$ approaches one, the familiar slowing of Gibbs samplers for hierarchical models. A singular A_0 without observations gives $\varrho_\star = 1$.

Strict flat coarse-graining against holonomy. For a cut between trivialized clusters the completed square (MT Lemma 10.22) is

$$\sum_e (z_I - \Theta_e z_J)^\top W_e (z_I - \Theta_e z_J) = (z_I - \bar{\Theta} z_J)^\top \Lambda_{II} (z_I - \bar{\Theta} z_J) + z_J^\top \Delta z_J,$$

so the bare link required by the strict family (MT Definition 10.21) exists iff the excess Δ vanishes, that is, iff all cut twists coincide. Nested flat clusters ending in one cluster or a bare two-cluster cut then gauge away every link, and every closed walk has trivial holonomy (Proposition G10.9). The default family admits Δ, exactly the anchor the pair needed; no law-level presentation of a cut with excess is balanced (MT Proposition 10.24), and balanced links have no continuous extension to singular cuts (MT Proposition 10.26, a proved no-go).

17.6 The Decision Metatheory (GA)

17.6.1 The question, the answer, and the picture

The Kron chapter proves specific closure theorems; the appendix asks what holds for every question of the same kind. When fine families, coarse presentation classes, and compatibility conditions are finite semialgebraic descriptions (finitely many polynomial equalities and inequalities, algebraic coefficients, rational matrix entries, finite graph inventories), whether every admitted fine Gaussian has an admitted presentation of its exact marginal is decidable, and the algorithm constructs the exact good and failure loci (Theorem GA.4); the same holds for singular laws on a declared quotient reference (Theorem GA.12). Three limits are proved: no algorithm handles families given by programs that enumerate them (Proposition GA.7); Borel selectors of presentations exist but continuous or gauge-equivariant ones need not (Proposition GA.6); and feasibility across a rank change gives neither total-variation continuity nor quadratic-mean differentiability (Proposition GA.13).

So the appendix is a completeness statement for the regime table: finite, algebraically specified closure questions are never open in principle. The engine is quantifier elimination for the real field (Proposition GA.2), due to Tarski [68] and made algorithmic in Basu et al. [5]; the supplement's own work is the polynomial encoding of the exact marginal through $M^{-1} = \operatorname{adj} M / \det M$ and leading-minor positivity tests (Lemma GA.3). It is also the supplement's only treatment of singular Gaussian laws.

Physics analogue (structural)

A singular Gaussian needs a declared slice and measure, and the law changes unless the reference metric transforms with the data (Theorem GA.10), as a gauge-fixed zero mode depends on its slice and measure [29]. The rank-change obstruction is a singular massless limit: along the precision path $\operatorname{diag}(1, t^2)$ a variance diverges, the Fisher information $2/t^2$ blows up, and the laws have no limit.

17.6.2 Objects and formulas

The good and failure loci are

$$\mathrm{Good}(x) \iff \mathrm{Adm}(x) \wedge \exists y \big[\mathrm{Rep}(x, y)\big], \qquad \mathrm{Fail}(x) = \mathrm{Adm}(x) \wedge \neg \mathrm{Good}(x),$$

with x the fine parameters, y a candidate presentation, and Rep the polynomial statement that y is admitted and reproduces the exact marginal, $(\det M)\Lambda_{\mathrm{c}}(y) = (\det M)\Lambda_{\mathrm{rr}} - \Lambda_{\mathrm{rh}}(\mathrm{adj}\, M)\Lambda_{\mathrm{hr}}$, with the source likewise. Nonemptiness is decided separately, so a vacuous closure is not mistaken for a construction, and the transcendental scalar $\frac{1}{2}\log\det M$ is carried outside the formula. For singular data the covariance is the unique weighted Moore–Penrose inverse (Proposition GA.9), and unboundedly many positive pair summands repack into at most $n(n+1)/2$ by Carathéodory (Proposition GA.5).

At 512 agents the appendix is a statement of principle: a fixed graph with fixed K and algebraic weights has a definite answer, but no complexity bound useful at any practical scale is claimed, and general pair data involve of order N^2K^2 real variables. The thermodynamic limit is outside: changing-dimension limits and smooth, equivariant, or functorial selection are open (Open Problem GA.14).

17.7 What is proved, and how solid it is

Result	What it says	Status	Scope and caveats
G6. Score tangents and the Hermite spectrum			
Theorem G6.4	Block-channel Fisher loss is an expected conditional variance	SOLID	Standard in substance
Proposition G6.3	Replication is a lift of norm $\sqrt{b}$, not a kernel	SOLID	Part (ii) a no-go
Remark G6.6	Gaussian tangent meets bounded actions only at zero	SOLID	Any nondegenerate Gaussian
Theorem G6.8	Normalized-sum spectrum $b^{1-k/2}$	TOY	Scalar independent copies
G7. Aggregation and Kron reduction			
Hypothesis G3.13	Precision is a spring-plus-pin network	CHOICE	Null set for $K, N \geq 2$
Proposition G7.3	Aggregation adds pins and cut weights, drops internal edges	SOLID	Standard (Galerkin)
Proposition G7.4	Aggregated law conditions on zero internal disagreement	SOLID	Linear statistic material
Proposition G7.5	Pushforward by P_Λ; no kernel across precisions	SOLID	Part (v) a no-go
Proposition G7.10	Kron is the exact marginal	SOLID	Standard
Proposition G7.6	Unrestricted matrix Kron leaves the family	SOLID	Refutation
Theorem G7.11	Fixed-pin pair-tested families lie in one cone $\mathcal{W}_H$	SOLID	SPD hidden pin
Theorem G7.14	Universal closure iff at most two terminals	SOLID	Flat, independent, $K \geq 2$

Result	What it says	Status	Scope and caveats
Corollary G7.15	With priors: iff one retained neighbor; else generic failure	SOLID	Pins on hidden agents
Proposition G7.16	A marginal with no positive two-node form	SOLID	Existence only
Theorem G7.20	Degree-two paths close, holonomy kept	SOLID	Zero hidden pin
G8. Nonflat links, relaxation, and quotient volume			
Proposition G8.1	Twisted internal edge leaves a self term	SOLID	Any links
Proposition G8.2	Exponential relaxation; exact natural gradient	SOLID	Constant metric
Proposition G8.3	Forward-KL parent is the moment match	SOLID	Standard
Corollary G8.4	Invariant parent is the Haar-projected match	COND	Compact holonomy
Proposition G8.5	No kernel realizes the two-normal barycenter	SOLID	No-go
Proposition G8.6	Membership Jacobian cancels in normalized laws	SOLID	Any partition
G9. The operator arrow			
Definition G9.1	Aggregate, then rescale by declared ζ_ℓ	CHOICE	No ζ_ℓ selected
Proposition G9.2	Positive, non-injective semigroup	SOLID	Finite levels
Proposition G9.3	Interior law step; kernel iff $\zeta_\ell \leq 1$	SOLID	Interior only
Proposition G9.5	Biadditive family: closed, unique positive additive	SOLID	Not attraction
G10. Reciprocal pairs, anchored stars, and loop holonomy			
Corollary G10.2	Flat unanchored fold is singular	SOLID	One flat potential
Corollary G10.7	Nonflat holonomy shrinks the kernel	SOLID	Odd $\mathrm{SO}(K)$ no-go
Proposition G10.8	Links enter $\det \Lambda$ via $\det(I - \mathrm{Hol})$	SOLID	Two-residual pair
Proposition G10.3	Normalizer varies with declared links	SOLID	$K = 1$ witness
Theorem G10.5	Star schedule rate is a canonical correlation	SOLID	Exact schedule
Proposition G10.9	Strict flat coarse-graining forces trivial holonomy	SOLID	No-go
GA. Decision metatheory			
Theorem GA.4	Finite semialgebraic Kron questions decidable	SOLID	No complexity bound
Proposition GA.6	No continuous or equivariant selector in general	SOLID	No-go
Proposition GA.7	Enumerated families undecidable	SOLID	No-go

Result	What it says	Status	Scope and caveats
Theorem GA.12	Singular laws decidable across ranks	SOLID	Declared reference
Proposition GA.13	Rank change can break regularity	SOLID	No-go

Many SOLID entries are negative: every row marked a no-go or refutation, the necessity halves of the Kron classifications, and the non-identifiability of folded holonomy marks. Each refutes a universal statement, so a small witness suffices, and none is an open question.

Conditions: which ones matter

Conditions that matter: membership in the interaction family, a modeling choice excluding almost every precision when $K, N \geq 2$, on which all closure statements of G7 and G9 rest; $K \geq 2$ for Kron failure; the "every independent assignment on flat links" quantifier of the graph criterion, so a specific correlated network may close outside it; a positive-definite hidden pin for the fixed-anchor classification, degree two and zero pin for series closure, and pins on all hidden agents for the anchored corollary; compact holonomy for the Haar-projected parent, which generic $\mathrm{GL}(K, \mathbb{R})$ loops violate; $\Lambda \succ 0$ for law steps; fixed precision for Fisher contraction along aggregation; the exact schedule for the star; $\Gamma_b \supseteq \mathrm{GL}^+(K)$ for the rank-only label. Supplement G is real throughout. Hygiene: standard Borel spaces, Lebesgue references, continuous conditional versions, domination for local Fisher expansions, and encodings of algebraic coefficients.

17.8 Toy versus general, and the 512-agent target

Most of the part is general in N and K. Its small instances are counterexamples, which are not toy results: the three-agent Kron witness refutes unrestricted closure at every size, and the graph theorem lifts the three-terminal witness to every graph with a three-terminal hidden component. Only the Hermite spectrum is instance-bound: its mechanism, that apparent growth comes from declared extensive normalization because exact arrows contract, rests on general contraction statements, while $b^{1-k/2}$ belongs to scalar independent copies. The one-dimensional normalizer witness proves a negative whose mechanism, the determinant factorization, holds for the pair in every dimension and instances the general kernel-holonomy isomorphism (MT Proposition 10.17). For 512 or more agents with $\mathrm{GL}(K, \mathbb{R})$ links the stated hypotheses give the following; the table adds no result.

Result	At $N = 512$ with $\mathrm{GL}(K, \mathbb{R})$ links
Aggregation, conditioning, Galerkin law	Holds for every hard partition, one step and finite depth
Twisted internal edge as self term	Holds for any invertible links
Biadditive closure, rank-only M	Holds, since $\mathrm{GL}^+(K) \subseteq \mathrm{GL}(K, \mathbb{R})$

Result	At $N = 512$ with $\mathrm{GL}(K, \mathbb{R})$ links
Kron closure, flat positive family	Fails generically for relays when every agent has a prior and $K \geq 2$
Nonflat series closure	Holds on degree-two paths; branching can defeat every positive pair form
Holonomy-constrained parent	Empty off a null set of links; compact groups covered
Reciprocal pair	Generic nonflat pairs proper; odd $\mathrm{SO}(K)$ always singular
Star contraction	Holds at any N; read without observations, the rate tends to one as N grows
Hermite spectrum; decision metatheory	Not applicable; decidable in principle only

Two gaps separate this part from a thermodynamic-limit reading of renormalization with Kadanoff blocks. It supplies the exact one-step block map, the analogue of a recursion relation, and its finite compositions, but no hierarchical identification, rescaling rule, or limiting protocol, all deferred to Supplement B. And its fibers are real: $\mathrm{GL}(K, \mathbb{C})$ and $\mathrm{SU}(N)$ enter only through real representations, and no statement here is specialized to them.

17.9 Open ends

The supplement's ledger is Table GB.2; every row bears on this part.

Item	What is unknown	What would settle it
Open Problem G9.4	Monotonicity under varying precision, at the boundary, at unbounded depth	Lyapunov functional; boundary repair; projective limit
Open Problem GB.3	Running aggregation backward	Priors for discarded weights and composable inverse kernels, or a no-go
Open Problem GB.1	Relation to Bayesian renormalization [7]	A common state space and a jointly monotone functional, or a counterexample
Open Problem GB.2	Model-space Fisher scale for a recognition precision	Coarse mean made generative, with nondegeneracy
Open Problem G1.14	Scaling of a retained mass coordinate	Induced mass law with an orbit theorem
Open Problem G1.18	Information projection versus gauge or coarse maps	Intertwining theorem or counterexample
Open Problem GA.14	Kron classification beyond finite encodings	Proof for a stated family, or finite reduction
Open Problem G10.6	Star under delayed, noisy, or asynchronous updates	Theorem with a rate, or counterexample

Behind these sit a natural partition selector (MT Open Problem 9.19) and an intrinsic criterion for recognition structure (MT Open Problem 5.14). Fixed points, flows, and the thermodynamic limit are declined and deferred to Supplement B. For large N the first three rows matter most: without them the exact step says nothing about many steps.

17.10 Verdict

Does it deliver what it claims? DELIVERED

Claimed. Closure of hard aggregation, identified with conditioning on zero within-cluster disagreement, without selecting the family or a partition; refutation of unrestricted Kron closure, the congruence-diagonal and fixed-anchor classifications, and the two-terminal criterion; an operator arrow that is a positive semigroup needing a declared rescaling, with interior law steps and biadditive closure; reciprocal pairs, anchored stars, and loop holonomy; a decision metatheory; and no fixed point or asymptotic scaling.

Proved. All of it, at the stated generality. G6 DELIVERED (the spectrum is the scalar case it announces); G7 DELIVERED (explicit witnesses for every negative); G8 DELIVERED (compactness is in the statement); G9 DELIVERED (no flow asserted); G10 DELIVERED (singularity confined to one flat potential); GA DELIVERED (decidability in principle).

Gaps. The part is narrower than the physicist's "renormalization" and says so: one exact step and its finite compositions, hard partitions, a declared measure-zero family of real matrix-spring networks. Its positive content is mostly standard mathematics in new typing; the new content is the Kron classification, the holonomy bookkeeping, and the no-go results. For the 512-agent program, integrating out relays with matrix weights generically leaves the flat positive family, holonomy-constrained Gaussian parents do not exist for generic $\mathrm{GL}(K,\mathbb{R})$ loops, and monotonicity under varying precision, stochastic inversion, iteration, and limits remain open or deferred.

Chapter 18

Supplement B: Base Transport, Infinite Networks, and the Lattice Cluster

18.1 The question and the answer

The main manuscript works at one base point c_*: every frame is a point of one principal fiber, the independent links on the edges of the agent network are the only transports, and the design is $\{c_*\}$. Supplement B collects what the theory would need if contexts varied over a base, if the network became infinite, or if agents sat on a lattice. For each extension it asks which single-point statements survive and exactly which additional data the extension consumes: a connection, a cover, a design, a base measure, a scale identification, or a lattice realization (Chapter B0). Nothing in the main manuscript depends on it. This chapter reads it at overview depth with the scope gate in force: it reports what the supplement proves and where it stops, and proposes nothing.

Part B-I answers the base question with kinematics. Relative frames, chosen connections, covariant defects of cross maps, four independent notions of flatness, finite designs, a Fisher geometry pulled back to the base, gauge-invariant durations for field histories, and a descent criterion are all typed and proved, with counterexamples wherever an expected implication fails. No connection, base measure, or probability law on fields is derived (Open Problem B1.22), and graph links become base transports only by declaration (Hypothesis B1.13). Part B-II answers what Wilsonian vocabulary requires: comparison maps that the coarse channels do not supply. A finite network at c_* carries no asymptotics (Proposition B5.15), beta functions are scheme dependent, and on Gaussian hierarchies attraction is proved only under a fixed primitive identification whose existence is open (Proposition B6.22). No infinite agent network, exponent, or universality class is constructed. Part B-III constructs one infinite-volume family completely: the square-lattice $\mathbb{Z}_2$ gauge–matter law at small plaquette coupling, with a coarse specification at every exterior for every block size on a block-independent domain (Theorem B7.7), a differentiable renormalization map, and proved volume limits of pressure, label selection, specific VFE, and Fisher densities (Corollary B9.7). Whether its sites are agents at c_* or base points is open (Open Problem B7.1).

The supplement does what its introduction says. Its title words promise more: no base transport of beliefs is derived from inference, no infinite agent network is built, and the one constructed infinite-volume family is untyped, abelian, and flows to the trivial fixed point (section 18.9).

18.2 Objects and notation

The supplement adds its own symbol table for Parts B-I and B-II (Appendix BD) and a local table for the lattice part (Table B7.1). The decoder covers what a reader meets first.

Term or symbol	Plain meaning	Physics analogue
Support $\mathcal{C}_i$; design $D = \{c_d\}$	Contexts where agent i exists; a finite set of sampled contexts, not a discretization of the base.	Local field support; probe points.
Relative frame h_i	Group element comparing belief and model frames, $u_i^m = u_i^b \cdot h_i$.	Bifundamental field of two gauge sectors.
Connections ω_b, ω_m; transports Ω_γ	Chosen rules for comparing fibers along base curves.	Gauge potential and Wilson line.
Covariant defect $\mathcal{D}\Phi$	Failure of a belief-to-model cross map to commute with transport.	$\nabla\Phi$ for a Hom-valued field.
Graph flatness	Every closed walk of the agent graph has trivial link product.	Pure-gauge lattice configuration.
Covariant jet $D^\omega s$; pullback h_s^ω	Rate of change of a belief field along the base with frame rotation removed; its squared Fisher length.	$D_\mu\phi$; gauged sigma-model metric.
Horizontal defect A_Ψ	Mismatch of fine and coarse connections under a coarse map over a base map.	Mismatched gauge field under blocking.
Comparison data J_ℓ	Declared identification of each blocked theory with a reference.	Rescaling step of a Wilsonian RG map.
Relevant, marginal, irrelevant	Sign of a growth rate per unit log block scale, relative to declared norms and normalizations.	Sign of an RG exponent y.
$z_e = \phi_x \psi_e \phi_{x+e_\iota}$	Gauge-invariant link sign of the lattice model.	Unitary-gauge link of the $\mathbb{Z}_2$ Higgs model.
$b = \tanh h$, $t = \tanh j$	Strengths of matter hopping h and plaquette coupling j.	High-temperature expansion variables.
Specification at every exterior	Coarse conditional laws defined and continuous at every boundary condition, including null ones.	Gibbsianness of the renormalized measure.
Label γ, bulk score s_γ	A member of a finite catalog of lattice laws and its extensive log-mass density.	Coexisting phases and their free energies.

Glyphs change meaning between parts. The letter b is the belief channel in Part B-I, a block ratio $b > 1$ in Part B-II, and $\tanh h$ on the lattice; t is a scale coordinate in Chapter B6 and $\tanh j$ on the lattice; beta functions are $\mathfrak{b}$, not the attention rows β_{ij}^x; on the lattice $\rho = \tanh(h + 2j)$ and τ is a decay rate. Words are narrower than a physicist expects. "Pullback" means the connection-relative pullback of fiber tensors to the base, not Supplement G's restriction along a parameter map. "Flat" has four meanings (section 18.3.2). "Connection" can be a principal connection, the scale connection of Definition B6.1, or a projected connection. "Relevant" is never a property of a flow

alone (Definition B5.9), and "fixed point" is typed by tier: a fixed object at one tier need not be fixed at another; six specific non-implications are proved in Proposition B6.10. "Declared" means chosen rather than derived, and usually marks where modeling enters.

18.3 Part B-I: base transport and multi-point designs

18.3.1 Frames, connections, and cross maps

Over a base, the belief and model channels of agent i may carry separately chosen local sections of the one principal bundle, and the torsor structure makes their comparison canonical:

$$u_i^m = u_i^b \cdot h_i, \qquad h_i \longmapsto g_i^b h_i (g_i^m)^{-1}, \qquad \Theta_e^m = h_i^{-1} \Theta_e^b h_j \quad (e : j \to i).$$

The relative frame h_i exists for any pair of representations, of any dimensions, and maps no representation space into another (Proposition B1.1). It transforms on the left under belief rechoices and on the right under model rechoices, like a bifundamental field. The two channels carry one link on e exactly when the third equation holds (Proposition B1.4); the main manuscript's literal equality $\Theta_e^b = \Theta_e^m$ is its gauge $h_i = e_G$. The second atlas does not enlarge the gauge group, whose active image at a context is still one copy of G (Proposition B1.2); a genuine $G_b \times G_m$ is an optional extension (Definition B1.16).

Connections are chosen data, with the textbook frame law $A' = \mathrm{Ad}_g A - (\mathrm{d}g)g^{-1}$ (Proposition B1.7). The new object is the covariant defect of a cross map,

$$(\mathcal{D}\Phi)_e(X) = T_e\Phi\big(H_e^b X\big) - H^m_{\Phi(e)} X, \qquad \text{in the linear case} \qquad D\Phi = \nabla^m \circ \Phi - \Phi \circ \nabla^b.$$

The first term carries the belief connection's horizontal direction through Φ, the second is the model connection's horizontal direction at the image. Their difference is vertical and vanishes exactly when Φ commutes with parallel transport along every base curve (Proposition B1.9). At c_* there is no base vector and no defect. Across scales, a coarse-graining map between principal bundles over a base map exists only when a topological condition holds, as for the Hopf bundle, which no homomorphism $z \mapsto z^n$ with $n \neq 0$ maps into a trivial bundle (Proposition B1.31).

18.3.2 Four kinds of flatness

At c_* every bundle is trivial and every connection vanishes, so "flat" can only mean graph flatness. Over a base four conditions separate completely (Proposition B1.15), with the first decided by one Čech class common to both atlases (Proposition B1.10).

Condition	Gauge-theory meaning	Separating witness
(a) Bundle triviality	A global gauge exists.	A flat S_3-bundle over a circle, glued by a 3-cycle, is nontrivial.
(b) Graph flatness	Agent-graph links are pure gauge.	Declared independently of (a), (c), (d).
(c) Connection flatness	$F = 0$.	$x\mathrm{d}y$ on $\mathbb{R}^2$: uniform field, not flat.
(d) Trivial holonomy on one loop	One Wilson loop is trivial.	$x\mathrm{d}y$ around a circle enclosing flux 2π; conversely an Aharonov–Bohm vortex is flat with nontrivial phase.

For the program this means that flat links at c_*, condition (b), say nothing about (a), (c), or (d). The supplement also records that one principal class does not make the belief and model bundles isomorphic (a cylinder and a Möbius band share one $\{\pm 1\}$-bundle), and that a global section of an associated bundle need not trivialize the principal bundle (Proposition B1.10).

18.3.3 When is a graph link a base transport?

When it is declared to be one. The standing hypothesis Hypothesis B1.13 chooses a context per agent and a base curve per edge copy and requires $\widehat{\rho}_b(\Theta_e^b) = \Omega_{\gamma_e}$ and $\widehat{\rho}_m(\Theta_e^m) = \widetilde{\Omega}_{\gamma_e}$. Under it, graph loop products are base holonomies (Proposition B1.14), a one-line consequence of the homomorphism property. The supplement is explicit that the hypothesis is satisfiable by definition, that no result realizes it for independently specified links (Open Problem B4.14), that recovering a connection from records is open (Open Problem B4.15), and that even an exact realization concerns connection holonomy, not bundle topology.

Physics analogue (structural)

At c_* the links are lattice gauge variables over a zero-dimensional base (chapter 2). The hypothesis is Wilson's discretization read backwards: a lattice link is declared to be the path-ordered exponential of a continuum potential along an assigned curve. In lattice gauge theory that is the definition of the discretization, and here too it is a definition, not a theorem.

18.3.4 Finite designs and shared links

A finite design is the multi-point version of $\{c_*\}$ (Definition B1.18). Under the design-point product hypothesis, distinct design points are conditionally independent given the structural data (Hypothesis B1.23), and the free energy decomposes over the design (Corollary B1.24); the hypothesis is substantive, since it excludes correlation of an agent's states at two contexts beyond the structural data. If a submodel ties the copies of one link across design points, each tied link ranging over all of G, the surviving gauge freedom is

$$g^x_{d,i} = \bar{g}^x_i z_d, \qquad g^x_{d,j} = \bar{g}^x_j z_d, \qquad z_d \in Z(G),$$

a context-independent rechoice times a central twist shared by both endpoints (Proposition B1.26). Tying a link across contexts turns local freedom along the design into global freedom up to the center, which is $\mathbb{R}^{\times} I$ for $\mathrm{GL}(K,\mathbb{R})$, $\mathbb{Z}_N$ for $\mathrm{SU}(N)$, and trivial for $\mathrm{SO}(3)$.

The finite construction supplies no probability law on a space of sections (Open Problem B1.22), and matching sections to finite laws at the design points determines neither the joint nor the sections off the design (Proposition B1.20). The named obstruction is the one familiar from constructive field theory: reference measures are finite products, and infinitely many non-probability factors give no measure, so a continuum theory must declare a measure on fields directly. Until then every evidence-bound statement of the main manuscript and of Supplement B is a statement about a finite design.

18.3.5 Base-direction information geometry

Once contexts vary, a belief field s changes along the base, and the connection separates the part of that change that is only frame rotation:

$$D^{\omega} s = \mathrm{ver}^{\omega} \circ T s, \qquad h^{\omega}_{s}(X,Y) = g^{F}_{s(c)}\big(D^{\omega} s(X), D^{\omega} s(Y)\big)$$

(Definition B2.1). The first is the analogue of $D_\mu \phi$; the second is the squared Fisher speed of the belief along the base direction X, whose square root is the Fisher–Rao distance per unit displacement between the belief at c and the belief transported back from a neighbor. It is invariant under passive frame changes (Theorem B2.2) but not under a change of connection: for $\omega' = \omega + \mathfrak{d}$,

$$h^{\omega'}_{s}(X,Y) = h^{\omega}_{s}(X,Y) + g^{F}\big(\mathsf{J}^{s}_{\mathfrak{d}} X, D^{\omega} s Y\big) + g^{F}\big(D^{\omega} s X, \mathsf{J}^{s}_{\mathfrak{d}} Y\big) + g^{F}\big(\mathsf{J}^{s}_{\mathfrak{d}} X, \mathsf{J}^{s}_{\mathfrak{d}} Y\big),$$

with $\mathsf{J}^{s}_{\mathfrak{d}}$ the vertical field generated by the connection difference (Proposition B2.4). A belief constant over the base has zero pullback under the zero connection and $\mathrm{c}_1^2 \mathrm{d}x^2$ under $A' = \mathrm{c}_1 \mathrm{d}x$, because the new connection says the fiber is being shifted underneath it. This answers what a "metric on the base built from beliefs" is: a metric relative to a chosen connection, and the documents choose none (Open Problem BD.1). It is also only a semimetric, with radical the directions of covariant constancy; constant rank gives a metric on a quotient vector bundle (Theorem B2.7), and a quotient manifold can fail even then, since the null directions can form a contact distribution (Proposition B2.9). Combining the belief and model channels needs declared weights (Definition B2.5).

Under a coarse map Ψ over a base map f, fine and coarse base geometries obey the exact identity

$$h^{\omega}_{s} - f^{*} \bar{h}^{\bar{\omega}}_{\bar{s}} = \delta_{\Psi} - \mathcal{X}_{\Psi} - \mathcal{Q}_{\Psi}$$

(Theorem B2.19). Here δ_Ψ is the Fisher information lost through the fiber channel, nonnegative for a Markov channel, the main manuscript's data-processing loss read along base directions; $\mathcal{X}_\Psi$ is a sign-indefinite cross term and $\mathcal{Q}_\Psi \geq 0$ a quadratic term, both produced by the horizontal defect. With zero defect and a Markov channel the comparison is a contraction and a meta-agent's perceived base geometry is smaller than its children's (Corollary B2.18, Corollary B2.22); with

nonzero defect it can be negative at zero information loss (Theorem B2.20). For maps induced by a principal scale map the defect is the fundamental field of $\mathfrak{A}_{\mathcal{P}} = \mathcal{P}^*\bar{\omega} - \mathrm{d}\kappa \circ \omega$ and vanishes exactly when $\mathfrak{A}_{\mathcal{P}}$ takes values in the isotropy algebra of the coarse section value, with $\mathcal{Q}_\Psi$ the orbit Fisher form evaluated on $\mathfrak{A}_{\mathcal{P}}$ (Theorem B2.25). Composed defects obey an ordered cocycle law (Theorem B2.27), and the base Fisher defect composes up to an exact residual (Theorem B2.29) whose relaxed pointwise problems are completely parameterized.

Physics analogue (structural)

The pullback is the kinetic metric $|D_\mu s|^2$ of a gauged sigma model with the statistical manifold as target, and its connection dependence is that of $|D_\mu \phi|^2$ on A_μ. In field theory the gauge field is dynamical or fixed by an equation of motion; the supplement supplies neither. The isotropy criterion has the form of a Higgs mass term: a quadratic form in the gauge field whose radical is the algebra fixing the vacuum value. The correspondence is of form only.

18.3.6 Section-space histories

When an agent's state is a whole field, its inference history is a curve of sections, and it changes pointwise vertically: at each context the belief moves within its fiber, and a base curve is not an agent history (Proposition B3.5). On a strong Riemannian Hilbert manifold with a free, proper, isometric gauge action, the quotient speed

$$\|d\pi_Q v\|^2 = \min_{\xi \in \mathfrak{g}} \mathrm{g}_Q(v - A_Q\xi, v - A_Q\xi)$$

has a unique minimizer, so gauge-invariant durations are well defined (Theorem B3.10); the minimization removes the velocity along gauge orbits, as a Coulomb-type condition does. It fails for weak metrics: the L^2 Fisher metric on H^1 fields on a circle gives a smooth proper quotient on which every tangent has zero speed (Proposition B3.11), and that weak metric is itself the Fisher metric of a natural observation family, so the obstruction is not an artifact of an exotic metric. Averaging a field over the fibers of a base map loses a nonnegative amount of integrated Fisher information, given a Markov fiber map, nonincreasing weights, and a jointly convex fiber chart; when the coarse metric is moreover constant in the chart, the loss is exactly the sum of a channel loss, a weight gap, and a context variance (Theorem B3.23). Without joint convexity the conclusion fails: in the covariance sector of the Gaussian chart the loss is negative on an explicit datum (Proposition B3.24).

18.3.7 Covers, availability, and descent

Chapter B4 is the part closest to the original picture of agents sprinkled over a base with overlapping supports. Four agents on a square of contexts have strip supports that overlap in pairs and triples with no common point (Theorem B4.2), and each is present in a record with a smooth availability probability $w_i(c)$ that vanishes off its support (Theorem B4.3). Attention then behaves as one would want over overlaps, with one qualification: $\beta^x_{ij}(c) = \eta^x_{ij}(c)/\alpha^x_i(c)$ is smooth where the receiver is present, while the event law η is smooth everywhere and imposes no row where the receiver

is absent (Proposition B4.4). The main manuscript's parent needs all its children present, so a four-child parent here has empty domain; an availability parent weighting locally present children by $w_i/\sum w$ exists on the union of supports (Proposition B4.8). The weights are chosen, and every compatibility statement holds by construction from one assumed ambient law (Hypothesis B4.5); gluing independently supplied local laws is Open Problem B4.9.

The substantive result is the descent criterion: along a surjective submersion, a section descends exactly when its image is constant on fibers, and a connection form exactly when its two pullbacks to the fiber product agree (Theorem B4.20). Around it, a coarse map with context-dependent weights has defect $O_{\mathrm{c}}(\mathrm{d}L)O_{\mathrm{f}}^{-1}$ even when all connections agree (Theorem B4.11); projecting a flat connection onto a moving quotient creates curvature $\bar{\mathrm{g}}^{-1}\mathrm{II}^{\top}\wedge\mathrm{II}$ from the second fundamental form, and staged projections agree with the direct one when the intermediate quotient metric is propagated, while resetting it can change the connection class (Theorem B4.17, Theorem B4.18). Finally, a declared record $o_1=(\mathrm{Hol}_{\iota}-I)P_{\mathrm{pr}}Y+\varepsilon$ distinguishes two base-holonomy classes by a frame-invariant statistic (Theorem B4.13), the base companion of the main manuscript's network-loop witness (MT Theorem 10.28). The record reads the holonomy by construction; the theorem shows that such a normalized experiment exists, not that the model's own records are holonomy sensitive.

Physics analogue (exact)

The projected-connection curvature is the Gauss equation for an induced connection; with a flat ambient connection and a moving rank-one image it is Berry curvature. The composition result says iterated adiabatic projections agree with the direct one when each stage uses the metric induced by the last.

18.3.8 How Part B-I connects to the base-manifold picture

Part B-I supplies the typed skeleton of the original picture: supports and Čech data, relative frames, chosen connections, covariant defects, finite designs, a pulled-back base geometry, gauge-invariant durations, availability-weighted attention over overlaps, and descent along base maps. It supplies no dynamics for any base datum. The connection is chosen, not determined by inference; base measures and weights are declared; the ambient law on a cover is assumed; and without a law on fields the evidence bound over a base exists only on finite designs. The single-point graph links and the base transports are related only by declaration. Part B-I is a precise list of what must be supplied when base transport is taken up, not yet a theory of agents over a base.

18.4 Part B-II: infinite networks and Wilsonian limits

18.4.1 What Wilsonian words require, and why finite depth has none

The main manuscript coarse-grains finite networks by exact channels and keeps an effective action, a closure residual, and finite-step multipliers (chapter 11). A Wilsonian step is "block, then identify

the blocked theory with the original", and the identification is extra structure: declared comparison isomorphisms J_ℓ, giving the reference cocycle $\widehat{\mathfrak{R}}_\ell = J_{\ell+1}^{-1} \circ C_{\ell,\ell+1} \circ J_\ell$ (Definition B5.1). Exponents are growth rates per unit of cumulative log block scale $\varsigma_{n\leftarrow\ell} = \sum_k \log b_k$,

$$\chi_{n\leftarrow\ell}(u) = \frac{\log\|D_{n\leftarrow\ell}u\|_n - \log\|u\|_\ell}{\varsigma_{n\leftarrow\ell}},$$

with $D_{n\leftarrow\ell}$ the main manuscript's multipliers (Definition B5.9). Where defined, their canonical content is the forward Lyapunov filtration (Proposition B5.10); they are invariant under tempered comparisons (Theorem B5.12), while a superexponential comparison can change the apparent exponent (Proposition B5.13); and no Oseledets splitting is claimed, since a deterministic blocking sequence meets none of the hypotheses of the multiplicative ergodic theorem [58]. Even uniform projective contraction can leave an attracting two-cycle rather than a fixed ray (Proposition B5.3).

Proposition B5.15 makes precise what every physicist knows about blocking a finite system. With blocks of at least two agents the depth is at most $\lfloor \log_2 |V_0| \rfloor$, so the limit defining an exponent does not exist; levels of different sizes admit no comparison isomorphism; and exponent signs depend on the block schedule unless it is two-sided bounded: a mode doubled at each step has exponent 1 under $b_k = 2$ and $2/(n+1) \to 0$ under $b_k = 2^{k+1}$. This does not conflict with renormalization at c_* as a thermodynamic limit; it says that asymptotic vocabulary needs one, an infinite network or a family in which size and depth grow together (section 18.4.4), whereas the main manuscript's interaction multigraph is finite by definition (MT Definition 2.11).

The exact step contributes no growth of its own: along an exact interaction orbit $\chi_{n\leftarrow\ell}(u) \le |V_n| \log 3/\varsigma_{n\leftarrow\ell}$, so no direction is relevant in the native norms when $|V_n|/\varsigma \to 0$, and the same holds in the reference norm when the comparisons are also tempered (Corollary B5.18). The condition $|V_n|/\varsigma \to 0$ is substantive: the bound grows with the number of coarse agents, and it is vacuous for blocking by vertex ratios on a finite network. Relevance must come from a declared extensive normalization. In the scalar Gaussian block each Hermite function of degree k has coefficient $b^{1-k/2}$, so degree one is relevant, two marginal, and higher degrees irrelevant, and the growth of degree one comes entirely from the extensive replication normalization, of norm $\sqrt{b}$, since the conditional expectation contracts (Definition B5.19). Correlated coordinates change this spectrum, a general $\mathrm{GL}(d)$ action leaves the setting, and no link or holonomy enters (Proposition B5.20).

Physics analogue (structural)

In Wilson's renormalization group the blocking step, a conditional expectation, contracts; eigenvalues above one come from the rescaling of fields that accompanies it [79]. Part B-II proves this division of labor for the exact step of the main manuscript. The Hermite spectrum is the Gaussian fixed point of the central limit theorem read as a renormalization map, whose binary case is in Jona-Lasinio [42].

18.4.2 Beta functions and fixed points

A beta function is a difference quotient on one reference space. For the action, $\mathfrak{b}_b^H[H;\varrho] = (\mathfrak{R}_b^H[H;\varrho] - H)/\log b$, and a change of reference $\varrho' = e^{-\Delta}\varrho$, $H' = H - \Delta$, which leaves the weighted measure unchanged, shifts it by the inhomogeneous term $-\mathfrak{b}_b^H[\Delta;\varrho]$ (Proposition B5.24). A beta function is thus a joint statement about the action and the reference trajectory. The retained beta of a truncated family differs from the exact one by the transported closure residual over $\log b$ (Proposition B5.26); a moving comparison frame gives even the identity step a nonzero beta, and what survives smooth conjugacy are projective eigenvalue ratios at a fixed ray (Proposition B5.27); and integer endpoints do not determine a continuous beta (Proposition B6.3).

A fixed theory is a measure pair, reference and weighted measure, fixed by the rescaled kernel. If the kernel has a unique invariant law, every fixed pair has constant action (Proposition B6.6), so a fixed theory with nontrivial action needs several invariant laws; this depends on the typing choice that the reference is pushed by the same kernel. The essential-spectrum theorem of the chapter has a hypothesis no exact arrow satisfies (Theorem B6.12), and one exact kernel, the circle doubling map, has spectral radius 1 on bounded functions and 2^α on Hölder functions (Proposition B6.13), so a spectrum belongs to a declared topology.

18.4.3 The homogeneous Gaussian hierarchy

On a homogeneous complete graph with equal blocks of size b, common self term A, and common coupling W, hard aggregation acts by

$$A \longmapsto bA, \qquad W \longmapsto b^2 W,$$

informally, because a block has b self terms and two blocks have b^2 cross couplings. On $\mathbb{S}^K \oplus \mathbb{S}^K$ the spectrum is b and b^2, each with multiplicity $K(K+1)/2$, so the ratio outside the dominant sector is $1/b$ (Proposition B6.20). Normalizing by b^2 fixes the coupling and contracts the self term toward a limiting ray with $A = 0$, a boundary operator whose consensus zero modes make the Gaussian non-normalizable, so pinning, quotienting, or a small mass are different infrared repairs. On general graphs the exponents follow the scaling of the cut: $\xi = 2$ for the complete graph, $\xi = 0$ for a chain. This is mean-field power counting for Gaussian couplings.

It does not show attraction. Birkhoff's cone contraction cannot reach the boundary ray, since sum-only aggregation is not primitive on the full cone and does not mix matrix directions (Proposition B6.21). Attraction is proved under one fixed primitive identification $B \otimes \mathrm{id}$ on edge indices: $r_B^{-\ell}(B^\ell W^{(0)})_f \to v_f M_\infty$ with $M_\infty = \sum_e u_e W_e^{(0)}$ and u, v the Perron vectors (Proposition B6.22). The spatial profile is attracted; the $K \times K$ fiber matrix is a Perron-weighted average of the initial couplings and is not selected. Whether any graph family admits such an identification is Open Problem B6.23, and nonautonomous identifications are not covered. Attraction on an infinite hierarchy is therefore not established.

18.4.4 Two-index limits, and what the absence of universality means

A thermodynamic scale statement has two indices, size n and depth ℓ, and three limits: fixed depth after size, maximal depth, and a diagonal $\ell(n)$ (Definition B6.14). They need not agree: for the blocked path graph the fixed-depth limit of the spectral count is the arccos law, the maximal-depth limit a single site (Proposition B6.15). A countable index set carries a joint law by Kolmogorov extension (Theorem B6.17), but consistency gives no ELBO limit and no scale flow. That no universality class is identified (Open Problem B6.19) means that the supplement states only what one would need: a triangular family of finite agent networks at c_* with consistency at fixed depth, uniform control of the coarse error, a diagonal scaling, and proved convergence of a scheme-invariant observable (Open Problem B6.19). Raw couplings and beta components are scheme-dependent coordinates; a spectral exponent is a label only after convergence of the integrated density of states is proved (Definition B6.24), and a finite graph has an atomic spectrum. For the view of renormalization at c_* as a thermodynamic limit, the two-index family is the typed home, and it is open.

18.4.5 The numerical statements

Supplement B has three NUMERICAL statements, all in Part B-II: a declared rescaling family on nested six-agent cycles fails the semigroup law, with total variation 0.042 between direct and staged blocking (Remark B5.5); a frozen-attention projection scheme on a six-cycle and a nine-cycle reaches fixed structures with vanishing pairwise block (Remark B6.7); and a row-regeneration scheme on the same instances keeps a nonzero pairwise block (Remark B6.8). The schemes are declared approximations, not the exact step, and the supplement says nothing here shows the exact pushforward maintains interaction. The implementation has been retired and the values were not recomputed (Appendix BD.3). The other rows of that index are exact rational checks of lattice statements that rest on their proofs.

18.5 Part B-III: the lattice cluster

18.5.1 The model and the flow

The source law lives on the edges of the square lattice, with matter signs ϕ_x, raw links ψ_e, and dressed signs $z_e = \phi_x \psi_e \phi_{x+e_\iota}$. On a finite region with exterior η its weight relative to counting measure is

$$r_\Lambda^\eta(z) = \prod_{e\in\Lambda}(1+bz_e) \prod_{p\cap\Lambda\neq\varnothing} \big(1+t\chi_p(z\eta)\big), \qquad b=\tanh h, \quad t=\tanh j$$

(Definition B7.2). Since $e^{hz} = \cosh h(1+bz)$ and $e^{j\chi} = \cosh j(1+t\chi)$, this is $\exp(h\sum_e z_e + j\sum_p \chi_p)$, the $\mathbb{Z}_2$ gauge theory of Wegner [76] coupled to a $\mathbb{Z}_2$ Higgs field as in Fradkin and Shenker [31], in unitary gauge: the first product is matter hopping, a field on each link, and χ_p is the plaquette holonomy. Small t is strong gauge coupling, where Osterwalder and Seiler [59] proved

existence and analyticity. For $0 < b < 1$ and $0 < t < 1/6$ the Gibbs law is unique and translation invariant, the matter signs are independent fair coins, and the law depends on them only through z (Theorem B7.3).

The channel C_B keeps every B-th matter sign and replaces each coarse link by the product of the B fine links along the straight path it covers; every fine link off the coarse grid lines is integrated out. It is gauge equivariant, composes exactly, $C_a C_B = C_{aB}$, and a coarse plaquette holonomy is the product of the fine ones inside it (Proposition B7.5). The central theorem (Theorem B7.7) says that for $0 < b < 1$, $0 < t < 1/12$, and every B, the pushed law has a unique positive specification, defined and continuous at every coarse exterior including null ones, whose dependence on the exterior beyond radius R decays like $B^2(R+1)e^{-\tau(R+1)}$, and that the domain $12t < 1$ does not depend on B. The renormalized measure is Gibbsian, with a renormalized Hamiltonian, at every block size on one fixed window, so the non-Gibbsianness catalogued by van Enter et al. [72] does not occur there. The generated interaction is complete with norm tending to zero in B (Theorem B7.9); for the integer factors $a \in \{2, 3, 6\}$, on a ball of character coefficients, the renormalization map $\mathscr{R}_a$ is Fréchet C^1 with contracting linearization, so zero is its only fixed point there (Theorem B7.11); and for a compact box of rational source parameters and rational norm parameters, the source law enters that ball once B exceeds an explicitly computable B_0 (Theorem B7.14).

The flow is to the trivial fixed point, and this should be read plainly. As B grows the blocked laws converge on fixed cylinders to the fair product law, for every Gibbs law at any finite couplings (Proposition B7.6); informally, because a long Wilson-line parity multiplies many signs whose conditional means stay away from ± 1. Interacting fixed points, critical exponents, and a continuum gauge-field measure lie outside the construction (Open Problem B7.17). What is nontrivial is the control: Gibbsianness uniform in block size, a complete interaction with explicit bounds, and a differentiable map, all constructed. At each fixed depth the Fisher information per coarse cell is a C^2 density that is a regular metric on a nonempty open region (Theorem B7.15), and $\mathcal{I}_{aB} \preceq a^2 \mathcal{I}_B$: per fine cell information decreases, the data-processing inequality (Proposition B7.16).

Physics analogue (exact)

The source is the $\mathbb{Z}_2$ gauge–Higgs model in unitary gauge, and the straight-path channel is blocking of gauge links by straight Wilson lines along the coarse grid, a product map rather than decimation.

Physics analogue (structural)

The open-ball construction is of the kind Israel [40] gave for decimation and Kadanoff maps of Ising-spin interactions at high temperature; here the channel and the interaction space are different, and the supplement separates its results from those maps.

18.5.2 Pressure, label selection, and the specific VFE

Chapter B8 places a finite catalog of such laws on one lattice; each label is a value of the structural mark of the main manuscript's partition catalog (MT Definition 12.12, chapter 12). Pressures exist

with explicit surface rates, $|\mathfrak{p} - n^{-2}\log Z_n| \le 4\mathsf{m}_1/n$ (Theorem B8.8); existence is standard [33] and the rates are the addition. Under global normalization the label with the largest bulk score takes all the weight exponentially fast in the volume, and at a tie the outcome is decided by subextensive terms, not by pressures (Theorem B8.11). Forgetting the label can destroy Gibbsianness outright: a mixture of two Bernoulli product laws has no continuous conditional at any exterior (Theorem B8.5).

Observation is handled for one record: every edge sensor reports $+$ (Hypothesis B8.13). It shifts each branch field from b_γ to $(b_\gamma+\eta_\gamma)/(1+b_\gamma\eta_\gamma)$, and the specific VFE of a label-retaining recognition satisfies $\min_{\mathbb{Q}} \mathfrak{f} = -\bar{e}$, attained on the maximal posterior-score labels with the shifted Gibbs laws (Proposition B8.19): the Gibbs variational principle in the thermodynamic limit, with a finite label. Along a fixed blocking word $B^2\mathfrak{f}_{\mathrm{fine}} = \mathfrak{f}_{\mathrm{coarse,induced}} + \mathfrak{d}_{\mathrm{fib}}$ with $\mathfrak{d}_{\mathrm{fib}} \ge 0$ (Theorem B8.24), the volume-limit form of the main manuscript's exact coarse free-energy identity. A general record produces random fields of both signs; its quenched evidence density is Open Problem B8.15.

18.5.3 Thermodynamic Fisher geometry

On compact source boxes with $h > 0$ and $0 < j < \operatorname{atanh}(1/12)$, the native pressure is C^4, and since the native law is an exponential family in $\mathsf{g} = (h, j)$ its Fisher density is the Hessian of the log-partition density, $\frac{1}{N} I \to \mathsf{H} = \nabla^2_{\mathsf{g}} \mathsf{A}$ in C^2 (Theorem B9.3, Proposition B9.4); pushes along a fixed word have the same C^2 density per coarse cell as the infinite-law marginals (Theorem B9.5). The main manuscript's finite-mixture asymptotics (MT Section 9.8) then hold with $N = L^2$ cells (Corollary B9.7): with a strict winner the Fisher density converges to the winner's; at a tie $I/N^2 \to \mathrm{Cov}_w(\mathsf{v}_\gamma)$, the covariance of the bulk-score slopes, which is information about the label; and in a window of width $1/N$ the label law is logistic.

Physics analogue (exact)

For an exponential family the Fisher metric is the Hessian of the log-partition function, the thermodynamic metric of fluctuation theory [18], per cell the susceptibility matrix.

Physics analogue (structural)

The tie and window statements have the form of finite-size rounding at a first-order transition, which Borgs and Kotecký [11] derive from the lattice model itself. Here the phase mixture is prescribed through the label weights, which makes the statement elementary; the correspondence is one of form.

18.5.4 The typing question and the thermodynamic-limit reading

No lattice statement declares what a site is (Open Problem B7.1). In the first reading the sites are agents at c_*, $G = \mathbb{Z}_2$, the raw link is an independent link, the dressed sign is the per-edge twist, and a plaquette is the holonomy of an agent four-cycle. In the second the sites are points of a discretized base, the raw link is the transport of a $\mathbb{Z}_2$ connection, a plaquette is a discrete curvature, and the channel is a scale map over a base map. Every statement holds under either reading.

The first reading is the reader's own: links as lattice gauge theory over a zero-dimensional base, and renormalization at c_* as a thermodynamic limit. Under it the lattice cluster is the one place in the three documents where a thermodynamic limit of a link law on an agent graph is actually constructed. The open problem states what is missing before the reading can be adopted: the lattice law must be the link marginal of a normalized joint with belief and model channels, a recognition law, and an ELBO on a countably infinite agent set, whereas the main manuscript's multigraph is finite and the lattice family carries none of these. The lattice part also fixes every blocking word before its volume limit; joint growth of depth and volume is part of Open Problem B7.17, and where depth and volume do grow together the limit is the trivial one. Even under the first reading the group is finite and abelian, so the per-edge twist is gauge invariant and nonabelian holonomy classes never appear, far from the target of 512 or more agents with $\mathrm{GL}(K,\mathbb{R})$, $\mathrm{GL}(K,\mathbb{C})$, or $\mathrm{SU}(N)$ links; the domain is a single-phase strong-coupling region, with a unique Gibbs law and a smooth pressure; and observation enters through one constant record. The lattice cluster proves that the coarse-channel calculus can be run to completion in infinite volume for one abelian family. It is not evidence about formation in the target class.

18.5.5 What is new, and the classical antecedents

The model and tools are classical: Wegner [76], Fradkin and Shenker [31], and Osterwalder and Seiler [59] for the model; Dobrushin uniqueness [25] and boundary comparison [33]; Kolmogorov extension; Kozlov's Gibbs representation [47]; Israel's high-temperature renormalization maps [40]; and van Enter et al. [72] for Gibbsian pathologies, of which Theorem B8.5 is a Bernoulli case in sharper form. The principal candidate for a new result is the block-size uniformity of $12t < 1$ in Theorem B7.7: the surveyed high-temperature results concern decimation, Kadanoff, and averaging maps of Ising-spin interactions, not this gauge–matter path channel, and the proof avoids a block estimate whose constant grows like Bt. The literature check is limited to the works the chapter cites, and priority is not claimed.

18.6 What is proved, and how solid it is

Established no-go statements and separating counterexamples count as solid. The scope column says how far each result's generality is from the program's target.

Result	What it says	Status	Scope and caveats
Proposition B1.1, Proposition B1.4	A unique relative frame; one shared link means $\Theta_e^m = h_i^{-1}\Theta_e^b h_j$.	SOLID	Standard in substance (torsor).
Proposition B1.15	Bundle triviality, graph flatness, connection flatness, and one-loop holonomy are independent.	SOLID	$\mathrm{U}(1)$ and S_3 witnesses; at c_* (a), (c), (d) hold trivially and only (b) has content.
Hypothesis B1.13	A graph link is a base transport only by declaration.	CHOICE	Realization for independent links is open.

Result	What it says	Status	Scope and caveats
Proposition B1.26	Tying a link across contexts leaves constant rechoices times a central twist.	SOLID	Admissible rechoices form a groupoid.
Open Problem B1.22	No law on fields; evidence bounds are finite-design statements.	OPEN	No infinite product of reference measures.
Theorem B2.2, Proposition B2.4	The base Fisher pullback is passively covariant and changes with the connection by exact terms.	SOLID	No connection-free base metric; only a semimetric.
Theorem B2.19	Fine minus coarse base Fisher is channel loss minus two anomaly terms.	SOLID	Exact; can be negative at zero loss.
Theorem B2.25	The horizontal defect vanishes exactly when the defect form lies in the isotropy algebra.	SOLID	Pointwise; no compatible connection claimed.
Proposition B4.8	An ambient law gives compatible local data; availability parents glue.	COND	Holds by construction (functoriality) from one assumed ambient law; gluing from local data open.
Theorem B4.20	Sections and connections descend exactly under fiber constancy and pullback agreement.	SOLID	Standard in substance.
Theorem B4.13	A declared record distinguishes two base-holonomy classes.	TOY	Base-loop instance of the MT witness, whose mechanism is general (SOLID in chapter 10); one loop, one connection family, Gaussian record; recovery beyond it is Open Problem B4.15.
Proposition B5.15	A finite network has bounded depth, no comparisons, and schedule-dependent exponents.	SOLID	A no-go for finite N, not for $N \to \infty$.
Corollary B5.18	The exact step has no relevant direction; relevance needs an extensive normalization.	COND	Needs $\lvert V_n\rvert/\varsigma \to 0$: the bound is exponential in $\lvert V_n\rvert$, so it is silent when coarse levels keep many agents, and vacuous for finite vertex-ratio blocking. Tempered comparisons are needed only for the reference-norm clause.

Result	What it says	Status	Scope and caveats
Proposition B6.20	Complete graph: $A \mapsto bA$, $W \mapsto b^2W$; dense normalization contracts the self term by $1/b$.	SOLID	Exact; limiting ray is a boundary operator.
Proposition B6.22	Under a fixed primitive identification orbits converge; the fiber matrix is not selected.	COND	Standard in substance (Perron–Frobenius on a primitive B); existence of the identification is open.
Remark B5.5, Remark B6.7, Remark B6.8	A rescaling family fails the semigroup law; frozen attention kills the pairwise block, regeneration keeps it.	NUM	Six- and nine-agent schemes; not recomputed.
Open Problem B6.19	Joint size and depth limits and universality for agent networks.	OPEN	Limits need not commute.
Theorem B7.7	Every straight-path blocking of the $\mathbb{Z}_2$ gauge–matter law at $12t < 1$ is Gibbsian at every exterior, on a domain independent of block size.	SOLID	Principal candidate for a new result; $\mathbb{Z}_2$, strong coupling.
Theorem B7.9, Theorem B7.11	Complete interaction tending to zero; for $a \in \{2, 3, 6\}$, a C^1 map on a ball whose only fixed point is zero.	SOLID	Israel-type in form, different channel; trivial fixed point.
Proposition B8.19	Specific VFE minimum is minus the evidence density.	COND	Only for the constant all-positive record.
Corollary B9.7	C^2 Fisher densities; strict, tie (N^2), and $1/N$-window limits.	SOLID	Compact boxes; blocking word fixed before volume.
Open Problem B7.1	Are lattice sites agents at c_* or base points?	OPEN	No lattice statement is yet about agents.

18.7 Conditions that matter

Conditions: which ones matter

Substantive conditions, which could fail in the cases the program cares about: chosen connections (Definition B1.6), relative to which every base-direction statement is made; the declaration Hypothesis B1.13; conditional independence of design points (Hypothesis B1.23); declared base measures, channel weights, and availability weights; constant rank of the covariant jet (which can jump, as in the rank-jump witness of Proposition B2.9), integrable null directions, and basic tensors for quotient geometries; chart joint convexity (Hypothesis B3.22), which fails for Gaussian covariances; a strong metric on a declared manifold of fields (Hypothesis B3.4), which the Fisher metric of the weak-metric witness is not; one ambient law on a cover (Hypothesis B4.5); declared comparison data, infinite blocking sequences, and tempered comparisons for every asymptotic statement; one fixed primitive identification for attraction; and on the lattice

the group $\mathbb{Z}_2$, small plaquette coupling with positive field, compact source boxes, blocking words fixed before volume limits, controlled label growth (Hypothesis B8.10), and the constant all-positive record (Hypothesis B8.13).
Hygiene a physicist can take for granted on a first reading: standard Borel spaces, measurability of section responses (Hypothesis B1.19), smoothness classes C^1 to C^4, paracompactness and the constant-rank bookkeeping that restricts the defect classifications to regular strata (a technical restriction, unlike the rank of the jet itself), integrability of the integrals, the submersion hypothesis of descent, boundary conventions, and centering of squares in local limits.

Toy content is limited and labeled: the four-agent cover, the holonomy witness, the scalar Hermite spectrum, and the numerical six- and nine-agent schemes. The many small counterexamples of Part B-I are not toys; each refutes a general implication and is a solid negative result. The lattice is infinite but special in its group, abelian and finite, and in its single-phase coupling region. Its mechanism, keeping a renormalization map Gibbsian uniformly in block size by a transport argument, is specific to strong coupling and unshown for nonabelian or non-compact links; its constants $(12, \mathrm{q}, \mathrm{c}_1)$ say nothing about the target class.

18.8 Open ends

The supplement lists twenty open problems (Table BD.2). For Part B-I, the unknowns are a probability law on fields with continuum limits of the bound (Open Problem B1.22); the orbit space of the two-atlas coordinate action (Open Problem B1.30); a base geometry fixed by the model rather than a chosen connection (Open Problem BD.1); whether declared recognition families have the structure the history theorems use (Open Problem B3.14) and whether descending fields form a smooth submanifold (Open Problem B3.19); gluing local laws on a cover (Open Problem B4.9); realizing independent graph links as base transports (Open Problem B4.14); recovering a connection from records (Open Problem B4.15); quotients across rank changes (Open Problem B4.19); and informative averaging along a base map (Open Problem B4.21). Each is settled by a construction for a declared class or by an obstruction.

For Part B-II, the unknowns are rescaling kernels on agent networks obeying the semigroup law (Open Problem B5.6); a multiplicative ergodic structure for scale cocycles (Open Problem B5.14); relevance spectra of correlated or gauge-mixed Gaussian blocks (Open Problem B5.21) and a basin for the Gaussian block law (Open Problem B5.22); a canonical continuous beta (Open Problem B6.4); joint size and depth limits with a convergent scheme-invariant observable (Open Problem B6.19); and a fixed primitive identification for some graph family (Open Problem B6.23). For Part B-III, they are the typing (Open Problem B7.1), settled by proving one reading; critical points, depth-uniform Fisher floors, joint depth and volume limits, and self-consistent curvature actions (Open Problem B7.17); and the quenched evidence density for random records (Open Problem B8.15), which needs every-exterior control of random fields of both signs.

Three gates carry most of the weight. Without Open Problem B1.22 there is no evidence bound over a base. Without Open Problem B4.14 the agent graph's links and the base's connection stay separate objects joined only by declaration. And Open Problem B6.19, with Open Problem B7.1, gates renormalization as a thermodynamic limit: the typed statement exists, the one constructed instance is an untyped abelian lattice at strong coupling, and no agent-network instance with its recognition law exists.

18.9 Verdict

Does it deliver what it claims? NARROWER THAN IT SOUNDS

What it says it does. It collects what the theory requires once contexts vary, networks become infinite, and agents sit on a lattice, stating for each extension which single-point statements survive and which additional data it consumes (Chapter B0). It declines to identify the base with physical space, graph links with base transports, or any continuum law, and declines asymptotic vocabulary for finite networks.

What the statements give. Part B-I gives the kinematics of a base extension, with exact identities and separating counterexamples: relative frames, covariant defects, four independent flatness notions, the shared-link stabilizer, an exact fine-to-coarse base Fisher comparison with a sharp isotropy criterion, durations on strong metrics, and descent. Part B-II gives a typing of Wilsonian vocabulary, no-go and scheme-dependence results, exact Gaussian power counting, and one conditional attraction theorem. Part B-III gives a complete constructive renormalization of one $\mathbb{Z}_2$ family at strong coupling, with pressure, label selection, specific VFE, and Fisher densities in the volume limit.

The gap. "Base transport" is transport by a chosen connection, not transport derived from inference; base measures, weights, and ambient laws are declared; no law on fields exists; and graph links become base transports only by declaration. "Infinite networks" names the apparatus an infinite network would need; no infinite agent network with channels, recognition law, and ELBO is constructed, no exponent or fixed point of the agent flow is established, and universality is a stated program. The lattice cluster is untyped, abelian, confined to strong coupling, observed through one record, and flows to the trivial fixed point. Each part does what its opening paragraph says; the supplement as a whole is a precise map of what base transport, infinite networks, and spatial realization would require, not a theory of any of them.

Part III

Back Matter

Glossary

The technical documents use many words in senses of their own. This glossary gives each one a plain meaning and, where one exists, the nearest physics analogue, with the companion chapter that explains it. An analogue listed here is a pointer, not an identification: the chapter says whether the correspondence is exact, structural, or loose.

Geometry and gauge at one base point

Term or symbol	Plain meaning	Physics analogue
Base point c_*	The single context at which the whole main manuscript works (chapter 1).	A zero-dimensional base.
Frame u_i	The agent's chosen point of the one principal fiber over c_*, used for both channels (chapter 2).	Local gauge choice; kinematically a group-valued Higgs field.
Passive re-expression g_i	A per-agent change of frame; coordinates change, nothing physical does (chapter 2).	Local gauge transformation read as a change of trivialization.
Active gauge group	Automorphisms of the one shared fiber: a single copy of G inside the product of per-agent groups (chapter 2).	A global symmetry.
Frame comparison T_{ij}	The unique group element relating two agents' frames in the same fiber; no path involved (chapter 2).	Canonical pure-gauge link.
Link Θ_e^x	An independent group element on an oriented edge copy, per channel (chapter 2).	Lattice link variable.
Per-edge twist	A link written against one reference point; its conjugacy class is invariant (chapter 2).	A unitary-gauge (dressed) link.
Holonomy $\mathrm{Hol}^x(p)$	Ordered product of links around a closed walk (chapter 2).	Wilson loop before the trace.
Graph flat; frame aligned	Every loop product is the identity; links equal to the frame comparisons (chapter 2).	Pure gauge; canonical pure-gauge background.
Law coherence	Agents' laws equal each other's transports on every internal edge, up to holonomy stabilizers (chapter 2).	Covariantly constant matter field.
Cross morphism Φ	A map between belief and model fibers; extra structure, not implied by the bundles (chapter 2).	Coupling between two matter species.

Term or symbol	Plain meaning	Physics analogue
Tree transport, rooted holonomy	Link products along a spanning tree to a block root, and all loop products at the root (chapter 10).	Maximal-tree (axial) gauge fixing.
Block-dressed edge	A boundary link with tree transports attached at both ends (chapter 10).	Covariant link between block roots.
Connection Laplacian; fixed fiber	$\sum_e (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j)$ and its kernel (chapter 10).	Covariant lattice Laplacian and its zero modes.
Frame sector	The meta-agent's frame, a free orbit invisible to its members (chapter 10).	Pure gauge.

Probability, the joint law, and the evidence bound

Term or symbol	Plain meaning	Physics analogue
Record, record kernel K_a	An observation or interaction factor attached to a set of agents, normalized as a conditional law (chapters 3 and 5).	A bond or plaquette Boltzmann factor.
Evidence	Marginal likelihood of the records (chapter 3).	Partition function.
Recognition law; selected coupling	The population's approximate posterior, a joint law with the agents' local laws as marginals, chosen by a declared rule (chapters 3 and 5).	Variational trial state, correlated in general.
Posterior slice	The exact posterior given the observation and the structural data (chapter 1).	Exact conditioned Gibbs state.
Law point	A belief treated as a latent random variable in its own right (chapter 3).	A one-body distribution treated as a field.
Evaluator	The map from a model presentation to a normalized generative kernel; a model is not itself a law (chapters 3 and 9).	Coupling-to-Hamiltonian map.
Typing prohibition	No generative kernel may read a recognition law or the posterior (chapter 4).	A Hamiltonian independent of the trial state.
Extended evidence bound	$\log p(o) - D_{\mathrm{KL}}(\mathbb{Q} \Vert \mathbf{\Pi})$, defined for every recognition law and equal to $-\infty$ on the singular branch (chapter 5).	Minus the Gibbs trial free energy.
Singular branch	The recognition law charges a set the posterior cannot see, so the bound is $-\infty$ (chapter 5).	A delta-function state on a continuum.
Total correlation	Relative entropy of a joint law from the product of its marginals (chapter 5).	Multi-information; the mean-field entropy error.
Local conditional free energy	A block's free energy with the rest of the population held fixed (chapter 5).	Finite-volume free energy with a boundary condition.

Term or symbol	Plain meaning	Physics analogue
Exact block E-coordinate	Replacing a block's law by its exact local posterior (chapter 5).	Heat-bath update, in law.
Accepted M-coordinate	A parameter step that does not lower the bound at fixed recognition (chapter 5).	The M-step of expectation maximization.
Operational environment node	An environment rewritten as a node that emits records (chapter 5).	A boundary degree of freedom.

Attention and selective communication

Term or symbol	Plain meaning	Physics analogue
Source label (pointer)	A latent categorical variable naming which neighbor a receiver listens to (chapter 6).	A Potts-like hidden assignment.
Source prior π_{ij}; attention row β_{ij}	Prior over sources, and the posterior (generative) or recognition row over them (chapter 6).	Prior weight; Boltzmann weight over sources.
Receiver occupancy α_i; event law η	Which receiver an event belongs to, and the joint receiver–source law $\eta = \alpha\beta$ (chapter 6).	Joint bond occupation.
Attention temperature τ_i	Scale of the record energy; in the Gaussian record only W/τ matters (chapter 6).	Canonical temperature.
Constant-row gap	The exact cost of using one attention row for every configuration (chapter 6).	Mean-field decoupling error.
Query and key maps S_e^Q, S_e^K	Linear maps from the receiver's and the source's internal spaces into a common comparison space (chapter 6).	None standard.
Communication map L_e	Encoder, transmission, and decoder composed; may be singular, and it is not the link (chapter 6).	A general linear link field.
Value map	The matrix by which the exact receiver update moves toward a source: a Kalman gain times the key map (chapter 6).	Gain times observation matrix.

Information geometry and dynamics

Term or symbol	Plain meaning	Physics analogue
Fisher metric	Covariance of the score (chapter 7).	Susceptibility matrix.
History	An oriented curve of configurations with no time label (chapter 7).	A trajectory without its clock.

Term or symbol	Plain meaning	Physics analogue
Fisher duration (information clock)	Fisher arc length along a history, invariant under reparameterization (chapter 7).	Thermodynamic length.
Oriented semiconjugacy	A coarse map carrying the fine flow to the coarse flow up to an increasing time change (chapter 7).	Exact dynamical coarse-graining.
Parity amplitude $\varkappa$	Strength of a six-bit correlation invisible to every proper marginal (chapter 7).	A six-spin coupling.
Fast and slow blocks	A declared split of coordinates by rate (chapter 8).	Electrons and nuclei.
Reduced functional F_{red}	The free energy minimized over the fast block (chapter 8).	Born–Oppenheimer surface.
Instantaneous refresh	Exact resampling of the fine state given the parent at every step (chapter 8).	Adiabatic elimination at infinite separation.
Partial refresh	Resampling with probability below one, which makes the parent non-Markov (chapter 8).	Finite separation with a memory kernel.

Coarse-graining

Term or symbol	Plain meaning	Physics analogue
Coarse channel	A Markov kernel on latent coordinates that reads neither recognition nor observation (chapter 9).	Kadanoff block-spin weight.
Energy precomposition; Galerkin restriction	Substituting a restricted configuration into an energy; compressing an operator (chapter 9).	Constrained configurations; the multigrid coarse operator.
Discarded conditional information	Expected relative entropy between recognition and posterior inside the blocks, given the coarse value (chapter 9).	Information lost inside blocks.
Pointwise parent datum	The pushforward of the generative, posterior, and recognition laws of a block through one channel (chapter 9).	The block-spin ensemble.
Extensive labeled family	A mixture over finitely many labels whose log masses grow with the number of agents (chapter 9).	A multiphase ensemble; ties are phase coexistence.
Natural selector	A relabeling-natural, gauge-compatible rule choosing a partition from the data; open (chapter 9).	A choice of blocking scheme.
Holonomy-conditioned score	Least weighted relative entropy from the transported marginals to a holonomy-invariant parent (chapter 10).	A twirled barycenter.

Term or symbol	Plain meaning	Physics analogue
Effective action	Minus the log density of the pushed evidence measure (chapter 11).	Block Hamiltonian; integrating out.
Hoeffding coordinates	One interaction component per set of agents, centered in each agent (chapter 11).	Cluster or ANOVA expansion.
Hyperedge	An interaction component on three or more agents, generated by coarse-graining (chapter 11).	A multi-spin coupling from decimation.
Bounded tier	The regime of essentially bounded interactions in which the coupling calculus is proved (chapter 11).	Finite or compact spins.
Native growth bound	The multiplier norm is at most $3^{\lvert V_n\rvert} - 1$ in the target's size (chapter 11).	No growth without rescaling.
Finite-network obstruction	Coupling spaces cannot be identified across a step that reduces the number of agents (chapter 11).	Finite-volume renormalization terminates.
Strong lumpability	A coarse process that is Markov for every initial law (chapter 11).	Kemeny–Snell lumpability.
Memory kernel; finite-memory defect	Exact non-Markov corrections, and the relative-entropy cost of a memory-k approximation (chapter 11).	Nakajima–Zwanzig.
Membership kernel; structural mark	Soft assignment of an agent to a block, and the random partition a structural channel produces (chapters 11 and 12).	Fuzzy block spin; random blocking.
Declared root	A parentless top latent with a supplied prior (chapter 12).	A boundary condition at the top.
Lifted joint	Parameters and record copies made latent so that delayed feedback is one joint law (chapter 12).	State augmentation in filtering.

Formation

Term or symbol	Plain meaning	Physics analogue
Coherence episode	Coherence held by a named restoring interaction and observed on a horizon that separates it from equilibrium (chapter 13; rule R15).	A dynamical ordering event.
Restored equilibrium coherence	Coherence the stationary law already has with probability tending to one; not formation (chapter 14).	Equilibrium order.

Term or symbol	Plain meaning	Physics analogue
Supplied-scale sequence	A second coalescence that happens regardless of the first stage (chapter 13; rule R14).	A hierarchy put into the Hamiltonian.
Frozen arm	The same model with learning switched off; the null for a selection claim (chapter 14; rule R13).	A control without feedback.
Energy strengths $\mathrm{J}_\ell = \epsilon^{-\mathrm{Q}_\ell}$	The supplied stiffness of each level of the formation chain (chapter 13).	A coupling hierarchy.
Preparation	The chain's initial state: a displaced common mean with thermal spread (chapter 13).	A Gibbs state in an applied field switched off at time zero.
Marked parent	The pushforward onto the retained space, keeping every label and loop (chapter 13).	Exact decimation.
Source gating	An edge carries energy only while one of its endpoints attends to it (chapter 13).	Bonds slaved to labels.
Sheet	The slice of state space at one fixed attention assignment (chapter 13).	A sector.
Minimum edge cover	The fewest edges touching every agent; the stationary blocks of the gated family (chapter 13).	None.
Matching seed	An initial assignment that is a perfect matching, with the initial means in a small ball about one of two specified configurations $\pm x_\circ$ (chapter 14).	A favorable initial condition.
Source learning, mobility ϵ_{L}	Prior rows relax toward the recognition marginal at a slow rate (chapter 14).	A slow variable.
Transient imprint	The time-integrated early asymmetry that learning amplifies (chapter 14).	A vanishing symmetry-breaking field.
Discarded conditional relative entropy	Mean relative entropy of the coordinates a parent drops, given the ones it keeps (chapters 13 and 14).	Error of adiabatic elimination.
Activation clock	The time at which a parent is switched on (chapter 14).	Waiting for fast modes to thermalize.

The Gaussian realization

Term or symbol	Plain meaning	Physics analogue
Interaction family	Precision $\Lambda_{ii} = A_i + \sum_j W_{ij}$, $\Lambda_{ij} = -W_{ij}$ with positive semidefinite blocks (chapter 16).	Massive matrix-weighted lattice Laplacian.
Self term A_i	The on-site block of the precision (chapter 16).	Mass or pinning term.
Gain absorption	Frames that turn every transition on a forest into the identity (chapter 16).	Axial gauge.

Term or symbol	Plain meaning	Physics analogue
Orientation criterion	$(-1)^K \det \Lambda_{12} > 0$: when a pair's coupling can be gauged to the family (chapter 16).	A ferromagnetic bond sign ($K = 1$).
Determinant gap	The loss of the bound from factorizing a Gaussian by blocks (chapter 16).	Mean-field free-energy error.
Interaction pencil	The pair (L, Λ) and its generalized eigenvalues (chapter 16).	A normal-mode problem.
Hard aggregation; Galerkin law	Identifying each cluster with one coordinate; the fine law conditioned on zero disagreement inside clusters (chapter 17).	Rigid-block Kadanoff block spin; the algebraic-multigrid operator.
Kron reduction	Integrating out hidden agents by a Schur complement (chapter 17).	Integrating out; star–mesh transform.
Internal residue	The term $(I - \Theta)^\top W (I - \Theta)$ a twisted internal edge leaves on a cluster (chapter 17).	Mass generated by a flux.
Operator arrow	Aggregation followed by a declared rescaling of the coarse precision (chapter 17).	Field rescaling of a renormalization step.
Declared repair	A pin, quotient, or retained mass at a singular member of the family (chapters 16 and 17).	Infrared regulator.
Reciprocal fold	Two agents joined by opposite edges with no anchor; its precision is singular when flat (chapter 17).	Two masses on springs with no ground.

Base transport and the lattice cluster

Term or symbol	Plain meaning	Physics analogue
Relative frame h_i	The group element comparing belief and model frames when they differ (chapter 18).	A bifundamental field of two gauge sectors.
Covariant defect	The failure of a belief-to-model map to commute with parallel transport along the base (chapter 18).	A covariant derivative of a Hom-valued field.
Four notions of flatness	Bundle triviality, graph flatness, connection flatness, trivial holonomy on a loop; at c_* only graph flatness has content (chapter 18).	Topological sector, pure gauge, zero field strength, trivial Wilson loop.
Fisher pullback	Squared Fisher speed of a belief along the base, relative to a chosen connection (chapter 18).	Gauged sigma-model kinetic metric.
Comparison data	A declared identification of each blocked theory with a reference (chapter 18).	The rescaling step of a Wilsonian flow.

Term or symbol	Plain meaning	Physics analogue
Relevant, marginal, irrelevant	Sign of a growth rate per logarithmic block scale, relative to declared norms (chapter 18).	Sign of a renormalization-group exponent.
Specification at every exterior	Coarse conditional laws continuous at every boundary condition (chapter 18).	Gibbsianness of the renormalized measure.
Specific VFE	Variational free energy per cell in the volume limit (chapter 18).	Free-energy density.

Open Ends of the Theory

The main manuscript lists its open problems in one table (MT Appendix E), each with a settling condition: a class of models and two outcomes, a proof or a counterexample. The individual chapters of this companion explain them where they arise. This chapter groups them by the question a physicist would ask, says in which class each one is posed, and ends with the questions the program cares about that the technical documents do not pose at all.

One feature of the settling conditions shapes everything below. Rule R16 of the authoring specification requires both outcomes to be nontrivial within a stated class, so the problems are posed in classes narrow enough to be decidable. Several of the central ones are posed on the four-cycle kinetics of MT Definition 13.21 or on switching diffusions of that type. Settling such a problem settles it for that class; it would ground the question at small size but would not answer it for a network of hundreds of agents.

The parent's own dynamics

Exact coarse-graining produces the law of a parent at every instant (MT Corollary 9.10). Whether that law evolves as an autonomous process, a parent with dynamics of its own, is MT Open Problem 11.38, which the introduction calls the defining open problem of the phase. It asks whether a time-homogeneous Markov parent, or one carrying a finite filter from a declared regular class, reproduces the native path law of the retained coordinates within a total variation that vanishes as $\epsilon \to 0$, both on windows of fixed length and over the complete history. It is posed for switching diffusions of the four-cycle type. The four-cycle continuation results of chapter 14 answer special cases on the positive side: a parent activated at a suitable time, after a coherence prefix, has vanishing discarded information and a retained path law close to the native one on short windows. They do not settle the problem, because they fix the activation event and the horizon. MT Open Problem 6.27 asks the same question for a block that communicates through learned low-rank interfaces: is there a parent of fixed lower dimension whose path law of the communicated coordinates is asymptotically exact on the learning time scale?

Formation that is not supplied

The formation theorem of chapter 13 relaxes a prepared state through a hierarchy of interaction strengths supplied by hand. Four open problems ask for more. MT Open Problem 13.16 asks whether the doubling recursion $\mathrm{Q}_\ell = 2\mathrm{Q}_{\ell+1} + 20$ of the strengths can be relaxed to inner base $4/3$, a separation that is still doubly exponential in the depth, with constants independent of the depth. MT Open Problem 13.20 asks, on the four-cycle, for parents with vanishing conditional and

dynamical errors together with a margin, bounded below, by which learning or the loop coordinate of the initial mean changes the law of the first formed block; or for a proof that for almost every initial mean both differences vanish. MT Open Problem 13.25 asks for repeated endogenous formation on general finite graphs with temperature-independent data: a second coalescence that depends on the first by a margin, is held by a restoring interaction, is observed on dynamical horizons, and has well-behaved parents; or a proof that none exists in that class. MT Open Problem 12.27 asks for the full package: a lineage of at least two levels formed with probability bounded below from every initial law with a density and a finite second moment, an intrinsic partition selector, and a slow manifold on which the upper level feeds back consistently. These are the questions on which the program's claim that collectives form through coupled inference rests. None of the four takes a limit of many agents; the last three are posed at fixed finite size in the low-temperature limit, while the regime-sharpness problem asks for constants uniform in the depth, and hence in the number of agents of the family.

Selecting structure

No functional analyzed in the documents selects a nontrivial partition into meta-agents: the costs studied either prefer an endpoint (every agent separate, or everything merged) or need an external coefficient. MT Open Problem 5.14 and MT Open Problem 9.19 ask for a functional of the laws, natural under relabeling and compatible with the gauge action, not monotone under refinement, with a nondegenerate optimizer, or for an impossibility theorem in a stated class. MT Open Problem 11.25 asks whether deterministic block channels generically admit retained coordinates beyond the trivial families of chapter 11. Structural dynamics of the graph itself is open too: MT Open Problem 6.9 asks for a normalized joint in which edges are created and deleted by a threshold rule while the collective free energy does not increase, and MT Open Problem 6.22 asks for an exact recursion for multi-head attention at the level of source labels.

Convergence and time scales

The alternating scheme of inference and learning is not known to converge (MT Open Problem 5.32). The reduction of fast beliefs onto a slow manifold is proved for held assignments; tracking along assignment paths with jumps is MT Open Problem 8.7, posed on the four-cycle. Delayed feedback across learning episodes has an exact joint law, and whether the per-episode accounting reproduces the aggregate update to first order is MT Open Problem 12.26. How a meta-agent's information clock compares with its constituents' is MT Open Problem 7.43.

Foundations

Five open problems concern the measure-theoretic and geometric footing: likelihood families with no common dominating measure (MT Open Problem 3.9), a normalized prior over law points

(MT Open Problem 3.23), equivariant posterior versions for noncompact stabilizers (MT Open Problem 9.18), global Fisher quotients (MT Open Problem 7.15), and correlated lifts beyond the finite parity witness (MT Open Problem 7.29). The equivariant-versions problem is open only when a law's stabilizer is noncompact and its posterior has no weakly continuous version (MT Proposition 9.16, MT Corollary 9.17). Noncompactness of $\mathrm{GL}(K)$ alone does not put the program's Gaussian laws in the open case: under a faithful linear representation a nondegenerate Gaussian law has a compact stabilizer, conjugate into $\mathrm{O}(K)$, so the open case concerns degenerate laws, nonfaithful actions, and non-Gaussian fibers.

Contact with the world

Two items concern testability rather than mathematics. MT Open Problem 14.7 asks for held-out evidence, under a protocol fixed in advance, that an object of the coarse-graining matches a named target system beyond a baseline. MT Open Problem 14.8 asks for a test at one base point that could falsify nontrivial loop holonomy or the formation of a block, or a proof that the records cannot distinguish them. The operational holonomy theorem of chapter 10 shows that records sensitive to network-loop holonomy exist in principle.

The supplements

Supplement G keeps its own open rows: the scaling of a retained mass sector (Open Problem G1.14), commutation of the information projection with coarse maps (Open Problem G1.18), monotonicity of Galerkin aggregation beyond one fixed precision (Open Problem G9.4), robustness of the anchored-star update (Open Problem G10.6), Kron classification beyond finite encodings (Open Problem GA.14), a bridge to Bayesian renormalization (Open Problem GB.1), transfer of a model-space Fisher scale (Open Problem GB.2), and stochastic inversion of aggregation (Open Problem GB.3). Chapters 16 and 17 explain them. Supplement B holds the open problems of the deferred program, among them continuum law theory (Open Problem B1.22), graph-to-base identification (Open Problem B4.14), compatible rescaling kernels on agent networks (Open Problem B5.6), two-index limits and universality (Open Problem B6.19), and the typing of lattice sites (Open Problem B7.1); chapter 18 explains them.

Questions the documents do not pose

Some questions the program cares about appear in none of the open-problem tables, either because they were deferred by decision or because no one has yet posed them in a decidable form. Listing them is not a criticism of the documents, which state their scope; it is a map of where the theory stops.

None of the main manuscript's open problems takes the limit of many agents. The limits taken are low temperature, slow learning, and small separation parameters; two problems concern size

without a limit (uniformity in the depth for the regime-sharpness problem, and the dependence of a clock ratio on the number of agents in MT Open Problem 7.43). The main manuscript's only large-size statements are the retained-label Fisher theorems of chapter 9. They assume that the log masses of a finite labeled mixture grow in proportion to an extensive size such as the number of agents, and derive the Fisher geometry from that assumption; near a tie between labels the natural coordinate is $N(\theta - \theta_0)$ (MT Theorem 9.30), the form of the finite-size rounding of a first-order transition, here in a retained label. They do not derive the extensive growth from a network of interacting agents, and they say nothing about formation. A thermodynamic limit of the agent network at one base point, with nonabelian and noncompact links, is therefore neither proved nor posed there; Supplement B holds the infinite-network and asymptotic apparatus. For the same reason, phase transitions, condensation, and critical slowing down are neither ruled in nor ruled out as formation mechanisms: at finite size, with a unique law-level dynamics, they cannot occur in the sense of statistical mechanics, and the main manuscript contains no statement about formation in the limit in which they could.

A normalized law on noncompact links is not constructed. The loop penalties of chapter 2 give a finite normalizer for compact groups with unitary representations; for $\mathrm{GL}(K)$ the question is declined in the scope paragraph of MT Chapter 2, and no open problem poses it. Relatedly, the holonomy-conditioned parent score of chapter 10 is infinite for generic $\mathrm{GL}(K)$ links (MT Corollary 10.11). That score is one particular construction, a projection of marginal laws onto a holonomy-fixed family, not the exact coarse channel, whose parent law always exists; but any use of it with the program's target groups needs a restriction of the links or a different parent family.

Base-manifold transport, the setting of the program's original picture of agents spread over a base with attention varying smoothly over overlaps, is deferred by the author's decision until the single-point theory is settled; its apparatus and open problems are in Supplement B.

Bibliography

[1] P.-A. Absil, R. Mahony, and B. Andrews. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization*, 16(2):531–547, 2005. doi: 10.1137/040605266.

[2] S. Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10:251–276, 1998.

[3] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2): 251–276, 1998.

[4] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, 2000.

[5] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*, volume 10 of *Algorithms and Computation in Mathematics*. Springer, Berlin, Heidelberg, 2nd edition, 2006. doi: 10.1007/3-540-33099-2.

[6] Cédric Bény and Tobias J. Osborne. Information-geometric approach to the renormalization group. *Physical Review A*, 92(2):022330, 2015. doi: 10.1103/PhysRevA.92.022330.

[7] David S. Berman, Marc S. Klinger, and Alexander G. Stapleton. Bayesian renormalization. *Machine Learning: Science and Technology*, 4(4):045011, 2023. doi: 10.1088/2632-2153/ad0102.

[8] R. Bhatia. *Positive Definite Matrices*. Princeton Series in Applied Mathematics. Princeton University Press, 2007.

[9] C.M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[10] D.M. Blei, A. Kucukelbir, and J.D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112:859–877, 2017.

[11] Christian Borgs and Roman Kotecký. A rigorous theory of finite-size scaling at first-order phase transitions. *Journal of Statistical Physics*, 61(1–2):79–119, 1990. doi: 10.1007/BF01013955.

[12] M. Born and R. Oppenheimer. Zur Quantentheorie der Molekeln. *Annalen der Physik*, 389: 457–484, 1927. English translation: On the quantum theory of molecules.

[13] J. Cardy. *Scaling and Renormalization in Statistical Physics*. Cambridge Lecture Notes in Physics. Cambridge University Press, Cambridge, 1996.

[14] Taco S. Cohen, Maurice Weiler, Berkay Kicanaoglu, and Max Welling. Gauge equivariant convolutional networks and the icosahedral CNN. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2019. URL https://proceedings.mlr.press/v97/cohen19d.html.

[15] T.M. Cover and J.A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, 2nd edition, 2006.

[16] M. Creutz. *Quarks, Gluons and Lattices*. Cambridge University Press, Cambridge, 1983. ISBN 978-0521244053.

[17] Michael Creutz. Gauge fixing, the transfer matrix, and confinement on a lattice. *Physical Review D*, 15(4):1128–1136, 1977. doi: 10.1103/PhysRevD.15.1128.

[18] Gavin E. Crooks. Measuring thermodynamic length. *Physical Review Letters*, 99(10):100602, 2007. doi: 10.1103/PhysRevLett.99.100602.

[19] James P. Crutchfield and David P. Feldman. Regularities unseen, randomness observed: Levels of entropy convergence. *Chaos*, 13(1):25–54, 2003. doi: 10.1063/1.1530990.

[20] I. Csiszár. Information-type measures of difference of probability distributions and indirect observations. *Studia Scientiarum Mathematicarum Hungarica*, 2:299–318, 1967.

[21] John M. Danskin. *The Theory of Max-Min and its Application to Weapons Allocation Problems*, volume 5 of *Ökonometrie und Unternehmensforschung / Econometrics and Operations Research*. Springer-Verlag, Berlin, Heidelberg, 1967. doi: 10.1007/978-3-642-46092-0. Chapter III, "The directional derivative", pp. 19–32.

[22] A.P. Dempster, N.M. Laird, and D.B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society Series B*, 39:1–38, 1977.

[23] Yuntian Deng, Yoon Kim, Justin Chiu, Demi Guo, and Alexander M. Rush. Latent alignment and variational attention. In *Advances in Neural Information Processing Systems*, volume 31, 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/b691334ccf10d4ab144d672f7783c8a3-Abstract.html.

[24] R. L. Dobrushin. Central limit theorem for nonstationary Markov chains. I. *Theory of Probability and Its Applications*, 1(1):65–80, 1956. doi: 10.1137/1101006.

[25] R. L. Dobrushin. The description of a random field by means of conditional probabilities and conditions of its regularity. *Theory of Probability and Its Applications*, 13(2):197–224, 1968. doi: 10.1137/1113026.

[26] Florian Dörfler and Francesco Bullo. Kron reduction of graphs with applications to electrical networks. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 60(1):150–163, 2013. doi: 10.1109/TCSI.2012.2215780.

[27] Freeman J. Dyson. Existence of a phase-transition in a one-dimensional Ising ferromagnet. *Communications in Mathematical Physics*, 12(2):91–107, 1969.

[28] S. Elitzur. Impossibility of spontaneously breaking local symmetries. *Physical Review D*, 12 (12):3978–3982, 1975.

[29] L. D. Faddeev and V. N. Popov. Feynman diagrams for the Yang–Mills field. *Physics Letters B*, 25(1):29–30, 1967. doi: 10.1016/0370-2693(67)90067-6.

[30] N. Fenichel. Geometric singular perturbation theory for ordinary differential equations. *Journal of Differential Equations*, 31(1):53–98, 1979.

[31] Eduardo Fradkin and Stephen H. Shenker. Phase diagrams of lattice gauge theories with Higgs fields. *Physical Review D*, 19(12):3682–3697, 1979. doi: 10.1103/PhysRevD.19.3682.

[32] Elena Garuccio, Margherita Lalli, and Diego Garlaschelli. Multiscale network renormalization: scale-invariance without geometry. *Physical Review Research*, 5(4):043101, 2023. doi: 10.1103/PhysRevResearch.5.043101.

[33] Hans-Otto Georgii. *Gibbs Measures and Phase Transitions*, volume 9 of *de Gruyter Studies in Mathematics*. Walter de Gruyter, Berlin, 1988.

[34] Michèle Giry. A categorical approach to probability theory. In *Categorical Aspects of Topology and Analysis*, volume 915 of *Lecture Notes in Mathematics*, pages 68–85. Springer, Berlin, Heidelberg, 1982. doi: 10.1007/BFb0092872.

[35] Nigel Goldenfeld. *Lectures on phase transitions and the renormalization group*. Addison-Wesley, 1992.

[36] Te Sun Han. Nonnegative entropy measures of multivariate symmetric correlations. *Information and Control*, 36(2):133–156, 1978. doi: 10.1016/S0019-9958(78)90275-9.

[37] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019. doi: 10.1007/s41468-019-00038-7.

[38] Matthew D. Hoffman and Matthew J. Johnson. ELBO surgery: Yet another way to carve up the variational evidence lower bound. In *Workshop in Advances in Approximate Bayesian Inference, 30th Conference on Neural Information Processing Systems (NIPS 2016)*, Barcelona, Spain, 2016. Checked against the author PDF; the average-KL decomposition is Eqs. (11)–(12).

[39] R.A. Horn and C.R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.

[40] Robert B. Israel. Banach algebras and Kadanoff transformations. In J. Fritz, J. L. Lebowitz, and D. Szász, editors, *Random Fields (Esztergom, 1979), Vol. II*, volume 27 of *Colloquia Mathematica Societatis János Bolyai*, pages 593–608. North-Holland, Amsterdam, 1981.

[41] Edwin T Jaynes. *Probability theory: The logic of science*. Cambridge University Press, 2003.

[42] G. Jona-Lasinio. Renormalization group and probability theory. *Physics Reports*, 352(4–6): 439–458, 2001. doi: 10.1016/S0370-1573(01)00042-4.

[43] Leo P. Kadanoff. Scaling laws for Ising models near t_c. *Physics Physique Fizika*, 2(6):263–272, 1966.

[44] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021.

[45] Hermann Karcher. Riemannian center of mass and mollifier smoothing. *Communications on Pure and Applied Mathematics*, 30(5):509–541, 1977. doi: 10.1002/cpa.3160300502.

[46] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Undergraduate Texts in Mathematics. Springer, New York, 1976. Reprint of the 1960 Van Nostrand edition.

[47] O. K. Kozlov. Gibbs description of a system of random variables. *Problems of Information Transmission*, 10(3):258–265, 1974. Translated from Problemy Peredachi Informatsii 10(3):94–103, 1974.

[48] S. Kullback and R.A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*, 22:79–86, 1951.

[49] S.L. Lauritzen. *Graphical Models*. Oxford Statistical Science Series. Oxford University Press, 1996.

[50] Oren E. Livne and Achi Brandt. Lean algebraic multigrid (LAMG): Fast graph Laplacian linear solver. *SIAM Journal on Scientific Computing*, 34(4):B499–B522, 2012. doi: 10.1137/110843563.

[51] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society Series B*, 44(2):226–233, 1982. doi: 10.1111/j.2517-6161.1982.tb01203.x.

[52] Marc Mezard, Giorgio Parisi, and Miguel Angel Virasoro. *Spin Glass Theory and Beyond: An Introduction to the Replica Method and Its Applications*, volume 9 of *World Scientific Lecture Notes in Physics*. World Scientific, Singapore, 1987. ISBN 978-9971-50-116-7. doi: 10.1142/0271.

[53] Dov Monderer and Lloyd S. Shapley. Potential games. *Games and Economic Behavior*, 14(1):124–143, 1996. doi: 10.1006/game.1996.0044.

[54] Sadao Nakajima. On quantum theory of transport phenomena. *Progress of Theoretical Physics*, 20(6):948–959, 1958. doi: 10.1143/PTP.20.948.

[55] Radford M. Neal and Geoffrey E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In Michael I. Jordan, editor, *Learning in Graphical Models*, volume 89 of *NATO ASI Series*, pages 355–368. Springer, Dordrecht, 1998. doi: 10.1007/978-94-011-5014-9_12. URL https://link.springer.com/chapter/10.1007/978-94-011-5014-9_12.

[56] Roger B. Nelsen. *An Introduction to Copulas*. Springer Series in Statistics. Springer, 2 edition, 2006.

[57] Yvan Notay. An aggregation-based algebraic multigrid method. *Electronic Transactions on Numerical Analysis*, 37:123–146, 2010.

[58] V. I. Oseledets. A multiplicative ergodic theorem: Lyapunov characteristic numbers for dynamical systems. *Transactions of the Moscow Mathematical Society*, 19:197–231, 1968.

[59] K. Osterwalder and E. Seiler. Gauge field theories on a lattice. *Annals of Physics*, 110(2): 440–471, 1978. doi: 10.1016/0003-4916(78)90039-8.

[60] Dénes Petz. Sufficiency of channels over von Neumann algebras. *The Quarterly Journal of Mathematics*, 39(1):97–108, 1988. doi: 10.1093/qmath/39.1.97.

[61] Julianna Pinele, João E. Strapasson, and Sueli I. R. Costa. The Fisher–Rao distance between multivariate normal distributions: Special cases, bounds and applications. *Entropy*, 22(4):404, 2020. doi: 10.3390/e22040404.

[62] David Pollard. A note on insufficiency and the preservation of fisher information. In *From Probability to Statistics and Back: High-Dimensional Models and Processes*, volume 9 of *IMS Collections*, pages 266–275. Institute of Mathematical Statistics, Beachwood, OH, 2013. doi: 10.1214/12-IMSCOLL919. URL https://doi.org/10.1214/12-IMSCOLL919.

[63] Herbert Robbins. An empirical Bayes approach to statistics. In Jerzy Neyman, editor, *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, pages 157–163. University of California Press, Berkeley, 1956. Checked against the Project Euclid chapter record.

[64] Gareth O. Roberts and Sujit K. Sahu. Updating schemes, correlation structure, blocking and parameterization for the Gibbs sampler. *Journal of the Royal Statistical Society: Series B (Methodological)*, 59(2):291–317, 1997. doi: 10.1111/1467-9868.00070. Record checked on the Oxford Academic article page. Lemma 1 (Gaussian AR(1) form of the Gibbs chain) and Theorem 1 (rate equals the spectral radius of its mean-iteration matrix) checked through their statement in Papaspiliopoulos, Roberts and Zanella, arXiv:1803.09460, Sec. 5.3; the primary full text was not accessed.

[65] C.E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27: 379–423, 623–656, 1948.

[66] Amit Singer and Hau-Tieng Wu. Vector diffusion maps and the connection Laplacian. *Communications on Pure and Applied Mathematics*, 65(8):1067–1144, 2012. doi: 10.1002/cpa.21395.

[67] L. T. Skovgaard. A Riemannian geometry of the multivariate normal model. *Scandinavian Journal of Statistics*, 11(4):211–223, 1984.

[68] Alfred Tarski. *A Decision Method for Elementary Algebra and Geometry*. University of California Press, Berkeley and Los Angeles, 2nd, revised edition, 1951. doi: 10.1525/9780520348097. Prepared for publication with the assistance of J. C. C. McKinsey.

[69] A. N. Tikhonov. Systems of differential equations containing small parameters in the derivatives. *Matematicheskii Sbornik (N.S.)*, 31(73)(3):575–586, 1952. In Russian.

[70] Jakub Tomczak and Max Welling. VAE with a VampPrior. In Amos Storkey and Fernando Perez-Cruz, editors, *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 1214–1223. PMLR, 2018. URL https://proceedings.mlr.press/v84/tomczak18a.html. Checked against the PMLR page and PDF; the aggregated posterior as ELBO-optimal prior is Eqs. (7)–(8).

[71] Minh Hoang Trinh, Chuong Van Nguyen, Young-Hun Lim, and Hyo-Sung Ahn. Matrix-weighted consensus and its applications. *Automatica*, 89:415–419, 2018. doi: 10.1016/j.automatica.2017.12.024.

[72] Aernout C. D. van Enter, Roberto Fernández, and Alan D. Sokal. Regularity properties and pathologies of position-space renormalization-group transformations: Scope and limitations of Gibbsian theory. *Journal of Statistical Physics*, 72(5–6):879–1167, 1993. doi: 10.1007/BF01048183. Section, proposition, and corollary numbers cited from arXiv:hep-lat/9210032.

[73] Nikolai Nikolaevich Čencov. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1982. ISBN 978-0821830512. Translated from Russian by Lev J. Leifman; originally published as *Statisticheskie reshayushchie pravila i optimalnye resheniya*, Moscow, Nauka, 1972.

[74] Pablo Villegas, Tommaso Gili, Guido Caldarelli, and Andrea Gabrielli. Laplacian renormalization group for heterogeneous networks. *Nature Physics*, 19(3):445–450, 2023. doi: 10.1038/s41567-022-01866-8.

[75] M.J. Wainwright and M.I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning*, 1(1–2):1–305, 2008. doi: 10.1561/2200000001.

[76] Franz J. Wegner. Duality in generalized Ising models and phase transitions without local order parameters. *Journal of Mathematical Physics*, 12(10):2259–2272, 1971. doi: 10.1063/1.1665530.

[77] Christof Wetterich. Linear lattice gauge theory. *Nuclear Physics B*, 884:44–65, 2014. doi: 10.1016/j.nuclphysb.2014.04.002.

[78] John Archibald Wheeler. Information, physics, quantum: The search for links. In Wojciech H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 3–28. Addison-Wesley, Redwood City, CA, 1990.

[79] Kenneth G. Wilson and John Kogut. The renormalization group and the ϵ expansion. *Physics Reports*, 12(2):75–199, 1974.

[80] Kenneth G Wilson and John Kogut. The renormalization group and the ϵ expansion. *Physics Reports*, 12(2):75–199, 1975.

[81] K.G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974.

[82] K.G. Wilson and J. Kogut. The renormalization group and the ϵ expansion. *Physics Reports*, 12:75–199, 1974.

[83] J. Worrall. Structural realism: The best of both worlds? *Dialectica*, 43(1-2):99–124, 1989.

[84] C. F. Jeff Wu. On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11(1):95–103, 1983.

[85] Jonathan S. Yedidia, William T. Freeman, and Yair Weiss. Constructing free-energy approximations and generalized belief propagation algorithms. *IEEE Transactions on Information Theory*, 51(7):2282–2312, 2005. doi: $10.1109/\mathrm{TIT}.2005.850085$.

[86] Robert Zwanzig. Ensemble method in the theory of irreversibility. *The Journal of Chemical Physics*, 33(5):1338–1341, 1960. doi: $10.1063/1.1731409$.